HL7 FHIR R4 — Chapter 9: Clinical Module (sections 9 – 9.11.13)

Source: HL7 FHIR R4 (v4.0.1) specification, https://hl7.org/fhir/R4/ — text-only rendition for AI ingestion; page order follows toc.html.

9 Clinical Module

URL: https://hl7.org/fhir/R4/clinicalsummary-module.html
Work Group Patient Care | Standards Status: Informative

9.0 Clinical Module

9.0.1 Introduction

This Clinical Module focuses on the FHIR Resources that represent core clinical information for a patient. The information contained in these Resources are those frequently documented, created or retrieved by healthcare providers during the course of clinical care. Resources generated during the course of diagnostic studies can be found in the Diagnostics Module, whereas the Resources related to medication ordering and administration process can be found in the Medications Module.

As an introduction to FHIR APIs and Resources, please see the Developer's Introduction or Clinical Introduction in the Overview section of the Foundation Module.

9.0.2 Index

The Clinical Module covers the following resources:

AllergyIntolerance Condition (Problem) Procedure FamilyMemberHistory | CarePlan Goal CareTeam ClinicalImpression | AdverseEvent DetectedIssue RiskAssessment

FHIR Resources have a low, moderate or high levels of complexity with respect to the number of primary and child elements as well as the number of referenced Resources, found in this module and others. To better understand the relationships between Resources, we recommend beginning with the lower complexity, core Resources such as Patient, Condition, and FamilyMemberHistory before addressing a high complexity Resource such as CarePlan.

9.0.3 Security and Privacy

The clinical resources often represent patient-related data, and as such are susceptible to data breaching. Necessary privacy and security provision must be in place for searching and fetching this information. For more general considerations, see the Security and Privacy module.

9.0.4 Common use Cases

• Documenting a patient's condition - The Condition Resource is used extensively throughout FHIR Resources to associate information and activities with specific conditions. The Condition Resource is broadly defined to include problems, diagnoses and health concerns.

• Retrieving the patient's problems

• Documenting and retrieving the patient's allergies - The AllergyIntolerance Resource is used to represent the patient's allergy or intolerance to a substance. There is vibrant debate within clinical community regarding what is appropriate to document as an allergy or intolerance. These terms are used both formally by the Allergy and Immunology community as well as informally by patients leading to confusion. Readers are referred to the Resource definition for the Scope and Usage of this resource. The AllergyIntolerance Resource also supports the documenting of the absence of an allergy.

• Family History - The FamilyMemberHistory Resource can be used to document known conditions of family members and support the creation of pedigrees.

• Care Plans - The CarePlan resource supports a problem based care plan with references to other Resources including CareTeam, Condition, Goal, and activities such as ServiceRequest

9.0.5 Developmental Roadmap

Over the next 18 months, we will continue to advance the resources through the Maturity Levels through the process of development and testing of the Resources. We anticipate more widespread implementation of core Resources such as Condition. Complex Resources such as CarePlan are dependent on the maturation of its referred Resources and are expected to mature more gradually. The clinical community will need to develop use cases to test and further mature the ServiceRequest Resource at opportunities such as the Clinicians on FHIR sessions at the HL7 Working Group Meetings.

9.1 Resource AllergyIntolerance - Content

URL: https://hl7.org/fhir/R4/allergyintolerance.html

Risk of harmful or undesirable, physiological response which is unique to an individual and associated with exposure to a substance.

9.1.1 Scope and Usage

A record of a clinical assessment of an allergy or intolerance; a propensity, or a potential risk to an individual, to have an adverse reaction on future exposure to the specified substance, or class of substance.

Where a propensity is identified, to record information or evidence about a reaction event that is characterized by any harmful or undesirable physiological response that is specific to the individual and triggered by exposure of an individual to the identified substance or class of substance.

Substances include, but are not limited to: a therapeutic substance administered correctly at an appropriate dosage for the individual; food; material derived from plants or animals; or venom from insect stings.

Note for Reviewers

Presently open issues for this resource:

• This resource represents a condition of susceptibility to a substance, with a list of supporting events and/or symptoms, and has no direct relationship to an event reporting framework; this will be re-assessed when adverse event reporting resource(s) and/or profiles are added to FHIR

• Other HL7 models and the openEHR archetype have "exposure date" but this is not found in any surveyed systems, so this is left as an extension (more appropriate for adverse event reporting)

9.1.2 Boundaries and Relationships

This resource is used to provide a single place within the health record to document a range of clinical statements about adverse reactions to substances/products, including:

• record a clinical assessment of the individual's propensity to a potential future reaction upon re-exposure

• record cumulative information about the reaction to each exposure, including 'no reaction' if appropriate

Use to record information about the positive presence of the risk of an adverse reaction:

• to support direct clinical care of an individual

• as part of a managed adverse reaction or allergy/intolerance list

• to support exchange of information about the propensity and events related to adverse reactions

• to inform adverse reaction reporting

• to assist computerized knowledge-based activities such as clinical decision support and alerts

Use to record information about adverse reactions to a broad range of substances, including: biological & blood products; incipients and excipients in medicinal preparations; foods; metal salts; and organic chemical compounds.

Adverse reactions may be:

• an allergy (typically type I hypersensitivity, plus other "allergy-like" reactions, including pseudoallergy)

• an intolerance (typically non-immune adverse reactions that are not determined or perceived to be allergic or "allergy-like", and are to some degree idiosyncratic and/or individually specific [i.e. are not a reaction that is expected to occur with most or all patients given similar circumstances])

In clinical practice distinguishing between allergy and intolerance is difficult and might not be practical. Often the term "allergy" is used rather generically and may overlap with "intolerance", and the boundaries between these concepts might not be well-defined or understood. As noted above, the term "intolerance" should generally be applied to a propensity for adverse reactions which is either determined (to the extent that is possible) or perceived to not be allergic or "allergy-like". If it is not possible to determine whether a particular propensity condition is an allergy or an intolerance, then the type element should be omitted from the resource. Identification of the type of reaction is not a proxy for seriousness or risk of harm to the patient, which is better expressed in the documentation of the clinical manifestation and the assessment of criticality.

The sensitivity in the case of either an allergy or intolerance is unique to the individual, and is distinguished from those reactions that are a property of the circumstance, such as toxicity of a food or drug, overdose, drug-drug, drug-food, or drug-disease interaction (which are reactions that would be expected to occur for any individual given the same circumstances).

The risk of an adverse reaction event or manifestation should not be recorded without identifying a proposed causative substance (including pharmaceutical products) or class of substance. If there is uncertainty that a specific substance is the cause, this uncertainty can be recorded using the 'verificationStatus' data element. If there are multiple possible substances that may have caused a reaction/manifestation, each substance should be recorded using a separate instance of this resource with the 'verificationStatus' set to an initial state of 'unconfirmed' so that adverse reaction checking can be supported in clinical systems. If a substance, agent or class is later proven not to be the cause for a given reaction then the 'verificationStatus' can be modified to 'refuted'.

This resource has been designed to allow recording of information about a specific substance (e.g., amoxicillin, oysters, or bee sting venom) or pharmaceutical product or, alternatively, a class of substance (e.g., penicillins). If a class of substance is recorded, then identification of the exact substance can be recorded on a per exposure basis.

The scope of this FHIR resource has deliberately focused on identifying a pragmatic data set that is used in most clinical systems or will be suitable for most common clinical scenarios; extensions can be used to add additional detail if required. Examples of clinical situations where the extension may be required include: a detailed allergist/immunologist assessment, for reporting to regulatory bodies or use in a clinical trial.

The act of recording any adverse reaction in a health record involves the clinical assessment that a potential hazard exists for an individual if they are exposed to the same substance/product/class in the future - that is, a relative contraindication - and, in the absence of additional information indicating a higher level of potential risk, the default 'criticality' value should be set to 'Low Risk'. If a clinician considers that it is not safe for the individual to be deliberately re-exposed to the substance/product again, for example, following a manifestation of a life-threatening anaphylaxis, then the 'criticality' data element should be amended to 'High Risk'.

A formal adverse event report to regulatory bodies is a document that will contain a broad range of information in addition to the specific details about the adverse reaction. The report could utilize parts of this resource plus include additional data as required per jurisdiction.

An adverse reaction or allergy/intolerance list is a record of all identified propensities for an adverse reaction for the individual upon future exposure to the substance/product or class, plus provides potential access to the evidence provided by details about each reaction event, such as manifestation.

Valuable first-level information that could be presented to the clinician when they need to assess propensity for future reactions are:

• statements about previous clinical manifestations following exposure

• source of the information/reporter

• the 'criticality' flag

Second-level information can be drawn from each exposure event and links to additional detailed information such as history, examination and diagnoses stored elsewhere in the record, if it is available.

AllergyIntolerance and RiskAssessment

AllergyIntolerance describes a specific type of risk - propensity to reaction to a substance/product while RiskAssessment describes general risks to a subject, not generally based on a reaction.

AllergyIntolerance and Immunization.reaction

Immunization.reaction may be an indication of an allergy or intolerance. If this is deemed to be the case, a separate AllergyIntolerance record should be created to indicate it, as most systems will not query against past immunization.reactions.

Misuse

• The allergy/intolerance list exists as a patient safety tool to inform decision support around ordering of medications and nutrition and to guide clinical treatment. Other reactions triggered by physical stimuli -- light, heat, cold, pressure, vibration, which may mimic allergic or intolerance reactions, should be recorded as Condition on the problem list, not using AllergyIntolerance.

• Not to be used to record adverse events, including failures of clinical process, interventions or products. For example, abnormal use or mistakes/errors made in maladministration of an agent or substance, incorrect dosage, mislabeling, harm or injury caused by an intervention or procedure, overdose/poisoning, etc.

• Not to be used as a proxy for an adverse event report. See above for how it may be used as one component of an adverse event report.

• Not to be used for recording alerts. Alerts are handled using Flag or - where event-specific, DetectedIssue.

• Not to be used for recording failed therapy.

Trial-Use Note: Requests have been received (GF#10369) to add codes to the extension-allergyintolerance-certainty value set (reaction-event-certainty), which is a required binding. The requested codes include "unknown", "ruled out" and "possible". The Patient Care WG has voted to add "unknown" to the value set, but recommends that if other terms (including "ruled out" and "possible") are desired for use in a specific setting, an extension or profile should be used. During the STU period feedback is solicited regarding (1) the need and desirability of adding the code "unknown" to the value set, vs. omitting the element if the certainty is not known; and (2) whether or not additional codes besides "unknown" should be added to the value set for the core specification, or whether additional codes, if needed, should be added in an extension/profile.

Feedback is welcome here .

This resource is referenced by AdverseEvent, ClinicalImpression, FamilyMemberHistory and NutritionOrder

9.1.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | ITU | | DomainResource | Allergy or Intolerance (generally: Risk of adverse reaction to a substance) + Rule: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. + Rule: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External ids for this item
clinicalStatus | ?!ΣI | 0..1 | CodeableConcept | active | inactive | resolved AllergyIntolerance Clinical Status Codes (Required)
verificationStatus | ?!ΣI | 0..1 | CodeableConcept | unconfirmed | confirmed | refuted | entered-in-error AllergyIntolerance Verification Status Codes (Required)
type | Σ | 0..1 | code | allergy | intolerance - Underlying mechanism (if known) AllergyIntoleranceType (Required)
category | Σ | 0..* | code | food | medication | environment | biologic AllergyIntoleranceCategory (Required)
criticality | Σ | 0..1 | code | low | high | unable-to-assess AllergyIntoleranceCriticality (Required)
code | Σ | 0..1 | CodeableConcept | Code that identifies the allergy or intolerance AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
patient | Σ | 1..1 | Reference(Patient) | Who the sensitivity is for
encounter | | 0..1 | Reference(Encounter) | Encounter when the allergy or intolerance was asserted
onset[x] | | 0..1 | | When allergy or intolerance was identified
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
recordedDate | | 0..1 | dateTime | Date first version of the resource instance was recorded
recorder | | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Who recorded the sensitivity
asserter | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Source of the information about the allergy
lastOccurrence | | 0..1 | dateTime | Date(/time) of last known occurrence of a reaction
note | | 0..* | Annotation | Additional text not captured in other fields
reaction | | 0..* | BackboneElement | Adverse Reaction Events linked to exposure to substance
substance | | 0..1 | CodeableConcept | Specific substance or pharmaceutical product considered to be responsible for event Substance Code (Example)
manifestation | | 1..* | CodeableConcept | Clinical symptoms/signs associated with the Event SNOMED CT Clinical Findings (Example)
description | | 0..1 | string | Description of the event as a whole
onset | | 0..1 | dateTime | Date(/time) when manifestations showed
severity | | 0..1 | code | mild | moderate | severe (of event as a whole) AllergyIntoleranceSeverity (Required)
exposureRoute | | 0..1 | CodeableConcept | How the subject was exposed to the substance SNOMED CT Route Codes (Example)
note | | 0..* | Annotation | Text about event not captured in other fields
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
 <clinicalStatus><!--  0..1 CodeableConcept active | inactive | resolved --></clinicalStatus>
 <verificationStatus><!--  0..1 CodeableConcept unconfirmed | confirmed | refuted | entered-in-error
    --></verificationStatus>
 <type value="[code]"/><!-- 0..1 allergy | intolerance - Underlying mechanism (if known) -->
 <category value="[code]"/><!-- 0..* food | medication | environment | biologic -->
 <criticality value="[code]"/><!-- 0..1 low | high | unable-to-assess -->
 <code><!-- 0..1 CodeableConcept Code that identifies the allergy or intolerance --></code>
 <patient><!-- 1..1 Reference(Patient) Who the sensitivity is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter when the allergy or intolerance was asserted --></encounter>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string When allergy or intolerance was identified --></onset[x]>
 <recordedDate value="[dateTime]"/><!-- 0..1 Date first version of the resource instance was recorded -->
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Who recorded the sensitivity --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Source of the information about the allergy --></asserter>
 <lastOccurrence value="[dateTime]"/><!-- 0..1 Date(/time) of last known occurrence of a reaction -->
 <note><!-- 0..* Annotation Additional text not captured in other fields --></note>
 <reaction>  <!-- 0..* Adverse Reaction Events linked to exposure to substance -->
  <substance><!-- 0..1 CodeableConcept Specific substance or pharmaceutical product considered to be responsible for
    event --></substance>
  <manifestation><!-- 1..* CodeableConcept Clinical symptoms/signs associated with the Event --></manifestation>
  <description value="[string]"/><!-- 0..1 Description of the event as a whole -->
```

```
  <onset value="[dateTime]"/><!-- 0..1 Date(/time) when manifestations showed -->
  <severity value="[code]"/><!-- 0..1 mild | moderate | severe (of event as a whole) -->
  <exposureRoute><!-- 0..1 CodeableConcept How the subject was exposed to the substance --></exposureRoute>
  <note><!-- 0..* Annotation Text about event not captured in other fields --></note>
 </reaction>
</AllergyIntolerance>
```

JSON Template

```
{
  "resourceType" : "AllergyIntolerance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "clinicalStatus" : { CodeableConcept }, // C? active | inactive | resolved
  "verificationStatus" : { CodeableConcept }, // C? unconfirmed | confirmed | refuted | entered-in-error
  "type" : "<code>", // allergy | intolerance - Underlying mechanism (if known)
  "category" : ["<code>"], // food | medication | environment | biologic
  "criticality" : "<code>", // low | high | unable-to-assess
  "code" : { CodeableConcept }, // Code that identifies the allergy or intolerance
  "patient" : { Reference(Patient) }, // R!  Who the sensitivity is for
  "encounter" : { Reference(Encounter) }, // Encounter when the allergy or intolerance was asserted
  // onset[x]: When allergy or intolerance was identified. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  "recordedDate" : "<dateTime>", // Date first version of the resource instance was recorded
  "recorder" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Who recorded the sensitivity
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Source of the information about the allergy
  "lastOccurrence" : "<dateTime>", // Date(/time) of last known occurrence of a reaction
  "note" : [{ Annotation }], // Additional text not captured in other fields
  "reaction" : [{ // Adverse Reaction Events linked to exposure to substance
    "substance" : { CodeableConcept }, // Specific substance or pharmaceutical product considered to be responsible for
    event
    "manifestation" : [{ CodeableConcept }], // R!  Clinical symptoms/signs associated with the Event
    "description" : "<string>", // Description of the event as a whole
    "onset" : "<dateTime>", // Date(/time) when manifestations showed
    "severity" : "<code>", // mild | moderate | severe (of event as a whole)
    "exposureRoute" : { CodeableConcept }, // How the subject was exposed to the substance
    "note" : [{ Annotation }] // Text about event not captured in other fields
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AllergyIntolerance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AllergyIntolerance.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:AllergyIntolerance.clinicalStatus [ CodeableConcept ]; # 0..1 active | inactive | resolved
  fhir:AllergyIntolerance.verificationStatus [ CodeableConcept ]; # 0..1 unconfirmed | confirmed | refuted | entered-in-
    error
  fhir:AllergyIntolerance.type [ code ]; # 0..1 allergy | intolerance - Underlying mechanism (if known)
  fhir:AllergyIntolerance.category [ code ], ... ; # 0..* food | medication | environment | biologic
  fhir:AllergyIntolerance.criticality [ code ]; # 0..1 low | high | unable-to-assess
  fhir:AllergyIntolerance.code [ CodeableConcept ]; # 0..1 Code that identifies the allergy or intolerance
  fhir:AllergyIntolerance.patient [ Reference(Patient) ]; # 1..1 Who the sensitivity is for
  fhir:AllergyIntolerance.encounter [ Reference(Encounter) ]; # 0..1 Encounter when the allergy or intolerance was
    asserted
  # AllergyIntolerance.onset[x] : 0..1 When allergy or intolerance was identified. One of these 5
    fhir:AllergyIntolerance.onsetDateTime [ dateTime ]
    fhir:AllergyIntolerance.onsetAge [ Age ]
    fhir:AllergyIntolerance.onsetPeriod [ Period ]
    fhir:AllergyIntolerance.onsetRange [ Range ]
    fhir:AllergyIntolerance.onsetString [ string ]
  fhir:AllergyIntolerance.recordedDate [ dateTime ]; # 0..1 Date first version of the resource instance was recorded
  fhir:AllergyIntolerance.recorder [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Who
    recorded the sensitivity
  fhir:AllergyIntolerance.asserter [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Source of
    the information about the allergy
  fhir:AllergyIntolerance.lastOccurrence [ dateTime ]; # 0..1 Date(/time) of last known occurrence of a reaction
  fhir:AllergyIntolerance.note [ Annotation ], ... ; # 0..* Additional text not captured in other fields
  fhir:AllergyIntolerance.reaction [ # 0..* Adverse Reaction Events linked to exposure to substance
    fhir:AllergyIntolerance.reaction.substance [ CodeableConcept ]; # 0..1 Specific substance or pharmaceutical product
    considered to be responsible for event
    fhir:AllergyIntolerance.reaction.manifestation [ CodeableConcept ], ... ; # 1..* Clinical symptoms/signs associated
    with the Event
    fhir:AllergyIntolerance.reaction.description [ string ]; # 0..1 Description of the event as a whole
    fhir:AllergyIntolerance.reaction.onset [ dateTime ]; # 0..1 Date(/time) when manifestations showed
    fhir:AllergyIntolerance.reaction.severity [ code ]; # 0..1 mild | moderate | severe (of event as a whole)
    fhir:AllergyIntolerance.reaction.exposureRoute [ CodeableConcept ]; # 0..1 How the subject was exposed to the
    substance
```

```
    fhir:AllergyIntolerance.reaction.note [ Annotation ], ... ; # 0..* Text about event not captured in other fields
  ], ...;
]
```

Changes since R3

AllergyIntolerance
AllergyIntolerance.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-clinical-status to http://hl7.org/fhir/ValueSet/allergyintolerance-clinical|4.0.1
AllergyIntolerance.verificationStatus | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-verification-status to http://hl7.org/fhir/ValueSet/allergyintolerance-verification|4.0.1
AllergyIntolerance.type | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-type to http://hl7.org/fhir/ValueSet/allergy-intolerance-type|4.0.1
AllergyIntolerance.category | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-category to http://hl7.org/fhir/ValueSet/allergy-intolerance-category|4.0.1
AllergyIntolerance.criticality | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality to http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality|4.0.1
AllergyIntolerance.encounter | Added Element
AllergyIntolerance.recordedDate | Renamed from assertedDate to recordedDate
AllergyIntolerance.recorder | Type Reference: Added Target Types PractitionerRole, RelatedPerson
AllergyIntolerance.asserter | Type Reference: Added Target Type PractitionerRole
AllergyIntolerance.reaction.severity | Change value set from http://hl7.org/fhir/ValueSet/reaction-event-severity to http://hl7.org/fhir/ValueSet/reaction-event-severity|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
AllergyIntolerance | ITU | | DomainResource | Allergy or Intolerance (generally: Risk of adverse reaction to a substance) + Rule: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. + Rule: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External ids for this item
clinicalStatus | ?!ΣI | 0..1 | CodeableConcept | active | inactive | resolved AllergyIntolerance Clinical Status Codes (Required)
verificationStatus | ?!ΣI | 0..1 | CodeableConcept | unconfirmed | confirmed | refuted | entered-in-error AllergyIntolerance Verification Status Codes (Required)
type | Σ | 0..1 | code | allergy | intolerance - Underlying mechanism (if known) AllergyIntoleranceType (Required)
category | Σ | 0..* | code | food | medication | environment | biologic AllergyIntoleranceCategory (Required)
criticality | Σ | 0..1 | code | low | high | unable-to-assess AllergyIntoleranceCriticality (Required)
code | Σ | 0..1 | CodeableConcept | Code that identifies the allergy or intolerance AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
patient | Σ | 1..1 | Reference(Patient) | Who the sensitivity is for
encounter | | 0..1 | Reference(Encounter) | Encounter when the allergy or intolerance was asserted
onset[x] | | 0..1 | | When allergy or intolerance was identified
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
recordedDate | | 0..1 | dateTime | Date first version of the resource instance was recorded
recorder | | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Who recorded the sensitivity
asserter | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Source of the information about the allergy
lastOccurrence | | 0..1 | dateTime | Date(/time) of last known occurrence of a reaction
note | | 0..* | Annotation | Additional text not captured in other fields
reaction | | 0..* | BackboneElement | Adverse Reaction Events linked to exposure to substance
substance | | 0..1 | CodeableConcept | Specific substance or pharmaceutical product considered to be responsible for event Substance Code (Example)
manifestation | | 1..* | CodeableConcept | Clinical symptoms/signs associated with the Event SNOMED CT Clinical Findings (Example)
description | | 0..1 | string | Description of the event as a whole
onset | | 0..1 | dateTime | Date(/time) when manifestations showed
severity | | 0..1 | code | mild | moderate | severe (of event as a whole) AllergyIntoleranceSeverity (Required)
exposureRoute | | 0..1 | CodeableConcept | How the subject was exposed to the substance SNOMED CT Route Codes (Example)
note | | 0..* | Annotation | Text about event not captured in other fields
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External ids for this item --></identifier>
```

```
 <clinicalStatus><!--  0..1 CodeableConcept active | inactive | resolved --></clinicalStatus>
 <verificationStatus><!--  0..1 CodeableConcept unconfirmed | confirmed | refuted | entered-in-error
    --></verificationStatus>
 <type value="[code]"/><!-- 0..1 allergy | intolerance - Underlying mechanism (if known) -->
 <category value="[code]"/><!-- 0..* food | medication | environment | biologic -->
 <criticality value="[code]"/><!-- 0..1 low | high | unable-to-assess -->
 <code><!-- 0..1 CodeableConcept Code that identifies the allergy or intolerance --></code>
 <patient><!-- 1..1 Reference(Patient) Who the sensitivity is for --></patient>
 <encounter><!-- 0..1 Reference(Encounter) Encounter when the allergy or intolerance was asserted --></encounter>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string When allergy or intolerance was identified --></onset[x]>
 <recordedDate value="[dateTime]"/><!-- 0..1 Date first version of the resource instance was recorded -->
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Who recorded the sensitivity --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Source of the information about the allergy --></asserter>
 <lastOccurrence value="[dateTime]"/><!-- 0..1 Date(/time) of last known occurrence of a reaction -->
 <note><!-- 0..* Annotation Additional text not captured in other fields --></note>
 <reaction>  <!-- 0..* Adverse Reaction Events linked to exposure to substance -->
  <substance><!-- 0..1 CodeableConcept Specific substance or pharmaceutical product considered to be responsible for
    event --></substance>
  <manifestation><!-- 1..* CodeableConcept Clinical symptoms/signs associated with the Event --></manifestation>
  <description value="[string]"/><!-- 0..1 Description of the event as a whole -->
  <onset value="[dateTime]"/><!-- 0..1 Date(/time) when manifestations showed -->
  <severity value="[code]"/><!-- 0..1 mild | moderate | severe (of event as a whole) -->
  <exposureRoute><!-- 0..1 CodeableConcept How the subject was exposed to the substance --></exposureRoute>
  <note><!-- 0..* Annotation Text about event not captured in other fields --></note>
 </reaction>
</AllergyIntolerance>
```

JSON Template

```
{
  "resourceType" : "AllergyIntolerance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External ids for this item
  "clinicalStatus" : { CodeableConcept }, // C? active | inactive | resolved
  "verificationStatus" : { CodeableConcept }, // C? unconfirmed | confirmed | refuted | entered-in-error
  "type" : "<code>", // allergy | intolerance - Underlying mechanism (if known)
  "category" : ["<code>"], // food | medication | environment | biologic
  "criticality" : "<code>", // low | high | unable-to-assess
  "code" : { CodeableConcept }, // Code that identifies the allergy or intolerance
  "patient" : { Reference(Patient) }, // R!  Who the sensitivity is for
  "encounter" : { Reference(Encounter) }, // Encounter when the allergy or intolerance was asserted
  // onset[x]: When allergy or intolerance was identified. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  "recordedDate" : "<dateTime>", // Date first version of the resource instance was recorded
  "recorder" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Who recorded the sensitivity
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Source of the information about the allergy
  "lastOccurrence" : "<dateTime>", // Date(/time) of last known occurrence of a reaction
  "note" : [{ Annotation }], // Additional text not captured in other fields
  "reaction" : [{ // Adverse Reaction Events linked to exposure to substance
    "substance" : { CodeableConcept }, // Specific substance or pharmaceutical product considered to be responsible for
    event
    "manifestation" : [{ CodeableConcept }], // R!  Clinical symptoms/signs associated with the Event
    "description" : "<string>", // Description of the event as a whole
    "onset" : "<dateTime>", // Date(/time) when manifestations showed
    "severity" : "<code>", // mild | moderate | severe (of event as a whole)
    "exposureRoute" : { CodeableConcept }, // How the subject was exposed to the substance
    "note" : [{ Annotation }] // Text about event not captured in other fields
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AllergyIntolerance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AllergyIntolerance.identifier [ Identifier ], ... ; # 0..* External ids for this item
  fhir:AllergyIntolerance.clinicalStatus [ CodeableConcept ]; # 0..1 active | inactive | resolved
  fhir:AllergyIntolerance.verificationStatus [ CodeableConcept ]; # 0..1 unconfirmed | confirmed | refuted | entered-in-
    error
  fhir:AllergyIntolerance.type [ code ]; # 0..1 allergy | intolerance - Underlying mechanism (if known)
  fhir:AllergyIntolerance.category [ code ], ... ; # 0..* food | medication | environment | biologic
  fhir:AllergyIntolerance.criticality [ code ]; # 0..1 low | high | unable-to-assess
  fhir:AllergyIntolerance.code [ CodeableConcept ]; # 0..1 Code that identifies the allergy or intolerance
  fhir:AllergyIntolerance.patient [ Reference(Patient) ]; # 1..1 Who the sensitivity is for
  fhir:AllergyIntolerance.encounter [ Reference(Encounter) ]; # 0..1 Encounter when the allergy or intolerance was
    asserted
  # AllergyIntolerance.onset[x] : 0..1 When allergy or intolerance was identified. One of these 5
```

```
    fhir:AllergyIntolerance.onsetDateTime [ dateTime ]
    fhir:AllergyIntolerance.onsetAge [ Age ]
    fhir:AllergyIntolerance.onsetPeriod [ Period ]
    fhir:AllergyIntolerance.onsetRange [ Range ]
    fhir:AllergyIntolerance.onsetString [ string ]
  fhir:AllergyIntolerance.recordedDate [ dateTime ]; # 0..1 Date first version of the resource instance was recorded
  fhir:AllergyIntolerance.recorder [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Who
    recorded the sensitivity
  fhir:AllergyIntolerance.asserter [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Source of
    the information about the allergy
  fhir:AllergyIntolerance.lastOccurrence [ dateTime ]; # 0..1 Date(/time) of last known occurrence of a reaction
  fhir:AllergyIntolerance.note [ Annotation ], ... ; # 0..* Additional text not captured in other fields
  fhir:AllergyIntolerance.reaction [ # 0..* Adverse Reaction Events linked to exposure to substance
    fhir:AllergyIntolerance.reaction.substance [ CodeableConcept ]; # 0..1 Specific substance or pharmaceutical product
    considered to be responsible for event
    fhir:AllergyIntolerance.reaction.manifestation [ CodeableConcept ], ... ; # 1..* Clinical symptoms/signs associated
    with the Event
    fhir:AllergyIntolerance.reaction.description [ string ]; # 0..1 Description of the event as a whole
    fhir:AllergyIntolerance.reaction.onset [ dateTime ]; # 0..1 Date(/time) when manifestations showed
    fhir:AllergyIntolerance.reaction.severity [ code ]; # 0..1 mild | moderate | severe (of event as a whole)
    fhir:AllergyIntolerance.reaction.exposureRoute [ CodeableConcept ]; # 0..1 How the subject was exposed to the
    substance
    fhir:AllergyIntolerance.reaction.note [ Annotation ], ... ; # 0..* Text about event not captured in other fields
  ], ...;
]
```

Changes since Release 3

AllergyIntolerance

AllergyIntolerance.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-clinical-status to http://hl7.org/fhir/ValueSet/allergyintolerance-clinical|4.0.1

AllergyIntolerance.verificationStatus | Min Cardinality changed from 1 to 0 Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/allergy-verification-status to http://hl7.org/fhir/ValueSet/allergyintolerance-verification|4.0.1

AllergyIntolerance.type | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-type to http://hl7.org/fhir/ValueSet/allergy-intolerance-type|4.0.1

AllergyIntolerance.category | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-category to http://hl7.org/fhir/ValueSet/allergy-intolerance-category|4.0.1

AllergyIntolerance.criticality | Change value set from http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality to http://hl7.org/fhir/ValueSet/allergy-intolerance-criticality|4.0.1

AllergyIntolerance.encounter | Added Element

AllergyIntolerance.recordedDate | Renamed from assertedDate to recordedDate

AllergyIntolerance.recorder | Type Reference: Added Target Types PractitionerRole, RelatedPerson

AllergyIntolerance.asserter | Type Reference: Added Target Type PractitionerRole

AllergyIntolerance.reaction.severity | Change value set from http://hl7.org/fhir/ValueSet/reaction-event-severity to http://hl7.org/fhir/ValueSet/reaction-event-severity|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.1.3.1 Terminology Bindings

Path | Definition | Type | Reference

AllergyIntolerance.clinicalStatus | The clinical status of the allergy or intolerance. | Required | AllergyIntoleranceClinicalStatusCodes

AllergyIntolerance.verificationStatus | Assertion about certainty associated with a propensity, or potential risk, of a reaction to the identified substance. | Required | AllergyIntoleranceVerificationStatusCodes

AllergyIntolerance.type | Identification of the underlying physiological mechanism for a Reaction Risk. | Required | AllergyIntoleranceType

AllergyIntolerance.category | Category of an identified substance associated with allergies or intolerances. | Required | AllergyIntoleranceCategory

AllergyIntolerance.criticality | Estimate of the potential clinical harm, or seriousness, of a reaction to an identified substance. | Required | AllergyIntoleranceCriticality

AllergyIntolerance.code | Type of the substance/product, allergy or intolerance condition, or negation/exclusion codes for reporting no known allergies. | Example | AllergyIntoleranceSubstance/Product,ConditionAndNegationCodes

AllergyIntolerance.reaction.substance | Codes defining the type of the substance (including pharmaceutical products). | Example | SubstanceCode

AllergyIntolerance.reaction.manifestation | Clinical symptoms and/or signs that are observed or associated with an Adverse Reaction Event. | Example | SNOMEDCTClinicalFindings

AllergyIntolerance.reaction.severity | Clinical assessment of the severity of a reaction event as a whole, potentially considering multiple different manifestations. | Required | AllergyIntoleranceSeverity

AllergyIntolerance.reaction.exposureRoute | A coded concept describing the route or physiological path of administration of a therapeutic agent into or onto the body of a subject. | Example | SNOMEDCTRouteCodes

9.1.3.2 Constraints

id | Level | Location | Description | Expression

ait-1 | Rule | (base) | AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists()

ait-2 | Rule | (base) | AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()

9.1.3.3 Negated Allergies and Intolerances

It is important to differentiate between affirmatively stating that a patient has no known allergies versus either not including allergies in the record (for example an episodic document where the allergies are not considered relevant to the document); or asserting that allergies were not reviewed and are unknown.

Allergies with the verificationStatus "entered-in-error" indicates that the allergy or intolerance statement is entered by mistake and hence invalid.

Allergies with the verificationStatus "refuted" must be displayed to indicate that a reaction to a substance has been ruled out with the high level of clinical certainty (e.g. additional testing, re-challenging).

Prior to adding a new allergy/intolerance, a list of existing negated and refuted reactions should be reviewed and reconciled.

Allergies Not Reviewed, Not Asked

When a sending system does not have any information about allergies being reviewed or the statement is about allergies not being asked yet, then the List resource should be used to indicate the List.emptyReason.code="notasked".

Allergies Reviewed, None Identified

Systems may use the List.emptyReason when a statement is about the full scope of the list (i.e. the patient has no known allergies or intolerances of any type). However, it is generally preferred to use a code for "No known allergies" (e.g., SNOMED CT: 716186003 |No known allergy (situation)|), so that all allergy data will be available and queryable from AllergyIntolerance resource instances. Negated AllergyIntolerance instances are also typically used when the record is more fine-grained (e.g. no drug allergies, no food allergies, no nut allergies, etc.).

However, it is possible to include negation statements that apply at the level of the whole list and it is also possible to have separate lists for things like medication allergies vs. food allergies, where that is appropriate to the architecture. Also note that care should be used when adding new AllergyIntolerances to a list to ensure that any negation statements that are voided by the addition of a new record are removed from the list. E.g. If the list contains a "no known food allergies" record and you add an "intolerance to grape flavor" record, then be sure you remove the "no known food allergies" record.

The substanceExposureRisk extension is also available for use as a more completely structured and flexible alternative to the 'code' element for representing positive and negative allergy and intolerance statements (either the 'code' element or the substanceExposureRisk extension may be used, but not both).

Trial-Use Note: There are two primary ways of reporting "no known allergies" in the current specification: using the CodeableConcept, as described above, or using the List resource with emptyReason. The third available option is using the substanceExposureRisk extension. During the STU period, it is not recommended to use the List resource for "no known allergies" reporting purposes. The principal reason for this is to allow all allergy or intolerance data to be found and to be consistently queryable from the single location of the AllergyIntolerance resource.

Provide feedback here .

No Known Allergies, using List.emptyReason (discouraged)

```
  <List xmlns="http://hl7.org/fhir" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://hl7.org/fhir ..\..\schema\list.xsd">
 <id value="example-empty-allergy"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p> The patient is not aware of any allergies.</p>
    </div>
  </text>
<code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="52472-8"/>
      <display value="Allergies and Adverse Drug Reactions"/>
    </coding>
    <text value="Current Allergy List"/>
  </code>
  <source>
    <reference value="Patient/example"/>
  </source>
  <status value="current"/>
  <date value="2012-11-26T07:30:23+11:00"/>
  <mode value="snapshot"/>
  <emptyReason>
    <coding>
```

```
      <system value="http://hl7.org/fhir/special-values"/>
      <code value="nil-known"/>
      <display value="Nil Known"/>
    </coding>
    <text value="The patient is not aware of any allergies."/>
  </emptyReason>
</List>
```

No Known Food Allergies and Medication Allergy List

```
  <?xml version="1.0" encoding="UTF-8"?>
<List xmlns="http://hl7.org/fhir" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://hl7.org/fhir ..\..\schema\list.xsd">
  <id value="current-allergies"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Patient Peter Chalmers, DOB = Dec 25, 1974, MRN = 12345 (Acme Healthcare) has the following allergies</p>
      <ul>
        <li>No known food allergies</li>
        <li>Allergenic extract, penicillin (high)</li>
      </ul>
    </div>
  </text>
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="52472-8"/>
      <display value="Allergies and Adverse Drug Reactions"/>
    </coding>
    <text value="Current Allergy List"/>
  </code>
  <source>
    <reference value="Patient/example"/>
  </source>
  <status value="current"/>
  <date value="2015-07-14T23:10:23+11:00"/>
  <mode value="snapshot"/>
  <entry>
    <item>
      <reference value="AllergyIntolerance/nofoodallergies"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="AllergyIntolerance/penicillin"/>
    </item>
  </entry>
</List>
```

If a new allergy is discovered, the negated allergy record must be updated with the "refuted" verificationStatus - to ensure that systems referring to this record are aware that this is no longer true.

9.1.3.4 Use of AllergyIntolerance.criticality

Systems that only support one notion will have to determine whether what they're capturing is criticality or severity and map to the appropriate place. Criticality refers to the likelihood the allergy/intolerance could result in significant harm. Severity refers to the degree of manifestation of the reaction symptom. Moderate breathing difficulty would have high criticality while a severe rash would have low criticality. Severity is specific to a particular reaction occurrence.

For systems that only track generic reaction characteristics rather than a specific reaction will provide guidance to use the "reaction" structure and simply provide no date.

9.1.3.5 References

• Adverse Reaction, draft archetype, National eHealth Transition Authority [Internet]. NEHTA Clinical Knowledge Manager. Authored: 08 Nov 2010. Available at: http://dcm.nehta.org.au/ckm/OKM.html#showarchetype_1013.1.868_7 (accessed Jan 16, 2012).

• Allergy and Intolerance Domain Analysis Model, Release 1, HL7 [Internet]. Publication pending, expected August 2014; Available at http://www.hl7.org/implement/standards/product_brief.cfm?product_id=308 (accessed 06 July 2014).

• Allergy, clinical element model, GE/Intermountain Healthcare. Clinical Element Model Search. Available at: http://intermountainhealthcare.org/cem/Pages/Detail.aspx?NCID=520861661&k=allergy (accessed Jan 16, 2012).

• Edwards IR, Aronson JK. Adverse drug reactions: definitions, diagnosis, and management. Lancet. 2000 Oct 7;356(9237):1255-9. PubMed PMID: 11072960.

• Horsfield P, Sibeko S. Representation in Electronic Patient Records of Allergic Reactions, Adverse Reactions, and Intolerance of Pharmaceutical Products [Internet]. London, UK: National Health Service; 2006 Sep 07 [cited 2011 Jun 21]; Available at https://svn.connectingforhealth.nhs.uk/svn/public/nhscontentmodels/TRUNK/ref/NPfIT/Allergy_ADR_Intolerance%20v%201.2Final.doc .

• Long R, Bentley S. SCG Guidance on the Representation of Allergies and Adverse Reaction Information Using NHS Message Templates [Internet]. London, UK: National Health Service; 2008 Apr 30 [cited 2011 Jun 21]; Available at http://www.connectingforhealth.nhs.uk/systemsandservices/data/scg/scg0001.pdf .

• Microsoft. Design Guidance: Displaying Adverse Drug Reaction Risks [Internet]. 2009 January 28 [cited 2011 Jun 21]; Available at www.mscui.net/DesignGuide/DisplayingAllergies.aspx .

• Microsoft. Design Guidance: Recording Adverse Drug Reaction Risks [Internet]. 2009 March 27 [cited 2011 Jun 21]; Available at www.mscui.net/DesignGuide/RecordingAllergies.aspx .

• Mosby. Mosby's Pocket Dictionary of Medicine, Nursing and Health Professions. 6th Edition. USA: Mosby Elsevier; 2010

• National E-Health Transition Authority. Adverse Reactions (Data Specifications) Version 1.1 [Internet]. Sydney, Australia: NEHTA; 2008 Feb 29 [cited 2011 Jun 21]; Available at http://www.nehta.gov.au/component/docman/doc_download/453-adverse-reaction-data-specification-v11

• Riedl MA, Casillas AM. Adverse drug reactions: types and treatment options. Am Fam Physician. 2003 Nov 1;68(9):1781-90. Review. PubMed PMID: 14620598.

• Royal Australian College of General Practitioners. Fact Sheet: Allergies & Adverse Reactions (Draft). 2010.

• Thien FC. Drug hypersensitivity. Med J Aust. 2006 Sep 18;185(6):333-8. Review. PubMed PMID: 16999678.

• Uppsala Monitoring Centre (WHO): http://www.who-umc.org/

• European Medicines Agency: http://www.ema.europa.eu/ema/

• DIRECTIVE 2010/84/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL, of 15 December 2010, amending, as regards pharmacovigilance, Directive 2001/83/EC on the Community code relating to medicinal products for human use: http://ec.europa.eu/health/files/eudralex/vol-1/dir_2010_84/dir_2010_84_en.pdf

9.1.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
asserter | reference | Source of the information about the allergy | AllergyIntolerance.asserter (Practitioner, Patient, PractitionerRole, RelatedPerson)
category | token | food | medication | environment | biologic | AllergyIntolerance.category
clinical-status | token | active | inactive | resolved | AllergyIntolerance.clinicalStatus
code | token | Code that identifies the allergy or intolerance | AllergyIntolerance.code | AllergyIntolerance.reaction.substance | 13 Resources
criticality | token | low | high | unable-to-assess | AllergyIntolerance.criticality
date | date | Date first version of the resource instance was recorded | AllergyIntolerance.recordedDate | 17 Resources
identifier | token | External ids for this item | AllergyIntolerance.identifier | 30 Resources
last-date | date | Date(/time) of last known occurrence of a reaction | AllergyIntolerance.lastOccurrence
manifestation | token | Clinical symptoms/signs associated with the Event | AllergyIntolerance.reaction.manifestation
onset | date | Date(/time) when manifestations showed | AllergyIntolerance.reaction.onset
patient | reference | Who the sensitivity is for | AllergyIntolerance.patient (Patient) | 33 Resources
recorder | reference | Who recorded the sensitivity | AllergyIntolerance.recorder (Practitioner, Patient, PractitionerRole, RelatedPerson)
route | token | How the subject was exposed to the substance | AllergyIntolerance.reaction.exposureRoute
severity | token | mild | moderate | severe (of event as a whole) | AllergyIntolerance.reaction.severity
type | token | allergy | intolerance - Underlying mechanism (if known) | AllergyIntolerance.type | 5 Resources
verification-status | token | unconfirmed | confirmed | refuted | entered-in-error | AllergyIntolerance.verificationStatus

9.1.5 Resource AllergyIntolerance - Examples

URL: https://hl7.org/fhir/R4/allergyintolerance-examples.html

9.1.5 Resource AllergyIntolerance - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Example Name | id | Format
A typical Food Allergy | example | XML | JSON | Turtle
A typical Medication allergy from a clinical system | medication | XML | JSON | Turtle
An allergy to a substance with additional details in notes | fishallergy | XML | JSON | Turtle
No Known Allergy | nka | XML | JSON | Turtle
No Known Drug Allergy | nkda | XML | JSON | Turtle
No Known Latext Allergy | nkla | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.1.6 Resource AllergyIntolerance - Detailed Descriptions

URL: https://hl7.org/fhir/R4/allergyintolerance-definitions.html

9.1.6 Resource AllergyIntolerance - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AllergyIntolerance resource.

AllergyIntolerance

Element Id | AllergyIntolerance

Definition | Risk of harmful or undesirable, physiological response which is unique to an individual and associated with exposure to a substance.

Cardinality | 0..*

Type | DomainResource

Requirements | To record a clinical assessment of a propensity, or potential risk to an individual, of an adverse reaction upon future exposure to the specified substance, or class of substance.

Alternate Names | Allergy; Intolerance; Adverse Reaction

Comments | Substances include, but are not limited to: a therapeutic substance administered correctly at an appropriate dosage for the individual; food; material derived from plants or animals; or venom from insect stings.

Invariants | Defined on this element ait-1 Rule AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists() ait-2 Rule AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()

Defined on this element

ait-1 | Rule | AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists()

ait-2 | Rule | AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()

AllergyIntolerance.identifier

Element Id | AllergyIntolerance.identifier

Definition | Business identifiers assigned to this AllergyIntolerance by the performer or other systems which remain constant as the resource is updated and propagates from server to server.

Note | This is a business identifier, not a resource identifier (see discussion)

Cardinality | 0..*

Type | Identifier

Requirements | Allows identification of the AllergyIntolerance as it is known by various participating systems and in a way that remains consistent across servers.

Summary | true

Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.

AllergyIntolerance.clinicalStatus

Element Id | AllergyIntolerance.clinicalStatus

Definition | The clinical status of the allergy or intolerance.

Cardinality | 0..1

Terminology Binding | AllergyIntolerance Clinical Status Codes (Required)

Type | CodeableConcept

Is Modifier | true (Reason: This element is labeled as a modifier because the status contains the codes inactive and resolved that mark the AllergyIntolerance as no longer active.)

Summary | true

Comments | Refer to discussion if clincalStatus is missing data. The data type is CodeableConcept because clinicalStatus has some clinical judgment involved, such that there might need to be more specificity than the required FHIR value set allows. For example, a SNOMED coding might allow for additional specificity.

Invariants | Affect this element ait-1 Rule AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists() ait-2 Rule AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()

Affect this element

ait-1 | Rule | AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists()

ait-2 | Rule | AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()

AllergyIntolerance.verificationStatus

Element Id | AllergyIntolerance.verificationStatus

Definition | Assertion about certainty associated with the propensity, or potential risk, of a reaction to the identified substance (including pharmaceutical product).

Cardinality | 0..1

Terminology Binding | AllergyIntolerance Verification Status Codes (Required)
Type | CodeableConcept
Is Modifier | true (Reason: This element is labeled as a modifier because the status contains the codes refuted and entered-in-error that mark the AllergyIntolerance as not currently valid.)
Summary | true
Comments | The data type is CodeableConcept because verificationStatus has some clinical judgment involved, such that there might need to be more specificity than the required FHIR value set allows. For example, a SNOMED coding might allow for additional specificity.
Invariants | Affect this element ait-1 Rule AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists() ait-2 Rule AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()
Affect this element
ait-1 | Rule | AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error. | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').exists() or clinicalStatus.exists()
ait-2 | Rule | AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' and code = 'entered-in-error').empty() or clinicalStatus.empty()
AllergyIntolerance.type
Element Id | AllergyIntolerance.type
Definition | Identification of the underlying physiological mechanism for the reaction risk.
Cardinality | 0..1
Terminology Binding | AllergyIntoleranceType (Required)
Type | code
Alternate Names | Category; Class
Summary | true
Comments | Allergic (typically immune-mediated) reactions have been traditionally regarded as an indicator for potential escalation to significant future risk. Contemporary knowledge suggests that some reactions previously thought to be immune-mediated are, in fact, non-immune, but in some cases can still pose a life threatening risk. It is acknowledged that many clinicians might not be in a position to distinguish the mechanism of a particular reaction. Often the term "allergy" is used rather generically and may overlap with the use of "intolerance" - in practice the boundaries between these two concepts might not be well-defined or understood. This data element is included nevertheless, because many legacy systems have captured this attribute. Immunologic testing may provide supporting evidence for the basis of the reaction and the causative substance, but no tests are 100% sensitive or specific for sensitivity to a particular substance. If, as is commonly the case, it is unclear whether the reaction is due to an allergy or an intolerance, then the type element should be omitted from the resource.
AllergyIntolerance.category
Element Id | AllergyIntolerance.category
Definition | Category of the identified substance.
Cardinality | 0..*
Terminology Binding | AllergyIntoleranceCategory (Required)
Type | code
Alternate Names | Category; Type; Reaction Type; Class
Summary | true
Comments | This data element has been included because it is currently being captured in some clinical systems. This data can be derived from the substance where coding systems are used, and is effectively redundant in that situation. When searching on category, consider the implications of AllergyIntolerance resources without a category. For example, when searching on category = medication, medication allergies that don't have a category valued will not be returned. Refer to search for more information on how to search category with a :missing modifier to get allergies that don't have a category. Additionally, category should be used with caution because category can be subjective based on the sender.
AllergyIntolerance.criticality
Element Id | AllergyIntolerance.criticality
Definition | Estimate of the potential clinical harm, or seriousness, of the reaction to the identified substance.
Cardinality | 0..1
Terminology Binding | AllergyIntoleranceCriticality (Required)
Type | code
Alternate Names | Severity; Seriousness; Contra-indication; Risk
Summary | true
Comments | The default criticality value for any propensity to an adverse reaction should be 'Low Risk', indicating at the very least a relative contraindication to deliberate or voluntary exposure to the substance. 'High Risk' is flagged if the clinician has identified a propensity for a more serious or potentially life-threatening reaction, such as anaphylaxis, and implies an absolute contraindication to deliberate or voluntary exposure to the substance. If this element is missing, the criticality is unknown (though it may be known elsewhere). Systems that capture a severity at the condition level are actually representing the concept of criticality whereas the severity documented at the reaction level is representing the true reaction severity. Existing systems that are capturing both condition criticality and reaction severity may use the term "severity" to represent both. Criticality is the worst it could be in the future (i.e. situation-agnostic) whereas severity is situation-dependent.
AllergyIntolerance.code
Element Id | AllergyIntolerance.code
Definition | Code for an allergy or intolerance statement (either a positive or a negated/excluded statement). This may be a code for a substance or pharmaceutical product that is considered to be responsible for the adverse reaction risk (e.g., "Latex"), an allergy or intolerance condition (e.g., "Latex allergy"), or a negated/excluded code for a specific substance or class (e.g., "No latex allergy") or a general or categorical negated statement (e.g., "No known allergy", "No known drug allergies"). Note: the substance for a specific reaction may be different from the substance identified as the cause of the risk, but it must be consistent with it. For instance, it may be a more specific substance (e.g. a brand medication) or a composite product that includes the identified

substance. It must be clinically safe to only process the 'code' and ignore the 'reaction.substance'. If a receiving system is unable to confirm that AllergyIntolerance.reaction.substance falls within the semantic scope of AllergyIntolerance.code, then the receiving system should ignore AllergyIntolerance.reaction.substance.
Cardinality | 0..1
Terminology Binding | AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
Type | CodeableConcept
Alternate Names | Code
Summary | true
Comments | It is strongly recommended that this element be populated using a terminology, where possible. For example, some terminologies used include RxNorm, SNOMED CT, DM+D, NDFRT, ICD-9, IDC-10, UNII, and ATC. Plain text should only be used if there is no appropriate terminology available. Additional details can be specified in the text. When a substance or product code is specified for the 'code' element, the "default" semantic context is that this is a positive statement of an allergy or intolerance (depending on the value of the 'type' element, if present) condition to the specified substance/product. In the corresponding SNOMED CT allergy model, the specified substance/product is the target (destination) of the "Causative agent" relationship. The 'substanceExposureRisk' extension is available as a structured and more flexible alternative to the 'code' element for making positive or negative allergy or intolerance statements. This extension provides the capability to make "no known allergy" (or "no risk of adverse reaction") statements regarding any coded substance/product (including cases when a pre-coordinated "no allergy to x" concept for that substance/product does not exist). If the 'substanceExposureRisk' extension is present, the AllergyIntolerance.code element SHALL be omitted.
AllergyIntolerance.patient
Element Id | AllergyIntolerance.patient
Definition | The patient who has the allergy or intolerance.
Cardinality | 1..1
Type | Reference(Patient)
Alternate Names | Patient
Summary | true
AllergyIntolerance.encounter
Element Id | AllergyIntolerance.encounter
Definition | The encounter when the allergy or intolerance was asserted.
Cardinality | 0..1
Type | Reference(Encounter)
AllergyIntolerance.onset[x]
Element Id | AllergyIntolerance.onset[x]
Definition | Estimated or actual date, date-time, or age when allergy or intolerance was identified.
Cardinality | 0..1
Type | dateTime|Age|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
AllergyIntolerance.recordedDate
Element Id | AllergyIntolerance.recordedDate
Definition | The recordedDate represents when this particular AllergyIntolerance record was created in the system, which is often a system-generated date.
Cardinality | 0..1
Type | dateTime
AllergyIntolerance.recorder
Element Id | AllergyIntolerance.recorder
Definition | Individual who recorded the record and takes responsibility for its content.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)
Alternate Names | Author
AllergyIntolerance.asserter
Element Id | AllergyIntolerance.asserter
Definition | The source of the information about the allergy that is recorded.
Cardinality | 0..1
Type | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole)
Alternate Names | Source; Informant
Summary | true
Comments | The recorder takes responsibility for the content, but can reference the source from where they got it.
AllergyIntolerance.lastOccurrence
Element Id | AllergyIntolerance.lastOccurrence
Definition | Represents the date and/or time of the last known occurrence of a reaction event.
Cardinality | 0..1
Type | dateTime
Comments | This date may be replicated by one of the Onset of Reaction dates. Where a textual representation of the date of last occurrence is required e.g. 'In Childhood, '10 years ago' the Comment element should be used.
AllergyIntolerance.note
Element Id | AllergyIntolerance.note
Definition | Additional narrative about the propensity for the Adverse Reaction, not captured in other fields.
Cardinality | 0..*
Type | Annotation
Comments | For example: including reason for flagging a seriousness of 'High Risk'; and instructions related to future exposure or administration of the substance, such as administration within an Intensive Care Unit or under corticosteroid cover. The notes should be related to an allergy or intolerance as a condition in general and not related to any particular episode of it. For episode notes and descriptions, use AllergyIntolerance.event.description and AllergyIntolerance.event.notes.
AllergyIntolerance.reaction

Element Id | AllergyIntolerance.reaction
Definition | Details about each adverse reaction event linked to exposure to the identified substance.
Cardinality | 0..*
AllergyIntolerance.reaction.substance
Element Id | AllergyIntolerance.reaction.substance
Definition | Identification of the specific substance (or pharmaceutical product) considered to be responsible for the Adverse Reaction event. Note: the substance for a specific reaction may be different from the substance identified as the cause of the risk, but it must be consistent with it. For instance, it may be a more specific substance (e.g. a brand medication) or a composite product that includes the identified substance. It must be clinically safe to only process the 'code' and ignore the 'reaction.substance'. If a receiving system is unable to confirm that AllergyIntolerance.reaction.substance falls within the semantic scope of AllergyIntolerance.code, then the receiving system should ignore AllergyIntolerance.reaction.substance.
Cardinality | 0..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept
Comments | Coding of the specific substance (or pharmaceutical product) with a terminology capable of triggering decision support should be used wherever possible. The 'code' element allows for the use of a specific substance or pharmaceutical product, or a group or class of substances. In the case of an allergy or intolerance to a class of substances, (for example, "penicillins"), the 'reaction.substance' element could be used to code the specific substance that was identified as having caused the reaction (for example, "amoxycillin"). Duplication of the value in the 'code' and 'reaction.substance' elements is acceptable when a specific substance has been recorded in 'code'.
AllergyIntolerance.reaction.manifestation
Element Id | AllergyIntolerance.reaction.manifestation
Definition | Clinical symptoms and/or signs that are observed or associated with the adverse reaction event.
Cardinality | 1..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Alternate Names | Symptoms; Signs
Comments | Manifestation can be expressed as a single word, phrase or brief description. For example: nausea, rash or no reaction. It is preferable that manifestation should be coded with a terminology, where possible. The values entered here may be used to display on an application screen as part of a list of adverse reactions, as recommended in the UK NHS CUI guidelines. Terminologies commonly used include, but are not limited to, SNOMED CT or ICD10.
AllergyIntolerance.reaction.description
Element Id | AllergyIntolerance.reaction.description
Definition | Text description about the reaction as a whole, including details of the manifestation if required.
Cardinality | 0..1
Type | string
Alternate Names | Narrative; Text
Comments | Use the description to provide any details of a particular event of the occurred reaction such as circumstances, reaction specifics, what happened before/after. Information, related to the event, but not describing a particular care should be captured in the comment field. For example: at the age of four, the patient was given penicillin for strep throat and subsequently developed severe hives.
AllergyIntolerance.reaction.onset
Element Id | AllergyIntolerance.reaction.onset
Definition | Record of the date and/or time of the onset of the Reaction.
Cardinality | 0..1
Type | dateTime
AllergyIntolerance.reaction.severity
Element Id | AllergyIntolerance.reaction.severity
Definition | Clinical assessment of the severity of the reaction event as a whole, potentially considering multiple different manifestations.
Cardinality | 0..1
Terminology Binding | AllergyIntoleranceSeverity (Required)
Type | code
Comments | It is acknowledged that this assessment is very subjective. There may be some specific practice domains where objective scales have been applied. Objective scales can be included in this model as extensions.
AllergyIntolerance.reaction.exposureRoute
Element Id | AllergyIntolerance.reaction.exposureRoute
Definition | Identification of the route by which the subject was exposed to the substance.
Cardinality | 0..1
Terminology Binding | SNOMED CT Route Codes (Example)
Type | CodeableConcept
Comments | Coding of the route of exposure with a terminology should be used wherever possible.
AllergyIntolerance.reaction.note
Element Id | AllergyIntolerance.reaction.note
Definition | Additional text about the adverse reaction event not captured in other fields.
Cardinality | 0..*
Type | Annotation
Comments | Use this field to record information indirectly related to a particular event and not captured in the description. For example: Clinical records are no longer available, recorded based on information provided to the patient by her mother and her mother is deceased.

9.1.7 Resource AllergyIntolerance - Mappings

URL: https://hl7.org/fhir/R4/allergyintolerance-mappings.html

9.1.7 Resource AllergyIntolerance - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the allergyintolerance resource (see Mappings to Other Standards for further information & status).

9.1.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

AllergyIntolerance
identifier | FiveWs.identifier
clinicalStatus | FiveWs.status
verificationStatus | FiveWs.status
type | FiveWs.class
category | FiveWs.class
criticality | FiveWs.grade
code | FiveWs.what[x]
patient | FiveWs.subject[x]
encounter | FiveWs.context
onset[x] | FiveWs.init
recordedDate | FiveWs.recorded
recorder | FiveWs.author
asserter | FiveWs.source

9.1.7.2 HL7 v2 Mapping (http://hl7.org/v2)

AllergyIntolerance
identifier | IAM-7
clinicalStatus
verificationStatus
type | IAM-9
category | AL1-2
criticality | AL1-4
code | AL1-3 / IAM-3
patient | (PID-3)
encounter
onset[x]
recordedDate | IAM-13
recorder
asserter | IAM-14 (if patient) / IAM-18 (if practitioner)
lastOccurrence
note
reaction
substance
manifestation | AL1-5
description
onset | AL1-6
severity
exposureRoute
note

9.1.7.3 RIM Mapping (http://hl7.org/v3)

AllergyIntolerance | Observation[classCode=OBS, moodCode=EVN]
identifier | id
clinicalStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="clinicalStatus", moodCode=EVN].value
verificationStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="verificationStatus", moodCode=EVN].value
type | code
category | value < IntoleranceValue (Agent)
criticality | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV, value <= SeverityObservation (Severity Level)]
code | substance/product: .participation[typeCode=CAGNT].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType] negated/excluded substance/product: .participation[typeCode=CAGNT, negationInd=true].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType] positive or negated/excluded condition/situation: Observation.code=ASSERTION; Observation.value
patient | .participation[typeCode=SBJ].role[classCode=PAT]

encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
onset[x] | effectiveTime.low
recordedDate | .participation[typeCode=AUT].time
recorder | .participation[typeCode=AUT].role
asserter | .participation[typeCode=INF].role
lastOccurrence | outBoundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN, code <= CommonClinicalObservationType, value <= ObservationValue (Reaction Type)].effectiveTime
note | subjectOf.observationEvent[code="annotation"].value
reaction | outBoundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN, code <= CommonClinicalObservationType, value <= ObservationValue (Reaction Type)]
substance | outBoundRelationship[typeCode=SAS].target[classCode=SBADM, code <= ExposureCode].participation[typeCode=CSM].role[classCode=ADMM].player[classCode=MAT, determinerCode=KIND, code <= ExposureAgentEntityType]
manifestation | code
description | text
onset | effectiveTime.low
severity | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV, value <= SeverityObservation (Severity Level)]
exposureRoute | outBoundRelationship[typeCode=SAS].target[classCode=SBADM, code <= ExposureCode].routeCode
note | subjectOf.observationEvent[code="annotation"].value

9.1.8 Resource AllergyIntolerance - Extensions & Profiles

URL: https://hl7.org/fhir/R4/allergyintolerance-profiles.html

9.1.8 Resource AllergyIntolerance - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the AllergyIntolerance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.1.8.1 Profiles

No Profiles defined for this resource

9.1.8.2 Extensions

allergyintolerance-assertedDate | AllergyIntolerance | assertedDate | for AllergyIntolerance HL7 Extensions
allergyintolerance-certainty | AllergyIntolerance.reaction | certainty | for AllergyIntolerance HL7 Extensions
allergyintolerance-duration | AllergyIntolerance.reaction | duration | for AllergyIntolerance HL7 Extensions
allergyintolerance-reasonRefuted | AllergyIntolerance | reasonRefuted | for AllergyIntolerance HL7 Extensions
allergyintolerance-resolutionAge | AllergyIntolerance | resolutionAge | for AllergyIntolerance HL7 Extensions
allergyintolerance-substanceExposureRisk | AllergyIntolerance | substanceExposureRisk | for AllergyIntolerance HL7 Extensions
openEHR-administration | AllergyIntolerance.reaction | administration | for openEHR Archetype Profile
openEHR-careplan | AllergyIntolerance.reaction | careplan | for openEHR Archetype Profile
openEHR-exposureDate | AllergyIntolerance.reaction | exposureDate | for openEHR Archetype Profile
openEHR-exposureDescription | AllergyIntolerance.reaction | exposureDescription | for openEHR Archetype Profile
openEHR-exposureDuration | AllergyIntolerance.reaction | exposureDuration | for openEHR Archetype Profile
openEHR-location | AllergyIntolerance.reaction | location | for openEHR Archetype Profile
openEHR-management | AllergyIntolerance.reaction | management | for openEHR Archetype Profile
openEHR-test | AllergyIntolerance | test | for openEHR Archetype Profile
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.1.8.3 Search Extensions

No Search Extensions defined for this resource

9.1.9 Resource AllergyIntolerance - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/allergyintolerance-version-maps.html

No validation errors - all conversions are clean

9.1.9 Resource AllergyIntolerance - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for AllergyIntolerance.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.1.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/AllergyIntolerance3to4" = "R3 to R4 Conversions for AllergyIntolerance"

uses "http://hl7.org/fhir/3.0/StructureDefinition/AllergyIntolerance" alias AllergyIntoleranceR3 as source
uses "http://hl7.org/fhir/StructureDefinition/AllergyIntolerance" alias AllergyIntolerance as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group AllergyIntolerance(source src : AllergyIntoleranceR3, target tgt : AllergyIntolerance) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.clinicalStatus as vs ->  tgt.clinicalStatus as vt,  vt.coding as c,  c.system =
    'http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical',  c.code = vs;
  src.verificationStatus as vs ->  tgt.verificationStatus as vt,  vt.coding as c,  c.system =
    'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification',  c.code = vs;
  src.type -> tgt.type;
  src.category -> tgt.category;
  src.criticality -> tgt.criticality;
  src.code -> tgt.code;
  src.patient -> tgt.patient;
  src.onset : dateTime as vs -> tgt.onset = create('dateTime') as vt then dateTime(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.assertedDate -> tgt.recordedDate "recordedDate";
  src.recorder -> tgt.recorder;
  src.asserter -> tgt.asserter;
  src.lastOccurrence -> tgt.lastOccurrence;
  src.note -> tgt.note;
  src.reaction as vs0 -> tgt.reaction as vt0 then {
    vs0.substance -> vt0.substance;
    vs0.manifestation -> vt0.manifestation;
    vs0.description -> vt0.description;
    vs0.onset -> vt0.onset;
    vs0.severity -> vt0.severity;
    vs0.exposureRoute -> vt0.exposureRoute;
    vs0.note -> vt0.note;
  };
}
```

9.1.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/AllergyIntolerance4to3" = "R4 to R3 Conversion for AllergyIntolerance"

uses "http://hl7.org/fhir/StructureDefinition/AllergyIntolerance" alias AllergyIntolerance as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/AllergyIntolerance" alias AllergyIntoleranceR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group AllergyIntolerance(source src : AllergyIntolerance, target tgt : AllergyIntoleranceR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.clinicalStatus as vs then {
    vs.coding as c where system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical' then {
      c.code -> tgt.clinicalStatus;
    };
  };
  src.verificationStatus as vs then {
    vs.coding as c where system = 'http://terminology.hl7.org/CodeSystem/allergyintolerance-verification' then {
      c.code -> tgt.verificationStatus;
    };
  };
  src.type -> tgt.type;
  src.category -> tgt.category;
  src.criticality -> tgt.criticality;
  src.code -> tgt.code;
  src.patient -> tgt.patient;
  src.onset : dateTime as vs -> tgt.onset = create('dateTime') as vt then dateTime(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.recordedDate -> tgt.assertedDate "assertedDate";
  src.recorder -> tgt.recorder;
  src.asserter -> tgt.asserter;
  src.lastOccurrence -> tgt.lastOccurrence;
```

```
  src.note -> tgt.note;
  src.reaction as vs0 -> tgt.reaction as vt0 then {
    vs0.substance -> vt0.substance;
    vs0.manifestation -> vt0.manifestation;
    vs0.description -> vt0.description;
    vs0.onset -> vt0.onset;
    vs0.severity -> vt0.severity;
    vs0.exposureRoute -> vt0.exposureRoute;
    vs0.note -> vt0.note;
  };
}
```

9.1.9.3 R4 Validation Errors

9.1.10 openEHR Archetype Profile

URL: https://hl7.org/fhir/R4/allergyintolerance-openehr.html

A set of extensions that define how the parts of the openEHR RiskOfAdverseReaction archetype that are not generaly encountered by implementers are represented

9.1.10 openEHR Archetype Profile

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.1.10.1 Content

Extensions:

openEHR-test | test : Observations that confirm or refute the risk and/or the substance.

openEHR-location | location : The anatomical location / body site(s) where the symptoms manifested.

openEHR-exposureDate | exposureDate : Record of the date and/or time of the first exposure to the Substance for this Reaction Event.

openEHR-exposureDuration | exposureDuration : The amount of time the individual was exposed to the Substance.

openEHR-exposureDescription | exposureDescription : Text description about exposure to the Substance.

openEHR-administration | administration : Link to an actual medication administration record with the full details of the administration, if a link is known.

openEHR-management | management : Text description about the clinical management provided.

openEHR-careplan | careplan : Additional details about the clinical management provided for this Reaction Event.

9.1.11 AllergyIntolerance HL7 Extensions

URL: https://hl7.org/fhir/R4/allergyintolerance-extensions.html

Defines common extensions used with or related to the AllergyIntolerance resource

9.1.11 AllergyIntolerance HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.1.11.1 Content

Extensions:

allergyintolerance-resolutionAge | resolutionAge : The estimated patient age at which the allergy or intolerance resolved. Should be specified only if the status is resolved.

allergyintolerance-reasonRefuted | reasonRefuted : A code capturing the explanation of why the allergy or intolerance has been refuted. Should be specified only if the status is refuted.

allergyintolerance-duration | duration : The amount of time that the Adverse Reaction persisted.

allergyintolerance-substanceExposureRisk | substanceExposureRisk : A complex extension allowing structured capture of the exposure risk of the patient for an adverse reaction (allergy or intolerance) to the specified substance/product.

allergyintolerance-certainty | certainty : Statement about the degree of clinical certainty that the specific substance was the cause of the manifestation in this reaction event.

allergyintolerance-assertedDate | assertedDate : The date on which the existence of the AllergyIntolerance was first asserted or acknowledged.

9.1.12 JSON Schema for AllergyIntolerance

URL: https://hl7.org/fhir/R4/allergyintolerance.schema.json.html

9.1.12 JSON Schema for AllergyIntolerance

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is a representation of the json schema for AllergyIntolerance, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/AllergyIntolerance",
  "$ref": "#/definitions/AllergyIntolerance",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "AllergyIntolerance": {
```

```
  "description": "Risk of harmful or undesirable, physiological response which is unique to an individual and
associated with exposure to a substance.",
  "properties": {
    "resourceType": {
      "description": "This is a AllergyIntolerance resource",
      "const": "AllergyIntolerance"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this AllergyIntolerance by the performer or other systems
which remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "clinicalStatus": {
      "description": "The clinical status of the allergy or intolerance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "verificationStatus": {
      "description": "Assertion about certainty associated with the propensity, or potential risk, of a reaction to
the identified substance (including pharmaceutical product).",
```

```
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "type": {
      "description": "Identification of the underlying physiological mechanism for the reaction risk.",
      "enum": [
        "allergy",
        "intolerance"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Category of the identified substance.",
      "items": {
        "enum": [
          "food",
          "medication",
          "environment",
          "biologic"
        ]
      },
      "type": "array"
    },
    "_category": {
      "description": "Extensions for category",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "criticality": {
      "description": "Estimate of the potential clinical harm, or seriousness, of the reaction to the identified
substance.",
      "enum": [
        "low",
        "high",
        "unable-to-assess"
      ]
    },
    "_criticality": {
      "description": "Extensions for criticality",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Code for an allergy or intolerance statement (either a positive or a negated/excluded
statement).  This may be a code for a substance or pharmaceutical product that is considered to be responsible for
the adverse reaction risk (e.g., \"Latex\"), an allergy or intolerance condition (e.g., \"Latex allergy\"), or a
negated/excluded code for a specific substance or class (e.g., \"No latex allergy\") or a general or categorical
negated statement (e.g.,  \"No known allergy\", \"No known drug allergies\").  Note: the substance for a specific
reaction may be different from the substance identified as the cause of the risk, but it must be consistent with it.
 For instance, it may be a more specific substance (e.g. a brand medication) or a composite product that includes
the identified substance. It must be clinically safe to only process the \u0027code\u0027 and ignore the
\u0027reaction.substance\u0027.  If a receiving system is unable to confirm that
AllergyIntolerance.reaction.substance falls within the semantic scope of AllergyIntolerance.code, then the receiving
 system should ignore AllergyIntolerance.reaction.substance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "patient": {
      "description": "The patient who has the allergy or intolerance.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The encounter when the allergy or intolerance was asserted.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onsetDateTime": {
      "description": "Estimated or actual date,  date-time, or age when allergy or intolerance was identified.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_onsetDateTime": {
      "description": "Extensions for onsetDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "onsetAge": {
      "description": "Estimated or actual date,  date-time, or age when allergy or intolerance was identified.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "onsetPeriod": {
      "description": "Estimated or actual date,  date-time, or age when allergy or intolerance was identified.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "onsetRange": {
      "description": "Estimated or actual date,  date-time, or age when allergy or intolerance was identified.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
```

```
    "onsetString": {
      "description": "Estimated or actual date,  date-time, or age when allergy or intolerance was identified.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_onsetString": {
      "description": "Extensions for onsetString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recordedDate": {
      "description": "The recordedDate represents when this particular AllergyIntolerance record was created in the
system, which is often a system-generated date.",
      "$ref": "#/definitions/dateTime"
    },
    "_recordedDate": {
      "description": "Extensions for recordedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recorder": {
      "description": "Individual who recorded the record and takes responsibility for its content.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "asserter": {
      "description": "The source of the information about the allergy that is recorded.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "lastOccurrence": {
      "description": "Represents the date and/or time of the last known occurrence of a reaction event.",
      "$ref": "#/definitions/dateTime"
    },
    "_lastOccurrence": {
      "description": "Extensions for lastOccurrence",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Additional narrative about the propensity for the Adverse Reaction, not captured in other
fields.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "reaction": {
      "description": "Details about each adverse reaction event linked to exposure to the identified substance.",
      "items": {
        "$ref": "#/definitions/AllergyIntolerance_Reaction"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "resourceType"
  ]
},
"AllergyIntolerance_Reaction": {
  "description": "Risk of harmful or undesirable, physiological response which is unique to an individual and
associated with exposure to a substance.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
        "substance": {
          "description": "Identification of the specific substance (or pharmaceutical product) considered to be
    responsible for the Adverse Reaction event. Note: the substance for a specific reaction may be different from the
    substance identified as the cause of the risk, but it must be consistent with it. For instance, it may be a more
    specific substance (e.g. a brand medication) or a composite product that includes the identified substance. It must
    be clinically safe to only process the \u0027code\u0027 and ignore the \u0027reaction.substance\u0027.  If a
    receiving system is unable to confirm that AllergyIntolerance.reaction.substance falls within the semantic scope of
    AllergyIntolerance.code, then the receiving system should ignore AllergyIntolerance.reaction.substance.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "manifestation": {
          "description": "Clinical symptoms and/or signs that are observed or associated with the adverse reaction
    event.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "description": {
          "description": "Text description about the reaction as a whole, including details of the manifestation if
    required.",
          "$ref": "#/definitions/string"
        },
        "_description": {
          "description": "Extensions for description",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "onset": {
          "description": "Record of the date and/or time of the onset of the Reaction.",
          "$ref": "#/definitions/dateTime"
        },
        "_onset": {
          "description": "Extensions for onset",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "severity": {
          "description": "Clinical assessment of the severity of the reaction event as a whole, potentially considering
    multiple different manifestations.",
          "enum": [
            "mild",
            "moderate",
            "severe"
          ]
        },
        "_severity": {
          "description": "Extensions for severity",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "exposureRoute": {
          "description": "Identification of the route by which the subject was exposed to the substance.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "note": {
          "description": "Additional text about the adverse reaction event not captured in other fields.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "manifestation"
      ]
    }
  }
}
```


9.1.13 Resource AllergyIntolerance - Maturity Dependency View

URL: https://hl7.org/fhir/R4/allergyintolerance-dependencies.html

9.1.13 Resource AllergyIntolerance - Maturity Dependency View

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Dependency Graph for AllergyIntolerance FMM level 3

Name | Card. | Type | Dependency Analysis
AllergyIntolerance | | DomainResource
identifier | 0..* | Identifier
clinicalStatus | 0..1 | CodeableConcept
verificationStatus | 0..1 | CodeableConcept
type | 0..1 | code
category | 0..* | code

criticality | 0..1 | code
code | 0..1 | CodeableConcept
patient | 1..1 | Reference(Patient)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
onset[x] | 0..1
onsetDateTime | | dateTime
onsetAge | | Age
onsetPeriod | | Period
onsetRange | | Range
onsetString | | string
recordedDate | 0..1 | dateTime
recorder | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
asserter | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
lastOccurrence | 0..1 | dateTime
note | 0..* | Annotation
reaction | 0..* | BackboneElement
substance | 0..1 | CodeableConcept
manifestation | 1..* | CodeableConcept
description | 0..1 | string
onset | 0..1 | dateTime
severity | 0..1 | code
exposureRoute | 0..1 | CodeableConcept
note | 0..* | Annotation

9.1.14 Value Set Sc-valueset-allergyintolerance-clinical - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-allergyintolerance-clinical.html

9.1.14 Value Set Sc-valueset-allergyintolerance-clinical - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.1.15 AllergyIntoleranceClinicalStatusCodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-allergyintolerance-clinical)

Mapping from http://hl7.org/fhir/ValueSet/allergyintolerance-clinical to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "Preferred value set for AllergyIntolerance Clinical Status."

Source Code | Equivalence | Destination Code
active | equivalent | active
inactive | equivalent | inactive
resolved | equivalent | resolved

9.1.16 Value Set Sc-valueset-allergyintolerance-verification - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-allergyintolerance-verification.html

9.1.16 Value Set Sc-valueset-allergyintolerance-verification - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.1.17 AllergyIntoleranceVerificationStatusCodesCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-allergyintolerance-verification)

Mapping from http://hl7.org/fhir/ValueSet/allergyintolerance-verification to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by FHIR Project team (http://hl7.org/fhir).

Canonical Mapping for "Preferred value set for AllergyIntolerance Verification Status."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
unconfirmed	equivalent	unconfirmed
confirmed	equivalent	confirmed
refuted	equivalent	refuted

9.2 Resource Condition - Content

URL: https://hl7.org/fhir/R4/condition.html
Patient Care Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern.

9.2.1 Scope and Usage

Condition is one of the event resources in the FHIR workflow specification.

This resource is used to record detailed information about a condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern. The condition could be a point in time diagnosis in context of an encounter, it could be an item on the practitioner’s Problem List, or it could be a concern that doesn’t exist on the practitioner’s Problem List. Often times, a condition is about a clinician's assessment and assertion of a particular aspect of a patient's state of health. It can be used to record information about a disease/illness identified from application of clinical reasoning over the pathologic and pathophysiologic findings (diagnosis), or identification of health issues/situations that a practitioner considers harmful, potentially harmful and may be investigated and managed (problem), or other health issue/situation that may require ongoing monitoring and/or management (health issue/concern).

The condition resource may be used to record a certain health state of a patient which does not normally present a negative outcome, e.g. pregnancy. The condition resource may be used to record a condition following a procedure, such as the condition of Amputee-BKA following an amputation procedure.

While conditions are frequently a result of a clinician's assessment and assertion of a particular aspect of a patient's state of health, conditions can also be expressed by the patient, related person, or any care team member. A clinician may have a concern about a patient condition (e.g. anorexia) that the patient is not concerned about. Likewise, the patient may have a condition (e.g. hair loss) that does not rise to the level of importance such that it belongs on a practitioner’s Problem List.

For example, each of the following conditions could rise to the level of importance such that it belongs on a problem or concern list due to its direct or indirect impact on the patient’s health. These examples may also be represented using other resources, such as FamilyMemberHistory, Observation, or Procedure.

- Unemployed
- Without transportation (or other barriers)
- Susceptibility to falls
- Exposure to communicable disease
- Family History of cardiovascular disease
- Fear of cancer
- Cardiac pacemaker
- Amputee-BKA
- Risk of Zika virus following travel to a country
- Former smoker
- Travel to a country planned (that warrants immunizations)
- Motor Vehicle Accident
- Patient has had coronary bypass graft

9.2.2 Boundaries and Relationships

The condition resource may be referenced by other resources as "reasons" for an action (e.g. MedicationRequest, Procedure, ServiceRequest, etc.)

This resource is not typically used to record information about subjective and objective information that might lead to the recording of a Condition resource. Such signs and symptoms are typically captured using the Observation resource; although in some cases a persistent symptom, e.g. fever, headache may be captured as a condition before a definitive diagnosis can be discerned by a clinician. By contrast, headache may be captured as an Observation when it contributes to the establishment of a meningitis Condition.

Use the Observation resource when a symptom is resolved without long term management, tracking, or when a symptom contributes to the establishment of a condition.

Use Condition when a symptom requires long term management, tracking, or is used as a proxy for a diagnosis or problem that is not yet determined.

When the diagnosis is related to an allergy or intolerance, the Condition and AllergyIntolerance resources can both be used. However, to be actionable for decision support, using Condition alone is not sufficient as the allergy or intolerance condition needs to be represented as an AllergyIntolerance.

This resource is referenced by AdverseEvent, Appointment, CarePlan, CareTeam, Claim, ClinicalImpression, Communication, CommunicationRequest, Contract, CoverageEligibilityRequest, DeviceRequest, DeviceUseStatement, Encounter, EpisodeOfCare, ExplanationOfBenefit, FamilyMemberHistory, Goal, GuidanceResponse, ImagingStudy, Immunization, MedicationAdministration, MedicationRequest, MedicationStatement, Procedure, RequestGroup, RiskAssessment, ServiceRequest and SupplyRequest

9.2.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Condition | ITU | | DomainResource | Detailed information about conditions, problems or diagnoses + Guideline: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item + Rule: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission + Rule: Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this condition
clinicalStatus | ?!ΣI | 0..1 | CodeableConcept | active | recurrence | relapse | inactive | remission | resolved Condition Clinical Status Codes (Required)
verificationStatus | ?!ΣI | 0..1 | CodeableConcept | unconfirmed | provisional | differential | confirmed | refuted | entered-in-error ConditionVerificationStatus (Required)
category | | 0..* | CodeableConcept | problem-list-item | encounter-diagnosis Condition Category Codes (Extensible)
severity | | 0..1 | CodeableConcept | Subjective severity of condition Condition/Diagnosis Severity (Preferred)
code | Σ | 0..1 | CodeableConcept | Identification of the condition, problem or diagnosis Condition/Problem/Diagnosis Codes (Example)
bodySite | Σ | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
subject | Σ | 1..1 | Reference(Patient | Group) | Who has the condition?
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
onset[x] | Σ | 0..1 | | Estimated or actual date, date-time, or age
onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
abatement[x] | I | 0..1 | | When in resolution/remission
abatementDateTime | | | dateTime
abatementAge | | | Age
abatementPeriod | | | Period
abatementRange | | | Range
abatementString | | | string
recordedDate | Σ | 0..1 | dateTime | Date record was first recorded
recorder | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Who recorded the condition
asserter | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Person who asserts this condition
stage | I | 0..* | BackboneElement | Stage/grade, usually assessed formally + Rule: Stage SHALL have summary or assessment
summary | I | 0..1 | CodeableConcept | Simple summary (disease specific) Condition Stage (Example)
assessment | I | 0..* | Reference(ClinicalImpression | DiagnosticReport | Observation) | Formal record of assessment
type | | 0..1 | CodeableConcept | Kind of staging Condition Stage Type (Example)
evidence | I | 0..* | BackboneElement | Supporting evidence + Rule: evidence SHALL have code or details
code | ΣI | 0..* | CodeableConcept | Manifestation/symptom Manifestation and Symptom Codes (Example)
detail | ΣI | 0..* | Reference(Any) | Supporting information found elsewhere
note | | 0..* | Annotation | Additional information about the Condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Condition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this condition --></identifier>
 <clinicalStatus><!--  0..1 CodeableConcept active | recurrence | relapse | inactive | remission | resolved
    --></clinicalStatus>
 <verificationStatus><!--  0..1 CodeableConcept unconfirmed | provisional | differential | confirmed | refuted |
    entered-in-error --></verificationStatus>
 <category><!-- 0..* CodeableConcept problem-list-item | encounter-diagnosis --></category>
 <severity><!-- 0..1 CodeableConcept Subjective severity of condition --></severity>
 <code><!-- 0..1 CodeableConcept Identification of the condition, problem or diagnosis --></code>
 <bodySite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodySite>
 <subject><!-- 1..1 Reference(Patient|Group) Who has the condition? --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string Estimated or actual date,  date-time, or age --></onset[x]>
 <abatement[x]><!--  0..1 dateTime|Age|Period|Range|string When in resolution/remission --></abatement[x]>
 <recordedDate value="[dateTime]"/><!-- 0..1 Date record was first recorded -->
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Who recorded the condition --></recorder>
 <asserter><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Person who asserts this condition --></asserter>
 <stage>  <!-- 0..* Stage/grade, usually assessed formally -->
  <summary><!--  0..1 CodeableConcept Simple summary (disease specific) --></summary>
  <assessment><!--  0..* Reference(ClinicalImpression|DiagnosticReport|Observation) Formal record of assessment
    --></assessment>
  <type><!-- 0..1 CodeableConcept Kind of staging --></type>
 </stage>
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!--  0..* CodeableConcept Manifestation/symptom --></code>
  <detail><!--  0..* Reference(Any) Supporting information found elsewhere --></detail>
 </evidence>
 <note><!-- 0..* Annotation Additional information about the Condition --></note>
</Condition>
```

JSON Template

```
{
  "resourceType" : "Condition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this condition
  "clinicalStatus" : { CodeableConcept }, // C? active | recurrence | relapse | inactive | remission | resolved
  "verificationStatus" : { CodeableConcept }, // C? unconfirmed | provisional | differential | confirmed | refuted |
    entered-in-error
  "category" : [{ CodeableConcept }], // problem-list-item | encounter-diagnosis
  "severity" : { CodeableConcept }, // Subjective severity of condition
  "code" : { CodeableConcept }, // Identification of the condition, problem or diagnosis
  "bodySite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "subject" : { Reference(Patient|Group) }, // R!  Who has the condition?
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // onset[x]: Estimated or actual date,  date-time, or age. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  // abatement[x]: When in resolution/remission. One of these 5:
  "abatementDateTime" : "<dateTime>",
  "abatementAge" : { Age },
  "abatementPeriod" : { Period },
  "abatementRange" : { Range },
  "abatementString" : "<string>",
  "recordedDate" : "<dateTime>", // Date record was first recorded
  "recorder" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Who recorded the condition
  "asserter" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Person who asserts this condition
  "stage" : [{ // Stage/grade, usually assessed formally
    "summary" : { CodeableConcept }, // C? Simple summary (disease specific)
    "assessment" : [{ Reference(ClinicalImpression|DiagnosticReport|Observation) }], // C? Formal record of assessment
    "type" : { CodeableConcept } // Kind of staging
  }],
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // C? Manifestation/symptom
    "detail" : [{ Reference(Any) }] // C? Supporting information found elsewhere
  }],
  "note" : [{ Annotation }] // Additional information about the Condition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Condition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Condition.identifier [ Identifier ], ... ; # 0..* External Ids for this condition
  fhir:Condition.clinicalStatus [ CodeableConcept ]; # 0..1 active | recurrence | relapse | inactive | remission |
```

```
    resolved
  fhir:Condition.verificationStatus [ CodeableConcept ]; # 0..1 unconfirmed | provisional | differential | confirmed |
    refuted | entered-in-error
  fhir:Condition.category [ CodeableConcept ], ... ; # 0..* problem-list-item | encounter-diagnosis
  fhir:Condition.severity [ CodeableConcept ]; # 0..1 Subjective severity of condition
  fhir:Condition.code [ CodeableConcept ]; # 0..1 Identification of the condition, problem or diagnosis
  fhir:Condition.bodySite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:Condition.subject [ Reference(Patient|Group) ]; # 1..1 Who has the condition?
  fhir:Condition.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # Condition.onset[x] : 0..1 Estimated or actual date,  date-time, or age. One of these 5
    fhir:Condition.onsetDateTime [ dateTime ]
    fhir:Condition.onsetAge [ Age ]
    fhir:Condition.onsetPeriod [ Period ]
    fhir:Condition.onsetRange [ Range ]
    fhir:Condition.onsetString [ string ]
  # Condition.abatement[x] : 0..1 When in resolution/remission. One of these 5
    fhir:Condition.abatementDateTime [ dateTime ]
    fhir:Condition.abatementAge [ Age ]
    fhir:Condition.abatementPeriod [ Period ]
    fhir:Condition.abatementRange [ Range ]
    fhir:Condition.abatementString [ string ]
  fhir:Condition.recordedDate [ dateTime ]; # 0..1 Date record was first recorded
  fhir:Condition.recorder [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Who recorded the
    condition
  fhir:Condition.asserter [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Person who asserts
    this condition
  fhir:Condition.stage [ # 0..* Stage/grade, usually assessed formally
    fhir:Condition.stage.summary [ CodeableConcept ]; # 0..1 Simple summary (disease specific)
    fhir:Condition.stage.assessment [ Reference(ClinicalImpression|DiagnosticReport|Observation) ], ... ; # 0..* Formal
    record of assessment
    fhir:Condition.stage.type [ CodeableConcept ]; # 0..1 Kind of staging
  ], ...;
  fhir:Condition.evidence [ # 0..* Supporting evidence
    fhir:Condition.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation/symptom
    fhir:Condition.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information found elsewhere
  ], ...;
  fhir:Condition.note [ Annotation ], ... ; # 0..* Additional information about the Condition
]
```

Changes since R3

Condition

Condition.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-clinical to http://hl7.org/fhir/ValueSet/condition-clinical|4.0.1

Condition.verificationStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-ver-status to http://hl7.org/fhir/ValueSet/condition-ver-status|4.0.1 Default Value "unknown" removed

Condition.category | Add Binding http://hl7.org/fhir/ValueSet/condition-category (extensible)

Condition.encounter | Added Element

Condition.abatement[x] | Remove Type boolean

Condition.recordedDate | Renamed from assertedDate to recordedDate

Condition.recorder | Added Element

Condition.asserter | Type Reference: Added Target Type PractitionerRole

Condition.stage | Max Cardinality changed from 1 to *

Condition.stage.type | Added Element

Condition.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 12 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

Condition | ITU | | DomainResource | Detailed information about conditions, problems or diagnoses + Guideline: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item + Rule: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission + Rule: Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | External Ids for this condition

clinicalStatus | ?!ΣI | 0..1 | CodeableConcept | active | recurrence | relapse | inactive | remission | resolved Condition Clinical Status Codes (Required)

verificationStatus | ?!ΣI | 0..1 | CodeableConcept | unconfirmed | provisional | differential | confirmed | refuted | entered-in-error ConditionVerificationStatus (Required)

category | | 0..* | CodeableConcept | problem-list-item | encounter-diagnosis Condition Category Codes (Extensible)

severity | | 0..1 | CodeableConcept | Subjective severity of condition Condition/Diagnosis Severity (Preferred)

code | Σ | 0..1 | CodeableConcept | Identification of the condition, problem or diagnosis Condition/Problem/Diagnosis Codes (Example)

bodySite | Σ | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)

subject | Σ | 1..1 | Reference(Patient | Group) | Who has the condition?

encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of

onset[x] | Σ | 0..1 | | Estimated or actual date, date-time, or age

onsetDateTime | | | dateTime
onsetAge | | | Age
onsetPeriod | | | Period
onsetRange | | | Range
onsetString | | | string
abatement[x] | I | 0..1 | | When in resolution/remission
abatementDateTime | | | dateTime
abatementAge | | | Age
abatementPeriod | | | Period
abatementRange | | | Range
abatementString | | | string
recordedDate | Σ | 0..1 | dateTime | Date record was first recorded
recorder | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Who recorded the condition
asserter | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Person who asserts this condition
stage | I | 0..* | BackboneElement | Stage/grade, usually assessed formally + Rule: Stage SHALL have summary or assessment
summary | I | 0..1 | CodeableConcept | Simple summary (disease specific) Condition Stage (Example)
assessment | I | 0..* | Reference(ClinicalImpression | DiagnosticReport | Observation) | Formal record of assessment
type | | 0..1 | CodeableConcept | Kind of staging Condition Stage Type (Example)
evidence | I | 0..* | BackboneElement | Supporting evidence + Rule: evidence SHALL have code or details
code | ΣI | 0..* | CodeableConcept | Manifestation/symptom Manifestation and Symptom Codes (Example)
detail | ΣI | 0..* | Reference(Any) | Supporting information found elsewhere
note | | 0..* | Annotation | Additional information about the Condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Condition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this condition --></identifier>
 <clinicalStatus><!--  0..1 CodeableConcept active | recurrence | relapse | inactive | remission | resolved
    --></clinicalStatus>
 <verificationStatus><!--  0..1 CodeableConcept unconfirmed | provisional | differential | confirmed | refuted |
    entered-in-error --></verificationStatus>
 <category><!-- 0..* CodeableConcept problem-list-item | encounter-diagnosis --></category>
 <severity><!-- 0..1 CodeableConcept Subjective severity of condition --></severity>
 <code><!-- 0..1 CodeableConcept Identification of the condition, problem or diagnosis --></code>
 <bodySite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodySite>
 <subject><!-- 1..1 Reference(Patient|Group) Who has the condition? --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <onset[x]><!-- 0..1 dateTime|Age|Period|Range|string Estimated or actual date,  date-time, or age --></onset[x]>
 <abatement[x]><!--  0..1 dateTime|Age|Period|Range|string When in resolution/remission --></abatement[x]>
 <recordedDate value="[dateTime]"/><!-- 0..1 Date record was first recorded -->
 <recorder><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Who recorded the condition --></recorder>
 <asserter><!-- 0..1 Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) Person who asserts this condition --></asserter>
 <stage>  <!-- 0..* Stage/grade, usually assessed formally -->
  <summary><!--  0..1 CodeableConcept Simple summary (disease specific) --></summary>
  <assessment><!--  0..* Reference(ClinicalImpression|DiagnosticReport|Observation) Formal record of assessment
    --></assessment>
  <type><!-- 0..1 CodeableConcept Kind of staging --></type>
 </stage>
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!--  0..* CodeableConcept Manifestation/symptom --></code>
  <detail><!--  0..* Reference(Any) Supporting information found elsewhere --></detail>
 </evidence>
 <note><!-- 0..* Annotation Additional information about the Condition --></note>
</Condition>
```

JSON Template

```
{
  "resourceType" : "Condition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this condition
  "clinicalStatus" : { CodeableConcept }, // C? active | recurrence | relapse | inactive | remission | resolved
  "verificationStatus" : { CodeableConcept }, // C? unconfirmed | provisional | differential | confirmed | refuted |
    entered-in-error
  "category" : [{ CodeableConcept }], // problem-list-item | encounter-diagnosis
  "severity" : { CodeableConcept }, // Subjective severity of condition
  "code" : { CodeableConcept }, // Identification of the condition, problem or diagnosis
  "bodySite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "subject" : { Reference(Patient|Group) }, // R!  Who has the condition?
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // onset[x]: Estimated or actual date,  date-time, or age. One of these 5:
  "onsetDateTime" : "<dateTime>",
  "onsetAge" : { Age },
  "onsetPeriod" : { Period },
  "onsetRange" : { Range },
  "onsetString" : "<string>",
  // abatement[x]: When in resolution/remission. One of these 5:
  "abatementDateTime" : "<dateTime>",
```

```
  "abatementAge" : { Age },
  "abatementPeriod" : { Period },
  "abatementRange" : { Range },
  "abatementString" : "<string>",
  "recordedDate" : "<dateTime>", // Date record was first recorded
  "recorder" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Who recorded the condition
  "asserter" : { Reference(Practitioner|PractitionerRole|Patient|
   RelatedPerson) }, // Person who asserts this condition
  "stage" : [{ // Stage/grade, usually assessed formally
    "summary" : { CodeableConcept }, // C? Simple summary (disease specific)
    "assessment" : [{ Reference(ClinicalImpression|DiagnosticReport|Observation) }], // C? Formal record of assessment
    "type" : { CodeableConcept } // Kind of staging
  }],
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // C? Manifestation/symptom
    "detail" : [{ Reference(Any) }] // C? Supporting information found elsewhere
  }],
  "note" : [{ Annotation }] // Additional information about the Condition
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Condition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Condition.identifier [ Identifier ], ... ; # 0..* External Ids for this condition
  fhir:Condition.clinicalStatus [ CodeableConcept ]; # 0..1 active | recurrence | relapse | inactive | remission |
    resolved
  fhir:Condition.verificationStatus [ CodeableConcept ]; # 0..1 unconfirmed | provisional | differential | confirmed |
    refuted | entered-in-error
  fhir:Condition.category [ CodeableConcept ], ... ; # 0..* problem-list-item | encounter-diagnosis
  fhir:Condition.severity [ CodeableConcept ]; # 0..1 Subjective severity of condition
  fhir:Condition.code [ CodeableConcept ]; # 0..1 Identification of the condition, problem or diagnosis
  fhir:Condition.bodySite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:Condition.subject [ Reference(Patient|Group) ]; # 1..1 Who has the condition?
  fhir:Condition.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # Condition.onset[x] : 0..1 Estimated or actual date,  date-time, or age. One of these 5
    fhir:Condition.onsetDateTime [ dateTime ]
    fhir:Condition.onsetAge [ Age ]
    fhir:Condition.onsetPeriod [ Period ]
    fhir:Condition.onsetRange [ Range ]
    fhir:Condition.onsetString [ string ]
  # Condition.abatement[x] : 0..1 When in resolution/remission. One of these 5
    fhir:Condition.abatementDateTime [ dateTime ]
    fhir:Condition.abatementAge [ Age ]
    fhir:Condition.abatementPeriod [ Period ]
    fhir:Condition.abatementRange [ Range ]
    fhir:Condition.abatementString [ string ]
  fhir:Condition.recordedDate [ dateTime ]; # 0..1 Date record was first recorded
  fhir:Condition.recorder [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Who recorded the
    condition
  fhir:Condition.asserter [ Reference(Practitioner|PractitionerRole|Patient|RelatedPerson) ]; # 0..1 Person who asserts
    this condition
  fhir:Condition.stage [ # 0..* Stage/grade, usually assessed formally
    fhir:Condition.stage.summary [ CodeableConcept ]; # 0..1 Simple summary (disease specific)
    fhir:Condition.stage.assessment [ Reference(ClinicalImpression|DiagnosticReport|Observation) ], ... ; # 0..* Formal
    record of assessment
    fhir:Condition.stage.type [ CodeableConcept ]; # 0..1 Kind of staging
  ], ...;
  fhir:Condition.evidence [ # 0..* Supporting evidence
    fhir:Condition.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation/symptom
    fhir:Condition.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information found elsewhere
  ], ...;
  fhir:Condition.note [ Annotation ], ... ; # 0..* Additional information about the Condition
]
```

Changes since Release 3

Condition

Condition.clinicalStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-clinical to http://hl7.org/fhir/ValueSet/condition-clinical|4.0.1

Condition.verificationStatus | Type changed from code to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/condition-ver-status to http://hl7.org/fhir/ValueSet/condition-ver-status|4.0.1 Default Value "unknown" removed

Condition.category | Add Binding http://hl7.org/fhir/ValueSet/condition-category (extensible)

Condition.encounter | Added Element

Condition.abatement[x] | Remove Type boolean

Condition.recordedDate | Renamed from assertedDate to recordedDate

Condition.recorder | Added Element

Condition.asserter | Type Reference: Added Target Type PractitionerRole

Condition.stage | Max Cardinality changed from 1 to *

Condition.stage.type | Added Element

Condition.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 12 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
Condition.clinicalStatus | The clinical status of the condition or diagnosis. | Required | ConditionClinicalStatusCodes
Condition.verificationStatus | The verification status to support or decline the clinical status of the condition or diagnosis. | Required | ConditionVerificationStatus
Condition.category | A category assigned to the condition. | Extensible | ConditionCategoryCodes
Condition.severity | A subjective assessment of the severity of the condition as evaluated by the clinician. | Preferred | Condition/DiagnosisSeverity
Condition.code | Identification of the condition or diagnosis. | Example | Condition/Problem/DiagnosisCodes
Condition.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
Condition.stage.summary | Codes describing condition stages (e.g. Cancer stages). | Example | ConditionStage
Condition.stage.type | Codes describing the kind of condition staging (e.g. clinical or pathological). | Example | ConditionStageType
Condition.evidence.code | Codes that describe the manifestation or symptoms of a condition. | Example | ManifestationAndSymptomCodes

9.2.3.2 Constraints

id | Level | Location | Description | Expression
con-1 | Rule | Condition.stage | Stage SHALL have summary or assessment | summary.exists() or assessment.exists()
con-2 | Rule | Condition.evidence | evidence SHALL have code or details | code.exists() or detail.exists()
con-3 | Guideline | (base) | Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item | clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis.
con-4 | Rule | (base) | If condition is abated, then clinicalStatus must be either inactive, resolved, or remission | abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists()
con-5 | Rule | (base) | Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()

9.2.3.3 Use of Condition.code

Many of the code systems used for coding conditions will provide codes that define not only the condition itself, but may also specify a particular stage, location, or causality as part of the code. This is particularly true if SNOMED CT is used for the condition, and especially if expressions are allowed.

The Condition.code may also include such concepts as "history of X" and "good health", where it is useful or appropriate to make such assertions. It can also be used to capture "risk of" and "fear of", in addition to physical conditions, as well as "no known problems" or "negated" conditions (e.g., "no X" or "no history of X" - see the following section for "No Known Problems" and Negated Conditions).

When the Condition.code specifies additional properties of the condition, the other properties are not given a value - instead, the value must be understood from the Condition.code.

9.2.3.4 "No Known Problems" and Negated Conditions

Conditions/Problems Not Reviewed, Not Asked

When a sending system does not have any information about conditions/problems being reviewed or the statement is about conditions/problems not yet being asked, then the List resource should be used to indicate the List.emptyReason.code="notasked".

Conditions/Problems Reviewed, None Identified

Systems may use the List.emptyReason when a statement is about the full scope of the list (i.e. the patient has no conditions/problems of any type). However, it may be preferred to use a code for "no known problems" (e.g., SNOMED CT: 160245001 |No current problems or disability (situation)|), so that all condition/problem data will be available and queryable from Condition resource instances.

Also note that care should be used when adding new Condition resources to a list to ensure that any negation statements that are voided by the addition of a new record are removed from the list. E.g. If the list contains a "no known problems" record and you add a "diabetes" condition record, then be sure that you remove the "no known problems" record.

Trial-Use Note: There are two primary ways of reporting "no known problems" in the current specification: using the CodeableConcept, as described above, or using the List resource with emptyReason. During the STU period, feedback is sought regarding the preferred approach.

Provide feedback here .

Patient Denies Condition

When the patient denies a condition, that can be annotated in the Condition.note element.

9.2.3.5 Assertions of Condition Absence

Generally, electronic records do not contain assertions of conditions that a patient does not have. There are however two exceptions:

• It is appropriate to capture a "refuted" Condition record if the patient or anyone else had reason to believe that a patient did have a condition for a period of time and subsequent evidence has demonstrated that belief was mistaken. In this case, a concrete statement acknowledging the belief as well as the refutation of it is useful.

• It is common as part of checklists prior to admission, surgery, enrollment in trials, etc. to ask questions such as "are you pregnant", "do you have a history of hypertension", etc. This information should NOT be captured using the Condition resource but should instead be captured using QuestionnaireResponse or Observation. In this case, the combination of the question and answer would convey that a particular condition was not present.

9.2.3.6 Use of Condition.evidence

The Condition.evidence provides the basis for whatever is present in Condition.code.

9.2.3.7 Use of Condition.abatementRange

A range is used to communicate age period of subject at time of abatement.

9.2.3.8 Use of Condition.asserter

If the data enterer is different from the asserter and needs to be known, this could be captured using a Provenance instance pointing to the Condition. For example, it is possible that a nurse records the condition on behalf of a physician. The physician is taking responsibility, despite the nurse entering it into the medical record.

9.2.3.9 Use of Condition.clinicalStatus

The Condition.stage and Condition.clinicalStatus may have interdependencies. For example, some "stages" of cancer, etc. will be different for a remission than for the initial occurrence.

9.2.3.10 Diagnosis Role and Rank within an Encounter

To represent the role of the diagnosis within an encounter, such as admission diagnosis or discharge diagnosis, use Encounter.diagnosis.role.

To represent the numeric ranking of the diagnosis within an encounter, such as primary, secondary, or tertiary, use Encounter.diagnosis.rank.

9.2.3.11 Known Issue

A known issue exists with circular references between Condition and ClinicalImpression, which is due to the low maturity level of ClinicalImpression. The Patient Care work group intends to address this issue when ClinicalImpression is considered substantially complete and ready for implementation.

9.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
abatement-age | quantity | Abatement as age or age range | Condition.abatement.as(Age) | Condition.abatement.as(Range)
abatement-date | date | Date-related abatements (dateTime and period) | Condition.abatement.as(dateTime) | Condition.abatement.as(Period)
abatement-string | string | Abatement as a string | Condition.abatement.as(string)
asserter | reference | Person who asserts this condition | Condition.asserter (Practitioner, Patient, PractitionerRole, RelatedPerson)
body-site | token | Anatomical location, if relevant | Condition.bodySite
category | token | The category of the condition | Condition.category
clinical-status | token | The clinical status of the condition | Condition.clinicalStatus
code | token | Code for the condition | Condition.code | 13 Resources
encounter | reference | Encounter created as part of | Condition.encounter (Encounter)
evidence | token | Manifestation/symptom | Condition.evidence.code
evidence-detail | reference | Supporting information found elsewhere | Condition.evidence.detail (Any)
identifier | token | A unique identifier of the condition record | Condition.identifier | 30 Resources
onset-age | quantity | Onsets as age or age range | Condition.onset.as(Age) | Condition.onset.as(Range)
onset-date | date | Date related onsets (dateTime and Period) | Condition.onset.as(dateTime) | Condition.onset.as(Period)

onset-info | string | Onsets as a string | Condition.onset.as(string)
patient | reference | Who has the condition? | Condition.subject.where(resolve() is Patient) (Patient) | 33 Resources
recorded-date | date | Date record was first recorded | Condition.recordedDate
severity | token | The severity of the condition | Condition.severity
stage | token | Simple summary (disease specific) | Condition.stage.summary
subject | reference | Who has the condition? | Condition.subject (Group, Patient)
verification-status | token | unconfirmed | provisional | differential | confirmed | refuted | entered-in-error | Condition.verificationStatus

9.2.5 Resource Condition - Examples

URL: https://hl7.org/fhir/R4/condition-examples.html

9.2.5 Resource Condition - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Condition Example | example | XML | JSON | Turtle
2nd Example | example2 | XML | JSON | Turtle
Real-word condition example (fever) | f201 | XML | JSON | Turtle
Real-word condition example (malignancy) | f202 | XML | JSON | Turtle
Real-word condition example (sepsis) | f203 | XML | JSON | Turtle
Real-word condition example (renal insufficiency) | f204 | XML | JSON | Turtle
Real-word condition example (Bacterial Infection) | f205 | XML | JSON | Turtle
Real-word condition example (heart) | f001 | XML | JSON | Turtle
Real-word condition example (lung) | f002 | XML | JSON | Turtle
Real-word condition example (abscess) | f003 | XML | JSON | Turtle
Stroke example (for prognosis) | stroke | XML | JSON | Turtle
Family history concern | family-history | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.2.6 Resource Condition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/condition-definitions.html

9.2.6 Resource Condition - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Condition resource.

Condition
Element Id | Condition
Definition | A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept that has risen to a level of concern.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element con-3 Guideline Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis. con-4 Rule If condition is abated, then clinicalStatus must be either inactive, resolved, or remission abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists() con-5 Rule Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Defined on this element
con-3 | Guideline | Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item | clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis.
con-4 | Rule | If condition is abated, then clinicalStatus must be either inactive, resolved, or remission | abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists()
con-5 | Rule | Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Condition.identifier
Element Id | Condition.identifier

Definition | Business identifiers assigned to this condition by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the condition as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
Condition.clinicalStatus
Element Id | Condition.clinicalStatus
Definition | The clinical status of the condition.
Cardinality | 0..1
Terminology Binding | Condition Clinical Status Codes (Required)
Type | CodeableConcept
Is Modifier | true (Reason: This element is labeled as a modifier because the status contains codes that mark the condition as no longer active.)
Summary | true
Comments | The data type is CodeableConcept because clinicalStatus has some clinical judgment involved, such that there might need to be more specificity than the required FHIR value set allows. For example, a SNOMED coding might allow for additional specificity.
Invariants | Affect this element con-3 Guideline Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis. con-4 Rule If condition is abated, then clinicalStatus must be either inactive, resolved, or remission abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists() con-5 Rule Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Affect this element
con-3 | Guideline | Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item | clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis.
con-4 | Rule | If condition is abated, then clinicalStatus must be either inactive, resolved, or remission | abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists()
con-5 | Rule | Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Condition.verificationStatus
Element Id | Condition.verificationStatus
Definition | The verification status to support the clinical status of the condition.
Cardinality | 0..1
Terminology Binding | ConditionVerificationStatus (Required)
Type | CodeableConcept
Is Modifier | true (Reason: This element is labeled as a modifier because the status contains the code refuted and entered-in-error that mark the Condition as not currently valid.)
Summary | true
Comments | verificationStatus is not required. For example, when a patient has abdominal pain in the ED, there is not likely going to be a verification status. The data type is CodeableConcept because verificationStatus has some clinical judgment involved, such that there might need to be more specificity than the required FHIR value set allows. For example, a SNOMED coding might allow for additional specificity.
Invariants | Affect this element con-3 Guideline Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis. con-5 Rule Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Affect this element
con-3 | Guideline | Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item | clinicalStatus.exists() or verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code = 'entered-in-error').exists() or category.select($this='problem-list-item').empty() This is (only) a best practice guideline because: Most systems will expect a clinicalStatus to be valued for problem-list-items that are managed over time, but might not need a clinicalStatus for point in time encounter-diagnosis.

con-5 | Rule | Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error | verificationStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-ver-status' and code='entered-in-error').empty() or clinicalStatus.empty()
Condition.category
Element Id | Condition.category
Definition | A category assigned to the condition.
Cardinality | 0..*
Terminology Binding | Condition Category Codes (Extensible)
Type | CodeableConcept
Comments | The categorization is often highly contextual and may appear poorly differentiated or not very useful in other contexts.
Condition.severity
Element Id | Condition.severity
Definition | A subjective assessment of the severity of the condition as evaluated by the clinician.
Cardinality | 0..1
Terminology Binding | Condition/Diagnosis Severity (Preferred)
Type | CodeableConcept
Comments | Coding of the severity with a terminology is preferred, where possible.
Condition.code
Element Id | Condition.code
Definition | Identification of the condition, problem or diagnosis.
Cardinality | 0..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Requirements | 0..1 to account for primarily narrative only resources.
Alternate Names | type
Summary | true
Condition.bodySite
Element Id | Condition.bodySite
Definition | The anatomical location where this condition manifests itself.
Cardinality | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | Only used if not implicit in code found in Condition.code. If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension bodySite. May be a summary code, or a reference to a very precise definition of the location, or both.
Condition.subject
Element Id | Condition.subject
Definition | Indicates the patient or group who the condition record is associated with.
Cardinality | 1..1
Type | Reference(Patient | Group)
Requirements | Group is typically used for veterinary or public health use cases.
Alternate Names | patient
Summary | true
Condition.encounter
Element Id | Condition.encounter
Definition | The Encounter during which this Condition was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter. This record indicates the encounter this particular record is associated with. In the case of a "new" diagnosis reflecting ongoing/revised information about the condition, this might be distinct from the first encounter in which the underlying condition was first "known".
Condition.onset[x]
Element Id | Condition.onset[x]
Definition | Estimated or actual date or date-time the condition began, in the opinion of the clinician.
Cardinality | 0..1
Type | dateTime|Age|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Age is generally used when the patient reports an age at which the Condition began to occur.
Condition.abatement[x]
Element Id | Condition.abatement[x]
Definition | The date or estimated date that the condition resolved or went into remission. This is called "abatement" because of the many overloaded connotations associated with "remission" or "resolution" - Conditions are never really resolved, but they can abate.
Cardinality | 0..1
Type | dateTime|Age|Period|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | There is no explicit distinction between resolution and remission because in many cases the distinction is not clear. Age is generally used when the patient reports an age at which the Condition abated. If there is no abatement element, it is

unknown whether the condition has resolved or entered remission; applications and users should generally assume that the condition is still valid. When abatementString exists, it implies the condition is abated.
Invariants | Affect this element con-4 Rule If condition is abated, then clinicalStatus must be either inactive, resolved, or remission abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists()
Affect this element
con-4 | Rule | If condition is abated, then clinicalStatus must be either inactive, resolved, or remission | abatement.empty() or clinicalStatus.coding.where(system='http://terminology.hl7.org/CodeSystem/condition-clinical' and (code='resolved' or code='remission' or code='inactive')).exists()
Condition.recordedDate
Element Id | Condition.recordedDate
Definition | The recordedDate represents when this particular Condition record was created in the system, which is often a system-generated date.
Cardinality | 0..1
Type | dateTime
Summary | true
Condition.recorder
Element Id | Condition.recorder
Definition | Individual who recorded the record and takes responsibility for its content.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)
Summary | true
Condition.asserter
Element Id | Condition.asserter
Definition | Individual who is making the condition statement.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)
Summary | true
Condition.stage
Element Id | Condition.stage
Definition | Clinical stage or grade of a condition. May include formal severity assessments.
Cardinality | 0..*
Invariants | Defined on this element con-1 Rule Stage SHALL have summary or assessment summary.exists() or assessment.exists()
Defined on this element
con-1 | Rule | Stage SHALL have summary or assessment | summary.exists() or assessment.exists()
Condition.stage.summary
Element Id | Condition.stage.summary
Definition | A simple summary of the stage such as "Stage 3". The determination of the stage is disease-specific.
Cardinality | 0..1
Terminology Binding | Condition Stage (Example)
Type | CodeableConcept
Invariants | Affect this element con-1 Rule Stage SHALL have summary or assessment summary.exists() or assessment.exists()
Affect this element
con-1 | Rule | Stage SHALL have summary or assessment | summary.exists() or assessment.exists()
Condition.stage.assessment
Element Id | Condition.stage.assessment
Definition | Reference to a formal record of the evidence on which the staging assessment is based.
Cardinality | 0..*
Type | Reference(ClinicalImpression | DiagnosticReport | Observation)
Invariants | Affect this element con-1 Rule Stage SHALL have summary or assessment summary.exists() or assessment.exists()
Affect this element
con-1 | Rule | Stage SHALL have summary or assessment | summary.exists() or assessment.exists()
Condition.stage.type
Element Id | Condition.stage.type
Definition | The kind of staging, such as pathological or clinical staging.
Cardinality | 0..1
Terminology Binding | Condition Stage Type (Example)
Type | CodeableConcept
Condition.evidence
Element Id | Condition.evidence
Definition | Supporting evidence / manifestations that are the basis of the Condition's verification status, such as evidence that confirmed or refuted the condition.
Cardinality | 0..*
Comments | The evidence may be a simple list of coded symptoms/manifestations, or references to observations or formal assessments, or both.
Invariants | Defined on this element con-2 Rule evidence SHALL have code or details code.exists() or detail.exists()
Defined on this element
con-2 | Rule | evidence SHALL have code or details | code.exists() or detail.exists()
Condition.evidence.code
Element Id | Condition.evidence.code
Definition | A manifestation or symptom that led to the recording of this condition.
Cardinality | 0..*

Terminology Binding | Manifestation and Symptom Codes (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element con-2 Rule evidence SHALL have code or details code.exists() or detail.exists()
Affect this element
con-2 | Rule | evidence SHALL have code or details | code.exists() or detail.exists()
Condition.evidence.detail
Element Id | Condition.evidence.detail
Definition | Links to other relevant information, including pathology reports.
Cardinality | 0..*
Type | Reference(Any)
Summary | true
Invariants | Affect this element con-2 Rule evidence SHALL have code or details code.exists() or detail.exists()
Affect this element
con-2 | Rule | evidence SHALL have code or details | code.exists() or detail.exists()
Condition.note
Element Id | Condition.note
Definition | Additional information about the Condition. This is a general notes/comments entry for description of the Condition, its diagnosis and prognosis.
Cardinality | 0..*
Type | Annotation

9.2.7 Resource Condition - Mappings

URL: https://hl7.org/fhir/R4/condition-mappings.html

9.2.7 Resource Condition - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

SNOMED CT Attribute Binding

SNOMED CT Concept Domain Binding

RIM Mapping

Mappings for the condition resource (see Mappings to Other Standards for further information & status).

9.2.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Condition
identifier | FiveWs.identifier
clinicalStatus | FiveWs.status
verificationStatus | FiveWs.status
category | FiveWs.class
severity | FiveWs.grade
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
onset[x] | FiveWs.init
abatement[x] | FiveWs.done[x]
recordedDate | FiveWs.recorded
recorder | FiveWs.author
asserter | FiveWs.source
code | FiveWs.why[x]
detail | FiveWs.why[x]

9.2.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Condition | Event
identifier | Event.identifier
clinicalStatus | Event.status
verificationStatus | Event.status
code | Event.code
subject | Event.subject
encounter | Event.context
onset[x] | Event.occurrence[x]
code | Event.reasonCode
note | Event.note

9.2.7.3 HL7 v2 Mapping (http://hl7.org/v2)

Condition | PPR message
identifier
clinicalStatus | PRB-14
verificationStatus | PRB-13
category | 'problem' if from PRB-3. 'diagnosis' if from DG1 segment in PV1 message
severity | PRB-26 / ABS-3
code | PRB-3
bodySite
subject | PID-3
encounter | PV1-19 (+PV1-54)
onset[x] | PRB-16
abatement[x]
recordedDate | REL-11
recorder
asserter | REL-7.1 identifier + REL-7.12 type code
stage
summary | PRB-14
assessment
type
evidence
code
detail
note | NTE child of PRB

9.2.7.4 SNOMED CT Attribute Binding (http://snomed.org/attributebinding)

Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support:

• clarifying the intended meaning of the data element

• Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model

• Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element

Condition
identifier
clinicalStatus
verificationStatus | 408729009
category
severity | 246112005
code | 246090004
bodySite | 363698007
subject
encounter
onset[x]
abatement[x]
recordedDate
recorder
asserter
stage
summary
assessment
type
evidence
code
detail
note

9.2.7.5 SNOMED CT Concept Domain Binding (http://snomed.info/conceptdomain)

Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support:

• Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances

• Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains

Condition | < 243796009 |Situation with explicit context| : 246090004 |Associated finding| = ((< 404684003 |Clinical finding| MINUS (<< 420134006 |Propensity to adverse reactions| OR << 473010000 |Hypersensitivity condition| OR << 79899007 |Drug interaction| OR << 69449002 |Drug action| OR << 441742003 |Evaluation finding| OR << 307824009 |Administrative status| OR << 385356007 |Tumor stage finding|)) OR < 272379006 |Event|)
identifier
clinicalStatus | < 303105007 |Disease phases|
verificationStatus | < 410514004 |Finding context value|
category | < 404684003 |Clinical finding|
severity | < 272141005 |Severities|
code | code 246090004 |Associated finding| (< 404684003 |Clinical finding| MINUS << 420134006 |Propensity to adverse reactions| MINUS << 473010000 |Hypersensitivity condition| MINUS << 79899007 |Drug interaction| MINUS << 69449002 |Drug action| MINUS << 441742003 |Evaluation finding| MINUS << 307824009 |Administrative status| MINUS << 385356007 |Tumor stage finding|) OR < 413350009 |Finding with explicit context| OR < 272379006 |Event|
bodySite | < 442083009 |Anatomical or acquired body structure|
subject
encounter
onset[x]
abatement[x]
recordedDate
recorder
asserter
stage
summary | < 254291000 |Staging and scales|
assessment
type
evidence
code | < 404684003 |Clinical finding|
detail
note

9.2.7.6 RIM Mapping (http://hl7.org/v3)

Condition | Observation[classCode=OBS, moodCode=EVN, code=ASSERTION, value<Diagnosis]
identifier | .id
clinicalStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="clinicalStatus", moodCode=EVN].value
verificationStatus | Observation ACT .inboundRelationship[typeCode=COMP].source[classCode=OBS, code="verificationStatus", moodCode=EVN].value
category | .code
severity | Can be pre/post-coordinated into value. Or ./inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="severity"].value
code | .value
bodySite | .targetBodySiteCode
subject | .participation[typeCode=SBJ].role[classCode=PAT]
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
onset[x] | .effectiveTime.low or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age at onset"].value
abatement[x] | .effectiveTime.high or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age at remission"].value or .inboundRelationship[typeCode=SUBJ]source[classCode=CONC, moodCode=EVN].status=completed
recordedDate | .participation[typeCode=AUT].time
recorder | .participation[typeCode=AUT].role
asserter | .participation[typeCode=INF].role
stage | ./inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="stage/grade"]
summary | .value
assessment | .self
type | ./inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="stage type"]
evidence | .outboundRelationship[typeCode=SPRT].target[classCode=OBS, moodCode=EVN]
code | [code="diagnosis"].value
detail | .self
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

9.2.8 Resource Condition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/condition-profiles.html

9.2.8 Resource Condition - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Condition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.2.8.1 Profiles

No Profiles defined for this resource

9.2.8.2 Extensions

condition-assertedDate | Condition | assertedDate | for Condition HL7 Extensions
condition-dueTo | Condition | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | Condition | occurredFollowing | for Condition HL7 Extensions
condition-outcome | Condition | outcome | for Condition HL7 Extensions
condition-related | Condition | related | for Condition HL7 Extensions
condition-ruledOut | Condition | ruledOut | for Condition HL7 Extensions
event-basedOn | Condition | basedOn | for Event Pattern HL7 Extensions
event-partOf | Condition | partOf | for Event Pattern HL7 Extensions
workflow-episodeOfCare | Condition | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-instantiatesCanonical | Condition | instantiatesCanonical | for Workflow Pattern HL7 Extensions
workflow-instantiatesUri | Condition | instantiatesUri | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

DiagnosticReport-geneticsAssessedCondition | AssessedCondition | for DiagnosticReport-Genetics
condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
condition-related | related | for Condition HL7 Extensions
condition-ruledOut | ruledOut | for Condition HL7 Extensions
workflow-reasonReference | reasonReference | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.2.8.3 Search Extensions

No Search Extensions defined for this resource

9.2.9 Resource Condition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/condition-version-maps.html

No validation errors - all conversions are clean

9.2.9 Resource Condition - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Condition.

Functional status for this map: 12 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

9.2.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Condition3to4" = "R3 to R4 Conversions for Condition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Condition" alias ConditionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Condition" alias Condition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Condition(source src : ConditionR3, target tgt : Condition) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.clinicalStatus as vs ->  tgt.clinicalStatus as vt,  vt.coding as c,  c.system =
    'http://terminology.hl7.org/CodeSystem/condition-clinical',  c.code = vs;
  src.verificationStatus as vs ->  tgt.verificationStatus as vt,  vt.coding as c,  c.system =
    'http://terminology.hl7.org/CodeSystem/condition-ver-status',  c.code = vs;
  src.category -> tgt.category;
  src.severity -> tgt.severity;
  src.code -> tgt.code;
  src.bodySite -> tgt.bodySite;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.onset : dateTime as vs -> tgt.onset = create('dateTime') as vt then dateTime(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.abatement : dateTime as vs -> tgt.abatement = create('dateTime') as vt then dateTime(vs, vt);
```

```
  src.abatement : Age as vs -> tgt.abatement = create('Age') as vt then Age(vs, vt);
  src.abatement : Period as vs -> tgt.abatement = create('Period') as vt then Period(vs, vt);
  src.abatement : Range as vs -> tgt.abatement = create('Range') as vt then Range(vs, vt);
  src.abatement : string as vs -> tgt.abatement = create('string') as vt then string(vs, vt);
  src.assertedDate -> tgt.recordedDate;
  src.asserter -> tgt.asserter;
  src.stage as s -> tgt.stage as t then stage(s, t);
  src.evidence as s -> tgt.evidence as t then evidence(s, t);
  src.note -> tgt.note;
}

group stage(source src, target tgt) extends BackboneElement {
  src.summary -> tgt.summary;
  src.assessment -> tgt.assessment;
  src.type -> tgt.type;
}

group evidence(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.detail -> tgt.detail;
}
```

9.2.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Condition4to3" = "R4 to R3 Conversion for Condition"

uses "http://hl7.org/fhir/StructureDefinition/Condition" alias Condition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Condition" alias ConditionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Condition(source src : Condition, target tgt : ConditionR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.clinicalStatus as vs then {
    vs.coding as c where system = 'http://terminology.hl7.org/CodeSystem/condition-clinical' then {
      c.code -> tgt.clinicalStatus;
    };
  };
  src.verificationStatus as vs then {
    vs.coding as c where system = 'http://terminology.hl7.org/CodeSystem/condition-ver-status' then {
      c.code -> tgt.verificationStatus;
    };
  };
  src.category -> tgt.category;
  src.severity -> tgt.severity;
  src.code -> tgt.code;
  src.bodySite -> tgt.bodySite;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.onset : dateTime as vs -> tgt.onset = create('dateTime') as vt then dateTime(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.abatement : dateTime as vs -> tgt.abatement = create('dateTime') as vt then dateTime(vs, vt);
  src.abatement : Age as vs -> tgt.abatement = create('Age') as vt then Age(vs, vt);
  src.abatement : Period as vs -> tgt.abatement = create('Period') as vt then Period(vs, vt);
  src.abatement : Range as vs -> tgt.abatement = create('Range') as vt then Range(vs, vt);
  src.abatement : string as vs -> tgt.abatement = create('string') as vt then string(vs, vt);
  src.recordedDate -> tgt.assertedDate;
  src.recorder -> tgt.recorder;
  src.asserter -> tgt.asserter;
  src.stage as s -> tgt.stage as t then stage(s, t);
  src.evidence as s -> tgt.evidence as t then evidence(s, t);
  src.note -> tgt.note;
}

group stage(source src, target tgt) extends BackboneElement {
  src.summary -> tgt.summary;
  src.assessment -> tgt.assessment;
  src.type -> tgt.type;
}

group evidence(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.detail -> tgt.detail;
}
```

9.2.9.3 R4 Validation Errors

9.2.10 Condition HL7 Extensions

URL: https://hl7.org/fhir/R4/condition-extensions.html

Defines common extensions used with or related to the Condition resource

9.2.10 Condition HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.2.10.1 Content

Extensions:

condition-dueTo | dueTo : Further conditions, problems, diagnoses, procedures or events or the substance that caused/triggered this Condition.

condition-occurredFollowing | occurredFollowing : Further conditions, problems, diagnoses, procedures or events or the substance that preceded this Condition.

condition-outcome | outcome : A result of the condition. The "Cause of death" for a patient is typically captured as an Observation. The "outcome" doesn't imply causality. Some outcomes might not be assessable until the condition.clinicalStatus is no longer active.

condition-ruledOut | ruledOut : Identifies what potential diagnoses have been ruled out for this condition.

condition-related | related : This condition has an unspecified relationship with another condition.

condition-assertedDate | assertedDate : The date on which the existence of the Condition was first asserted or acknowledged.

9.2.11 JSON Schema for Condition

URL: https://hl7.org/fhir/R4/condition.schema.json.html

9.2.11 JSON Schema for Condition

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Condition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Condition",
  "$ref": "#/definitions/Condition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Condition": {
      "description": "A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept
    that has risen to a level of concern.",
      "properties": {
        "resourceType": {
          "description": "This is a Condition resource",
          "const": "Condition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this condition by the performer or other systems which remain
 constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "clinicalStatus": {
      "description": "The clinical status of the condition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "verificationStatus": {
      "description": "The verification status to support the clinical status of the condition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "A category assigned to the condition.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "severity": {
      "description": "A subjective assessment of the severity of the condition as evaluated by the clinician.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "code": {
      "description": "Identification of the condition, problem or diagnosis.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "bodySite": {
      "description": "The anatomical location where this condition manifests itself.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "subject": {
      "description": "Indicates the patient or group who the condition record is associated with.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this Condition was created or to which the creation of this record
is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onsetDateTime": {
      "description": "Estimated or actual date or date-time  the condition began, in the opinion of the clinician.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_onsetDateTime": {
      "description": "Extensions for onsetDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "onsetAge": {
      "description": "Estimated or actual date or date-time  the condition began, in the opinion of the clinician.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "onsetPeriod": {
      "description": "Estimated or actual date or date-time  the condition began, in the opinion of the clinician.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "onsetRange": {
      "description": "Estimated or actual date or date-time  the condition began, in the opinion of the clinician.",
      "$ref": "Range.schema.json#/definitions/Range"
```

```
    },
    "onsetString": {
      "description": "Estimated or actual date or date-time  the condition began, in the opinion of the clinician.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_onsetString": {
      "description": "Extensions for onsetString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "abatementDateTime": {
      "description": "The date or estimated date that the condition resolved or went into remission. This is called
\"abatement\" because of the many overloaded connotations associated with \"remission\" or \"resolution\" -
Conditions are never really resolved, but they can abate.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_abatementDateTime": {
      "description": "Extensions for abatementDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "abatementAge": {
      "description": "The date or estimated date that the condition resolved or went into remission. This is called
\"abatement\" because of the many overloaded connotations associated with \"remission\" or \"resolution\" -
Conditions are never really resolved, but they can abate.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "abatementPeriod": {
      "description": "The date or estimated date that the condition resolved or went into remission. This is called
\"abatement\" because of the many overloaded connotations associated with \"remission\" or \"resolution\" -
Conditions are never really resolved, but they can abate.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "abatementRange": {
      "description": "The date or estimated date that the condition resolved or went into remission. This is called
\"abatement\" because of the many overloaded connotations associated with \"remission\" or \"resolution\" -
Conditions are never really resolved, but they can abate.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "abatementString": {
      "description": "The date or estimated date that the condition resolved or went into remission. This is called
\"abatement\" because of the many overloaded connotations associated with \"remission\" or \"resolution\" -
Conditions are never really resolved, but they can abate.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_abatementString": {
      "description": "Extensions for abatementString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recordedDate": {
      "description": "The recordedDate represents when this particular Condition record was created in the system,
which is often a system-generated date.",
      "$ref": "#/definitions/dateTime"
    },
    "_recordedDate": {
      "description": "Extensions for recordedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recorder": {
      "description": "Individual who recorded the record and takes responsibility for its content.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "asserter": {
      "description": "Individual who is making the condition statement.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "stage": {
      "description": "Clinical stage or grade of a condition. May include formal severity assessments.",
      "items": {
        "$ref": "#/definitions/Condition_Stage"
      },
      "type": "array"
    },
    "evidence": {
      "description": "Supporting evidence / manifestations that are the basis of the Condition\u0027s verification
status, such as evidence that confirmed or refuted the condition.",
      "items": {
        "$ref": "#/definitions/Condition_Evidence"
      },
      "type": "array"
    },
    "note": {
      "description": "Additional information about the Condition. This is a general notes/comments entry  for
description of the Condition, its diagnosis and prognosis.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
```

```
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"Condition_Stage": {
  "description": "A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept
that has risen to a level of concern.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "summary": {
      "description": "A simple summary of the stage such as \"Stage 3\". The determination of the stage is disease-
specific.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "assessment": {
      "description": "Reference to a formal record of the evidence on which the staging assessment is based.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "type": {
      "description": "The kind of staging, such as pathological or clinical staging.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false
},
"Condition_Evidence": {
  "description": "A clinical condition, problem, diagnosis, or other event, situation, issue, or clinical concept
that has risen to a level of concern.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
```

```
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "code": {
          "description": "A manifestation or symptom that led to the recording of this condition.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "detail": {
          "description": "Links to other relevant information, including pathology reports.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


9.2.12 Resource Condition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/condition-dependencies.html

9.2.12 Resource Condition - Maturity Dependency View

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for Condition FMM level 3

Name | Card. | Type | Dependency Analysis
Condition | | DomainResource
identifier | 0..* | Identifier
clinicalStatus | 0..1 | CodeableConcept
verificationStatus | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
severity | 0..1 | CodeableConcept
code | 0..1 | CodeableConcept
bodySite | 0..* | CodeableConcept
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
onset[x] | 0..1
onsetDateTime | | dateTime
onsetAge | | Age
onsetPeriod | | Period
onsetRange | | Range
onsetString | | string
abatement[x] | 0..1
abatementDateTime | | dateTime
abatementAge | | Age
abatementPeriod | | Period
abatementRange | | Range
abatementString | | string
recordedDate | 0..1 | dateTime
recorder | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
asserter | 0..1 | Reference(Practitioner | PractitionerRole | Patient | RelatedPerson) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use)
stage | 0..* | BackboneElement
summary | 0..1 | CodeableConcept
assessment | 0..* | Reference(ClinicalImpression | DiagnosticReport | Observation) | Type Warning: (ClinicalImpression = FMM0-Trial Use vs. Element = FMM3-Trial Use)
type | 0..1 | CodeableConcept
evidence | 0..* | BackboneElement
code | 0..* | CodeableConcept
detail | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM3-Trial Use)
note | 0..* | Annotation

9.2.13 Value Set Sc-valueset-condition-ver-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-condition-ver-status.html

9.2.13 Value Set Sc-valueset-condition-ver-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.2.14 ConditionVerificationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-condition-ver-status)

Mapping from http://hl7.org/fhir/ValueSet/condition-ver-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The verification status to support or decline the clinical status of the condition or diagnosis."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
unknown | equivalent | unknown
unconfirmed, provisional | equivalent | unconfirmed
confirmed | equivalent | confirmed
refuted | equivalent | refuted
differential | equivalent | differential

9.3 Resource Procedure - Content

URL: https://hl7.org/fhir/R4/procedure.html
Patient Care Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

An action that is or was performed on or for a patient. This can be a physical intervention like an operation, or less invasive like long term services, counseling, or hypnotherapy.

9.3.1 Scope and Usage

Procedure is one of the event resources in the FHIR workflow specification.

This resource is used to record the details of current and historical procedures performed on or for a patient. A procedure is an activity that is performed on, with, or for a patient as part of the provision of care. Examples include surgical procedures, diagnostic procedures, endoscopic procedures, biopsies, counseling, physiotherapy, personal support services, adult day care services, non-emergency transportation, home modification, exercise, etc. Procedures may be performed by a healthcare professional, a service provider, a friend or relative or in some cases by the patient themselves.

This resource provides summary information about the occurrence of the procedure and is not intended to provide real-time snapshots of a procedure as it unfolds, though for long-running procedures such as psychotherapy, it could represent summary level information about overall progress. The creation of a resource to support detailed real-time procedure information awaits the identification of a specific implementation use-case to share such information.

9.3.2 Boundaries and Relationships

The Procedure resource should not be used to capture an event if a more specific resource already exists - i.e. immunizations, drug administrations and communications. The boundary between determining whether an action is a Procedure (training or counseling) as opposed to a Communication is based on whether there's a specific intent to change the mind-set of the patient. Mere disclosure of information would be considered a Communication. A process that involves verification of the patient's comprehension or to change the patient's mental state would be a Procedure.

Note that many diagnostic processes are procedures that generate Observations and DiagnosticReports. In many cases, such an observation does not require an explicit representation of the procedure used to create the observation, but where there are details of interest about how the diagnostic procedure was performed, the Procedure resource is used to describe the activity.

Some diagnostic procedures might not have a Procedure record. The Procedure record is only necessary when there is a need to capture information about the physical intervention that was performed to capture the diagnostic information (e.g. anesthetic, incision, scope size, etc.)

A Task is a workflow step such as cancelling an order, fulfilling an order, signing an order, merging a set of records, admitting a patient. Procedures are actions that are intended to result in a physical or mental change to or for the subject (e.g. surgery, physiotherapy, training, counseling). A Task resource often exists in parallel with clinical resources. For example, a Task might request fulfillment of a ServiceRequest ordering a Procedure.

This resource is referenced by AdverseEvent, Appointment, ChargeItem, Claim, DeviceUseStatement, Encounter, ExplanationOfBenefit, Flag, ImagingStudy, MedicationAdministration, MedicationDispense,

MedicationStatement, Observation, itself and QuestionnaireResponse

9.3.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Procedure | TU | | DomainResource | An action that is being or was performed on a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Identifiers for this procedure
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest) | A request for this procedure
partOf | Σ | 0..* | Reference(Procedure | Observation | MedicationAdministration) | Part of referenced event
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status Procedure Not Performed Reason (SNOMED-CT) (Example)
category | Σ | 0..1 | CodeableConcept | Classification of the procedure Procedure Category Codes (SNOMED CT) (Example)
code | Σ | 0..1 | CodeableConcept | Identification of the procedure Procedure Codes (SNOMED CT) (Example)
subject | Σ | 1..1 | Reference(Patient | Group) | Who the procedure was performed on
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
performed[x] | Σ | 0..1 | | When the procedure was performed
performedDateTime | | | dateTime
performedPeriod | | | Period
performedString | | | string
performedAge | | | Age
performedRange | | | Range
recorder | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Who recorded the procedure
asserter | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Person who asserts this procedure
performer | Σ | 0..* | BackboneElement | The people who performed the procedure
function | Σ | 0..1 | CodeableConcept | Type of performance Procedure Performer Role Codes (Example)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | The reference to the practitioner
onBehalfOf | | 0..1 | Reference(Organization) | Organization the device or practitioner was acting for
location | Σ | 0..1 | Reference(Location) | Where the procedure happened
reasonCode | Σ | 0..* | CodeableConcept | Coded reason procedure performed Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | Procedure | DiagnosticReport | DocumentReference) | The justification that the procedure was performed
bodySite | Σ | 0..* | CodeableConcept | Target body sites SNOMED CT Body Structures (Example)
outcome | Σ | 0..1 | CodeableConcept | The result of procedure Procedure Outcome Codes (SNOMED CT) (Example)
report | | 0..* | Reference(DiagnosticReport | DocumentReference | Composition) | Any report resulting from the procedure
complication | | 0..* | CodeableConcept | Complication following the procedure Condition/Problem/Diagnosis Codes (Example)
complicationDetail | | 0..* | Reference(Condition) | A condition that is a result of the procedure
followUp | | 0..* | CodeableConcept | Instructions for follow up Procedure Follow up Codes (SNOMED CT) (Example)
note | | 0..* | Annotation | Additional information about the procedure
focalDevice | | 0..* | BackboneElement | Manipulated, implanted, or removed device
action | | 0..1 | CodeableConcept | Kind of change to device Procedure Device Action Codes (Preferred)
manipulated | | 1..1 | Reference(Device) | Device that was changed
usedReference | | 0..* | Reference(Device | Medication | Substance) | Items used during procedure
usedCode | | 0..* | CodeableConcept | Coded items used during the procedure FHIR Device Types (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Procedure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifiers for this procedure --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest) A request for this procedure --></basedOn>
 <partOf><!-- 0..* Reference(Procedure|Observation|MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
```

```
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..1 CodeableConcept Classification of the procedure --></category>
 <code><!-- 0..1 CodeableConcept Identification of the procedure --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who the procedure was performed on --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <performed[x]><!-- 0..1 dateTime|Period|string|Age|Range When the procedure was performed --></performed[x]>
 <recorder><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Who recorded the procedure --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Person who asserts this procedure --></asserter>
 <performer>  <!-- 0..* The people who performed the procedure -->
  <function><!-- 0..1 CodeableConcept Type of performance --></function>
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
    RelatedPerson|Device) The reference to the practitioner --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization the device or practitioner was acting for --></onBehalfOf>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the procedure happened --></location>
 <reasonCode><!-- 0..* CodeableConcept Coded reason procedure performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|Procedure|
   DiagnosticReport|DocumentReference) The justification that the procedure was performed --></reasonReference>
 <bodySite><!-- 0..* CodeableConcept Target body sites --></bodySite>
 <outcome><!-- 0..1 CodeableConcept The result of procedure --></outcome>
 <report><!-- 0..* Reference(DiagnosticReport|DocumentReference|Composition) Any report resulting from the procedure
    --></report>
 <complication><!-- 0..* CodeableConcept Complication following the procedure --></complication>
 <complicationDetail><!-- 0..* Reference(Condition) A condition that is a result of the procedure
    --></complicationDetail>
 <followUp><!-- 0..* CodeableConcept Instructions for follow up --></followUp>
 <note><!-- 0..* Annotation Additional information about the procedure --></note>
 <focalDevice>  <!-- 0..* Manipulated, implanted, or removed device -->
  <action><!-- 0..1 CodeableConcept Kind of change to device --></action>
  <manipulated><!-- 1..1 Reference(Device) Device that was changed --></manipulated>
 </focalDevice>
 <usedReference><!-- 0..* Reference(Device|Medication|Substance) Items used during procedure --></usedReference>
 <usedCode><!-- 0..* CodeableConcept Coded items used during the procedure --></usedCode>
</Procedure>
```

JSON Template

```
{
  "resourceType" : "Procedure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifiers for this procedure
  "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|ServiceRequest) }], // A request for this procedure
  "partOf" : [{ Reference(Procedure|Observation|MedicationAdministration) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : { CodeableConcept }, // Classification of the procedure
  "code" : { CodeableConcept }, // Identification of the procedure
  "subject" : { Reference(Patient|Group) }, // R!  Who the procedure was performed on
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // performed[x]: When the procedure was performed. One of these 5:
  "performedDateTime" : "<dateTime>",
  "performedPeriod" : { Period },
  "performedString" : "<string>",
  "performedAge" : { Age },
  "performedRange" : { Range },
  "recorder" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Who recorded the procedure
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Person who asserts this procedure
  "performer" : [{ // The people who performed the procedure
    "function" : { CodeableConcept }, // Type of performance
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
    RelatedPerson|Device) }, // R!  The reference to the practitioner
    "onBehalfOf" : { Reference(Organization) } // Organization the device or practitioner was acting for
  }],
  "location" : { Reference(Location) }, // Where the procedure happened
  "reasonCode" : [{ CodeableConcept }], // Coded reason procedure performed
  "reasonReference" : [{ Reference(Condition|Observation|Procedure|
   DiagnosticReport|DocumentReference) }], // The justification that the procedure was performed
  "bodySite" : [{ CodeableConcept }], // Target body sites
  "outcome" : { CodeableConcept }, // The result of procedure
  "report" : [{ Reference(DiagnosticReport|DocumentReference|Composition) }], // Any report resulting from the procedure
  "complication" : [{ CodeableConcept }], // Complication following the procedure
  "complicationDetail" : [{ Reference(Condition) }], // A condition that is a result of the procedure
  "followUp" : [{ CodeableConcept }], // Instructions for follow up
  "note" : [{ Annotation }], // Additional information about the procedure
  "focalDevice" : [{ // Manipulated, implanted, or removed device
    "action" : { CodeableConcept }, // Kind of change to device
    "manipulated" : { Reference(Device) } // R!  Device that was changed
  }],
  "usedReference" : [{ Reference(Device|Medication|Substance) }], // Items used during procedure
  "usedCode" : [{ CodeableConcept }] // Coded items used during the procedure
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Procedure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Procedure.identifier [ Identifier ], ... ; # 0..* External Identifiers for this procedure
  fhir:Procedure.instantiatesCanonical [
    canonical(PlanDefinition|ActivityDefinition|Measure|OperationDefinition|Questionnaire) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:Procedure.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:Procedure.basedOn [ Reference(CarePlan|ServiceRequest) ], ... ; # 0..* A request for this procedure
  fhir:Procedure.partOf [ Reference(Procedure|Observation|MedicationAdministration) ], ... ; # 0..* Part of referenced
    event
  fhir:Procedure.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-
    in-error | unknown
  fhir:Procedure.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Procedure.category [ CodeableConcept ]; # 0..1 Classification of the procedure
  fhir:Procedure.code [ CodeableConcept ]; # 0..1 Identification of the procedure
  fhir:Procedure.subject [ Reference(Patient|Group) ]; # 1..1 Who the procedure was performed on
  fhir:Procedure.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # Procedure.performed[x] : 0..1 When the procedure was performed. One of these 5
    fhir:Procedure.performedDateTime [ dateTime ]
    fhir:Procedure.performedPeriod [ Period ]
    fhir:Procedure.performedString [ string ]
    fhir:Procedure.performedAge [ Age ]
    fhir:Procedure.performedRange [ Range ]
  fhir:Procedure.recorder [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Who recorded the
    procedure
  fhir:Procedure.asserter [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Person who asserts
    this procedure
  fhir:Procedure.performer [ # 0..* The people who performed the procedure
    fhir:Procedure.performer.function [ CodeableConcept ]; # 0..1 Type of performance
    fhir:Procedure.performer.actor [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device)
    ]; # 1..1 The reference to the practitioner
    fhir:Procedure.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization the device or practitioner was
    acting for
  ], ...;
  fhir:Procedure.location [ Reference(Location) ]; # 0..1 Where the procedure happened
  fhir:Procedure.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason procedure performed
  fhir:Procedure.reasonReference [ Reference(Condition|Observation|Procedure|DiagnosticReport|DocumentReference) ], ...
    ; # 0..* The justification that the procedure was performed
  fhir:Procedure.bodySite [ CodeableConcept ], ... ; # 0..* Target body sites
  fhir:Procedure.outcome [ CodeableConcept ]; # 0..1 The result of procedure
  fhir:Procedure.report [ Reference(DiagnosticReport|DocumentReference|Composition) ], ... ; # 0..* Any report resulting
     from the procedure
  fhir:Procedure.complication [ CodeableConcept ], ... ; # 0..* Complication following the procedure
  fhir:Procedure.complicationDetail [ Reference(Condition) ], ... ; # 0..* A condition that is a result of the procedure
  fhir:Procedure.followUp [ CodeableConcept ], ... ; # 0..* Instructions for follow up
  fhir:Procedure.note [ Annotation ], ... ; # 0..* Additional information about the procedure
  fhir:Procedure.focalDevice [ # 0..* Manipulated, implanted, or removed device
    fhir:Procedure.focalDevice.action [ CodeableConcept ]; # 0..1 Kind of change to device
    fhir:Procedure.focalDevice.manipulated [ Reference(Device) ]; # 1..1 Device that was changed
  ], ...;
  fhir:Procedure.usedReference [ Reference(Device|Medication|Substance) ], ... ; # 0..* Items used during procedure
  fhir:Procedure.usedCode [ CodeableConcept ], ... ; # 0..* Coded items used during the procedure
]
```

Changes since R3

Procedure
Procedure.instantiatesCanonical | Added Element
Procedure.instantiatesUri | Added Element
Procedure.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Procedure.statusReason | Added Element
Procedure.encounter | Added Element
Procedure.performed[x] | Add Types string, Age, Range
Procedure.recorder | Added Element
Procedure.asserter | Added Element
Procedure.performer.function | Added Element
Procedure.performer.actor | Type Reference: Added Target Type PractitionerRole
Procedure.reasonReference | Type Reference: Added Target Types Procedure, DiagnosticReport, DocumentReference
Procedure.report | Type Reference: Added Target Types DocumentReference, Composition
Procedure.definition | deleted
Procedure.notDone | deleted
Procedure.notDoneReason | deleted
Procedure.context | deleted
Procedure.performer.role | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 15 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Procedure | TU | | DomainResource | An action that is being or was performed on a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Identifiers for this procedure
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan | ServiceRequest) | A request for this procedure
partOf | Σ | 0..* | Reference(Procedure | Observation | MedicationAdministration) | Part of referenced event
status | ?!Σ | 1..1 | code | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown EventStatus (Required)
statusReason | Σ | 0..1 | CodeableConcept | Reason for current status Procedure Not Performed Reason (SNOMED-CT) (Example)
category | Σ | 0..1 | CodeableConcept | Classification of the procedure Procedure Category Codes (SNOMED CT) (Example)
code | Σ | 0..1 | CodeableConcept | Identification of the procedure Procedure Codes (SNOMED CT) (Example)
subject | Σ | 1..1 | Reference(Patient | Group) | Who the procedure was performed on
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
performed[x] | Σ | 0..1 | | When the procedure was performed
performedDateTime | | | dateTime
performedPeriod | | | Period
performedString | | | string
performedAge | | | Age
performedRange | | | Range
recorder | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Who recorded the procedure
asserter | Σ | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Person who asserts this procedure
performer | Σ | 0..* | BackboneElement | The people who performed the procedure
function | Σ | 0..1 | CodeableConcept | Type of performance Procedure Performer Role Codes (Example)
actor | Σ | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | The reference to the practitioner
onBehalfOf | | 0..1 | Reference(Organization) | Organization the device or practitioner was acting for
location | Σ | 0..1 | Reference(Location) | Where the procedure happened
reasonCode | Σ | 0..* | CodeableConcept | Coded reason procedure performed Procedure Reason Codes (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | Procedure | DiagnosticReport | DocumentReference) | The justification that the procedure was performed
bodySite | Σ | 0..* | CodeableConcept | Target body sites SNOMED CT Body Structures (Example)
outcome | Σ | 0..1 | CodeableConcept | The result of procedure Procedure Outcome Codes (SNOMED CT) (Example)
report | | 0..* | Reference(DiagnosticReport | DocumentReference | Composition) | Any report resulting from the procedure
complication | | 0..* | CodeableConcept | Complication following the procedure Condition/Problem/Diagnosis Codes (Example)
complicationDetail | | 0..* | Reference(Condition) | A condition that is a result of the procedure
followUp | | 0..* | CodeableConcept | Instructions for follow up Procedure Follow up Codes (SNOMED CT) (Example)
note | | 0..* | Annotation | Additional information about the procedure
focalDevice | | 0..* | BackboneElement | Manipulated, implanted, or removed device
action | | 0..1 | CodeableConcept | Kind of change to device Procedure Device Action Codes (Preferred)
manipulated | | 1..1 | Reference(Device) | Device that was changed
usedReference | | 0..* | Reference(Device | Medication | Substance) | Items used during procedure
usedCode | | 0..* | CodeableConcept | Coded items used during the procedure FHIR Device Types (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Procedure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Identifiers for this procedure --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan|ServiceRequest) A request for this procedure --></basedOn>
 <partOf><!-- 0..* Reference(Procedure|Observation|MedicationAdministration) Part of referenced event --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-
    error | unknown -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <category><!-- 0..1 CodeableConcept Classification of the procedure --></category>
 <code><!-- 0..1 CodeableConcept Identification of the procedure --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who the procedure was performed on --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <performed[x]><!-- 0..1 dateTime|Period|string|Age|Range When the procedure was performed --></performed[x]>
 <recorder><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Who recorded the procedure --></recorder>
 <asserter><!-- 0..1 Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) Person who asserts this procedure --></asserter>
 <performer>  <!-- 0..* The people who performed the procedure -->
  <function><!-- 0..1 CodeableConcept Type of performance --></function>
```

```
  <actor><!-- 1..1 Reference(Practitioner|PractitionerRole|Organization|Patient|
    RelatedPerson|Device) The reference to the practitioner --></actor>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization the device or practitioner was acting for --></onBehalfOf>
 </performer>
 <location><!-- 0..1 Reference(Location) Where the procedure happened --></location>
 <reasonCode><!-- 0..* CodeableConcept Coded reason procedure performed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|Procedure|
   DiagnosticReport|DocumentReference) The justification that the procedure was performed --></reasonReference>
 <bodySite><!-- 0..* CodeableConcept Target body sites --></bodySite>
 <outcome><!-- 0..1 CodeableConcept The result of procedure --></outcome>
 <report><!-- 0..* Reference(DiagnosticReport|DocumentReference|Composition) Any report resulting from the procedure
    --></report>
 <complication><!-- 0..* CodeableConcept Complication following the procedure --></complication>
 <complicationDetail><!-- 0..* Reference(Condition) A condition that is a result of the procedure
    --></complicationDetail>
 <followUp><!-- 0..* CodeableConcept Instructions for follow up --></followUp>
 <note><!-- 0..* Annotation Additional information about the procedure --></note>
 <focalDevice>  <!-- 0..* Manipulated, implanted, or removed device -->
  <action><!-- 0..1 CodeableConcept Kind of change to device --></action>
  <manipulated><!-- 1..1 Reference(Device) Device that was changed --></manipulated>
 </focalDevice>
 <usedReference><!-- 0..* Reference(Device|Medication|Substance) Items used during procedure --></usedReference>
 <usedCode><!-- 0..* CodeableConcept Coded items used during the procedure --></usedCode>
</Procedure>
```

JSON Template

```
{
  "resourceType" : "Procedure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Identifiers for this procedure
  "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
   Measure|OperationDefinition|Questionnaire) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan|ServiceRequest) }], // A request for this procedure
  "partOf" : [{ Reference(Procedure|Observation|MedicationAdministration) }], // Part of referenced event
  "status" : "<code>", // R!  preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error |
    unknown
  "statusReason" : { CodeableConcept }, // Reason for current status
  "category" : { CodeableConcept }, // Classification of the procedure
  "code" : { CodeableConcept }, // Identification of the procedure
  "subject" : { Reference(Patient|Group) }, // R!  Who the procedure was performed on
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // performed[x]: When the procedure was performed. One of these 5:
  "performedDateTime" : "<dateTime>",
  "performedPeriod" : { Period },
  "performedString" : "<string>",
  "performedAge" : { Age },
  "performedRange" : { Range },
  "recorder" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Who recorded the procedure
  "asserter" : { Reference(Patient|RelatedPerson|Practitioner|
   PractitionerRole) }, // Person who asserts this procedure
  "performer" : [{ // The people who performed the procedure
    "function" : { CodeableConcept }, // Type of performance
    "actor" : { Reference(Practitioner|PractitionerRole|Organization|Patient|
    RelatedPerson|Device) }, // R!  The reference to the practitioner
    "onBehalfOf" : { Reference(Organization) } // Organization the device or practitioner was acting for
  }],
  "location" : { Reference(Location) }, // Where the procedure happened
  "reasonCode" : [{ CodeableConcept }], // Coded reason procedure performed
  "reasonReference" : [{ Reference(Condition|Observation|Procedure|
   DiagnosticReport|DocumentReference) }], // The justification that the procedure was performed
  "bodySite" : [{ CodeableConcept }], // Target body sites
  "outcome" : { CodeableConcept }, // The result of procedure
  "report" : [{ Reference(DiagnosticReport|DocumentReference|Composition) }], // Any report resulting from the procedure
  "complication" : [{ CodeableConcept }], // Complication following the procedure
  "complicationDetail" : [{ Reference(Condition) }], // A condition that is a result of the procedure
  "followUp" : [{ CodeableConcept }], // Instructions for follow up
  "note" : [{ Annotation }], // Additional information about the procedure
  "focalDevice" : [{ // Manipulated, implanted, or removed device
    "action" : { CodeableConcept }, // Kind of change to device
    "manipulated" : { Reference(Device) } // R!  Device that was changed
  }],
  "usedReference" : [{ Reference(Device|Medication|Substance) }], // Items used during procedure
  "usedCode" : [{ CodeableConcept }] // Coded items used during the procedure
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Procedure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Procedure.identifier [ Identifier ], ... ; # 0..* External Identifiers for this procedure
  fhir:Procedure.instantiatesCanonical [
```

```
    canonical(PlanDefinition|ActivityDefinition|Measure|OperationDefinition|Questionnaire) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:Procedure.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:Procedure.basedOn [ Reference(CarePlan|ServiceRequest) ], ... ; # 0..* A request for this procedure
  fhir:Procedure.partOf [ Reference(Procedure|Observation|MedicationAdministration) ], ... ; # 0..* Part of referenced
    event
  fhir:Procedure.status [ code ]; # 1..1 preparation | in-progress | not-done | on-hold | stopped | completed | entered-
    in-error | unknown
  fhir:Procedure.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:Procedure.category [ CodeableConcept ]; # 0..1 Classification of the procedure
  fhir:Procedure.code [ CodeableConcept ]; # 0..1 Identification of the procedure
  fhir:Procedure.subject [ Reference(Patient|Group) ]; # 1..1 Who the procedure was performed on
  fhir:Procedure.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # Procedure.performed[x] : 0..1 When the procedure was performed. One of these 5
    fhir:Procedure.performedDateTime [ dateTime ]
    fhir:Procedure.performedPeriod [ Period ]
    fhir:Procedure.performedString [ string ]
    fhir:Procedure.performedAge [ Age ]
    fhir:Procedure.performedRange [ Range ]
  fhir:Procedure.recorder [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Who recorded the
    procedure
  fhir:Procedure.asserter [ Reference(Patient|RelatedPerson|Practitioner|PractitionerRole) ]; # 0..1 Person who asserts
    this procedure
  fhir:Procedure.performer [ # 0..* The people who performed the procedure
    fhir:Procedure.performer.function [ CodeableConcept ]; # 0..1 Type of performance
    fhir:Procedure.performer.actor [ Reference(Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device)
    ]; # 1..1 The reference to the practitioner
    fhir:Procedure.performer.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization the device or practitioner was
    acting for
  ], ...;
  fhir:Procedure.location [ Reference(Location) ]; # 0..1 Where the procedure happened
  fhir:Procedure.reasonCode [ CodeableConcept ], ... ; # 0..* Coded reason procedure performed
  fhir:Procedure.reasonReference [ Reference(Condition|Observation|Procedure|DiagnosticReport|DocumentReference) ], ...
    ; # 0..* The justification that the procedure was performed
  fhir:Procedure.bodySite [ CodeableConcept ], ... ; # 0..* Target body sites
  fhir:Procedure.outcome [ CodeableConcept ]; # 0..1 The result of procedure
  fhir:Procedure.report [ Reference(DiagnosticReport|DocumentReference|Composition) ], ... ; # 0..* Any report resulting
     from the procedure
  fhir:Procedure.complication [ CodeableConcept ], ... ; # 0..* Complication following the procedure
  fhir:Procedure.complicationDetail [ Reference(Condition) ], ... ; # 0..* A condition that is a result of the procedure
  fhir:Procedure.followUp [ CodeableConcept ], ... ; # 0..* Instructions for follow up
  fhir:Procedure.note [ Annotation ], ... ; # 0..* Additional information about the procedure
  fhir:Procedure.focalDevice [ # 0..* Manipulated, implanted, or removed device
    fhir:Procedure.focalDevice.action [ CodeableConcept ]; # 0..1 Kind of change to device
    fhir:Procedure.focalDevice.manipulated [ Reference(Device) ]; # 1..1 Device that was changed
  ], ...;
  fhir:Procedure.usedReference [ Reference(Device|Medication|Substance) ], ... ; # 0..* Items used during procedure
  fhir:Procedure.usedCode [ CodeableConcept ], ... ; # 0..* Coded items used during the procedure
]
```

Changes since Release 3

Procedure
Procedure.instantiatesCanonical | Added Element
Procedure.instantiatesUri | Added Element
Procedure.basedOn | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
Procedure.status | Change value set from http://hl7.org/fhir/ValueSet/event-status to http://hl7.org/fhir/ValueSet/event-status|4.0.1
Procedure.statusReason | Added Element
Procedure.encounter | Added Element
Procedure.performed[x] | Add Types string, Age, Range
Procedure.recorder | Added Element
Procedure.asserter | Added Element
Procedure.performer.function | Added Element
Procedure.performer.actor | Type Reference: Added Target Type PractitionerRole
Procedure.reasonReference | Type Reference: Added Target Types Procedure, DiagnosticReport, DocumentReference
Procedure.report | Type Reference: Added Target Types DocumentReference, Composition
Procedure.definition | deleted
Procedure.notDone | deleted
Procedure.notDoneReason | deleted
Procedure.context | deleted
Procedure.performer.role | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 15 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.3.3.1 Terminology Bindings

Path | Definition | Type | Reference

Procedure.status | A code specifying the state of the procedure. | Required | EventStatus
Procedure.statusReason | A code that identifies the reason a procedure was not performed. | Example | ProcedureNotPerformedReason(SNOMED-CT)
Procedure.category | A code that classifies a procedure for searching, sorting and display purposes. | Example | ProcedureCategoryCodes(SNOMEDCT)
Procedure.code | A code to identify a specific procedure . | Example | ProcedureCodes(SNOMEDCT)
Procedure.performer.function | A code that identifies the role of a performer of the procedure. | Example | ProcedurePerformerRoleCodes
Procedure.reasonCode | A code that identifies the reason a procedure is required. | Example | ProcedureReasonCodes
Procedure.bodySite | Codes describing anatomical locations. May include laterality. | Example | SNOMEDCTBodyStructures
Procedure.outcome | An outcome of a procedure - whether it was resolved or otherwise. | Example | ProcedureOutcomeCodes(SNOMEDCT)
Procedure.complication | Codes describing complications that resulted from a procedure. | Example | Condition/Problem/DiagnosisCodes
Procedure.followUp | Specific follow up required for a procedure e.g. removal of sutures. | Example | ProcedureFollowUpCodes(SNOMEDCT)
Procedure.focalDevice.action | A kind of change that happened to the device during the procedure. | Preferred | ProcedureDeviceActionCodes
Procedure.usedCode | Codes describing items used during a procedure. | Example | FHIRDeviceTypes

9.3.3.2 Use of Procedure properties

Many of the elements of Procedure have inherent relationships and may be conveyed by the Procedure.code or in the text element of the Procedure.code property. I.e. you may be able to infer category, bodySite and even indication. Whether these other properties will be populated may vary by implementation.

Care should be taken to avoid nonsensical combinations/statements; e.g. "name=amputation, bodySite=heart".

9.3.3.3 Use of Procedure.used

For devices, these are devices that are incidental to / or used to perform the procedure - scalpels, gauze, endoscopes, etc. Devices that are the focus of the procedure should appear in Procedure.device instead.

9.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | A request for this procedure | Procedure.basedOn (CarePlan, ServiceRequest)
category | token | Classification of the procedure | Procedure.category
code | token | A code to identify a procedure | Procedure.code | 13 Resources
date | date | When the procedure was performed | Procedure.performed | 17 Resources
encounter | reference | Encounter created as part of | Procedure.encounter (Encounter) | 12 Resources
identifier | token | A unique identifier for a procedure | Procedure.identifier | 30 Resources
instantiates-canonical | reference | Instantiates FHIR protocol or definition | Procedure.instantiatesCanonical (Questionnaire, Measure, PlanDefinition, OperationDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | Procedure.instantiatesUri
location | reference | Where the procedure happened | Procedure.location (Location)
part-of | reference | Part of referenced event | Procedure.partOf (Observation, Procedure, MedicationAdministration)
patient | reference | Search by subject - a patient | Procedure.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | The reference to the practitioner | Procedure.performer.actor (Practitioner, Organization, Device, Patient, PractitionerRole, RelatedPerson)
reason-code | token | Coded reason procedure performed | Procedure.reasonCode
reason-reference | reference | The justification that the procedure was performed | Procedure.reasonReference (Condition, Observation, Procedure, DiagnosticReport, DocumentReference)
status | token | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Procedure.status
subject | reference | Search by subject | Procedure.subject (Group, Patient)

9.3.5 Resource Procedure - Examples

URL: https://hl7.org/fhir/R4/procedure-examples.html

9.3.5 Resource Procedure - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
General Procedure Example | example | XML | JSON | Turtle
Example of a Biopsy | biopsy | XML | JSON | Turtle
Real-world procedure example | f201 | XML | JSON | Turtle
Real-world procedure example | f001 | XML | JSON | Turtle
Real-world procedure example | f002 | XML | JSON | Turtle
Real-world procedure example | f003 | XML | JSON | Turtle
Real-world procedure example | f004 | XML | JSON | Turtle

Example of a device manipulation | example-implant | XML | JSON | Turtle
Example of ambulation procedure | ambulation | XML | JSON | Turtle
Example of biopsy procedure that was part of a colonoscopy | colon-biopsy | XML | JSON | Turtle
Example of colonoscopy procedure with complication | colonoscopy | XML | JSON | Turtle
Example of an appendectomy procedure that is primarily narrative | appendectomy-narrative | XML | JSON | Turtle
Example of an obstetric procedure | ob | XML | JSON | Turtle
Example of a physical therapy evaluation procedure | physical-therapy | XML | JSON | Turtle
Example of patient education | education | XML | JSON | Turtle
Example of personal care services provided at person's home | HCBS | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.3.6 Resource Procedure - Detailed Descriptions

URL: https://hl7.org/fhir/R4/procedure-definitions.html

9.3.6 Resource Procedure - Detailed Descriptions

Patient Care Work Group | Maturity Level: 3 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Procedure resource.

Procedure
Element Id | Procedure
Definition | An action that is or was performed on or for a patient. This can be a physical intervention like an operation, or less invasive like long term services, counseling, or hypnotherapy.
Cardinality | 0..*
Type | DomainResource
Procedure.identifier
Element Id | Procedure.identifier
Definition | Business identifiers assigned to this procedure by the performer or other systems which remain constant as the resource is updated and is propagated from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the procedure as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and Person resource instances might share the same social insurance number.
Procedure.instantiatesCanonical
Element Id | Procedure.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, order set or other definition that is adhered to in whole or in part by this Procedure.
Cardinality | 0..*
Type | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire)
Summary | true
Procedure.instantiatesUri
Element Id | Procedure.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, order set or other definition that is adhered to in whole or in part by this Procedure.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
Procedure.basedOn
Element Id | Procedure.basedOn
Definition | A reference to a resource that contains details of the request for this procedure.
Cardinality | 0..*
Type | Reference(CarePlan | ServiceRequest)
Alternate Names | fulfills
Summary | true
Procedure.partOf
Element Id | Procedure.partOf
Definition | A larger event of which this particular procedure is a component or step.
Cardinality | 0..*
Type | Reference(Procedure | Observation | MedicationAdministration)
Alternate Names | container
Summary | true
Comments | The MedicationAdministration resource has a partOf reference to Procedure, but this is not a circular reference. For example, the anesthesia MedicationAdministration is part of the surgical Procedure (MedicationAdministration.partOf = Procedure). For example, the procedure to insert the IV port for an IV medication administration is part of the medication

administration (Procedure.partOf = MedicationAdministration).
Procedure.status
Element Id | Procedure.status
Definition | A code specifying the state of the procedure. Generally, this will be the in-progress or completed state.
Cardinality | 1..1
Terminology Binding | EventStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | The "unknown" code is not to be used to convey other statuses. The "unknown" code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the procedure. This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Procedure.statusReason
Element Id | Procedure.statusReason
Definition | Captures the reason for the current state of the procedure.
Cardinality | 0..1
Terminology Binding | Procedure Not Performed Reason (SNOMED-CT) (Example)
Type | CodeableConcept
Alternate Names | Suspended Reason; Cancelled Reason
Summary | true
Comments | This is generally only used for "exception" statuses such as "not-done", "suspended" or "aborted". The reason for performing the event at all is captured in reasonCode, not here.
Procedure.category
Element Id | Procedure.category
Definition | A code that classifies the procedure for searching, sorting and display purposes (e.g. "Surgical Procedure").
Cardinality | 0..1
Terminology Binding | Procedure Category Codes (SNOMED CT) (Example)
Type | CodeableConcept
Summary | true
Procedure.code
Element Id | Procedure.code
Definition | The specific procedure that is performed. Use text if the exact nature of the procedure cannot be coded (e.g. "Laparoscopic Appendectomy").
Cardinality | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | 0..1 to account for primarily narrative only resources.
Alternate Names | type
Summary | true
Procedure.subject
Element Id | Procedure.subject
Definition | The person, animal or group on which the procedure was performed.
Cardinality | 1..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
Procedure.encounter
Element Id | Procedure.encounter
Definition | The Encounter during which this Procedure was created or performed or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter.
Procedure.performed[x]
Element Id | Procedure.performed[x]
Definition | Estimated or actual date, date-time, period, or age when the procedure was performed. Allows a period to support complex procedures that span more than one date, and also allows for the length of the procedure to be captured.
Cardinality | 0..1
Type | dateTime|Period|string|Age|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Age is generally used when the patient reports an age at which the procedure was performed. Range is generally used when the patient reports an age range when the procedure was performed, such as sometime between 20-25 years old. dateTime supports a range of precision due to some procedures being reported as past procedures that might not have millisecond precision while other procedures performed and documented during the encounter might have more precise UTC timestamps with timezone.
Procedure.recorder
Element Id | Procedure.recorder
Definition | Individual who recorded the record and takes responsibility for its content.
Cardinality | 0..1

Type | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole)
Summary | true
Procedure.asserter
Element Id | Procedure.asserter
Definition | Individual who is making the procedure statement.
Cardinality | 0..1
Type | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole)
Summary | true
Procedure.performer
Element Id | Procedure.performer
Definition | Limited to "real" people rather than equipment.
Cardinality | 0..*
Summary | true
Procedure.performer.function
Element Id | Procedure.performer.function
Definition | Distinguishes the type of involvement of the performer in the procedure. For example, surgeon, anaesthetist, endoscopist.
Cardinality | 0..1
Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Requirements | Allows disambiguation of the types of involvement of different performers.
Summary | true
Procedure.performer.actor
Element Id | Procedure.performer.actor
Definition | The practitioner who was involved in the procedure.
Cardinality | 1..1
Type | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)
Requirements | A reference to Device supports use cases, such as pacemakers.
Summary | true
Procedure.performer.onBehalfOf
Element Id | Procedure.performer.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when performing the action.
Procedure.location
Element Id | Procedure.location
Definition | The location where the procedure actually happened. E.g. a newborn at home, a tracheostomy at a restaurant.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Ties a procedure to where the records are likely kept.
Summary | true
Procedure.reasonCode
Element Id | Procedure.reasonCode
Definition | The coded reason why the procedure was performed. This may be a coded entity of some type, or may simply be present as text.
Cardinality | 0..*
Terminology Binding | Procedure Reason Codes (Example)
Type | CodeableConcept
Summary | true
Comments | Use Procedure.reasonCode when a code sufficiently describes the reason. Use Procedure.reasonReference when referencing a resource, which allows more information to be conveyed, such as onset date. Procedure.reasonCode and Procedure.reasonReference are not meant to be duplicative. For a single reason, either Procedure.reasonCode or Procedure.reasonReference can be used. Procedure.reasonCode may be a summary code, or Procedure.reasonReference may be used to reference a very precise definition of the reason using Condition | Observation | Procedure | DiagnosticReport | DocumentReference. Both Procedure.reasonCode and Procedure.reasonReference can be used if they are describing different reasons for the procedure.
Procedure.reasonReference
Element Id | Procedure.reasonReference
Definition | The justification of why the procedure was performed.
Cardinality | 0..*
Type | Reference(Condition | Observation | Procedure | DiagnosticReport | DocumentReference)
Summary | true
Comments | It is possible for a procedure to be a reason (such as C-Section) for another procedure (such as an epidural). Other examples include endoscopy for dilatation and biopsy (a combination of diagnostic and therapeutic use). Use Procedure.reasonCode when a code sufficiently describes the reason. Use Procedure.reasonReference when referencing a resource, which allows more information to be conveyed, such as onset date. Procedure.reasonCode and Procedure.reasonReference are not meant to be duplicative. For a single reason, either Procedure.reasonCode or Procedure.reasonReference can be used. Procedure.reasonCode may be a summary code, or Procedure.reasonReference may be used to reference a very precise definition of the reason using Condition | Observation | Procedure | DiagnosticReport | DocumentReference. Both Procedure.reasonCode and Procedure.reasonReference can be used if they are describing different reasons for the procedure.
Procedure.bodySite

Element Id | Procedure.bodySite
Definition | Detailed and structured anatomical location information. Multiple locations are allowed - e.g. multiple punch biopsies of a lesion.
Cardinality | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension procedure-targetbodystructure.
Procedure.outcome
Element Id | Procedure.outcome
Definition | The outcome of the procedure - did it resolve the reasons for the procedure being performed?
Cardinality | 0..1
Terminology Binding | Procedure Outcome Codes (SNOMED CT) (Example)
Type | CodeableConcept
Summary | true
Comments | If outcome contains narrative text only, it can be captured using the CodeableConcept.text.
Procedure.report
Element Id | Procedure.report
Definition | This could be a histology result, pathology report, surgical report, etc.
Cardinality | 0..*
Type | Reference(DiagnosticReport | DocumentReference | Composition)
Comments | There could potentially be multiple reports - e.g. if this was a procedure which took multiple biopsies resulting in a number of anatomical pathology reports.
Procedure.complication
Element Id | Procedure.complication
Definition | Any complications that occurred during the procedure, or in the immediate post-performance period. These are generally tracked separately from the notes, which will typically describe the procedure itself rather than any 'post procedure' issues.
Cardinality | 0..*
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
Comments | If complications are only expressed by the narrative text, they can be captured using the CodeableConcept.text.
Procedure.complicationDetail
Element Id | Procedure.complicationDetail
Definition | Any complications that occurred during the procedure, or in the immediate post-performance period.
Cardinality | 0..*
Type | Reference(Condition)
Requirements | This is used to document a condition that is a result of the procedure, not the condition that was the reason for the procedure.
Procedure.followUp
Element Id | Procedure.followUp
Definition | If the procedure required specific follow up - e.g. removal of sutures. The follow up may be represented as a simple note or could potentially be more complex, in which case the CarePlan resource can be used.
Cardinality | 0..*
Terminology Binding | Procedure Follow up Codes (SNOMED CT) (Example)
Type | CodeableConcept
Procedure.note
Element Id | Procedure.note
Definition | Any other notes and comments about the procedure.
Cardinality | 0..*
Type | Annotation
Procedure.focalDevice
Element Id | Procedure.focalDevice
Definition | A device that is implanted, removed or otherwise manipulated (calibration, battery replacement, fitting a prosthesis, attaching a wound-vac, etc.) as a focal portion of the Procedure.
Cardinality | 0..*
Procedure.focalDevice.action
Element Id | Procedure.focalDevice.action
Definition | The kind of change that happened to the device during the procedure.
Cardinality | 0..1
Terminology Binding | Procedure Device Action Codes (Preferred)
Type | CodeableConcept
Procedure.focalDevice.manipulated
Element Id | Procedure.focalDevice.manipulated
Definition | The device that was manipulated (changed) during the procedure.
Cardinality | 1..1
Type | Reference(Device)
Procedure.usedReference
Element Id | Procedure.usedReference
Definition | Identifies medications, devices and any other substance used as part of the procedure.
Cardinality | 0..*
Type | Reference(Device | Medication | Substance)

Requirements | Used for tracking contamination, etc.
Comments | For devices actually implanted or removed, use Procedure.device.
Procedure.usedCode
Element Id | Procedure.usedCode
Definition | Identifies coded items that were used as part of the procedure.
Cardinality | 0..*
Terminology Binding | FHIR Device Types (Example)
Type | CodeableConcept
Comments | For devices actually implanted or removed, use Procedure.device.

9.3.7 Resource Procedure - Mappings

URL: https://hl7.org/fhir/R4/procedure-mappings.html

9.3.7 Resource Procedure - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the procedure resource (see Mappings to Other Standards for further information & status).

9.3.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Procedure
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
performed[x] | FiveWs.done[x]
recorder | FiveWs.author
asserter | FiveWs.source
actor | FiveWs.actor
location | FiveWs.where[x]
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

9.3.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Procedure | Event
identifier | Event.identifier
instantiatesCanonical | Event.instantiatesCanonical
instantiatesUri | Event.instantiatesUri
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
statusReason | Event.statusReason
code | Event.code
subject | Event.subject
encounter | Event.context
performed[x] | Event.occurrence[x]
performer | Event.performer
function | Event.performer.function
actor | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

9.3.7.3 HL7 v2 Mapping (http://hl7.org/v2)

Procedure
identifier | Some combination of ORC-2 / ORC-3 / OBR-2 / OBR-3 / IPC-1 / IPC-2 / IPC-3 / IPC-4
instantiatesCanonical
instantiatesUri
basedOn
partOf
status

statusReason
category
code | OBR-44/OBR-45
subject | PID-3
encounter | PV1-19
performed[x] | OBR-7
recorder
asserter
performer
function | Some combination of STF-18 / PRA-3 / PRT-4 / ROL-3 / ORC-12 / OBR-16 / PV1-7 / PV1-8 / PV1-9 / PV1-17 / OBX-25
actor | ORC-19/PRT-5
onBehalfOf
location
reasonCode
reasonReference
bodySite | OBX-20
outcome
report
complication
complicationDetail
followUp
note | NTE
focalDevice
action
manipulated
usedReference
usedCode

9.3.7.4 RIM Mapping (http://hl7.org/v3)

Procedure | Procedure[moodCode=EVN]
identifier | .id
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
basedOn | .outboundRelationship[typeCode=FLFS].target[classCode=(various e.g. PROC, OBS, PCPR, ACT, moodCode=RQO].code
partOf | .inboundRelationship[typeCode=COMP].source[classCode=SBADM or PROC or OBS, moodCode=EVN]
status | statusCode
statusReason | .reason.Observation.value
category | .outboundRelationship[typeCode="COMP].target[classCode="LIST", moodCode="EVN"].code
code | .code
subject | .participation[typeCode=SBJ].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
performed[x] | .effectiveTime
recorder | .participation[typeCode=AUT].role
asserter | .participation[typeCode=INF].role
performer | .participation[typeCode=PRF]
function | .functionCode
actor | .role
onBehalfOf | .scoper
location | .participation[typeCode=LOC].role[classCode=SDLOC]
reasonCode | .reasonCode
reasonReference | .reasonCode
bodySite | .targetSiteCode
outcome | .outboundRelationship[typeCode=OUT].target.text
report | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN]
complication | .outboundRelationship[typeCode=OUTC].target[classCode=OBS, code="complication", moodCode=EVN].value
complicationDetail | .outboundRelationship[typeCode=OUTC].target[classCode=OBS, code="complication", moodCode=EVN].value
followUp | .outboundRelationship[typeCode=COMP].target[classCode=ACT, moodCode=INT].code
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value
focalDevice | .participation[typeCode=DEV].role[classCode=MANU]
action | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="procedure device action"].value=:procedure device action codes
manipulated | .participation[typeCode=DEV].role[classCode=SDLOC]
usedReference | .participation[typeCode=DEV].role[classCode=MANU] or .participation[typeCode=CSM].role[classCode=ADMM] (for Medication or Substance)
usedCode | participation[typeCode=Dev].role[classCode=MANU]

9.3.8 Resource Procedure - Extensions & Profiles

URL: https://hl7.org/fhir/R4/procedure-profiles.html

9.3.8 Resource Procedure - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the Procedure resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.3.8.1 Profiles

No Profiles defined for this resource

9.3.8.2 Extensions

procedure-approachBodyStructure | Procedure | approachBodyStructure | for Procedure HL7 Extensions
procedure-causedBy | Procedure | causedBy | for Procedure HL7 Extensions
procedure-directedBy | Procedure | directedBy | for Procedure HL7 Extensions
procedure-incisionDateTime | Procedure | incisionDateTime | for Procedure HL7 Extensions
procedure-method | Procedure | method | for Procedure HL7 Extensions
procedure-progressStatus | Procedure | progressStatus | for Procedure HL7 Extensions
procedure-schedule | Procedure | schedule | for Procedure HL7 Extensions
procedure-targetBodyStructure | Procedure | targetBodyStructure | for Procedure HL7 Extensions
workflow-episodeOfCare | Procedure | episodeOfCare | for Workflow Pattern HL7 Extensions
workflow-researchStudy | Procedure | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

condition-dueTo | dueTo | for Condition HL7 Extensions
condition-occurredFollowing | occurredFollowing | for Condition HL7 Extensions
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.3.8.3 Search Extensions

No Search Extensions defined for this resource

9.3.9 Resource Procedure - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/procedure-version-maps.html

No validation errors - all conversions are clean

9.3.9 Resource Procedure - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for Procedure.

Functional status for this map: 15 tests that all execute ok. 3 fail round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

9.3.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Procedure3to4" = "R3 to R4 Conversions for Procedure"

conceptmap "EventStatus" {
  prefix s = "http://hl7.org/fhir/ValueSet/event-status"
  prefix t = "http://hl7.org/fhir/ValueSet/event-status"

  s:"'entered-in-error'" - t:"'entered-in-error'"
  s:"'in-progress'" - t:"'in-progress'"
  s:aborted - t:stopped
  s:completed - t:completed
  s:preparation - t:preparation
  s:suspended - t:suspended
  s:unknown - t:unknown
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/Procedure" alias ProcedureR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Procedure" alias Procedure as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Procedure(source src : ProcedureR3, target tgt : Procedure) extends DomainResource <<type+>> {
```

```
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.instantiatesCanonical;
  src.instantiatesUri -> tgt.instantiatesUri;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status as vs where (src.notDone = true).not() -> tgt.status = translate(vs, '#EventStatus', 'code');
  src.status where (src.notDone = true) ->  tgt.status = create('code') as vt,  vt.value = 'not-done';
  src.notDoneReason as vs -> tgt.statusReason = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context : Reference as vs -> tgt.encounter = create('Reference') as vt then Reference(vs, vt);
  src.performed -> tgt.performed;
  src.recorder -> tgt.recorder;
  src.asserter -> tgt.asserter;
  src.performer as s -> tgt.performer as t then ProcedurePerformer(s, t);
  src.location -> tgt.location;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.bodySite -> tgt.bodySite;
  src.outcome -> tgt.outcome;
  src.report -> tgt.report;
  src.complication -> tgt.complication;
  src.complicationDetail -> tgt.complicationDetail;
  src.followUp -> tgt.followUp;
  src.note -> tgt.note;
  src.focalDevice as s -> tgt.focalDevice as t then ProcedureFocalDevice(s, t);
  src.usedReference -> tgt.usedReference;
  src.usedCode -> tgt.usedCode;
}

group ProcedurePerformer(source src, target tgt) extends BackboneElement {
  src.role -> tgt.function;
  src.actor -> tgt.actor;
  src.onBehalfOf -> tgt.onBehalfOf;
}

group ProcedureFocalDevice(source src, target tgt) extends BackboneElement {
  src.action -> tgt.action;
  src.manipulated -> tgt.manipulated;
}
```

9.3.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Procedure4to3" = "R4 to R3 Conversion for Procedure"

conceptmap "EventStatus" {
  prefix s = "http://hl7.org/fhir/ValueSet/event-status"
  prefix t = "http://hl7.org/fhir/ValueSet/event-status"

  s:"'entered-in-error'" - t:"'entered-in-error'"
  s:"'in-progress'" - t:"'in-progress'"
  s:"'on-hold'" - t:suspended
  s:aborted - t:stopped
  s:completed - t:completed
  s:preparation - t:preparation
  s:stopped - t:aborted
  s:suspended - t:suspended
  s:unknown - t:unknown
}

uses "http://hl7.org/fhir/StructureDefinition/Procedure" alias Procedure as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Procedure" alias ProcedureR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Procedure(source src : ProcedureR3, target tgt : Procedure) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.instantiatesCanonical;
  src.instantiatesUri -> tgt.instantiatesUri;
  src.basedOn -> tgt.basedOn;
  src.partOf -> tgt.partOf;
  src.status where value = 'not-done' ->  tgt.status = create('code') as vt,  vt.value = 'suspended',  tgt.notDone =
    true;
  src.status as vs where (value = 'not-done').not() -> tgt.status = translate(vs, '#EventStatus', 'code');
  src.statusReason as vs where src.status = 'not-done' -> tgt.notDoneReason = create('CodeableConcept') as vt then
    CodeableConcept(vs, vt);
  src.category -> tgt.category;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter : Reference as vs -> tgt.context = create('Reference') as vt then Reference(vs, vt);
  src.performed -> tgt.performed;
  src.recorder -> tgt.recorder;
  src.asserter -> tgt.asserter;
  src.performer as s -> tgt.performer as t then ProcedurePerformer(s, t);
  src.location -> tgt.location;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.bodySite -> tgt.bodySite;
  src.outcome -> tgt.outcome;
```

```
  src.report -> tgt.report;
  src.complication -> tgt.complication;
  src.complicationDetail -> tgt.complicationDetail;
  src.followUp -> tgt.followUp;
  src.note -> tgt.note;
  src.focalDevice as s -> tgt.focalDevice as t then ProcedureFocalDevice(s, t);
  src.usedReference -> tgt.usedReference;
  src.usedCode -> tgt.usedCode;
}

group ProcedurePerformer(source src, target tgt) extends BackboneElement {
  src.function -> tgt.role;
  src.actor -> tgt.actor;
  src.onBehalfOf -> tgt.onBehalfOf;
}

group ProcedureFocalDevice(source src, target tgt) extends BackboneElement {
  src.action -> tgt.action;
  src.manipulated -> tgt.manipulated;
}
```

9.3.9.3 R4 Validation Errors

9.3.10 Procedure HL7 Extensions

URL: https://hl7.org/fhir/R4/procedure-extensions.html

Defines common extensions used with or related to the Procedure resource

9.3.10 Procedure HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.3.10.1 Content

Extensions:

procedure-targetBodyStructure | targetBodyStructure : The target body site used for this procedure. Multiple locations are allowed.

procedure-approachBodyStructure | approachBodyStructure : The approach body site used for this procedure. Multiple locations are allowed.

procedure-method | method : The method used to perform this procedure.

procedure-progressStatus | progressStatus : A code to track a detailed progress of a procedure (e.g. In Recovery, Prepared for Surgery).

procedure-incisionDateTime | incisionDateTime : The time of the first incision.

procedure-causedBy | causedBy : This procedure is because of the related item.

procedure-schedule | schedule : The schedule that was followed.

procedure-directedBy | directedBy : The target of the extension is a distinct actor from the requester and has decision-making authority over the service and takes direct responsibility to manage the service.

9.3.11 JSON Schema for Procedure

URL: https://hl7.org/fhir/R4/procedure.schema.json.html

9.3.11 JSON Schema for Procedure

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for Procedure, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Procedure",
  "$ref": "#/definitions/Procedure",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Procedure": {
      "description": "An action that is or was performed on or for a patient. This can be a physical intervention like
    an operation, or less invasive like long term services, counseling, or hypnotherapy.",
      "properties": {
        "resourceType": {
          "description": "This is a Procedure resource",
          "const": "Procedure"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
```

```
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this procedure by the performer or other systems which remain constant as the resource is updated and is propagated from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, order set or other definition that is adhered to in whole or in part by this Procedure.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, order set or other definition that is adhered to in whole or in part by this Procedure.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A reference to a resource that contains details of the request for this procedure.",
```

```
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger event of which this particular procedure is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "A code specifying the state of the procedure. Generally, this will be the in-progress or
completed state.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current state of the procedure.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "A code that classifies the procedure for searching, sorting and display purposes (e.g.
\"Surgical Procedure\").",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "code": {
      "description": "The specific procedure that is performed. Use text if the exact nature of the procedure cannot
 be coded (e.g. \"Laparoscopic Appendectomy\").",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The person, animal or group on which the procedure was performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this Procedure was created or performed or to which the creation of
 this record is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performedDateTime": {
      "description": "Estimated or actual date, date-time, period, or age when the procedure was performed.  Allows
a period to support complex procedures that span more than one date, and also allows for the length of the procedure
 to be captured.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_performedDateTime": {
      "description": "Extensions for performedDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performedPeriod": {
      "description": "Estimated or actual date, date-time, period, or age when the procedure was performed.  Allows
a period to support complex procedures that span more than one date, and also allows for the length of the procedure
 to be captured.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "performedString": {
      "description": "Estimated or actual date, date-time, period, or age when the procedure was performed.  Allows
a period to support complex procedures that span more than one date, and also allows for the length of the procedure
 to be captured.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_performedString": {
      "description": "Extensions for performedString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "performedAge": {
      "description": "Estimated or actual date, date-time, period, or age when the procedure was performed.  Allows
a period to support complex procedures that span more than one date, and also allows for the length of the procedure
 to be captured.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "performedRange": {
      "description": "Estimated or actual date, date-time, period, or age when the procedure was performed.  Allows
a period to support complex procedures that span more than one date, and also allows for the length of the procedure
 to be captured.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "recorder": {
      "description": "Individual who recorded the record and takes responsibility for its content.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
```

```
    "asserter": {
      "description": "Individual who is making the procedure statement.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performer": {
      "description": "Limited to \"real\" people rather than equipment.",
      "items": {
        "$ref": "#/definitions/Procedure_Performer"
      },
      "type": "array"
    },
    "location": {
      "description": "The location where the procedure actually happened.  E.g. a newborn at home, a tracheostomy at
 a restaurant.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "The coded reason why the procedure was performed. This may be a coded entity of some type, or
may simply be present as text.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "The justification of why the procedure was performed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "bodySite": {
      "description": "Detailed and structured anatomical location information. Multiple locations are allowed - e.g.
 multiple punch biopsies of a lesion.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "outcome": {
      "description": "The outcome of the procedure - did it resolve the reasons for the procedure being performed?",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "report": {
      "description": "This could be a histology result, pathology report, surgical report, etc.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "complication": {
      "description": "Any complications that occurred during the procedure, or in the immediate post-performance
period. These are generally tracked separately from the notes, which will typically describe the procedure itself
rather than any \u0027post procedure\u0027 issues.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "complicationDetail": {
      "description": "Any complications that occurred during the procedure, or in the immediate post-performance
period.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "followUp": {
      "description": "If the procedure required specific follow up - e.g. removal of sutures. The follow up may be
represented as a simple note or could potentially be more complex, in which case the CarePlan resource can be
used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "Any other notes and comments about the procedure.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "focalDevice": {
      "description": "A device that is implanted, removed or otherwise manipulated (calibration, battery
replacement, fitting a prosthesis, attaching a wound-vac, etc.) as a focal portion of the Procedure.",
      "items": {
        "$ref": "#/definitions/Procedure_FocalDevice"
      },
```

```
      "type": "array"
    },
    "usedReference": {
      "description": "Identifies medications, devices and any other substance used as part of the procedure.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "usedCode": {
      "description": "Identifies coded items that were used as part of the procedure.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"Procedure_Performer": {
  "description": "An action that is or was performed on or for a patient. This can be a physical intervention like
an operation, or less invasive like long term services, counseling, or hypnotherapy.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "function": {
      "description": "Distinguishes the type of involvement of the performer in the procedure. For example, surgeon,
 anaesthetist, endoscopist.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "actor": {
      "description": "The practitioner who was involved in the procedure.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onBehalfOf": {
      "description": "The organization the device or practitioner was acting on behalf of.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "actor"
  ]
},
"Procedure_FocalDevice": {
  "description": "An action that is or was performed on or for a patient. This can be a physical intervention like
an operation, or less invasive like long term services, counseling, or hypnotherapy.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
```

```
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "action": {
        "description": "The kind of change that happened to the device during the procedure.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "manipulated": {
        "description": "The device that was manipulated (changed) during the procedure.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      }
    },
    "additionalProperties": false,
    "required": [
      "manipulated"
    ]
  }
 }
}
```


9.3.12 Resource Procedure - Maturity Dependency View

URL: https://hl7.org/fhir/R4/procedure-dependencies.html

9.3.12 Resource Procedure - Maturity Dependency View

Patient Care Work Group | Maturity Level: 3 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for Procedure FMM level 3

Name | Card. | Type | Dependency Analysis
Procedure | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(PlanDefinition | ActivityDefinition | Measure | OperationDefinition | Questionnaire) | Type Warning: (PlanDefinition = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ActivityDefinition = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Measure = FMM2-Trial Use vs. Element = FMM3-Trial Use)
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(CarePlan | ServiceRequest) | Type Warning: (CarePlan = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (ServiceRequest = FMM2-Trial Use vs. Element = FMM3-Trial Use)
partOf | 0..* | Reference(Procedure | Observation | MedicationAdministration) | Type Warning: (MedicationAdministration = FMM2-Trial Use vs. Element = FMM3-Trial Use)
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
category | 0..1 | CodeableConcept
code | 0..1 | CodeableConcept
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM3-Trial Use)
encounter | 0..1 | Reference(Encounter) | Type Warning: (Encounter = FMM2-Trial Use vs. Element = FMM3-Trial Use)
performed[x] | 0..1
performedDateTime | | dateTime
performedPeriod | | Period
performedString | | string
performedAge | | Age
performedRange | | Range
recorder | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
asserter | 0..1 | Reference(Patient | RelatedPerson | Practitioner | PractitionerRole) | Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use)
performer | 0..* | BackboneElement
function | 0..1 | CodeableConcept
actor | 1..1 | Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device) | Type Warning: (PractitionerRole = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (RelatedPerson = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
onBehalfOf | 0..1 | Reference(Organization)
location | 0..1 | Reference(Location)
reasonCode | 0..* | CodeableConcept

reasonReference | 0..* | Reference(Condition | Observation | Procedure | DiagnosticReport | DocumentReference)
bodySite | 0..* | CodeableConcept
outcome | 0..1 | CodeableConcept
report | 0..* | Reference(DiagnosticReport | DocumentReference | Composition) | Type Warning: (Composition = FMM2-Trial Use vs. Element = FMM3-Trial Use)
complication | 0..* | CodeableConcept
complicationDetail | 0..* | Reference(Condition)
followUp | 0..* | CodeableConcept
note | 0..* | Annotation
focalDevice | 0..* | BackboneElement
action | 0..1 | CodeableConcept
manipulated | 1..1 | Reference(Device) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use)
usedReference | 0..* | Reference(Device | Medication | Substance) | Type Warning: (Device = FMM2-Trial Use vs. Element = FMM3-Trial Use) Type Warning: (Substance = FMM2-Trial Use vs. Element = FMM3-Trial Use)
usedCode | 0..* | CodeableConcept

9.4 Resource FamilyMemberHistory - Content

URL: https://hl7.org/fhir/R4/familymemberhistory.html
Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

Significant health conditions for a person related to the patient relevant in the context of care for the patient.

9.4.1 Scope and Usage

FamilyMemberHistory is one of the event resources in the FHIR workflow specification.

This resource records significant health conditions for a particular individual related to the subject. This information can be known to different levels of accuracy. Sometimes the exact condition ('asthma') is known, and sometimes it is less precise ('some sort of cancer'). Equally, sometimes the person can be identified ('my aunt Agatha') and sometimes all that is known is that the person was an uncle.

This resource represents a simple structure used to capture an 'elementary' family history for a particular family member. However, it can also be the basis for capturing a more rigorous history useful for genetic and other analysis - refer to the Genetic Pedigree profile for an example.

The entire family history for an individual can be represented by combining references to FamilyMemberHistory instances into a List resource instance.

This resource is referenced by AdverseEvent and ClinicalImpression

9.4.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | ITU | | DomainResource | Information about patient's relatives, relevant for patient + Rule: Can have age[x] or born[x], but not both + Rule: Can only have estimatedAge if age[x] is present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Id(s) for this record
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
status | ?!Σ | 1..1 | code | partial | completed | entered-in-error | health-unknown FamilyHistoryStatus (Required)
dataAbsentReason | Σ | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred FamilyHistoryAbsentReason (Example)
patient | Σ | 1..1 | Reference(Patient) | Patient history is about
date | Σ | 0..1 | dateTime | When history was recorded or last updated
name | Σ | 0..1 | string | The family member described
relationship | Σ | 1..1 | CodeableConcept | Relationship to the subject V3 Value SetFamilyMember (Example)
sex | Σ | 0..1 | CodeableConcept | male | female | other | unknown AdministrativeGender (Extensible)
born[x] | I | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x] | ΣI | 0..1 | | (approximate) age
ageAge | | | Age

ageRange | | | Range
ageString | | | string
estimatedAge | ΣI | 0..1 | boolean | Age is estimated?
deceased[x] | Σ | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
reasonCode | Σ | 0..* | CodeableConcept | Why was family member history performed? SNOMED CT Clinical Findings (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | | 0..* | BackboneElement | Condition that the related person had
code | | 1..1 | CodeableConcept | Condition suffered by relation Condition/Problem/Diagnosis Codes (Example)
outcome | | 0..1 | CodeableConcept | deceased | permanent disability | etc. Condition Outcome Codes (Example)
contributedToDeath | | 0..1 | boolean | Whether the condition contributed to the cause of death
onset[x] | | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..* | Annotation | Extra information about condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Id(s) for this record --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|Questionnaire|
   ActivityDefinition|Measure|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown -->
 <dataAbsentReason><!-- 0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred
    --></dataAbsentReason>
 <patient><!-- 1..1 Reference(Patient) Patient history is about --></patient>
 <date value="[dateTime]"/><!-- 0..1 When history was recorded or last updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject --></relationship>
 <sex><!-- 0..1 CodeableConcept male | female | other | unknown --></sex>
 <born[x]><!--  0..1 Period|date|string (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse|DiagnosticReport|DocumentReference) Why was family member history performed?
    --></reasonReference>
 <note><!-- 0..* Annotation General note about related person --></note>
 <condition>  <!-- 0..* Condition that the related person had -->
  <code><!-- 1..1 CodeableConcept Condition suffered by relation --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc. --></outcome>
  <contributedToDeath value="[boolean]"/><!-- 0..1 Whether the condition contributed to the cause of death -->
  <onset[x]><!-- 0..1 Age|Range|Period|string When condition first manifested --></onset[x]>
  <note><!-- 0..* Annotation Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{
  "resourceType" : "FamilyMemberHistory",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Id(s) for this record
  "instantiatesCanonical" : [{ canonical(PlanDefinition|Questionnaire|
   ActivityDefinition|Measure|OperationDefinition) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "status" : "<code>", // R!  partial | completed | entered-in-error | health-unknown
  "dataAbsentReason" : { CodeableConcept }, // subject-unknown | withheld | unable-to-obtain | deferred
  "patient" : { Reference(Patient) }, // R!  Patient history is about
  "date" : "<dateTime>", // When history was recorded or last updated
  "name" : "<string>", // The family member described
  "relationship" : { CodeableConcept }, // R!  Relationship to the subject
  "sex" : { CodeableConcept }, // male | female | other | unknown
  // born[x]: (approximate) date of birth. One of these 3:
  "bornPeriod" : { Period },
  "bornDate" : "<date>",
  "bornString" : "<string>",
  // age[x]: (approximate) age. One of these 3:
  "ageAge" : { Age },
  "ageRange" : { Range },
```

```
  "ageString" : "<string>",
  "estimatedAge" : <boolean>, // C? Age is estimated?
  // deceased[x]: Dead? How old/when?. One of these 5:
  "deceasedBoolean" : <boolean>,
  "deceasedAge" : { Age },
  "deceasedRange" : { Range },
  "deceasedDate" : "<date>",
  "deceasedString" : "<string>",
  "reasonCode" : [{ CodeableConcept }], // Why was family member history performed?
  "reasonReference" : [{ Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse|DiagnosticReport|DocumentReference) }], // Why was family member history performed?
  "note" : [{ Annotation }], // General note about related person
  "condition" : [{ // Condition that the related person had
    "code" : { CodeableConcept }, // R!  Condition suffered by relation
    "outcome" : { CodeableConcept }, // deceased | permanent disability | etc.
    "contributedToDeath" : <boolean>, // Whether the condition contributed to the cause of death
    // onset[x]: When condition first manifested. One of these 4:
    "onsetAge" : { Age },
    "onsetRange" : { Range },
    "onsetPeriod" : { Period },
    "onsetString" : "<string>",
    "note" : [{ Annotation }] // Extra information about condition
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:FamilyMemberHistory;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:FamilyMemberHistory.identifier [ Identifier ], ... ; # 0..* External Id(s) for this record
  fhir:FamilyMemberHistory.instantiatesCanonical [
    canonical(PlanDefinition|Questionnaire|ActivityDefinition|Measure|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:FamilyMemberHistory.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:FamilyMemberHistory.status [ code ]; # 1..1 partial | completed | entered-in-error | health-unknown
  fhir:FamilyMemberHistory.dataAbsentReason [ CodeableConcept ]; # 0..1 subject-unknown | withheld | unable-to-obtain |
    deferred
  fhir:FamilyMemberHistory.patient [ Reference(Patient) ]; # 1..1 Patient history is about
  fhir:FamilyMemberHistory.date [ dateTime ]; # 0..1 When history was recorded or last updated
  fhir:FamilyMemberHistory.name [ string ]; # 0..1 The family member described
  fhir:FamilyMemberHistory.relationship [ CodeableConcept ]; # 1..1 Relationship to the subject
  fhir:FamilyMemberHistory.sex [ CodeableConcept ]; # 0..1 male | female | other | unknown
  # FamilyMemberHistory.born[x] : 0..1 (approximate) date of birth. One of these 3
    fhir:FamilyMemberHistory.bornPeriod [ Period ]
    fhir:FamilyMemberHistory.bornDate [ date ]
    fhir:FamilyMemberHistory.bornString [ string ]
  # FamilyMemberHistory.age[x] : 0..1 (approximate) age. One of these 3
    fhir:FamilyMemberHistory.ageAge [ Age ]
    fhir:FamilyMemberHistory.ageRange [ Range ]
    fhir:FamilyMemberHistory.ageString [ string ]
  fhir:FamilyMemberHistory.estimatedAge [ boolean ]; # 0..1 Age is estimated?
  # FamilyMemberHistory.deceased[x] : 0..1 Dead? How old/when?. One of these 5
    fhir:FamilyMemberHistory.deceasedBoolean [ boolean ]
    fhir:FamilyMemberHistory.deceasedAge [ Age ]
    fhir:FamilyMemberHistory.deceasedRange [ Range ]
    fhir:FamilyMemberHistory.deceasedDate [ date ]
    fhir:FamilyMemberHistory.deceasedString [ string ]
  fhir:FamilyMemberHistory.reasonCode [ CodeableConcept ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.reasonReference [
    Reference(Condition|Observation|AllergyIntolerance|QuestionnaireResponse|DiagnosticReport|
  DocumentReference) ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.note [ Annotation ], ... ; # 0..* General note about related person
  fhir:FamilyMemberHistory.condition [ # 0..* Condition that the related person had
    fhir:FamilyMemberHistory.condition.code [ CodeableConcept ]; # 1..1 Condition suffered by relation
    fhir:FamilyMemberHistory.condition.outcome [ CodeableConcept ]; # 0..1 deceased | permanent disability | etc.
    fhir:FamilyMemberHistory.condition.contributedToDeath [ boolean ]; # 0..1 Whether the condition contributed to the
    cause of death
    # FamilyMemberHistory.condition.onset[x] : 0..1 When condition first manifested. One of these 4
      fhir:FamilyMemberHistory.condition.onsetAge [ Age ]
      fhir:FamilyMemberHistory.condition.onsetRange [ Range ]
      fhir:FamilyMemberHistory.condition.onsetPeriod [ Period ]
      fhir:FamilyMemberHistory.condition.onsetString [ string ]
    fhir:FamilyMemberHistory.condition.note [ Annotation ], ... ; # 0..* Extra information about condition
  ], ...;
]
```

Changes since R3

FamilyMemberHistory

FamilyMemberHistory.instantiatesCanonical | Added Element

FamilyMemberHistory.instantiatesUri | Added Element

FamilyMemberHistory.status | Change value set from http://hl7.org/fhir/ValueSet/history-status to http://hl7.org/fhir/ValueSet/history-status|4.0.1

FamilyMemberHistory.dataAbsentReason | Added Element

FamilyMemberHistory.sex | Added Element
FamilyMemberHistory.estimatedAge | No longer marked as Modifier
FamilyMemberHistory.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
FamilyMemberHistory.condition.contributedToDeath | Added Element
FamilyMemberHistory.definition | deleted
FamilyMemberHistory.notDone | deleted
FamilyMemberHistory.notDoneReason | deleted
FamilyMemberHistory.gender | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | ITU | | DomainResource | Information about patient's relatives, relevant for patient + Rule: Can have age[x] or born[x], but not both + Rule: Can only have estimatedAge if age[x] is present Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Id(s) for this record
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
status | ?!Σ | 1..1 | code | partial | completed | entered-in-error | health-unknown FamilyHistoryStatus (Required)
dataAbsentReason | Σ | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred FamilyHistoryAbsentReason (Example)
patient | Σ | 1..1 | Reference(Patient) | Patient history is about
date | Σ | 0..1 | dateTime | When history was recorded or last updated
name | Σ | 0..1 | string | The family member described
relationship | Σ | 1..1 | CodeableConcept | Relationship to the subject V3 Value SetFamilyMember (Example)
sex | Σ | 0..1 | CodeableConcept | male | female | other | unknown AdministrativeGender (Extensible)
born[x] | I | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x] | ΣI | 0..1 | | (approximate) age
ageAge | | | Age
ageRange | | | Range
ageString | | | string
estimatedAge | ΣI | 0..1 | boolean | Age is estimated?
deceased[x] | Σ | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
reasonCode | Σ | 0..* | CodeableConcept | Why was family member history performed? SNOMED CT Clinical Findings (Example)
reasonReference | Σ | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference) | Why was family member history performed?
note | | 0..* | Annotation | General note about related person
condition | | 0..* | BackboneElement | Condition that the related person had
code | | 1..1 | CodeableConcept | Condition suffered by relation Condition/Problem/Diagnosis Codes (Example)
outcome | | 0..1 | CodeableConcept | deceased | permanent disability | etc. Condition Outcome Codes (Example)
contributedToDeath | | 0..1 | boolean | Whether the condition contributed to the cause of death
onset[x] | | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..* | Annotation | Extra information about condition
Documentation for this format

UML Diagram (Legend)

XML Template

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Id(s) for this record --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|Questionnaire|
   ActivityDefinition|Measure|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <status value="[code]"/><!-- 1..1 partial | completed | entered-in-error | health-unknown -->
 <dataAbsentReason><!-- 0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred
    --></dataAbsentReason>
 <patient><!-- 1..1 Reference(Patient) Patient history is about --></patient>
```

```
 <date value="[dateTime]"/><!-- 0..1 When history was recorded or last updated -->
 <name value="[string]"/><!-- 0..1 The family member described -->
 <relationship><!-- 1..1 CodeableConcept Relationship to the subject --></relationship>
 <sex><!-- 0..1 CodeableConcept male | female | other | unknown --></sex>
 <born[x]><!--  0..1 Period|date|string (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!-- 0..1 boolean|Age|Range|date|string Dead? How old/when? --></deceased[x]>
 <reasonCode><!-- 0..* CodeableConcept Why was family member history performed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse|DiagnosticReport|DocumentReference) Why was family member history performed?
    --></reasonReference>
 <note><!-- 0..* Annotation General note about related person --></note>
 <condition>  <!-- 0..* Condition that the related person had -->
  <code><!-- 1..1 CodeableConcept Condition suffered by relation --></code>
  <outcome><!-- 0..1 CodeableConcept deceased | permanent disability | etc. --></outcome>
  <contributedToDeath value="[boolean]"/><!-- 0..1 Whether the condition contributed to the cause of death -->
  <onset[x]><!-- 0..1 Age|Range|Period|string When condition first manifested --></onset[x]>
  <note><!-- 0..* Annotation Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{
  "resourceType" : "FamilyMemberHistory",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Id(s) for this record
  "instantiatesCanonical" : [{ canonical(PlanDefinition|Questionnaire|
   ActivityDefinition|Measure|OperationDefinition) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "status" : "<code>", // R!  partial | completed | entered-in-error | health-unknown
  "dataAbsentReason" : { CodeableConcept }, // subject-unknown | withheld | unable-to-obtain | deferred
  "patient" : { Reference(Patient) }, // R!  Patient history is about
  "date" : "<dateTime>", // When history was recorded or last updated
  "name" : "<string>", // The family member described
  "relationship" : { CodeableConcept }, // R!  Relationship to the subject
  "sex" : { CodeableConcept }, // male | female | other | unknown
  // born[x]: (approximate) date of birth. One of these 3:
  "bornPeriod" : { Period },
  "bornDate" : "<date>",
  "bornString" : "<string>",
  // age[x]: (approximate) age. One of these 3:
  "ageAge" : { Age },
  "ageRange" : { Range },
  "ageString" : "<string>",
  "estimatedAge" : <boolean>, // C? Age is estimated?
  // deceased[x]: Dead? How old/when?. One of these 5:
  "deceasedBoolean" : <boolean>,
  "deceasedAge" : { Age },
  "deceasedRange" : { Range },
  "deceasedDate" : "<date>",
  "deceasedString" : "<string>",
  "reasonCode" : [{ CodeableConcept }], // Why was family member history performed?
  "reasonReference" : [{ Reference(Condition|Observation|AllergyIntolerance|
   QuestionnaireResponse|DiagnosticReport|DocumentReference) }], // Why was family member history performed?
  "note" : [{ Annotation }], // General note about related person
  "condition" : [{ // Condition that the related person had
    "code" : { CodeableConcept }, // R!  Condition suffered by relation
    "outcome" : { CodeableConcept }, // deceased | permanent disability | etc.
    "contributedToDeath" : <boolean>, // Whether the condition contributed to the cause of death
    // onset[x]: When condition first manifested. One of these 4:
    "onsetAge" : { Age },
    "onsetRange" : { Range },
    "onsetPeriod" : { Period },
    "onsetString" : "<string>",
    "note" : [{ Annotation }] // Extra information about condition
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:FamilyMemberHistory;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:FamilyMemberHistory.identifier [ Identifier ], ... ; # 0..* External Id(s) for this record
  fhir:FamilyMemberHistory.instantiatesCanonical [
    canonical(PlanDefinition|Questionnaire|ActivityDefinition|Measure|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:FamilyMemberHistory.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:FamilyMemberHistory.status [ code ]; # 1..1 partial | completed | entered-in-error | health-unknown
  fhir:FamilyMemberHistory.dataAbsentReason [ CodeableConcept ]; # 0..1 subject-unknown | withheld | unable-to-obtain |
    deferred
  fhir:FamilyMemberHistory.patient [ Reference(Patient) ]; # 1..1 Patient history is about
  fhir:FamilyMemberHistory.date [ dateTime ]; # 0..1 When history was recorded or last updated
```

```
  fhir:FamilyMemberHistory.name [ string ]; # 0..1 The family member described
  fhir:FamilyMemberHistory.relationship [ CodeableConcept ]; # 1..1 Relationship to the subject
  fhir:FamilyMemberHistory.sex [ CodeableConcept ]; # 0..1 male | female | other | unknown
  # FamilyMemberHistory.born[x] : 0..1 (approximate) date of birth. One of these 3
    fhir:FamilyMemberHistory.bornPeriod [ Period ]
    fhir:FamilyMemberHistory.bornDate [ date ]
    fhir:FamilyMemberHistory.bornString [ string ]
  # FamilyMemberHistory.age[x] : 0..1 (approximate) age. One of these 3
    fhir:FamilyMemberHistory.ageAge [ Age ]
    fhir:FamilyMemberHistory.ageRange [ Range ]
    fhir:FamilyMemberHistory.ageString [ string ]
  fhir:FamilyMemberHistory.estimatedAge [ boolean ]; # 0..1 Age is estimated?
  # FamilyMemberHistory.deceased[x] : 0..1 Dead? How old/when?. One of these 5
    fhir:FamilyMemberHistory.deceasedBoolean [ boolean ]
    fhir:FamilyMemberHistory.deceasedAge [ Age ]
    fhir:FamilyMemberHistory.deceasedRange [ Range ]
    fhir:FamilyMemberHistory.deceasedDate [ date ]
    fhir:FamilyMemberHistory.deceasedString [ string ]
  fhir:FamilyMemberHistory.reasonCode [ CodeableConcept ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.reasonReference [
    Reference(Condition|Observation|AllergyIntolerance|QuestionnaireResponse|DiagnosticReport|
  DocumentReference) ], ... ; # 0..* Why was family member history performed?
  fhir:FamilyMemberHistory.note [ Annotation ], ... ; # 0..* General note about related person
  fhir:FamilyMemberHistory.condition [ # 0..* Condition that the related person had
    fhir:FamilyMemberHistory.condition.code [ CodeableConcept ]; # 1..1 Condition suffered by relation
    fhir:FamilyMemberHistory.condition.outcome [ CodeableConcept ]; # 0..1 deceased | permanent disability | etc.
    fhir:FamilyMemberHistory.condition.contributedToDeath [ boolean ]; # 0..1 Whether the condition contributed to the
    cause of death
    # FamilyMemberHistory.condition.onset[x] : 0..1 When condition first manifested. One of these 4
      fhir:FamilyMemberHistory.condition.onsetAge [ Age ]
      fhir:FamilyMemberHistory.condition.onsetRange [ Range ]
      fhir:FamilyMemberHistory.condition.onsetPeriod [ Period ]
      fhir:FamilyMemberHistory.condition.onsetString [ string ]
    fhir:FamilyMemberHistory.condition.note [ Annotation ], ... ; # 0..* Extra information about condition
  ], ...;
]
```

Changes since Release 3

FamilyMemberHistory
FamilyMemberHistory.instantiatesCanonical | Added Element
FamilyMemberHistory.instantiatesUri | Added Element
FamilyMemberHistory.status | Change value set from http://hl7.org/fhir/ValueSet/history-status to http://hl7.org/fhir/ValueSet/history-status|4.0.1
FamilyMemberHistory.dataAbsentReason | Added Element
FamilyMemberHistory.sex | Added Element
FamilyMemberHistory.estimatedAge | No longer marked as Modifier
FamilyMemberHistory.reasonReference | Type Reference: Added Target Types DiagnosticReport, DocumentReference
FamilyMemberHistory.condition.contributedToDeath | Added Element
FamilyMemberHistory.definition | deleted
FamilyMemberHistory.notDone | deleted
FamilyMemberHistory.notDoneReason | deleted
FamilyMemberHistory.gender | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.4.2.1 Terminology Bindings

Path | Definition | Type | Reference
FamilyMemberHistory.status | A code that identifies the status of the family history record. | Required | FamilyHistoryStatus
FamilyMemberHistory.dataAbsentReason | Codes describing the reason why a family member's history is not available. | Example | FamilyHistoryAbsentReason
FamilyMemberHistory.relationship | The nature of the relationship between the patient and the related person being described in the family member history. | Example | v3.FamilyMember
FamilyMemberHistory.sex | Codes describing the sex assigned at birth as documented on the birth registration. | Extensible | AdministrativeGender
FamilyMemberHistory.reasonCode | Codes indicating why the family member history was done. | Example | SNOMEDCTClinicalFindings
FamilyMemberHistory.condition.code | Identification of the Condition or diagnosis. | Example | Condition/Problem/DiagnosisCodes
FamilyMemberHistory.condition.outcome | The result of the condition for the patient; e.g. death, permanent disability, temporary disability, etc. | Example | ConditionOutcomeCodes

9.4.2.2 Constraints

id | Level | Location | Description | Expression
fhs-1 | Rule | (base) | Can have age[x] or born[x], but not both | age.empty() or born.empty()
fhs-2 | Rule | (base) | Can only have estimatedAge if age[x] is present | age.exists() or estimatedAge.empty()

9.4.2.3 Processing information about the Family Member History

The Family Member History List may contain other than FamilyMemberHistory resources. For example, a full Family History could be a List that might include a mixture of FamilyMemberHistory records as well as Observation records of things like "maternal family history of breast cancer", "number of siblings", "number of female family members with breast cancer" etc.

The List representing a patient's "family history" can include Condition and Observation records that capture "family-history" relevant assertions about the patient themselves that would typically be captured as part of a family history.

9.4.2.4 No Known Patient History, No Known Problems, and Negated Conditions

Not Reviewed, Not Asked

When a sending system does not have family history about any family members or the statement is about family history not yet being asked, then the List resource should be used to indicate the List.emptyReason="notasked".

Reviewed, None Identified for Family

After reasonable investigation that there are no known items for the family member history list, then the List resource should be used to indicate the List.emptyReason="nilknown". The List.emptyReason represents a statement about the full scope of the list (i.e. the patient or patient's agent/guardian has asserted that there are no conditions or significant events for any family members to record).

Reviewed, None Identified for a Family Member

When an individual family member's history is not available, FamilyMemberHistory.dataAbsentReason can be used to indicate why that family member's history is not available (e.g. subject unknown).

Reviewed, No Known Problems or Negated Condition for a Family Member

The FamilyMemberHistory.condition.code can be used to capture "No Known Problems" or negated conditions, such as "No history of malignant tumor of breast", for an individual family member.

9.4.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | A search by a condition code | FamilyMemberHistory.condition.code | 13 Resources
date | date | When history was recorded or last updated | FamilyMemberHistory.date | 17 Resources
identifier | token | A search by a record identifier | FamilyMemberHistory.identifier | 30 Resources
instantiates-canonical | reference | Instantiates FHIR protocol or definition | FamilyMemberHistory.instantiatesCanonical (Questionnaire, Measure, PlanDefinition, OperationDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | FamilyMemberHistory.instantiatesUri
patient | reference | The identity of a subject to list family member history items for | FamilyMemberHistory.patient (Patient) | 33 Resources
relationship | token | A search by a relationship type | FamilyMemberHistory.relationship
sex | token | A search by a sex code of a family member | FamilyMemberHistory.sex
status | token | partial | completed | entered-in-error | health-unknown | FamilyMemberHistory.status

9.4.4 Resource FamilyMemberHistory - Examples

URL: https://hl7.org/fhir/R4/familymemberhistory-examples.html

9.4.4 Resource FamilyMemberHistory - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Examples of the Family History, using the List resource:

• Real-world patient example

• Simple genetic family member history

• Example for risk assessment (The source picture can be checked as a binary file in the source xml or json file)

• Example for double cousin relationship

Example Name | id | Format
Basic Example. Describes the fathers death at age 74 from a heart attack | father | XML | JSON | Turtle
Mother died from a stroke aged 56. Brother with diabetes. | mother | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.4.5 Resource FamilyMemberHistory - Detailed Descriptions

URL: https://hl7.org/fhir/R4/familymemberhistory-definitions.html

9.4.5 Resource FamilyMemberHistory - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the FamilyMemberHistory resource.

FamilyMemberHistory
Element Id | FamilyMemberHistory
Definition | Significant health conditions for a person related to the patient relevant in the context of care for the patient.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element fhs-1 Rule Can have age[x] or born[x], but not both age.empty() or born.empty() fhs-2 Rule Can only have estimatedAge if age[x] is present age.exists() or estimatedAge.empty()
Defined on this element
fhs-1 | Rule | Can have age[x] or born[x], but not both | age.empty() or born.empty()
fhs-2 | Rule | Can only have estimatedAge if age[x] is present | age.exists() or estimatedAge.empty()
FamilyMemberHistory.identifier
Element Id | FamilyMemberHistory.identifier
Definition | Business identifiers assigned to this family member history by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the family member history as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
FamilyMemberHistory.instantiatesCanonical
Element Id | FamilyMemberHistory.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this FamilyMemberHistory.
Cardinality | 0..*
Type | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition)
Summary | true
FamilyMemberHistory.instantiatesUri
Element Id | FamilyMemberHistory.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this FamilyMemberHistory.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
FamilyMemberHistory.status
Element Id | FamilyMemberHistory.status
Definition | A code specifying the status of the record of the family history of a specific family member.
Cardinality | 1..1
Terminology Binding | FamilyHistoryStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
FamilyMemberHistory.dataAbsentReason
Element Id | FamilyMemberHistory.dataAbsentReason
Definition | Describes why the family member's history is not available.
Cardinality | 0..1
Terminology Binding | FamilyHistoryAbsentReason (Example)
Type | CodeableConcept
Requirements | This is a separate element to allow it to have a distinct binding from reasonCode.
Summary | true
FamilyMemberHistory.patient
Element Id | FamilyMemberHistory.patient
Definition | The person who this history concerns.
Cardinality | 1..1
Type | Reference(Patient)
Alternate Names | Proband
Summary | true
FamilyMemberHistory.date
Element Id | FamilyMemberHistory.date
Definition | The date (and possibly time) when the family member history was recorded or last updated.

Cardinality | 0..1
Type | dateTime
Requirements | Allows determination of how current the summary is.
Summary | true
Comments | This should be captured even if the same as the date on the List aggregating the full family history.
FamilyMemberHistory.name
Element Id | FamilyMemberHistory.name
Definition | This will either be a name or a description; e.g. "Aunt Susan", "my cousin with the red hair".
Cardinality | 0..1
Type | string
Requirements | Allows greater ease in ensuring the same person is being talked about.
Summary | true
FamilyMemberHistory.relationship
Element Id | FamilyMemberHistory.relationship
Definition | The type of relationship this person has to the patient (father, mother, brother etc.).
Cardinality | 1..1
Terminology Binding | V3 Value SetFamilyMember (Example)
Type | CodeableConcept
Summary | true
FamilyMemberHistory.sex
Element Id | FamilyMemberHistory.sex
Definition | The birth sex of the family member.
Cardinality | 0..1
Terminology Binding | AdministrativeGender (Extensible)
Type | CodeableConcept
Requirements | Not all relationship codes imply sex and the relative's sex can be relevant for risk assessments.
Summary | true
Comments | This element should ideally reflect whether the individual is genetically male or female. However, as reported information based on the knowledge of the patient or reporting friend/relative, there may be situations where the reported sex might not be totally accurate. E.g. 'Aunt Sue' might be XY rather than XX. Questions soliciting this information should be phrased to encourage capture of genetic sex where known. However, systems performing analysis should also allow for the possibility of imprecision with this element.
FamilyMemberHistory.born[x]
Element Id | FamilyMemberHistory.born[x]
Definition | The actual or approximate date of birth of the relative.
Cardinality | 0..1
Type | Period|date|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows calculation of the relative's age.
Invariants | Affect this element fhs-1 Rule Can have age[x] or born[x], but not both age.empty() or born.empty()
Affect this element
fhs-1 | Rule | Can have age[x] or born[x], but not both | age.empty() or born.empty()
FamilyMemberHistory.age[x]
Element Id | FamilyMemberHistory.age[x]
Definition | The age of the relative at the time the family member history is recorded.
Cardinality | 0..1
Type | Age|Range|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | While age can be calculated from date of birth, sometimes recording age directly is more natural for clinicians.
Summary | true
Comments | use estimatedAge to indicate whether the age is actual or not.
Invariants | Affect this element fhs-1 Rule Can have age[x] or born[x], but not both age.empty() or born.empty()
Affect this element
fhs-1 | Rule | Can have age[x] or born[x], but not both | age.empty() or born.empty()
FamilyMemberHistory.estimatedAge
Element Id | FamilyMemberHistory.estimatedAge
Definition | If true, indicates that the age value specified is an estimated value.
Cardinality | 0..1
Type | boolean
Meaning if Missing | It is unknown whether the age is an estimate or not
Requirements | Clinicians often prefer to specify an estimaged age rather than an age range.
Summary | true
Comments | This element is labeled as a modifier because the fact that age is estimated can/should change the results of any algorithm that calculates based on the specified age.
Invariants | Affect this element fhs-2 Rule Can only have estimatedAge if age[x] is present age.exists() or estimatedAge.empty()
Affect this element
fhs-2 | Rule | Can only have estimatedAge if age[x] is present | age.exists() or estimatedAge.empty()
FamilyMemberHistory.deceased[x]
Element Id | FamilyMemberHistory.deceased[x]
Definition | Deceased flag or the actual or approximate age of the relative at the time of death for the family member history record.
Cardinality | 0..1
Type | boolean|Age|Range|date|string

[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
FamilyMemberHistory.reasonCode
Element Id | FamilyMemberHistory.reasonCode
Definition | Describes why the family member history occurred in coded or textual form.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be captured using reasonCode.text.
FamilyMemberHistory.reasonReference
Element Id | FamilyMemberHistory.reasonReference
Definition | Indicates a Condition, Observation, AllergyIntolerance, or QuestionnaireResponse that justifies this family member history event.
Cardinality | 0..*
Type | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference)
Summary | true
FamilyMemberHistory.note
Element Id | FamilyMemberHistory.note
Definition | This property allows a non condition-specific note to the made about the related person. Ideally, the note would be in the condition property, but this is not always possible.
Cardinality | 0..*
Type | Annotation
FamilyMemberHistory.condition
Element Id | FamilyMemberHistory.condition
Definition | The significant Conditions (or condition) that the family member had. This is a repeating section to allow a system to represent more than one condition per resource, though there is nothing stopping multiple resources - one per condition.
Cardinality | 0..*
FamilyMemberHistory.condition.code
Element Id | FamilyMemberHistory.condition.code
Definition | The actual condition specified. Could be a coded condition (like MI or Diabetes) or a less specific string like 'cancer' depending on how much is known about the condition and the capabilities of the creating system.
Cardinality | 1..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
FamilyMemberHistory.condition.outcome
Element Id | FamilyMemberHistory.condition.outcome
Definition | Indicates what happened following the condition. If the condition resulted in death, deceased date is captured on the relation.
Cardinality | 0..1
Terminology Binding | Condition Outcome Codes (Example)
Type | CodeableConcept
FamilyMemberHistory.condition.contributedToDeath
Element Id | FamilyMemberHistory.condition.contributedToDeath
Definition | This condition contributed to the cause of death of the related person. If contributedToDeath is not populated, then it is unknown.
Cardinality | 0..1
Type | boolean
FamilyMemberHistory.condition.onset[x]
Element Id | FamilyMemberHistory.condition.onset[x]
Definition | Either the age of onset, range of approximate age or descriptive string can be recorded. For conditions with multiple occurrences, this describes the first known occurrence.
Cardinality | 0..1
Type | Age|Range|Period|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Age of onset of a condition in relatives is predictive of risk for the patient.
FamilyMemberHistory.condition.note
Element Id | FamilyMemberHistory.condition.note
Definition | An area where general notes can be placed about this specific condition.
Cardinality | 0..*
Type | Annotation

9.4.6 Resource FamilyMemberHistory - Mappings

URL: https://hl7.org/fhir/R4/familymemberhistory-mappings.html

9.4.6 Resource FamilyMemberHistory - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the familymemberhistory resource (see Mappings to Other Standards for further information & status).

9.4.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

FamilyMemberHistory
identifier | FiveWs.identifier
status | FiveWs.status
patient | FiveWs.subject[x]
date | FiveWs.recorded
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

9.4.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

FamilyMemberHistory | Event
identifier | Event.identifier
instantiatesCanonical | Event.instantiatesCanonical
instantiatesUri | Event.instantiatesUri
status | Event.status
patient | Event.subject
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

9.4.6.3 HL7 v2 Mapping (http://hl7.org/v2)

FamilyMemberHistory | Not in scope for v2
identifier
instantiatesCanonical
instantiatesUri
status
dataAbsentReason
patient
date
name
relationship
sex
born[x]
age[x]
estimatedAge
deceased[x]
reasonCode
reasonReference
note
condition
code
outcome
contributedToDeath
onset[x]
note

9.4.6.4 RIM Mapping (http://hl7.org/v3)

FamilyMemberHistory | Observation[classCode=OBS, moodCode=EVN]
identifier | id
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
dataAbsentReason | .reasonCode
patient | participation[typeCode=SBJ].role
date | participation[typeCode=AUT].time
name | name
relationship | code
born[x] | player[classCode=LIV, determinerCode=INSTANCE]. birthDate (could be URG)
age[x] | participation[typeCode=SBJ].act[classCode=OBS,moodCode=EVN, code="age"].value
deceased[x] | player[classCode=LIV, determinerCode=INSTANCE].deceasedInd, deceasedDate (could be URG) For age, you'd hang an observation off the role
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
note | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ANNGEN].value
condition | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ASSERTION, value<Diagnosis]

code | .value
outcome | outboundRelationship[typeCode=OUTC)].target[classCode=OBS, moodCode=EVN, code=ActCode#ASSERTION].value
onset[x] | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="Subject Age at measurement", value<Diagnosis].value[@xsi:typeCode='TS' or 'IVL_TS'] Use originalText for string
note | inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=ActCode#ANNGEN].value

9.4.7 Resource FamilyMemberHistory - Extensions & Profiles

URL: https://hl7.org/fhir/R4/familymemberhistory-profiles.html

9.4.7 Resource FamilyMemberHistory - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the FamilyMemberHistory resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.4.7.1 Profiles

Profile | Description | Context
FamilyMemberHistory-Genetic | Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients | Family member history for genetics analysis

9.4.7.2 Extensions

family-member-history-genetics-observation | FamilyMemberHistory | observation | for Family member history for genetics analysis
family-member-history-genetics-parent | FamilyMemberHistory | parent | for Family member history for genetics analysis
family-member-history-genetics-sibling | FamilyMemberHistory | sibling | for Family member history for genetics analysis
familymemberhistory-abatement | FamilyMemberHistory.condition | abatement | for Family Member History HL7 Extensions
familymemberhistory-patient-record | FamilyMemberHistory | patient-record | for Family Member History HL7 Extensions
familymemberhistory-severity | FamilyMemberHistory.condition | severity | for Family Member History HL7 Extensions
familymemberhistory-type | FamilyMemberHistory | type | for Family Member History HL7 Extensions
workflow-researchStudy | FamilyMemberHistory | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

DiagnosticReport-geneticsFamilyMemberHistory | FamilyMemberHistory | for DiagnosticReport-Genetics
family-member-history-genetics-parent | parent | for Family member history for genetics analysis
family-member-history-genetics-sibling | sibling | for Family member history for genetics analysis
procedure-causedBy | causedBy | for Procedure HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.4.7.3 Search Extensions

No Search Extensions defined for this resource

9.4.8 Resource FamilyMemberHistory - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/familymemberhistory-version-maps.html

No validation errors - all conversions are clean

9.4.8 Resource FamilyMemberHistory - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for FamilyMemberHistory.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

9.4.8.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/FamilyMemberHistory3to4" = "R3 to R4 Conversions for FamilyMemberHistory"

uses "http://hl7.org/fhir/3.0/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistoryR3 as source
uses "http://hl7.org/fhir/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistory as target
```

```
imports "http://hl7.org/fhir/StructureMap/*3to4"

group FamilyMemberHistory(source src : FamilyMemberHistoryR3, target tgt : FamilyMemberHistory) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiatesCanonical;
  src.status -> tgt.status;
  src.notDone as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    FamilyMemberHistory.notDone',  vt.value = v;
  src.notDoneReason -> tgt.dataAbsentReason;
  src.patient -> tgt.patient;
  src.date -> tgt.date;
  src.name -> tgt.name;
  src.relationship -> tgt.relationship;
  src.gender as v ->  tgt.sex as cc,  cc.coding as c,  c.system = 'http://hl7.org/fhir/administrative-gender',  c.code =
     v;
  src.born : Period as vs -> tgt.born = create('Period') as vt then Period(vs, vt);
  src.born : Date as vs -> tgt.born = create('date') as vt then Date(vs, vt);
  src.born : string as vs -> tgt.born = create('string') as vt then string(vs, vt);
  src.age : Age as vs -> tgt.age = create('Age') as vt then Age(vs, vt);
  src.age : Range as vs -> tgt.age = create('Range') as vt then Range(vs, vt);
  src.age : string as vs -> tgt.age = create('string') as vt then string(vs, vt);
  src.estimatedAge -> tgt.estimatedAge;
  src.deceased : boolean as vs -> tgt.deceased = create('boolean') as vt then boolean(vs, vt);
  src.deceased : Age as vs -> tgt.deceased = create('Age') as vt then Age(vs, vt);
  src.deceased : Range as vs -> tgt.deceased = create('Range') as vt then Range(vs, vt);
  src.deceased : Date as vs -> tgt.deceased = create('date') as vt then Date(vs, vt);
  src.deceased : string as vs -> tgt.deceased = create('string') as vt then string(vs, vt);
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.condition as s -> tgt.condition as t then condition(s, t);
}

group condition(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.outcome -> tgt.outcome;
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.note -> tgt.note;
}
```

9.4.8.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/FamilyMemberHistory4to3" = "R4 to R3 Conversion for FamilyMemberHistory"

uses "http://hl7.org/fhir/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistory as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/FamilyMemberHistory" alias FamilyMemberHistoryR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group FamilyMemberHistory(source src : FamilyMemberHistory, target tgt : FamilyMemberHistoryR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.definition;
  src.status -> tgt.status;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-FamilyMemberHistory.notDone'
    then {
    e.value -> tgt.notDone;
  };
  src.dataAbsentReason -> tgt.notDoneReason;
  src.dataAbsentReason where src.extension.where(url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    FamilyMemberHistory.notDone').empty() -> tgt.notDone = true;
  src.patient -> tgt.patient;
  src.date -> tgt.date;
  src.name -> tgt.name;
  src.relationship -> tgt.relationship;
  src.sex as cc then {
    cc.coding as c where system = 'http://hl7.org/fhir/administrative-gender' then {
      c.code -> tgt.gender;
    };
  };
  src.born : Period as vs -> tgt.born = create('Period') as vt then Period(vs, vt);
  src.born : Date as vs -> tgt.born = create('date') as vt then Date(vs, vt);
  src.born : string as vs -> tgt.born = create('string') as vt then string(vs, vt);
  src.age : Age as vs -> tgt.age = create('Age') as vt then Age(vs, vt);
  src.age : Range as vs -> tgt.age = create('Range') as vt then Range(vs, vt);
  src.age : string as vs -> tgt.age = create('string') as vt then string(vs, vt);
  src.estimatedAge -> tgt.estimatedAge;
  src.deceased : boolean as vs -> tgt.deceased = create('boolean') as vt then boolean(vs, vt);
  src.deceased : Age as vs -> tgt.deceased = create('Age') as vt then Age(vs, vt);
  src.deceased : Range as vs -> tgt.deceased = create('Range') as vt then Range(vs, vt);
  src.deceased : Date as vs -> tgt.deceased = create('date') as vt then Date(vs, vt);
  src.deceased : string as vs -> tgt.deceased = create('string') as vt then string(vs, vt);
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.note -> tgt.note;
  src.condition as s -> tgt.condition as t then condition(s, t);
```

```
}

group condition(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.outcome -> tgt.outcome;
  src.onset : string as vs -> tgt.onset = create('string') as vt then string(vs, vt);
  src.onset : Age as vs -> tgt.onset = create('Age') as vt then Age(vs, vt);
  src.onset : Range as vs -> tgt.onset = create('Range') as vt then Range(vs, vt);
  src.onset : Period as vs -> tgt.onset = create('Period') as vt then Period(vs, vt);
  src.note -> tgt.note;
}
```

9.4.8.3 R4 Validation Errors

9.4.9 Family member history for genetics analysis

URL: https://hl7.org/fhir/R4/family-member-history-genetics.html

Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

9.4.9 Family member history for genetics analysis

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.4.9.1 Content

Profiles:

FamilyMemberHistory-Genetic | Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

Extensions:

family-member-history-genetics-parent | parent : Identifies a parent of the relative.

family-member-history-genetics-sibling | sibling : Identifies a sibling of the relative.

family-member-history-genetics-observation | observation : Allows capturing risk-relevant observations about the relative that aren't themselves a specific health condition; e.g. Certain ethnic ancestries that are disease-relevant, presence of particular genetic markers, etc.

9.4.9.2 StructureDefinition: FamilyMemberHistory-Genetic

URL: https://hl7.org/fhir/R4/familymemberhistory-genetic.html

9.4.9.2 StructureDefinition: FamilyMemberHistory-Genetic

Patient Care Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/familymemberhistory-genetic`

Adds additional information to a family member history supporting both the capture of mother/father relationships as well as additional observations necessary to enable genetics-based risk analysis for patients

This profile was published on Wed, Oct 2, 2013 00:00+1000 as a draft by Health Level Seven International (Clinical Genomics).

9.4.9.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from FamilyMemberHistory.

Summary

Mandatory: 1 element (+1 nested mandatory element) Must-Support: 12 elements

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation

This structure is derived from FamilyMemberHistory.

Name | Flags | Card. | Type | Description & Constraints

FamilyMemberHistory | | 0..*

family-member-history-genetics-parent | S | 0..* | (Complex) | Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | S | 0..* | (Complex) | natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | S | 0..* | Reference(Observation) | Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
relationship:Relationship | S | 1..1 | CodeableConcept
sex:Sex | S | 0..1 | CodeableConcept
born[x]:BornAge | S | 0..1
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x]:Age | S | 0..1
ageAge | | | Age
ageRange | | | Range
ageString | | | string
deceased[x]:DeceasedAge | S | 0..1
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
condition:Condition | S | 0..* | BackboneElement
code | S | 1..1 | CodeableConcept
outcome | S | 0..1 | CodeableConcept
onset[x] | S | 0..1
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..1 | Annotation
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | 0..* | | Information about patient's relatives, relevant for patient
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
family-member-history-genetics-parent | SI | 0..* | (Complex) | Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | SI | 0..* | (Complex) | natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | SI | 0..* | Reference(Observation) | Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | External Id(s) for this record
instantiatesCanonical | ΣI | 0..* | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | ΣI | 0..* | uri | Instantiates external protocol or definition
status | ?!ΣI | 1..1 | code | partial | completed | entered-in-error | health-unknown Binding: FamilyHistoryStatus (required)
dataAbsentReason | ΣI | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred Binding: FamilyHistoryAbsentReason (example)
patient | ΣI | 1..1 | Reference(Patient) | Patient history is about
date | ΣI | 0..1 | dateTime | When history was recorded or last updated
name | ΣI | 0..1 | string | The family member described
relationship:Relationship | SΣI | 1..1 | CodeableConcept | Relationship to the subject Binding: FamilyMember (example)
sex:Sex | SΣI | 0..1 | CodeableConcept | male | female | other | unknown Binding: AdministrativeGender (extensible)
born[x]:BornAge | SI | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x]:Age | SΣI | 0..1 | | (approximate) age
ageAge | | | Age
ageRange | | | Range
ageString | | | string
estimatedAge | ΣI | 0..1 | boolean | Age is estimated?
deceased[x]:DeceasedAge | SΣI | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range

deceasedDate | | | date
deceasedString | | | string
reasonCode | ΣI | 0..* | CodeableConcept | Why was family member history performed? Binding: SNOMED CT Clinical Findings (example)
reasonReference | ΣI | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference) | Why was family member history performed?
note | I | 0..* | Annotation | General note about related person
condition:Condition | SI | 0..* | BackboneElement | Condition that the related person had
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SI | 1..1 | CodeableConcept | Condition suffered by relation Binding: Condition/Problem/Diagnosis Codes (example)
outcome | SI | 0..1 | CodeableConcept | deceased | permanent disability | etc. Binding: Condition Outcome Codes (example)
contributedToDeath | I | 0..1 | boolean | Whether the condition contributed to the cause of death
onset[x] | SI | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | I | 0..1 | Annotation | Extra information about condition
Documentation for this format

```
<!-- Family member history for genetics analysis -->

<FamilyMemberHistory xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     External Id(s) for this record --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  0..* Instantiates external protocol or definition -->
 <status value="[code]"/><!--  1..1 partial | completed | entered-in-error | health-unknown   -->
 <dataAbsentReason><!--  0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred
    --></dataAbsentReason>
 <patient><!--  1..1 Reference(Patient)
     Patient history is about --></patient>
 <date value="[dateTime]"/><!--  0..1 When history was recorded or last updated -->
 <name value="[string]"/><!--  0..1 The family member described -->
 <relationship><!--  1..1 CodeableConcept Relationship to the subject   --></relationship>
 <sex><!--  0..1 CodeableConcept male | female | other | unknown   --></sex>
 <born[x]><!--  0..1 Period|date|string
     (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string
     (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!--  0..1 boolean|Age|Range|date|string
     Dead? How old/when? --></deceased[x]>
 <reasonCode><!--  0..* CodeableConcept Why was family member history performed?   --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why was family member history performed? --></reasonReference>
 <note><!--  0..* Annotation
     General note about related person --></note>
 <condition>  0..* BackboneElement  <!--  0..* Condition that the related person had -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Condition suffered by relation   --></code>
  <outcome><!--  0..1 CodeableConcept deceased | permanent disability | etc.   --></outcome>
  <contributedToDeath value="[boolean]"/><!--  0..1
      Whether the condition contributed to the cause of death -->
  <onset[x]><!--  0..1 Age|Range|Period|string
      When condition first manifested --></onset[x]>
  <note><!--  0..1 Annotation
      Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

```
{ // Family member history for genetics analysis
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Mother(s) & Father(s) - genetic & other // C?
        // from Element: extension
      },
```

```
      { // natural brother(s) & natural sister(s) - genetic & other // C?
        // from Element: extension
      },
      { // Genetic markers, ethnicity, etc. // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?External Id(s) for this record
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
    "instantiatesUri" : ["<uri>"], // C?Instantiates external protocol or definition
    "status" : "<code>", // C? R! partial | completed | entered-in-error | health-unknown
    "dataAbsentReason" : { CodeableConcept }, // C?subject-unknown | withheld | unable-to-obtain | deferred
    "patient" : { Reference(Patient) }, // C? R! Patient history is about
    "date" : "<dateTime>", // C?When history was recorded or last updated
    "name" : "<string>", // C?The family member described
    "relationship" : { CodeableConcept }, // C? R! Relationship to the subject
    "sex" : { CodeableConcept }, // C?male | female | other | unknown
// value[x]: (approximate) date of birth. One of these 3:
    "bornPeriod" : { Period }, // C?(approximate) date of birth
    "bornDate" : "<date>", // C?(approximate) date of birth
    "bornString" : "<string>", // C?(approximate) date of birth
// value[x]: (approximate) age. One of these 3:
    "ageAge" : { Age }, // C?(approximate) age
    "ageRange" : { Range }, // C?(approximate) age
    "ageString" : "<string>", // C?(approximate) age
    "estimatedAge" : <boolean>, // C?Age is estimated?
// value[x]: Dead? How old/when?. One of these 5:
    "deceasedBoolean" : <boolean>, // C?Dead? How old/when?
    "deceasedAge" : { Age }, // C?Dead? How old/when?
    "deceasedRange" : { Range }, // C?Dead? How old/when?
    "deceasedDate" : "<date>" // C?Dead? How old/when?
    "deceasedString" : "<string>", // C?Dead? How old/when?
    "reasonCode" : [{ CodeableConcept }], // C?Why was family member history performed?
    "reasonReference" : [{ Reference(Condition) }], // C?Why was family member history performed?
    "note" : [{ Annotation }], // C?General note about related person
    "condition" : [{ BackboneElement }], // C?Condition that the related person had
  }
```

This structure is derived from FamilyMemberHistory.

Summary

Mandatory: 1 element (+1 nested mandatory element) Must-Support: 12 elements

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
- http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation

Differential View

This structure is derived from FamilyMemberHistory.

Name	Flags	Card.	Type	Description & Constraints
FamilyMemberHistory			0..*	
family-member-history-genetics-parent	S	0..*	(Complex)	Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling	S	0..*	(Complex)	natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation	S	0..*	Reference(Observation)	Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
relationship:Relationship	S	1..1	CodeableConcept	
sex:Sex	S	0..1	CodeableConcept	
born[x]:BornAge	S	0..1		
bornPeriod			Period	
bornDate			date	
bornString			string	
age[x]:Age	S	0..1		
ageAge			Age	
ageRange			Range	
ageString			string	
deceased[x]:DeceasedAge	S	0..1		
deceasedBoolean			boolean	
deceasedAge			Age	
deceasedRange			Range	
deceasedDate			date	
deceasedString			string	
condition:Condition	S	0..*	BackboneElement	
code	S	1..1	CodeableConcept	
outcome	S	0..1	CodeableConcept	

onset[x] | S | 0..1
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | | 0..1 | Annotation
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
FamilyMemberHistory | I | 0..* | | Information about patient's relatives, relevant for patient
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
family-member-history-genetics-parent | SI | 0..* | (Complex) | Mother(s) & Father(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-parent
family-member-history-genetics-sibling | SI | 0..* | (Complex) | natural brother(s) & natural sister(s) - genetic & other URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-sibling
family-member-history-genetics-observation | SI | 0..* | Reference(Observation) | Genetic markers, ethnicity, etc. URL: http://hl7.org/fhir/StructureDefinition/family-member-history-genetics-observation
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 0..* | Identifier | External Id(s) for this record
instantiatesCanonical | ΣI | 0..* | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | ΣI | 0..* | uri | Instantiates external protocol or definition
status | ?!ΣI | 1..1 | code | partial | completed | entered-in-error | health-unknown Binding: FamilyHistoryStatus (required)
dataAbsentReason | ΣI | 0..1 | CodeableConcept | subject-unknown | withheld | unable-to-obtain | deferred Binding: FamilyHistoryAbsentReason (example)
patient | ΣI | 1..1 | Reference(Patient) | Patient history is about
date | ΣI | 0..1 | dateTime | When history was recorded or last updated
name | ΣI | 0..1 | string | The family member described
relationship:Relationship | SΣI | 1..1 | CodeableConcept | Relationship to the subject Binding: FamilyMember (example)
sex:Sex | SΣI | 0..1 | CodeableConcept | male | female | other | unknown Binding: AdministrativeGender (extensible)
born[x]:BornAge | SI | 0..1 | | (approximate) date of birth
bornPeriod | | | Period
bornDate | | | date
bornString | | | string
age[x]:Age | SΣI | 0..1 | | (approximate) age
ageAge | | | Age
ageRange | | | Range
ageString | | | string
estimatedAge | ΣI | 0..1 | boolean | Age is estimated?
deceased[x]:DeceasedAge | SΣI | 0..1 | | Dead? How old/when?
deceasedBoolean | | | boolean
deceasedAge | | | Age
deceasedRange | | | Range
deceasedDate | | | date
deceasedString | | | string
reasonCode | ΣI | 0..* | CodeableConcept | Why was family member history performed? Binding: SNOMED CT Clinical Findings (example)
reasonReference | ΣI | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference) | Why was family member history performed?
note | I | 0..* | Annotation | General note about related person
condition:Condition | SI | 0..* | BackboneElement | Condition that the related person had
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | SI | 1..1 | CodeableConcept | Condition suffered by relation Binding: Condition/Problem/Diagnosis Codes (example)
outcome | SI | 0..1 | CodeableConcept | deceased | permanent disability | etc. Binding: Condition Outcome Codes (example)
contributedToDeath | I | 0..1 | boolean | Whether the condition contributed to the cause of death
onset[x] | SI | 0..1 | | When condition first manifested
onsetAge | | | Age
onsetRange | | | Range
onsetPeriod | | | Period
onsetString | | | string
note | I | 0..1 | Annotation | Extra information about condition
Documentation for this format

XML Template

```
<!-- Family member history for genetics analysis -->
```

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  0..* Identifier
     External Id(s) for this record --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  0..* Instantiates external protocol or definition -->
 <status value="[code]"/><!--  1..1 partial | completed | entered-in-error | health-unknown   -->
 <dataAbsentReason><!--  0..1 CodeableConcept subject-unknown | withheld | unable-to-obtain | deferred
    --></dataAbsentReason>
 <patient><!--  1..1 Reference(Patient)
     Patient history is about --></patient>
 <date value="[dateTime]"/><!--  0..1 When history was recorded or last updated -->
 <name value="[string]"/><!--  0..1 The family member described -->
 <relationship><!--  1..1 CodeableConcept Relationship to the subject   --></relationship>
 <sex><!--  0..1 CodeableConcept male | female | other | unknown   --></sex>
 <born[x]><!--  0..1 Period|date|string
     (approximate) date of birth --></born[x]>
 <age[x]><!--  0..1 Age|Range|string
     (approximate) age --></age[x]>
 <estimatedAge value="[boolean]"/><!--  0..1 Age is estimated? -->
 <deceased[x]><!--  0..1 boolean|Age|Range|date|string
     Dead? How old/when? --></deceased[x]>
 <reasonCode><!--  0..* CodeableConcept Why was family member history performed?   --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why was family member history performed? --></reasonReference>
 <note><!--  0..* Annotation
     General note about related person --></note>
 <condition>  0..* BackboneElement  <!--  0..* Condition that the related person had -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  1..1 CodeableConcept Condition suffered by relation   --></code>
  <outcome><!--  0..1 CodeableConcept deceased | permanent disability | etc.   --></outcome>
  <contributedToDeath value="[boolean]"/><!--  0..1
      Whether the condition contributed to the cause of death -->
  <onset[x]><!--  0..1 Age|Range|Period|string
      When condition first manifested --></onset[x]>
  <note><!--  0..1 Annotation
      Extra information about condition --></note>
 </condition>
</FamilyMemberHistory>
```

JSON Template

```
{ // Family member history for genetics analysis
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Mother(s) & Father(s) - genetic & other // C?
        // from Element: extension
      },
      { // natural brother(s) & natural sister(s) - genetic & other // C?
        // from Element: extension
      },
      { // Genetic markers, ethnicity, etc. // C?
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : [{ Identifier }], // C?External Id(s) for this record
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
    "instantiatesUri" : ["<uri>"], // C?Instantiates external protocol or definition
    "status" : "<code>", // C? R! partial | completed | entered-in-error | health-unknown
    "dataAbsentReason" : { CodeableConcept }, // C?subject-unknown | withheld | unable-to-obtain | deferred
    "patient" : { Reference(Patient) }, // C? R! Patient history is about
    "date" : "<dateTime>", // C?When history was recorded or last updated
    "name" : "<string>", // C?The family member described
    "relationship" : { CodeableConcept }, // C? R! Relationship to the subject
    "sex" : { CodeableConcept }, // C?male | female | other | unknown
// value[x]: (approximate) date of birth. One of these 3:
    "bornPeriod" : { Period }, // C?(approximate) date of birth
    "bornDate" : "<date>", // C?(approximate) date of birth
    "bornString" : "<string>", // C?(approximate) date of birth
// value[x]: (approximate) age. One of these 3:
    "ageAge" : { Age }, // C?(approximate) age
    "ageRange" : { Range }, // C?(approximate) age
    "ageString" : "<string>", // C?(approximate) age
```

```
    "estimatedAge" : <boolean>, // C?Age is estimated?
// value[x]: Dead? How old/when?. One of these 5:
    "deceasedBoolean" : <boolean>, // C?Dead? How old/when?
    "deceasedAge" : { Age }, // C?Dead? How old/when?
    "deceasedRange" : { Range }, // C?Dead? How old/when?
    "deceasedDate" : "<date>" // C?Dead? How old/when?
    "deceasedString" : "<string>", // C?Dead? How old/when?
    "reasonCode" : [{ CodeableConcept }], // C?Why was family member history performed?
    "reasonReference" : [{ Reference(Condition) }], // C?Why was family member history performed?
    "note" : [{ Annotation }], // C?General note about related person
    "condition" : [{ BackboneElement }], // C?Condition that the related person had
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

9.4.9.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet
FamilyMemberHistory.language | Common Languages | preferred | Common Languages
FamilyMemberHistory.status | FamilyHistoryStatus | required | FamilyHistoryStatus
FamilyMemberHistory.dataAbsentReason | FamilyHistoryAbsentReason | example | FamilyHistoryAbsentReason
FamilyMemberHistory.relationship | V3 Value SetFamilyMember | example | V3 Value SetFamilyMember
FamilyMemberHistory.sex | AdministrativeGender | extensible | AdministrativeGender
FamilyMemberHistory.reasonCode | SNOMED CT Clinical Findings | example | SNOMED CT Clinical Findings
FamilyMemberHistory.condition.code | Condition/Problem/Diagnosis Codes | example | Condition/Problem/Diagnosis Codes
FamilyMemberHistory.condition.outcome | Condition Outcome Codes | example | Condition Outcome Codes

9.4.10 Family Member History HL7 Extensions

URL: https://hl7.org/fhir/R4/familymemberhistory-extensions.html

Defines common extensions used with or related to the FamilyMemberHistory resource

9.4.10 Family Member History HL7 Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.4.10.1 Content

Extensions:
familymemberhistory-abatement | abatement : The approximate date, age, or flag indicating that the condition of the family member resolved. The abatement should only be specified if the condition is stated in the positive sense, i.e., the didNotHave flag is false.
familymemberhistory-severity | severity : A qualification of the seriousness or impact on health of the family member condition.
familymemberhistory-patient-record | patient-record : A link to one to more patient records for the relation.
familymemberhistory-type | type : Purpose of the family member history or why it was created, such as when family member history is targeted for cardiovascular health, mental health, or genetic counseling.

9.4.11 JSON Schema for FamilyMemberHistory

URL: https://hl7.org/fhir/R4/familymemberhistory.schema.json.html

9.4.11 JSON Schema for FamilyMemberHistory

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for FamilyMemberHistory, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/FamilyMemberHistory",
  "$ref": "#/definitions/FamilyMemberHistory",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "FamilyMemberHistory": {
      "description": "Significant health conditions for a person related to the patient relevant in the context of care
    for the patient.",
      "properties": {
        "resourceType": {
          "description": "This is a FamilyMemberHistory resource",
          "const": "FamilyMemberHistory"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
```

```
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which must be understood when processing the content. Often, this is a reference to an implementation guide that defines the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this family member history by the performer or other systems which remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this FamilyMemberHistory.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, orderset or other definition that is adhered to in whole or in part by this FamilyMemberHistory.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "status": {
      "description": "A code specifying the status of the record of the family history of a specific family
```

```
member.",
      "enum": [
        "partial",
        "completed",
        "entered-in-error",
        "health-unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "dataAbsentReason": {
      "description": "Describes why the family member\u0027s history is not available.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "patient": {
      "description": "The person who this history concerns.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date (and possibly time) when the family member history was recorded or last updated.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "This will either be a name or a description; e.g. \"Aunt Susan\", \"my cousin with the red
hair\".",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "relationship": {
      "description": "The type of relationship this person has to the patient (father, mother, brother etc.).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "sex": {
      "description": "The birth sex of the family member.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "bornPeriod": {
      "description": "The actual or approximate date of birth of the relative.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "bornDate": {
      "description": "The actual or approximate date of birth of the relative.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_bornDate": {
      "description": "Extensions for bornDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "bornString": {
      "description": "The actual or approximate date of birth of the relative.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_bornString": {
      "description": "Extensions for bornString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "ageAge": {
      "description": "The age of the relative at the time the family member history is recorded.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "ageRange": {
      "description": "The age of the relative at the time the family member history is recorded.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "ageString": {
      "description": "The age of the relative at the time the family member history is recorded.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_ageString": {
      "description": "Extensions for ageString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "estimatedAge": {
      "description": "If true, indicates that the age value specified is an estimated value.",
      "$ref": "#/definitions/boolean"
    },
```

```
    "_estimatedAge": {
      "description": "Extensions for estimatedAge",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deceasedBoolean": {
      "description": "Deceased flag or the actual or approximate age of the relative at the time of death for the
family member history record.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_deceasedBoolean": {
      "description": "Extensions for deceasedBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deceasedAge": {
      "description": "Deceased flag or the actual or approximate age of the relative at the time of death for the
family member history record.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "deceasedRange": {
      "description": "Deceased flag or the actual or approximate age of the relative at the time of death for the
family member history record.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "deceasedDate": {
      "description": "Deceased flag or the actual or approximate age of the relative at the time of death for the
family member history record.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_deceasedDate": {
      "description": "Extensions for deceasedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "deceasedString": {
      "description": "Deceased flag or the actual or approximate age of the relative at the time of death for the
family member history record.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_deceasedString": {
      "description": "Extensions for deceasedString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reasonCode": {
      "description": "Describes why the family member history occurred in coded or textual form.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates a Condition, Observation, AllergyIntolerance, or QuestionnaireResponse that
justifies this family member history event.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "This property allows a non condition-specific note to the made about the related person.
Ideally, the note would be in the condition property, but this is not always possible.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "condition": {
      "description": "The significant Conditions (or condition) that the family member had. This is a repeating
section to allow a system to represent more than one condition per resource, though there is nothing stopping
multiple resources - one per condition.",
      "items": {
        "$ref": "#/definitions/FamilyMemberHistory_Condition"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "patient",
    "relationship",
    "resourceType"
  ]
},
"FamilyMemberHistory_Condition": {
  "description": "Significant health conditions for a person related to the patient relevant in the context of care
for the patient.",
  "properties": {
```

```
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "code": {
        "description": "The actual condition specified. Could be a coded condition (like MI or Diabetes) or a less specific string like \u0027cancer\u0027 depending on how much is known about the condition and the capabilities of the creating system.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "outcome": {
        "description": "Indicates what happened following the condition.  If the condition resulted in death, deceased date is captured on the relation.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "contributedToDeath": {
        "description": "This condition contributed to the cause of death of the related person. If contributedToDeath is not populated, then it is unknown.",
        "$ref": "#/definitions/boolean"
      },
      "_contributedToDeath": {
        "description": "Extensions for contributedToDeath",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "onsetAge": {
        "description": "Either the age of onset, range of approximate age or descriptive string can be recorded.  For conditions with multiple occurrences, this describes the first known occurrence.",
        "$ref": "Age.schema.json#/definitions/Age"
      },
      "onsetRange": {
        "description": "Either the age of onset, range of approximate age or descriptive string can be recorded.  For conditions with multiple occurrences, this describes the first known occurrence.",
        "$ref": "Range.schema.json#/definitions/Range"
      },
      "onsetPeriod": {
        "description": "Either the age of onset, range of approximate age or descriptive string can be recorded.  For conditions with multiple occurrences, this describes the first known occurrence.",
        "$ref": "Period.schema.json#/definitions/Period"
      },
      "onsetString": {
        "description": "Either the age of onset, range of approximate age or descriptive string can be recorded.  For conditions with multiple occurrences, this describes the first known occurrence.",
        "pattern": "^[ \\r\\n\\t\\S]+$",
        "type": "string"
      },
      "_onsetString": {
        "description": "Extensions for onsetString",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "note": {
        "description": "An area where general notes can be placed about this specific condition.",
        "items": {
          "$ref": "Annotation.schema.json#/definitions/Annotation"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "code"
    ]
  }
 }
}
```

9.4.12 Resource FamilyMemberHistory - Maturity Dependency View

URL: https://hl7.org/fhir/R4/familymemberhistory-dependencies.html

9.4.12 Resource FamilyMemberHistory - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Dependency Graph for FamilyMemberHistory FMM level 2

Name | Card. | Type | Dependency Analysis
FamilyMemberHistory | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(PlanDefinition | Questionnaire | ActivityDefinition | Measure | OperationDefinition)
instantiatesUri | 0..* | uri
status | 1..1 | code
dataAbsentReason | 0..1 | CodeableConcept
patient | 1..1 | Reference(Patient)
date | 0..1 | dateTime
name | 0..1 | string
relationship | 1..1 | CodeableConcept
sex | 0..1 | CodeableConcept
born[x] | 0..1
bornPeriod | | Period
bornDate | | date
bornString | | string
age[x] | 0..1
ageAge | | Age
ageRange | | Range
ageString | | string
estimatedAge | 0..1 | boolean
deceased[x] | 0..1
deceasedBoolean | | boolean
deceasedAge | | Age
deceasedRange | | Range
deceasedDate | | date
deceasedString | | string
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | AllergyIntolerance | QuestionnaireResponse | DiagnosticReport | DocumentReference)
note | 0..* | Annotation
condition | 0..* | BackboneElement
code | 1..1 | CodeableConcept
outcome | 0..1 | CodeableConcept
contributedToDeath | 0..1 | boolean
onset[x] | 0..1
onsetAge | | Age
onsetRange | | Range
onsetPeriod | | Period
onsetString | | string
note | 0..* | Annotation

9.4.13 Value Set Sc-valueset-history-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-history-status.html

9.4.13 Value Set Sc-valueset-history-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.4.14 FamilyHistoryStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-history-status)

Mapping from http://hl7.org/fhir/ValueSet/history-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "A code that identifies the status of the family history record."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
completed | equivalent | complete

health-unknown | equivalent | unknown
partial | equivalent | partial

9.5 Resource CarePlan - Content

URL: https://hl7.org/fhir/R4/careplan.html
Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Describes the intention of how one or more practitioners intend to deliver care for a particular patient, group or community for a period of time, possibly limited to care for a specific condition or set of conditions.

9.5.1 Scope and Usage

CarePlan is one of the request resources in the FHIR workflow specification.

Care Plans are used in many areas of healthcare with a variety of scopes. They can be as simple as a general practitioner keeping track of when their patient is next due for a tetanus immunization through to a detailed plan for an oncology patient covering diet, chemotherapy, radiation, lab work and counseling with detailed timing relationships, pre-conditions and goals. They may be used in veterinary care or clinical research to describe the care of a herd or other collection of animals. In public health, they may describe education or immunization campaigns.

This resource takes an intermediate approach to complexity. It captures basic details about who is involved and what actions are intended without dealing in discrete data about dependencies and timing relationships. These can be supported where necessary using the extension mechanism.

The scope of care plans may vary widely. Examples include:

• Multi-disciplinary cross-organizational care plans; e.g. An oncology plan including the oncologist, home nursing staff, pharmacy and others

• Plans to manage specific disease/condition(s) (e.g. nutritional plan for a patient post bowel resection, neurological plan post head injury, pre-natal plan, post-partum plan, grief management plan, etc.)

• Decision support generated plans following specific practice guidelines (e.g. stroke care plan, diabetes plan, falls prevention, etc.)

• Self-maintained patient or care-giver authored plans identifying their goals and an integrated understanding of actions to be taken

This resource can be used to represent both proposed plans (for example, recommendations from a decision support engine or returned as part of a consult report) as well as active plans. The nature of the plan is communicated by the status. Some systems may need to filter CarePlans to ensure that only appropriate plans are exposed via a given user interface.

9.5.2 Boundaries and Relationships

For simplicity's sake, CarePlan allows the inline definition of activities as part of a plan using the activity.detail element. However, activities can also be defined using references to the various "request" resources. These references could be to resources with a status of "planned" or to an active order. It is possible for planned activities to exist (e.g. appointments) without needing a CarePlan at all. CarePlans are used when there's a need to group activities, goals and/or participants together to provide some degree of context.

CarePlans can be tied to specific Conditions, however they can also be condition-independent and instead focused on a particular type of care (e.g. psychological, nutritional) or the care delivered by a particular practitioner or group of practitioners.

An ImmunizationRecommendation can be interpreted as a narrow type of CarePlan dealing only with immunization events. Where such information could appear in either resource, the immunization-specific resource is preferred.

CarePlans represent a specific plan instance for a particular patient or group. It is not intended to be used to define generic plans or protocols that are independent of a specific individual or group. CarePlan represents a specific intent, not a general definition. Protocols and order sets are supported through PlanDefinition.

This resource is referenced by itself, DiagnosticReport, GuidanceResponse, ImagingStudy, Media, MedicationRequest, MedicationStatement, Observation, Procedure, QuestionnaireResponse and ServiceRequest

9.5.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle

- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
CarePlan | TU | | DomainResource | Healthcare plan for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this plan
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | Questionnaire | Measure | ActivityDefinition | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan) | Fulfills CarePlan
replaces | Σ | 0..* | Reference(CarePlan) | CarePlan replaced by this CarePlan
partOf | Σ | 0..* | Reference(CarePlan) | Part of referenced CarePlan
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | order | option Care Plan Intent (Required)
category | Σ | 0..* | CodeableConcept | Type of plan Care Plan Category (Example)
title | Σ | 0..1 | string | Human-friendly name for the care plan
description | Σ | 0..1 | string | Summary of nature of plan
subject | Σ | 1..1 | Reference(Patient | Group) | Who the care plan is for
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
period | Σ | 0..1 | Period | Time period plan covers
created | Σ | 0..1 | dateTime | Date record was first recorded
author | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam) | Who is the designated responsible party
contributor | | 0..* | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam) | Who provided the content of the care plan
careTeam | | 0..* | Reference(CareTeam) | Who's involved in plan?
addresses | Σ | 0..* | Reference(Condition) | Health issues this plan addresses
supportingInfo | | 0..* | Reference(Any) | Information considered as part of plan
goal | | 0..* | Reference(Goal) | Desired outcome of plan
activity | I | 0..* | BackboneElement | Action to occur as part of plan + Rule: Provide a reference or detail, not both
outcomeCodeableConcept | | 0..* | CodeableConcept | Results of the activity Care Plan Activity Outcome (Example)
outcomeReference | | 0..* | Reference(Any) | Appointment, Encounter, Procedure, etc.
progress | | 0..* | Annotation | Comments about the activity status/progress
reference | I | 0..1 | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription | RequestGroup) | Activity details defined in specific resource
detail | I | 0..1 | BackboneElement | In-line definition of activity
kind | | 0..1 | code | Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription Care Plan Activity Kind (Required)
instantiatesCanonical | | 0..* | canonical(PlanDefinition | ActivityDefinition | Questionnaire | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | | 0..* | uri | Instantiates external protocol or definition
code | | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
reasonCode | | 0..* | CodeableConcept | Why activity should be done or why activity was prohibited SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why activity is needed
goal | | 0..* | Reference(Goal) | Goals this activity relates to
status | ?! | 1..1 | code | not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped | unknown | entered-in-error CarePlanActivityStatus (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status
doNotPerform | ?! | 0..1 | boolean | If true, activity is prohibiting action
scheduled[x] | | 0..1 | | When activity is to occur
scheduledTiming | | | Timing
scheduledPeriod | | | Period
scheduledString | | | string
location | | 0..1 | Reference(Location) | Where it should happen
performer | | 0..* | Reference(Practitioner | PractitionerRole | Organization | RelatedPerson | Patient | CareTeam | HealthcareService | Device) | Who will be responsible?
product[x] | | 0..1 | | What is to be administered/supplied SNOMED CT Medication Codes (Example)
productCodeableConcept | | | CodeableConcept
productReference | | | Reference(Medication | Substance)
dailyAmount | | 0..1 | SimpleQuantity | How to consume/day?
quantity | | 0..1 | SimpleQuantity | How much to administer/supply/consume
description | | 0..1 | string | Extra info describing activity to perform
note | | 0..* | Annotation | Comments about the plan
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CarePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this plan --></identifier>
```

```
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|Questionnaire|Measure|
   ActivityDefinition|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan) Fulfills CarePlan --></basedOn>
 <replaces><!-- 0..* Reference(CarePlan) CarePlan replaced by this CarePlan --></replaces>
 <partOf><!-- 0..* Reference(CarePlan) Part of referenced CarePlan --></partOf>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | option -->
 <category><!-- 0..* CodeableConcept Type of plan --></category>
 <title value="[string]"/><!-- 0..1 Human-friendly name for the care plan -->
 <description value="[string]"/><!-- 0..1 Summary of nature of plan -->
 <subject><!-- 1..1 Reference(Patient|Group) Who the care plan is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <period><!-- 0..1 Period Time period plan covers --></period>
 <created value="[dateTime]"/><!-- 0..1 Date record was first recorded -->
 <author><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) Who is the designated responsible party --></author>
 <contributor><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) Who provided the content of the care plan --></contributor>
 <careTeam><!-- 0..* Reference(CareTeam) Who's involved in plan? --></careTeam>
 <addresses><!-- 0..* Reference(Condition) Health issues this plan addresses --></addresses>
 <supportingInfo><!-- 0..* Reference(Any) Information considered as part of plan --></supportingInfo>
 <goal><!-- 0..* Reference(Goal) Desired outcome of plan --></goal>
 <activity>  <!-- 0..* Action to occur as part of plan -->
  <outcomeCodeableConcept><!-- 0..* CodeableConcept Results of the activity --></outcomeCodeableConcept>
  <outcomeReference><!-- 0..* Reference(Any) Appointment, Encounter, Procedure, etc. --></outcomeReference>
  <progress><!-- 0..* Annotation Comments about the activity status/progress --></progress>
  <reference><!--  0..1 Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ServiceRequest|VisionPrescription|
    RequestGroup) Activity details defined in specific resource --></reference>
  <detail>  <!--  0..1 In-line definition of activity -->
   <kind value="[code]"/><!-- 0..1 Appointment | CommunicationRequest | DeviceRequest | MedicationRequest |
    NutritionOrder | Task | ServiceRequest | VisionPrescription -->
   <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
     Questionnaire|Measure|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
   <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
   <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
   <reasonCode><!-- 0..* CodeableConcept Why activity should be done or why activity was prohibited --></reasonCode>
   <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
     DocumentReference) Why activity is needed --></reasonReference>
   <goal><!-- 0..* Reference(Goal) Goals this activity relates to --></goal>
   <status value="[code]"/><!-- 1..1 not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped |
     unknown | entered-in-error -->
   <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
   <doNotPerform value="[boolean]"/><!-- 0..1 If true, activity is prohibiting action -->
   <scheduled[x]><!-- 0..1 Timing|Period|string When activity is to occur --></scheduled[x]>
   <location><!-- 0..1 Reference(Location) Where it should happen --></location>
   <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
     RelatedPerson|Patient|CareTeam|HealthcareService|Device) Who will be responsible? --></performer>
   <product[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance) What is to be administered/supplied
    --></product[x]>
   <dailyAmount><!-- 0..1 Quantity(SimpleQuantity) How to consume/day? --></dailyAmount>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) How much to administer/supply/consume --></quantity>
   <description value="[string]"/><!-- 0..1 Extra info describing activity to perform -->
  </detail>
 </activity>
 <note><!-- 0..* Annotation Comments about the plan --></note>
</CarePlan>
```

JSON Template

```
{
  "resourceType" : "CarePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this plan
  "instantiatesCanonical" : [{ canonical(PlanDefinition|Questionnaire|Measure|
   ActivityDefinition|OperationDefinition) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan) }], // Fulfills CarePlan
  "replaces" : [{ Reference(CarePlan) }], // CarePlan replaced by this CarePlan
  "partOf" : [{ Reference(CarePlan) }], // Part of referenced CarePlan
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | order | option
  "category" : [{ CodeableConcept }], // Type of plan
  "title" : "<string>", // Human-friendly name for the care plan
  "description" : "<string>", // Summary of nature of plan
  "subject" : { Reference(Patient|Group) }, // R!  Who the care plan is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "period" : { Period }, // Time period plan covers
  "created" : "<dateTime>", // Date record was first recorded
  "author" : { Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) }, // Who is the designated responsible party
  "contributor" : [{ Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) }], // Who provided the content of the care plan
  "careTeam" : [{ Reference(CareTeam) }], // Who's involved in plan?
  "addresses" : [{ Reference(Condition) }], // Health issues this plan addresses
  "supportingInfo" : [{ Reference(Any) }], // Information considered as part of plan
  "goal" : [{ Reference(Goal) }], // Desired outcome of plan
  "activity" : [{ // Action to occur as part of plan
```

```
    "outcomeCodeableConcept" : [{ CodeableConcept }], // Results of the activity
    "outcomeReference" : [{ Reference(Any) }], // Appointment, Encounter, Procedure, etc.
    "progress" : [{ Annotation }], // Comments about the activity status/progress
    "reference" : { Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ServiceRequest|VisionPrescription|
    RequestGroup) }, // C? Activity details defined in specific resource
    "detail" : { // C? In-line definition of activity
      "kind" : "<code>", // Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder |
    Task | ServiceRequest | VisionPrescription
      "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
     Questionnaire|Measure|OperationDefinition) }], // Instantiates FHIR protocol or definition
      "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
      "code" : { CodeableConcept }, // Detail type of activity
      "reasonCode" : [{ CodeableConcept }], // Why activity should be done or why activity was prohibited
      "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
     DocumentReference) }], // Why activity is needed
      "goal" : [{ Reference(Goal) }], // Goals this activity relates to
      "status" : "<code>", // R!  not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped |
    unknown | entered-in-error
      "statusReason" : { CodeableConcept }, // Reason for current status
      "doNotPerform" : <boolean>, // If true, activity is prohibiting action
      // scheduled[x]: When activity is to occur. One of these 3:
      "scheduledTiming" : { Timing },
      "scheduledPeriod" : { Period },
      "scheduledString" : "<string>",
      "location" : { Reference(Location) }, // Where it should happen
      "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
     RelatedPerson|Patient|CareTeam|HealthcareService|Device) }], // Who will be responsible?
      // product[x]: What is to be administered/supplied. One of these 2:
      "productCodeableConcept" : { CodeableConcept },
      "productReference" : { Reference(Medication|Substance) },
      "dailyAmount" : { Quantity(SimpleQuantity) }, // How to consume/day?
      "quantity" : { Quantity(SimpleQuantity) }, // How much to administer/supply/consume
      "description" : "<string>" // Extra info describing activity to perform
    }
  }],
  "note" : [{ Annotation }] // Comments about the plan
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CarePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CarePlan.identifier [ Identifier ], ... ; # 0..* External Ids for this plan
  fhir:CarePlan.instantiatesCanonical [
    canonical(PlanDefinition|Questionnaire|Measure|ActivityDefinition|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:CarePlan.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:CarePlan.basedOn [ Reference(CarePlan) ], ... ; # 0..* Fulfills CarePlan
  fhir:CarePlan.replaces [ Reference(CarePlan) ], ... ; # 0..* CarePlan replaced by this CarePlan
  fhir:CarePlan.partOf [ Reference(CarePlan) ], ... ; # 0..* Part of referenced CarePlan
  fhir:CarePlan.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:CarePlan.intent [ code ]; # 1..1 proposal | plan | order | option
  fhir:CarePlan.category [ CodeableConcept ], ... ; # 0..* Type of plan
  fhir:CarePlan.title [ string ]; # 0..1 Human-friendly name for the care plan
  fhir:CarePlan.description [ string ]; # 0..1 Summary of nature of plan
  fhir:CarePlan.subject [ Reference(Patient|Group) ]; # 1..1 Who the care plan is for
  fhir:CarePlan.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CarePlan.period [ Period ]; # 0..1 Time period plan covers
  fhir:CarePlan.created [ dateTime ]; # 0..1 Date record was first recorded
  fhir:CarePlan.author [ Reference(Patient|Practitioner|PractitionerRole|Device|RelatedPerson|Organization|CareTeam) ];
    # 0..1 Who is the designated responsible party
  fhir:CarePlan.contributor [
    Reference(Patient|Practitioner|PractitionerRole|Device|RelatedPerson|Organization|CareTeam) ], ... ; # 0..* Who
    provided the content of the care plan
  fhir:CarePlan.careTeam [ Reference(CareTeam) ], ... ; # 0..* Who's involved in plan?
  fhir:CarePlan.addresses [ Reference(Condition) ], ... ; # 0..* Health issues this plan addresses
  fhir:CarePlan.supportingInfo [ Reference(Any) ], ... ; # 0..* Information considered as part of plan
  fhir:CarePlan.goal [ Reference(Goal) ], ... ; # 0..* Desired outcome of plan
  fhir:CarePlan.activity [ # 0..* Action to occur as part of plan
    fhir:CarePlan.activity.outcomeCodeableConcept [ CodeableConcept ], ... ; # 0..* Results of the activity
    fhir:CarePlan.activity.outcomeReference [ Reference(Any) ], ... ; # 0..* Appointment, Encounter, Procedure, etc.
    fhir:CarePlan.activity.progress [ Annotation ], ... ; # 0..* Comments about the activity status/progress
    fhir:CarePlan.activity.reference [
    Reference(Appointment|CommunicationRequest|DeviceRequest|MedicationRequest|NutritionOrder|
  Task|ServiceRequest|VisionPrescription|RequestGroup) ]; # 0..1 Activity details defined in specific resource
    fhir:CarePlan.activity.detail [ # 0..1 In-line definition of activity
      fhir:CarePlan.activity.detail.kind [ code ]; # 0..1 Appointment | CommunicationRequest | DeviceRequest |
    MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription
      fhir:CarePlan.activity.detail.instantiatesCanonical [
    canonical(PlanDefinition|ActivityDefinition|Questionnaire|Measure|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
      fhir:CarePlan.activity.detail.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
      fhir:CarePlan.activity.detail.code [ CodeableConcept ]; # 0..1 Detail type of activity
```

```
    fhir:CarePlan.activity.detail.reasonCode [ CodeableConcept ], ... ; # 0..* Why activity should be done or why
  activity was prohibited
    fhir:CarePlan.activity.detail.reasonReference [
  Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; # 0..* Why activity is needed
    fhir:CarePlan.activity.detail.goal [ Reference(Goal) ], ... ; # 0..* Goals this activity relates to
    fhir:CarePlan.activity.detail.status [ code ]; # 1..1 not-started | scheduled | in-progress | on-hold | completed
  | cancelled | stopped | unknown | entered-in-error
    fhir:CarePlan.activity.detail.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
    fhir:CarePlan.activity.detail.doNotPerform [ boolean ]; # 0..1 If true, activity is prohibiting action
    # CarePlan.activity.detail.scheduled[x] : 0..1 When activity is to occur. One of these 3
      fhir:CarePlan.activity.detail.scheduledTiming [ Timing ]
      fhir:CarePlan.activity.detail.scheduledPeriod [ Period ]
      fhir:CarePlan.activity.detail.scheduledString [ string ]
    fhir:CarePlan.activity.detail.location [ Reference(Location) ]; # 0..1 Where it should happen
    fhir:CarePlan.activity.detail.performer [
  Reference(Practitioner|PractitionerRole|Organization|RelatedPerson|Patient|CareTeam|
HealthcareService|Device) ], ... ; # 0..* Who will be responsible?
    # CarePlan.activity.detail.product[x] : 0..1 What is to be administered/supplied. One of these 2
      fhir:CarePlan.activity.detail.productCodeableConcept [ CodeableConcept ]
      fhir:CarePlan.activity.detail.productReference [ Reference(Medication|Substance) ]
    fhir:CarePlan.activity.detail.dailyAmount [ Quantity(SimpleQuantity) ]; # 0..1 How to consume/day?
    fhir:CarePlan.activity.detail.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much to administer/supply/consume
    fhir:CarePlan.activity.detail.description [ string ]; # 0..1 Extra info describing activity to perform
  ];
], ...;
fhir:CarePlan.note [ Annotation ], ... ; # 0..* Comments about the plan
]
```

Changes since R3

CarePlan
CarePlan.instantiatesCanonical | Added Element
CarePlan.instantiatesUri | Added Element
CarePlan.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
CarePlan.intent | Change value set from http://hl7.org/fhir/ValueSet/care-plan-intent to http://hl7.org/fhir/ValueSet/care-plan-intent|4.0.1
CarePlan.encounter | Added Element
CarePlan.created | Added Element
CarePlan.author | Max Cardinality changed from * to 1 Type Reference: Added Target Types PractitionerRole, Device
CarePlan.contributor | Added Element
CarePlan.activity.reference | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
CarePlan.activity.detail.kind | Renamed from category to kind Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/care-plan-activity-kind|4.0.1 (required)
CarePlan.activity.detail.instantiatesCanonical | Added Element
CarePlan.activity.detail.instantiatesUri | Added Element
CarePlan.activity.detail.reasonReference | Type Reference: Added Target Types Observation, DiagnosticReport, DocumentReference
CarePlan.activity.detail.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-activity-status to http://hl7.org/fhir/ValueSet/care-plan-activity-status|4.0.1
CarePlan.activity.detail.statusReason | Type changed from string to CodeableConcept
CarePlan.activity.detail.doNotPerform | Renamed from prohibited to doNotPerform Default Value "false" removed
CarePlan.activity.detail.performer | Type Reference: Added Target Types PractitionerRole, HealthcareService, Device
CarePlan.definition | deleted
CarePlan.context | deleted
CarePlan.activity.detail.definition | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 10 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
CarePlan | TU | | DomainResource | Healthcare plan for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this plan
instantiatesCanonical | Σ | 0..* | canonical(PlanDefinition | Questionnaire | Measure | ActivityDefinition | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | Σ | 0..* | Reference(CarePlan) | Fulfills CarePlan
replaces | Σ | 0..* | Reference(CarePlan) | CarePlan replaced by this CarePlan
partOf | Σ | 0..* | Reference(CarePlan) | Part of referenced CarePlan
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | order | option Care Plan Intent (Required)
category | Σ | 0..* | CodeableConcept | Type of plan Care Plan Category (Example)
title | Σ | 0..1 | string | Human-friendly name for the care plan
description | Σ | 0..1 | string | Summary of nature of plan

subject | Σ | 1..1 | Reference(Patient | Group) | Who the care plan is for
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
period | Σ | 0..1 | Period | Time period plan covers
created | Σ | 0..1 | dateTime | Date record was first recorded
author | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam) | Who is the designated responsible party
contributor | | 0..* | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam) | Who provided the content of the care plan
careTeam | | 0..* | Reference(CareTeam) | Who's involved in plan?
addresses | Σ | 0..* | Reference(Condition) | Health issues this plan addresses
supportingInfo | | 0..* | Reference(Any) | Information considered as part of plan
goal | | 0..* | Reference(Goal) | Desired outcome of plan
activity | I | 0..* | BackboneElement | Action to occur as part of plan + Rule: Provide a reference or detail, not both
outcomeCodeableConcept | | 0..* | CodeableConcept | Results of the activity Care Plan Activity Outcome (Example)
outcomeReference | | 0..* | Reference(Any) | Appointment, Encounter, Procedure, etc.
progress | | 0..* | Annotation | Comments about the activity status/progress
reference | I | 0..1 | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription | RequestGroup) | Activity details defined in specific resource
detail | I | 0..1 | BackboneElement | In-line definition of activity
kind | | 0..1 | code | Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription Care Plan Activity Kind (Required)
instantiatesCanonical | | 0..* | canonical(PlanDefinition | ActivityDefinition | Questionnaire | Measure | OperationDefinition) | Instantiates FHIR protocol or definition
instantiatesUri | | 0..* | uri | Instantiates external protocol or definition
code | | 0..1 | CodeableConcept | Detail type of activity Procedure Codes (SNOMED CT) (Example)
reasonCode | | 0..* | CodeableConcept | Why activity should be done or why activity was prohibited SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why activity is needed
goal | | 0..* | Reference(Goal) | Goals this activity relates to
status | ?! | 1..1 | code | not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped | unknown | entered-in-error CarePlanActivityStatus (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status
doNotPerform | ?! | 0..1 | boolean | If true, activity is prohibiting action
scheduled[x] | | 0..1 | | When activity is to occur
scheduledTiming | | | Timing
scheduledPeriod | | | Period
scheduledString | | | string
location | | 0..1 | Reference(Location) | Where it should happen
performer | | 0..* | Reference(Practitioner | PractitionerRole | Organization | RelatedPerson | Patient | CareTeam | HealthcareService | Device) | Who will be responsible?
product[x] | | 0..1 | | What is to be administered/supplied SNOMED CT Medication Codes (Example)
productCodeableConcept | | | CodeableConcept
productReference | | | Reference(Medication | Substance)
dailyAmount | | 0..1 | SimpleQuantity | How to consume/day?
quantity | | 0..1 | SimpleQuantity | How much to administer/supply/consume
description | | 0..1 | string | Extra info describing activity to perform
note | | 0..* | Annotation | Comments about the plan
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CarePlan xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this plan --></identifier>
 <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|Questionnaire|Measure|
   ActivityDefinition|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(CarePlan) Fulfills CarePlan --></basedOn>
 <replaces><!-- 0..* Reference(CarePlan) CarePlan replaced by this CarePlan --></replaces>
 <partOf><!-- 0..* Reference(CarePlan) Part of referenced CarePlan --></partOf>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order | option -->
 <category><!-- 0..* CodeableConcept Type of plan --></category>
 <title value="[string]"/><!-- 0..1 Human-friendly name for the care plan -->
 <description value="[string]"/><!-- 0..1 Summary of nature of plan -->
 <subject><!-- 1..1 Reference(Patient|Group) Who the care plan is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <period><!-- 0..1 Period Time period plan covers --></period>
 <created value="[dateTime]"/><!-- 0..1 Date record was first recorded -->
 <author><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) Who is the designated responsible party --></author>
 <contributor><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) Who provided the content of the care plan --></contributor>
 <careTeam><!-- 0..* Reference(CareTeam) Who's involved in plan? --></careTeam>
 <addresses><!-- 0..* Reference(Condition) Health issues this plan addresses --></addresses>
 <supportingInfo><!-- 0..* Reference(Any) Information considered as part of plan --></supportingInfo>
 <goal><!-- 0..* Reference(Goal) Desired outcome of plan --></goal>
```

```
 <activity>  <!-- 0..* Action to occur as part of plan -->
  <outcomeCodeableConcept><!-- 0..* CodeableConcept Results of the activity --></outcomeCodeableConcept>
  <outcomeReference><!-- 0..* Reference(Any) Appointment, Encounter, Procedure, etc. --></outcomeReference>
  <progress><!-- 0..* Annotation Comments about the activity status/progress --></progress>
  <reference><!--  0..1 Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ServiceRequest|VisionPrescription|
    RequestGroup) Activity details defined in specific resource --></reference>
  <detail>  <!--  0..1 In-line definition of activity -->
   <kind value="[code]"/><!-- 0..1 Appointment | CommunicationRequest | DeviceRequest | MedicationRequest |
    NutritionOrder | Task | ServiceRequest | VisionPrescription -->
   <instantiatesCanonical><!-- 0..* canonical(PlanDefinition|ActivityDefinition|
     Questionnaire|Measure|OperationDefinition) Instantiates FHIR protocol or definition --></instantiatesCanonical>
   <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
   <code><!-- 0..1 CodeableConcept Detail type of activity --></code>
   <reasonCode><!-- 0..* CodeableConcept Why activity should be done or why activity was prohibited --></reasonCode>
   <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
     DocumentReference) Why activity is needed --></reasonReference>
   <goal><!-- 0..* Reference(Goal) Goals this activity relates to --></goal>
   <status value="[code]"/><!-- 1..1 not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped |
     unknown | entered-in-error -->
   <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
   <doNotPerform value="[boolean]"/><!-- 0..1 If true, activity is prohibiting action -->
   <scheduled[x]><!-- 0..1 Timing|Period|string When activity is to occur --></scheduled[x]>
   <location><!-- 0..1 Reference(Location) Where it should happen --></location>
   <performer><!-- 0..* Reference(Practitioner|PractitionerRole|Organization|
     RelatedPerson|Patient|CareTeam|HealthcareService|Device) Who will be responsible? --></performer>
   <product[x]><!-- 0..1 CodeableConcept|Reference(Medication|Substance) What is to be administered/supplied
    --></product[x]>
   <dailyAmount><!-- 0..1 Quantity(SimpleQuantity) How to consume/day? --></dailyAmount>
   <quantity><!-- 0..1 Quantity(SimpleQuantity) How much to administer/supply/consume --></quantity>
   <description value="[string]"/><!-- 0..1 Extra info describing activity to perform -->
  </detail>
 </activity>
 <note><!-- 0..* Annotation Comments about the plan --></note>
</CarePlan>
```

JSON Template

```
{
  "resourceType" : "CarePlan",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this plan
  "instantiatesCanonical" : [{ canonical(PlanDefinition|Questionnaire|Measure|
   ActivityDefinition|OperationDefinition) }], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(CarePlan) }], // Fulfills CarePlan
  "replaces" : [{ Reference(CarePlan) }], // CarePlan replaced by this CarePlan
  "partOf" : [{ Reference(CarePlan) }], // Part of referenced CarePlan
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | order | option
  "category" : [{ CodeableConcept }], // Type of plan
  "title" : "<string>", // Human-friendly name for the care plan
  "description" : "<string>", // Summary of nature of plan
  "subject" : { Reference(Patient|Group) }, // R!  Who the care plan is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "period" : { Period }, // Time period plan covers
  "created" : "<dateTime>", // Date record was first recorded
  "author" : { Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) }, // Who is the designated responsible party
  "contributor" : [{ Reference(Patient|Practitioner|PractitionerRole|Device|
   RelatedPerson|Organization|CareTeam) }], // Who provided the content of the care plan
  "careTeam" : [{ Reference(CareTeam) }], // Who's involved in plan?
  "addresses" : [{ Reference(Condition) }], // Health issues this plan addresses
  "supportingInfo" : [{ Reference(Any) }], // Information considered as part of plan
  "goal" : [{ Reference(Goal) }], // Desired outcome of plan
  "activity" : [{ // Action to occur as part of plan
    "outcomeCodeableConcept" : [{ CodeableConcept }], // Results of the activity
    "outcomeReference" : [{ Reference(Any) }], // Appointment, Encounter, Procedure, etc.
    "progress" : [{ Annotation }], // Comments about the activity status/progress
    "reference" : { Reference(Appointment|CommunicationRequest|DeviceRequest|
    MedicationRequest|NutritionOrder|Task|ServiceRequest|VisionPrescription|
    RequestGroup) }, // C? Activity details defined in specific resource
    "detail" : { // C? In-line definition of activity
      "kind" : "<code>", // Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder |
    Task | ServiceRequest | VisionPrescription
      "instantiatesCanonical" : [{ canonical(PlanDefinition|ActivityDefinition|
     Questionnaire|Measure|OperationDefinition) }], // Instantiates FHIR protocol or definition
      "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
      "code" : { CodeableConcept }, // Detail type of activity
      "reasonCode" : [{ CodeableConcept }], // Why activity should be done or why activity was prohibited
      "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
     DocumentReference) }], // Why activity is needed
      "goal" : [{ Reference(Goal) }], // Goals this activity relates to
      "status" : "<code>", // R!  not-started | scheduled | in-progress | on-hold | completed | cancelled | stopped |
    unknown | entered-in-error
      "statusReason" : { CodeableConcept }, // Reason for current status
      "doNotPerform" : <boolean>, // If true, activity is prohibiting action
      // scheduled[x]: When activity is to occur. One of these 3:
      "scheduledTiming" : { Timing },
```

```
      "scheduledPeriod" : { Period },
      "scheduledString" : "<string>",
      "location" : { Reference(Location) }, // Where it should happen
      "performer" : [{ Reference(Practitioner|PractitionerRole|Organization|
     RelatedPerson|Patient|CareTeam|HealthcareService|Device) }], // Who will be responsible?
      // product[x]: What is to be administered/supplied. One of these 2:
      "productCodeableConcept" : { CodeableConcept },
      "productReference" : { Reference(Medication|Substance) },
      "dailyAmount" : { Quantity(SimpleQuantity) }, // How to consume/day?
      "quantity" : { Quantity(SimpleQuantity) }, // How much to administer/supply/consume
      "description" : "<string>" // Extra info describing activity to perform
    }
  }],
  "note" : [{ Annotation }] // Comments about the plan
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CarePlan;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CarePlan.identifier [ Identifier ], ... ; # 0..* External Ids for this plan
  fhir:CarePlan.instantiatesCanonical [
    canonical(PlanDefinition|Questionnaire|Measure|ActivityDefinition|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
  fhir:CarePlan.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:CarePlan.basedOn [ Reference(CarePlan) ], ... ; # 0..* Fulfills CarePlan
  fhir:CarePlan.replaces [ Reference(CarePlan) ], ... ; # 0..* CarePlan replaced by this CarePlan
  fhir:CarePlan.partOf [ Reference(CarePlan) ], ... ; # 0..* Part of referenced CarePlan
  fhir:CarePlan.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:CarePlan.intent [ code ]; # 1..1 proposal | plan | order | option
  fhir:CarePlan.category [ CodeableConcept ], ... ; # 0..* Type of plan
  fhir:CarePlan.title [ string ]; # 0..1 Human-friendly name for the care plan
  fhir:CarePlan.description [ string ]; # 0..1 Summary of nature of plan
  fhir:CarePlan.subject [ Reference(Patient|Group) ]; # 1..1 Who the care plan is for
  fhir:CarePlan.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CarePlan.period [ Period ]; # 0..1 Time period plan covers
  fhir:CarePlan.created [ dateTime ]; # 0..1 Date record was first recorded
  fhir:CarePlan.author [ Reference(Patient|Practitioner|PractitionerRole|Device|RelatedPerson|Organization|CareTeam) ];
    # 0..1 Who is the designated responsible party
  fhir:CarePlan.contributor [
    Reference(Patient|Practitioner|PractitionerRole|Device|RelatedPerson|Organization|CareTeam) ], ... ; # 0..* Who
    provided the content of the care plan
  fhir:CarePlan.careTeam [ Reference(CareTeam) ], ... ; # 0..* Who's involved in plan?
  fhir:CarePlan.addresses [ Reference(Condition) ], ... ; # 0..* Health issues this plan addresses
  fhir:CarePlan.supportingInfo [ Reference(Any) ], ... ; # 0..* Information considered as part of plan
  fhir:CarePlan.goal [ Reference(Goal) ], ... ; # 0..* Desired outcome of plan
  fhir:CarePlan.activity [ # 0..* Action to occur as part of plan
    fhir:CarePlan.activity.outcomeCodeableConcept [ CodeableConcept ], ... ; # 0..* Results of the activity
    fhir:CarePlan.activity.outcomeReference [ Reference(Any) ], ... ; # 0..* Appointment, Encounter, Procedure, etc.
    fhir:CarePlan.activity.progress [ Annotation ], ... ; # 0..* Comments about the activity status/progress
    fhir:CarePlan.activity.reference [
    Reference(Appointment|CommunicationRequest|DeviceRequest|MedicationRequest|NutritionOrder|
  Task|ServiceRequest|VisionPrescription|RequestGroup) ]; # 0..1 Activity details defined in specific resource
    fhir:CarePlan.activity.detail [ # 0..1 In-line definition of activity
      fhir:CarePlan.activity.detail.kind [ code ]; # 0..1 Appointment | CommunicationRequest | DeviceRequest |
    MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription
      fhir:CarePlan.activity.detail.instantiatesCanonical [
    canonical(PlanDefinition|ActivityDefinition|Questionnaire|Measure|OperationDefinition) ], ... ; # 0..* Instantiates
    FHIR protocol or definition
      fhir:CarePlan.activity.detail.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
      fhir:CarePlan.activity.detail.code [ CodeableConcept ]; # 0..1 Detail type of activity
      fhir:CarePlan.activity.detail.reasonCode [ CodeableConcept ], ... ; # 0..* Why activity should be done or why
    activity was prohibited
      fhir:CarePlan.activity.detail.reasonReference [
    Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; # 0..* Why activity is needed
      fhir:CarePlan.activity.detail.goal [ Reference(Goal) ], ... ; # 0..* Goals this activity relates to
      fhir:CarePlan.activity.detail.status [ code ]; # 1..1 not-started | scheduled | in-progress | on-hold | completed
    | cancelled | stopped | unknown | entered-in-error
      fhir:CarePlan.activity.detail.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
      fhir:CarePlan.activity.detail.doNotPerform [ boolean ]; # 0..1 If true, activity is prohibiting action
      # CarePlan.activity.detail.scheduled[x] : 0..1 When activity is to occur. One of these 3
        fhir:CarePlan.activity.detail.scheduledTiming [ Timing ]
        fhir:CarePlan.activity.detail.scheduledPeriod [ Period ]
        fhir:CarePlan.activity.detail.scheduledString [ string ]
      fhir:CarePlan.activity.detail.location [ Reference(Location) ]; # 0..1 Where it should happen
      fhir:CarePlan.activity.detail.performer [
    Reference(Practitioner|PractitionerRole|Organization|RelatedPerson|Patient|CareTeam|
  HealthcareService|Device) ], ... ; # 0..* Who will be responsible?
      # CarePlan.activity.detail.product[x] : 0..1 What is to be administered/supplied. One of these 2
        fhir:CarePlan.activity.detail.productCodeableConcept [ CodeableConcept ]
        fhir:CarePlan.activity.detail.productReference [ Reference(Medication|Substance) ]
      fhir:CarePlan.activity.detail.dailyAmount [ Quantity(SimpleQuantity) ]; # 0..1 How to consume/day?
      fhir:CarePlan.activity.detail.quantity [ Quantity(SimpleQuantity) ]; # 0..1 How much to administer/supply/consume
      fhir:CarePlan.activity.detail.description [ string ]; # 0..1 Extra info describing activity to perform
```

```
    ];
  ], ...;
  fhir:CarePlan.note [ Annotation ], ... ; # 0..* Comments about the plan
]
```

Changes since Release 3

CarePlan
CarePlan.instantiatesCanonical | Added Element
CarePlan.instantiatesUri | Added Element
CarePlan.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
CarePlan.intent | Change value set from http://hl7.org/fhir/ValueSet/care-plan-intent to http://hl7.org/fhir/ValueSet/care-plan-intent|4.0.1
CarePlan.encounter | Added Element
CarePlan.created | Added Element
CarePlan.author | Max Cardinality changed from * to 1 Type Reference: Added Target Types PractitionerRole, Device
CarePlan.contributor | Added Element
CarePlan.activity.reference | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Types ProcedureRequest, ReferralRequest
CarePlan.activity.detail.kind | Renamed from category to kind Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/care-plan-activity-kind|4.0.1 (required)
CarePlan.activity.detail.instantiatesCanonical | Added Element
CarePlan.activity.detail.instantiatesUri | Added Element
CarePlan.activity.detail.reasonReference | Type Reference: Added Target Types Observation, DiagnosticReport, DocumentReference
CarePlan.activity.detail.status | Change value set from http://hl7.org/fhir/ValueSet/care-plan-activity-status to http://hl7.org/fhir/ValueSet/care-plan-activity-status|4.0.1
CarePlan.activity.detail.statusReason | Type changed from string to CodeableConcept
CarePlan.activity.detail.doNotPerform | Renamed from prohibited to doNotPerform Default Value "false" removed
CarePlan.activity.detail.performer | Type Reference: Added Target Types PractitionerRole, HealthcareService, Device
CarePlan.definition | deleted
CarePlan.context | deleted
CarePlan.activity.detail.definition | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 10 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.5.3.1 Terminology Bindings

Path | Definition | Type | Reference
CarePlan.status | Indicates whether the plan is currently being acted upon, represents future intentions or is now a historical record. | Required | RequestStatus
CarePlan.intent | Codes indicating the degree of authority/intentionality associated with a care plan. | Required | CarePlanIntent
CarePlan.category | Identifies what "kind" of plan this is to support differentiation between multiple co-existing plans; e.g. "Home health", "psychiatric", "asthma", "disease management", etc. | Example | CarePlanCategory
CarePlan.activity.outcomeCodeableConcept | Identifies the results of the activity. | Example | CarePlanActivityOutcome
CarePlan.activity.detail.kind | Resource types defined as part of FHIR that can be represented as in-line definitions of a care plan activity. | Required | CarePlanActivityKind
CarePlan.activity.detail.code | Detailed description of the type of activity; e.g. What lab test, what procedure, what kind of encounter. | Example | ProcedureCodes(SNOMEDCT)
CarePlan.activity.detail.reasonCode | Identifies why a care plan activity is needed. Can include any health condition codes as well as such concepts as "general wellness", prophylaxis, surgical preparation, etc. | Example | SNOMEDCTClinicalFindings
CarePlan.activity.detail.status | Codes that reflect the current state of a care plan activity within its overall life cycle. | Required | CarePlanActivityStatus
CarePlan.activity.detail.product[x] | A product supplied or administered as part of a care plan activity. | Example | SNOMEDCTMedicationCodes

9.5.3.2 Constraints

id | Level | Location | Description | Expression
cpl-3 | Rule | CarePlan.activity | Provide a reference or detail, not both | detail.empty() or reference.empty()

9.5.3.3 Notes

The Provenance resource can be used for detailed review information, such as when the care plan was last reviewed and by whom.

9.5.4 Open Issues

9.5.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
activity-code | token | Detail type of activity | CarePlan.activity.detail.code
activity-date | date | Specified date occurs within period specified by CarePlan.activity.detail.scheduled[x] | CarePlan.activity.detail.scheduled
activity-reference | reference | Activity details defined in specific resource | CarePlan.activity.reference (Appointment, MedicationRequest, Task, NutritionOrder, RequestGroup, VisionPrescription, DeviceRequest, ServiceRequest, CommunicationRequest)
based-on | reference | Fulfills CarePlan | CarePlan.basedOn (CarePlan)
care-team | reference | Who's involved in plan? | CarePlan.careTeam (CareTeam)
category | token | Type of plan | CarePlan.category
condition | reference | Health issues this plan addresses | CarePlan.addresses (Condition)
date | date | Time period plan covers | CarePlan.period | 17 Resources
encounter | reference | Encounter created as part of | CarePlan.encounter (Encounter)
goal | reference | Desired outcome of plan | CarePlan.goal (Goal)
identifier | token | External Ids for this plan | CarePlan.identifier | 30 Resources
instantiates-canonical | reference | Instantiates FHIR protocol or definition | CarePlan.instantiatesCanonical (Questionnaire, Measure, PlanDefinition, OperationDefinition, ActivityDefinition)
instantiates-uri | uri | Instantiates external protocol or definition | CarePlan.instantiatesUri
intent | token | proposal | plan | order | option | CarePlan.intent
part-of | reference | Part of referenced CarePlan | CarePlan.partOf (CarePlan)
patient | reference | Who the care plan is for | CarePlan.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | Matches if the practitioner is listed as a performer in any of the "simple" activities. (For performers of the detailed activities, chain through the activitydetail search parameter.) | CarePlan.activity.detail.performer (Practitioner, Organization, CareTeam, Device, Patient, HealthcareService, PractitionerRole, RelatedPerson)
replaces | reference | CarePlan replaced by this CarePlan | CarePlan.replaces (CarePlan)
status | token | draft | active | on-hold | revoked | completed | entered-in-error | unknown | CarePlan.status
subject | reference | Who the care plan is for | CarePlan.subject (Group, Patient)

9.5.6 Resource CarePlan - Examples

URL: https://hl7.org/fhir/R4/careplan-examples.html

9.5.6 Resource CarePlan - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Care plan to address obesity that has a goal of weight loss | example | XML | JSON | Turtle
Care plan to address pregnancy | preg | XML | JSON | Turtle
Integrated care plan with multiple goals & conditions (from Mayo) | integrate | XML | JSON | Turtle
Encounter-specific care plan for GP visit | gpvisit | XML | JSON | Turtle
Care plan for an operative procedure on heart | f001 | XML | JSON | Turtle
Care plan for a partial lobectomy of lung | f002 | XML | JSON | Turtle
Care plan with goal of retropharyngeal abscess removal | f003 | XML | JSON | Turtle
Care plan to address renal insufficiency | f201 | XML | JSON | Turtle
Care plan to address head-neck tumor by chemotherapy | f202 | XML | JSON | Turtle
Care plan to address sepsis | f203 | XML | JSON | Turtle
Primarily narrative care plan to address obesity | obesity-narrative | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.5.7 Resource CarePlan - Detailed Descriptions

URL: https://hl7.org/fhir/R4/careplan-definitions.html

9.5.7 Resource CarePlan - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CarePlan resource.

CarePlan
Element Id | CarePlan
Definition | Describes the intention of how one or more practitioners intend to deliver care for a particular patient, group or community for a period of time, possibly limited to care for a specific condition or set of conditions.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Care Team
CarePlan.identifier
Element Id | CarePlan.identifier

Definition | Business identifiers assigned to this care plan by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the care plan as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
CarePlan.instantiatesCanonical
Element Id | CarePlan.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan.
Cardinality | 0..*
Type | canonical(PlanDefinition | Questionnaire | Measure | ActivityDefinition | OperationDefinition)
Summary | true
CarePlan.instantiatesUri
Element Id | CarePlan.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan.
Cardinality | 0..*
Type | uri
Summary | true
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
CarePlan.basedOn
Element Id | CarePlan.basedOn
Definition | A care plan that is fulfilled in whole or in part by this care plan.
Cardinality | 0..*
Type | Reference(CarePlan)
Hierarchy | This reference is part of a strict Hierarchy
Requirements | Allows tracing of the care plan and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
CarePlan.replaces
Element Id | CarePlan.replaces
Definition | Completed or terminated care plan whose function is taken by this new care plan.
Cardinality | 0..*
Type | Reference(CarePlan)
Hierarchy | This reference is part of a strict Hierarchy
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple care plans.
Alternate Names | supersedes
Summary | true
Comments | The replacement could be because the initial care plan was immediately rejected (due to an issue) or because the previous care plan was completed, but the need for the action described by the care plan remains ongoing.
CarePlan.partOf
Element Id | CarePlan.partOf
Definition | A larger care plan of which this particular care plan is a component or step.
Cardinality | 0..*
Type | Reference(CarePlan)
Hierarchy | This reference is part of a strict Hierarchy
Summary | true
Comments | Each care plan is an independent request, such that having a care plan be part of another care plan can cause issues with cascading statuses. As such, this element is still being discussed.
CarePlan.status
Element Id | CarePlan.status
Definition | Indicates whether the plan is currently being acted upon, represents future intentions or is now a historical record.
Cardinality | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Allows clinicians to determine whether the plan is actionable or not.
Summary | true
Comments | The unknown code is not to be used to convey other statuses. The unknown code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the care plan. This element is labeled as a modifier because the status contains the code entered-in-error that marks the plan as not currently valid.
CarePlan.intent
Element Id | CarePlan.intent
Definition | Indicates the level of authority/intentionality associated with the care plan and where the care plan fits into the workflow chain.

Cardinality | 1..1
Terminology Binding | Care Plan Intent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request")
Requirements | Proposals/recommendations, plans and orders all use the same structure and can exist in the same fulfillment chain.
Summary | true
Comments | This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
CarePlan.category
Element Id | CarePlan.category
Definition | Identifies what "kind" of plan this is to support differentiation between multiple co-existing plans; e.g. "Home health", "psychiatric", "asthma", "disease management", "wellness plan", etc.
Cardinality | 0..*
Terminology Binding | Care Plan Category (Example)
Type | CodeableConcept
Requirements | Used for filtering what plan(s) are retrieved and displayed to different types of users.
Summary | true
Comments | There may be multiple axes of categorization and one plan may serve multiple purposes. In some cases, this may be redundant with references to CarePlan.concern.
CarePlan.title
Element Id | CarePlan.title
Definition | Human-friendly name for the care plan.
Cardinality | 0..1
Type | string
Summary | true
CarePlan.description
Element Id | CarePlan.description
Definition | A description of the scope and nature of the plan.
Cardinality | 0..1
Type | string
Requirements | Provides more detail than conveyed by category.
Summary | true
CarePlan.subject
Element Id | CarePlan.subject
Definition | Identifies the patient or group whose intended care is described by the plan.
Cardinality | 1..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
CarePlan.encounter
Element Id | CarePlan.encounter
Definition | The Encounter during which this CarePlan was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter. CarePlan activities conducted as a result of the care plan may well occur as part of other encounters.
CarePlan.period
Element Id | CarePlan.period
Definition | Indicates when the plan did (or is intended to) come into effect and end.
Cardinality | 0..1
Type | Period
Requirements | Allows tracking what plan(s) are in effect at a particular time.
Alternate Names | timing
Summary | true
Comments | Any activities scheduled as part of the plan should be constrained to the specified period regardless of whether the activities are planned within a single encounter/episode or across multiple encounters/episodes (e.g. the longitudinal management of a chronic condition).
CarePlan.created
Element Id | CarePlan.created
Definition | Represents when this particular CarePlan record was created in the system, which is often a system-generated date.
Cardinality | 0..1
Type | dateTime
Alternate Names | authoredOn
Summary | true
CarePlan.author
Element Id | CarePlan.author
Definition | When populated, the author is responsible for the care plan. The care plan is attributed to the author.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam)

Summary | true
Comments | The author may also be a contributor. For example, an organization can be an author, but not listed as a contributor.
CarePlan.contributor
Element Id | CarePlan.contributor
Definition | Identifies the individual(s) or organization who provided the contents of the care plan.
Cardinality | 0..*
Type | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam)
Comments | Collaborative care plans may have multiple contributors.
CarePlan.careTeam
Element Id | CarePlan.careTeam
Definition | Identifies all people and organizations who are expected to be involved in the care envisioned by this plan.
Cardinality | 0..*
Type | Reference(CareTeam)
Requirements | Allows representation of care teams, helps scope care plan. In some cases may be a determiner of access permissions.
CarePlan.addresses
Element Id | CarePlan.addresses
Definition | Identifies the conditions/problems/concerns/diagnoses/etc. whose management and/or mitigation are handled by this plan.
Cardinality | 0..*
Type | Reference(Condition)
Requirements | Links plan to the conditions it manages. The element can identify risks addressed by the plan as well as active conditions. (The Condition resource can include things like "at risk for hypertension" or "fall risk".) Also scopes plans - multiple plans may exist addressing different concerns.
Summary | true
Comments | When the diagnosis is related to an allergy or intolerance, the Condition and AllergyIntolerance resources can both be used. However, to be actionable for decision support, using Condition alone is not sufficient as the allergy or intolerance condition needs to be represented as an AllergyIntolerance.
CarePlan.supportingInfo
Element Id | CarePlan.supportingInfo
Definition | Identifies portions of the patient's record that specifically influenced the formation of the plan. These might include comorbidities, recent procedures, limitations, recent assessments, etc.
Cardinality | 0..*
Type | Reference(Any)
Requirements | Identifies barriers and other considerations associated with the care plan.
Comments | Use "concern" to identify specific conditions addressed by the care plan.
CarePlan.goal
Element Id | CarePlan.goal
Definition | Describes the intended objective(s) of carrying out the care plan.
Cardinality | 0..*
Type | Reference(Goal)
Requirements | Provides context for plan. Allows plan effectiveness to be evaluated by clinicians.
Comments | Goal can be achieving a particular change or merely maintaining a current state or even slowing a decline.
CarePlan.activity
Element Id | CarePlan.activity
Definition | Identifies a planned action to occur as part of the plan. For example, a medication to be used, lab tests to perform, self-monitoring, education, etc.
Cardinality | 0..*
Requirements | Allows systems to prompt for performance of planned activities, and validate plans against best practice.
Invariants | Defined on this element cpl-3 Rule Provide a reference or detail, not both detail.empty() or reference.empty()
Defined on this element
cpl-3 | Rule | Provide a reference or detail, not both | detail.empty() or reference.empty()
CarePlan.activity.outcomeCodeableConcept
Element Id | CarePlan.activity.outcomeCodeableConcept
Definition | Identifies the outcome at the point when the status of the activity is assessed. For example, the outcome of an education activity could be patient understands (or not).
Cardinality | 0..*
Terminology Binding | Care Plan Activity Outcome (Example)
Type | CodeableConcept
Comments | Note that this should not duplicate the activity status (e.g. completed or in progress).
CarePlan.activity.outcomeReference
Element Id | CarePlan.activity.outcomeReference
Definition | Details of the outcome or action resulting from the activity. The reference to an "event" resource, such as Procedure or Encounter or Observation, is the result/outcome of the activity itself. The activity can be conveyed using CarePlan.activity.detail OR using the CarePlan.activity.reference (a reference to a “request” resource).
Cardinality | 0..*
Type | Reference(Any)
Requirements | Links plan to resulting actions.
Comments | The activity outcome is independent of the outcome of the related goal(s). For example, if the goal is to achieve a target body weight of 150 lbs and an activity is defined to diet, then the activity outcome could be calories consumed whereas the goal outcome is an observation for the actual body weight measured.
CarePlan.activity.progress
Element Id | CarePlan.activity.progress

Definition | Notes about the adherence/status/progress of the activity.
Cardinality | 0..*
Type | Annotation
Requirements | Can be used to capture information about adherence, progress, concerns, etc.
Comments | This element should NOT be used to describe the activity to be performed - that occurs either within the resource pointed to by activity.detail.reference or in activity.detail.description.
CarePlan.activity.reference
Element Id | CarePlan.activity.reference
Definition | The details of the proposed activity represented in a specific resource.
Cardinality | 0..1
Type | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription | RequestGroup)
Requirements | Details in a form consistent with other applications and contexts of use.
Comments | Standard extension exists (resource-pertainsToGoal) that allows goals to be referenced from any of the referenced resources in CarePlan.activity.reference. The goal should be visible when the resource referenced by CarePlan.activity.reference is viewed independently from the CarePlan. Requests that are pointed to by a CarePlan using this element should not point to this CarePlan using the "basedOn" element. i.e. Requests that are part of a CarePlan are not "based on" the CarePlan.
Invariants | Affect this element cpl-3 Rule Provide a reference or detail, not both detail.empty() or reference.empty()
Affect this element
cpl-3 | Rule | Provide a reference or detail, not both | detail.empty() or reference.empty()
CarePlan.activity.detail
Element Id | CarePlan.activity.detail
Definition | A simple summary of a planned activity suitable for a general care plan system (e.g. form driven) that doesn't know about specific resources such as procedure etc.
Cardinality | 0..1
Requirements | Details in a simple form for generic care plan systems.
Invariants | Affect this element cpl-3 Rule Provide a reference or detail, not both detail.empty() or reference.empty()
Affect this element
cpl-3 | Rule | Provide a reference or detail, not both | detail.empty() or reference.empty()
CarePlan.activity.detail.kind
Element Id | CarePlan.activity.detail.kind
Definition | A description of the kind of resource the in-line definition of a care plan activity is representing. The CarePlan.activity.detail is an in-line definition when a resource is not referenced using CarePlan.activity.reference. For example, a MedicationRequest, a ServiceRequest, or a CommunicationRequest.
Cardinality | 0..1
Terminology Binding | Care Plan Activity Kind (Required)
Type | code
Requirements | May determine what types of extensions are permitted.
CarePlan.activity.detail.instantiatesCanonical
Element Id | CarePlan.activity.detail.instantiatesCanonical
Definition | The URL pointing to a FHIR-defined protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan activity.
Cardinality | 0..*
Type | canonical(PlanDefinition | ActivityDefinition | Questionnaire | Measure | OperationDefinition)
Requirements | Allows Questionnaires that the patient (or practitioner) should fill in to fulfill the care plan activity.
CarePlan.activity.detail.instantiatesUri
Element Id | CarePlan.activity.detail.instantiatesUri
Definition | The URL pointing to an externally maintained protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan activity.
Cardinality | 0..*
Type | uri
Requirements | Allows Questionnaires that the patient (or practitioner) should fill in to fulfill the care plan activity.
Comments | This might be an HTML page, PDF, etc. or could just be a non-resolvable URI identifier.
CarePlan.activity.detail.code
Element Id | CarePlan.activity.detail.code
Definition | Detailed description of the type of planned activity; e.g. what lab test, what procedure, what kind of encounter.
Cardinality | 0..1
Terminology Binding | Procedure Codes (SNOMED CT) (Example)
Type | CodeableConcept
Requirements | Allows matching performed to planned as well as validation against protocols.
Comments | Tends to be less relevant for activities involving particular products. Codes should not convey negation - use "prohibited" instead.
CarePlan.activity.detail.reasonCode
Element Id | CarePlan.activity.detail.reasonCode
Definition | Provides the rationale that drove the inclusion of this particular activity as part of the plan or the reason why the activity was prohibited.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Comments | This could be a diagnosis code. If a full condition record exists or additional detail is needed, use reasonCondition instead.
CarePlan.activity.detail.reasonReference
Element Id | CarePlan.activity.detail.reasonReference

Definition | Indicates another resource, such as the health condition(s), whose existence justifies this request and drove the inclusion of this particular activity as part of the plan.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
Comments | Conditions can be identified at the activity level that are not identified as reasons for the overall plan.
CarePlan.activity.detail.goal
Element Id | CarePlan.activity.detail.goal
Definition | Internal reference that identifies the goals that this activity is intended to contribute towards meeting.
Cardinality | 0..*
Type | Reference(Goal)
Requirements | So that participants know the link explicitly.
CarePlan.activity.detail.status
Element Id | CarePlan.activity.detail.status
Definition | Identifies what progress is being made for the specific activity.
Cardinality | 1..1
Terminology Binding | CarePlanActivityStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the activity should not be treated as valid)
Requirements | Indicates progress against the plan, whether the activity is still relevant for the plan.
Comments | Some aspects of status can be inferred based on the resources linked in actionTaken. Note that "status" is only as current as the plan was most recently updated. The unknown code is not to be used to convey other statuses. The unknown code should be used when one of the statuses applies, but the authoring system doesn't know the current state of the activity.
CarePlan.activity.detail.statusReason
Element Id | CarePlan.activity.detail.statusReason
Definition | Provides reason why the activity isn't yet started, is on hold, was cancelled, etc.
Cardinality | 0..1
Type | CodeableConcept
Comments | Will generally not be present if status is "complete". Be sure to prompt to update this (or at least remove the existing value) if the status is changed.
CarePlan.activity.detail.doNotPerform
Element Id | CarePlan.activity.detail.doNotPerform
Definition | If true, indicates that the described activity is one that must NOT be engaged in when following the plan. If false, or missing, indicates that the described activity is one that should be engaged in when following the plan.
Cardinality | 0..1
Type | boolean
Is Modifier | true (Reason: If true this element negates the specified action. For example, instead of a request for a procedure, it is a request for the procedure to not occur.)
Meaning if Missing | If missing indicates that the described activity is one that should be engaged in when following the plan.
Requirements | Captures intention to not do something that may have been previously typical.
Comments | This element is labeled as a modifier because it marks an activity as an activity that is not to be performed.
CarePlan.activity.detail.scheduled[x]
Element Id | CarePlan.activity.detail.scheduled[x]
Definition | The period, timing or frequency upon which the described activity is to occur.
Cardinality | 0..1
Type | Timing|Period|string
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Allows prompting for activities and detection of missed planned activities.
CarePlan.activity.detail.location
Element Id | CarePlan.activity.detail.location
Definition | Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic, etc.
Cardinality | 0..1
Type | Reference(Location)
Requirements | Helps in planning of activity.
Comments | May reference a specific clinical location or may identify a type of location.
CarePlan.activity.detail.performer
Element Id | CarePlan.activity.detail.performer
Definition | Identifies who's expected to be involved in the activity.
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Organization | RelatedPerson | Patient | CareTeam | HealthcareService | Device)
Requirements | Helps in planning of activity.
Comments | A performer MAY also be a participant in the care plan.
CarePlan.activity.detail.product[x]
Element Id | CarePlan.activity.detail.product[x]
Definition | Identifies the food, drug or other product to be consumed or supplied in the activity.
Cardinality | 0..1
Terminology Binding | SNOMED CT Medication Codes (Example)
Type | CodeableConcept|Reference(Medication | Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
CarePlan.activity.detail.dailyAmount
Element Id | CarePlan.activity.detail.dailyAmount
Definition | Identifies the quantity expected to be consumed in a given day.
Cardinality | 0..1

Type | SimpleQuantity
Requirements | Allows rough dose checking.
Alternate Names | daily dose
CarePlan.activity.detail.quantity
Element Id | CarePlan.activity.detail.quantity
Definition | Identifies the quantity expected to be supplied, administered or consumed by the subject.
Cardinality | 0..1
Type | SimpleQuantity
CarePlan.activity.detail.description
Element Id | CarePlan.activity.detail.description
Definition | This provides a textual description of constraints on the intended activity occurrence, including relation to other activities. It may also include objectives, pre-conditions and end-conditions. Finally, it may convey specifics about the activity such as body site, method, route, etc.
Cardinality | 0..1
Type | string
CarePlan.note
Element Id | CarePlan.note
Definition | General notes about the care plan not covered elsewhere.
Cardinality | 0..*
Type | Annotation
Requirements | Used to capture information that applies to the plan as a whole that doesn't fit into discrete elements.

9.5.8 Resource CarePlan - Mappings

URL: https://hl7.org/fhir/R4/careplan-mappings.html

9.5.8 Resource CarePlan - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the careplan resource (see Mappings to Other Standards for further information & status).

9.5.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CarePlan
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
description | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
period | FiveWs.planned
created | FiveWs.recorded
author | FiveWs.author
careTeam | FiveWs.actor
addresses | FiveWs.why[x]

9.5.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

CarePlan | Request
identifier | Request.identifier
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
basedOn | Request.basedOn
replaces | Request.replaces
status | Request.status {uses different ValueSet}
intent | Request.intent
subject | Request.subject
encounter | Request.context
period | Request.occurrence[x]
created | Request.authoredOn
author | Request.requester
careTeam | Request.performer {similar but does not entail CareTeam}
addresses | Request.reasonReference
supportingInfo | Request.supportingInfo
activity | {no mapping NOTE: This is a list of contained Request-Event tuples!}
outcomeReference | {Event that is outcome of Request in activity.reference}

reference | {Request that resulted in Event in activity.actionResulting}
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
code | Request.code
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
status | Request.status
statusReason | Request.statusReason
doNotPerform | Request.doNotPerform
scheduled[x] | Request.occurrence[x]
performer | Request.performer
note | Request.note

9.5.8.3 HL7 v2 Mapping (http://hl7.org/v2)

CarePlan
identifier | PTH-3
instantiatesCanonical
instantiatesUri
basedOn
replaces
partOf
status | PTH-5
intent
category
title
description
subject | PID-3
encounter | Associated PV1
period | GOL-7 / GOL-8
created
author
contributor
careTeam
addresses | PRB-4
supportingInfo
goal | GOL.1
activity
outcomeCodeableConcept
outcomeReference
progress | NTE?
reference
detail
kind
instantiatesCanonical
instantiatesUri
code | OBR-4 / RXE-2 / RXO-1 / RXD-2
reasonCode
reasonReference
goal
status | ORC-5?
statusReason
doNotPerform
scheduled[x] | TQ1
location | OBR-24(???!!)
performer | PRT-5 : (PRV-4 = (provider participations)); PRT-5 : (PRV-4 = (non-provider person participations)) ; PRT-5 : (PRV-4 = (patient non-subject of care)) ; PRT-8
product[x] | RXE-2 / RXO-1 / RXD-2
dailyAmount | RXO-23 / RXE-19 / RXD-12
quantity | RXO-11 / RXE-10 / RXD-4 / RXG-5 / RXA-6 / TQ1-2.1 *and* RXO-12 / RXE-11 / RXD-5 / RXG-7 / RXA-7 / TQ1-2.2
description | NTE?
note | NTE?

9.5.8.4 RIM Mapping (http://hl7.org/v3)

CarePlan | Act[classCode=PCPR, moodCode=INT]
identifier | .id
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
status | .statusCode planned = new active = active completed = completed
subject | .participation[typeCode=PAT].role[classCode=PAT]
encounter | .
period | .effectiveTime
created | .participation[typeCode=AUT].time

addresses | .actRelationship[typeCode=SUBJ].target[classCode=CONC, moodCode=EVN]
goal | .outboundRelationship[typeCode<=OBJ].
activity | .outboundRelationship[typeCode=COMP].target
outcomeReference | .inboundRelationship[typeCode=FLFS].source
progress | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value
reference | .outboundRelationship[typeCode=COMP].target
detail | .outboundRelationship[typeCode=COMP, subsetCode=SUMM].target
kind | .inboundRelationship[typeCode=COMP].source[classCode=LIST].code
instantiatesCanonical | .outboundRelationship[typeCode=DEFN].target
instantiatesUri | .outboundRelationship[typeCode=DEFN].target
code | .code
goal | .outboundRelationship[typeCode<=OBJ].
status | .statusCode not-started = new scheduled = not-started (and fulfillment relationship to appointent) in-progress = active on-hold = suspended completed = completed cancelled = aborted
doNotPerform | actionNegationInd
scheduled[x] | .effectiveTime
location | .participation[typeCode=LOC].role
performer | .participation[typeCode=PFM]
product[x] | .participation[typeCode=PRD].role
dailyAmount | .outboundRelationship[typeCode=COMP][classCode=SBADM].doseQuantity
quantity | .outboundRelationship[typeCode=COMP][classCode=SPLY].quantity
description | .text
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

9.5.9 Resource CarePlan - Extensions & Profiles

URL: https://hl7.org/fhir/R4/careplan-profiles.html

9.5.9 Resource CarePlan - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the CarePlan resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.5.9.1 Profiles

No Profiles defined for this resource

9.5.9.2 Extensions

careplan-activity-title | CarePlan.activity | activity-title | for Common Care Plan Extensions
workflow-episodeOfCare | CarePlan | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

openEHR-careplan | careplan | for openEHR Archetype Profile
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.5.9.3 Search Extensions

No Search Extensions defined for this resource

9.5.10 Resource CarePlan - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/careplan-version-maps.html

No validation errors - all conversions are clean

9.5.10 Resource CarePlan - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for CarePlan.

Functional status for this map: 11 tests that all execute ok. All tests pass round-trip testing and 10 r3 resources are invalid (0 errors). (see documentation)

9.5.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CarePlan3to4" = "R3 to R4 Conversions for CarePlan"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CarePlan" alias CarePlanR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CarePlan" alias CarePlan as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CarePlan(source src : CarePlanR3, target tgt : CarePlan) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiatesCanonical;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.category -> tgt.category;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.period -> tgt.period;
  src.author -> tgt.author;
  src.contributor -> tgt.contributor;
  src.careTeam -> tgt.careTeam;
  src.addresses -> tgt.addresses;
  src.supportingInfo -> tgt.supportingInfo;
  src.goal -> tgt.goal;
  src.activity as vs -> tgt.activity as vt then activity(vs, vt);
  src.note -> tgt.note;
}

group activity(source src, target tgt) extends BackboneElement {
  src.outcomeCodeableConcept -> tgt.outcomeCodeableConcept;
  src.outcomeReference -> tgt.outcomeReference;
  src.progress -> tgt.progress;
  src.reference -> tgt.reference;
  src.detail as vs -> tgt.detail as vt then detail(vs, vt);
}

group detail(source src, target tgt) extends BackboneElement {
  src.category as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CarePlan.activity.detail.category',  vt.value = v;
  src.definition -> tgt.instantiatesCanonical;
  src.code -> tgt.code;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.goal -> tgt.goal;
  src.status -> tgt.status;
  src.statusReason as s ->  tgt.statusReason as t,  t.text = s;
  src.prohibited -> tgt.doNotPerform;
  src.scheduled : Timing as s -> tgt.scheduled = create('Timing') as t then Timing(s, t);
  src.scheduled : Period as s -> tgt.scheduled = create('Period') as t then Period(s, t);
  src.scheduled : string as s -> tgt.scheduled = create('string') as t then string(s, t);
  src.location -> tgt.location;
  src.performer -> tgt.performer;
  src.product : CodeableConcept as s -> tgt.product = create('CodeableConcept') as t then CodeableConcept(s, t);
  src.product : Reference as s -> tgt.product = create('Reference') as t then Reference(s, t);
  src.dailyAmount -> tgt.dailyAmount;
  src.quantity -> tgt.quantity;
  src.description -> tgt.description;
}
```

9.5.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CarePlan4to3" = "R4 to R3 Conversion for CarePlan"

uses "http://hl7.org/fhir/StructureDefinition/CarePlan" alias CarePlan as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CarePlan" alias CarePlanR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CarePlan(source src : CarePlanR3, target tgt : CarePlan) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.definition;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.partOf -> tgt.partOf;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.category -> tgt.category;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
```

```
  src.period -> tgt.period;
  src.author -> tgt.author;
  src.contributor -> tgt.contributor;
  src.careTeam -> tgt.careTeam;
  src.addresses -> tgt.addresses;
  src.supportingInfo -> tgt.supportingInfo;
  src.goal -> tgt.goal;
  src.activity as vs -> tgt.activity as vt then activity(vs, vt);
  src.note -> tgt.note;
}

group activity(source src, target tgt) extends BackboneElement {
  src.outcomeCodeableConcept -> tgt.outcomeCodeableConcept;
  src.outcomeReference -> tgt.outcomeReference;
  src.progress -> tgt.progress;
  src.reference -> tgt.reference;
  src.detail as vs -> tgt.detail as vt then detail(vs, vt);
}

group detail(source src, target tgt) extends BackboneElement {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CarePlan.activity.detail.category' then {
    e.value -> tgt.category;
  };
  src.instantiatesCanonical -> tgt.definition;
  src.code -> tgt.code;
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.goal -> tgt.goal;
  src.status -> tgt.status;
  src.statusReason as s then {
    s.text -> tgt.statusReason;
  };
  src.doNotPerform -> tgt.prohibited;
  src.scheduled : Timing as s -> tgt.scheduled = create('Timing') as t then Timing(s, t);
  src.scheduled : Period as s -> tgt.scheduled = create('Period') as t then Period(s, t);
  src.scheduled : string as s -> tgt.scheduled = create('string') as t then string(s, t);
  src.location -> tgt.location;
  src.performer -> tgt.performer;
  src.product : CodeableConcept as s -> tgt.product = create('CodeableConcept') as t then CodeableConcept(s, t);
  src.product : Reference as s -> tgt.product = create('Reference') as t then Reference(s, t);
  src.dailyAmount -> tgt.dailyAmount;
  src.quantity -> tgt.quantity;
  src.description -> tgt.description;
}
```

9.5.10.3 R4 Validation Errors

9.5.11 Common Care Plan Extensions

URL: https://hl7.org/fhir/R4/careplan-extensions.html

A set of commonly used extensions that don't make the usage requirements for inclusion in the resource itself

9.5.11 Common Care Plan Extensions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.5.11.1 Content

Extensions:
careplan-activity-title | activity-title : Human-friendly name for the activity.

9.5.12 JSON Schema for CarePlan

URL: https://hl7.org/fhir/R4/careplan.schema.json.html

9.5.12 JSON Schema for CarePlan

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for CarePlan, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CarePlan",
  "$ref": "#/definitions/CarePlan",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CarePlan": {
      "description": "Describes the intention of how one or more practitioners intend to deliver care for a particular
    patient, group or community for a period of time, possibly limited to care for a specific condition or set of
    conditions.",
      "properties": {
        "resourceType": {
          "description": "This is a CarePlan resource",
```

```
      "const": "CarePlan"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this care plan by the performer or other systems which remain
 constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, questionnaire or other definition that
 is adhered to in whole or in part by this CarePlan.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, questionnaire or other
definition that is adhered to in whole or in part by this CarePlan.",
      "items": {
```

```
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A care plan that is fulfilled in whole or in part by this care plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "replaces": {
      "description": "Completed or terminated care plan whose function is taken by this new care plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "partOf": {
      "description": "A larger care plan of which this particular care plan is a component or step.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates whether the plan is currently being acted upon, represents future intentions or is
now a historical record.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "intent": {
      "description": "Indicates the level of authority/intentionality associated with the care plan and where the
care plan fits into the workflow chain.",
      "$ref": "#/definitions/code"
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Identifies what \"kind\" of plan this is to support differentiation between multiple co-
existing plans; e.g. \"Home health\", \"psychiatric\", \"asthma\", \"disease management\", \"wellness plan\", etc.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "title": {
      "description": "Human-friendly name for the care plan.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A description of the scope and nature of the plan.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "Identifies the patient or group whose intended care is described by the plan.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this CarePlan was created or to which the creation of this record
is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "period": {
      "description": "Indicates when the plan did (or is intended to) come into effect and end.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "created": {
      "description": "Represents when this particular CarePlan record was created in the system, which is often a
```

```
system-generated date.",
      "$ref": "#/definitions/dateTime"
    },
    "_created": {
      "description": "Extensions for created",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "When populated, the author is responsible for the care plan.  The care plan is attributed to
the author.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "contributor": {
      "description": "Identifies the individual(s) or organization who provided the contents of the care plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "careTeam": {
      "description": "Identifies all people and organizations who are expected to be involved in the care envisioned
 by this plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "addresses": {
      "description": "Identifies the conditions/problems/concerns/diagnoses/etc. whose management and/or mitigation
are handled by this plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "supportingInfo": {
      "description": "Identifies portions of the patient\u0027s record that specifically influenced the formation of
 the plan.  These might include comorbidities, recent procedures, limitations, recent assessments, etc.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "goal": {
      "description": "Describes the intended objective(s) of carrying out the care plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"

    },
    "activity": {
      "description": "Identifies a planned action to occur as part of the plan.  For example, a medication to be
used, lab tests to perform, self-monitoring, education, etc.",
      "items": {
        "$ref": "#/definitions/CarePlan_Activity"
      },
      "type": "array"
    },
    "note": {
      "description": "General notes about the care plan not covered elsewhere.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"CarePlan_Activity": {
  "description": "Describes the intention of how one or more practitioners intend to deliver care for a particular
patient, group or community for a period of time, possibly limited to care for a specific condition or set of
conditions.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "outcomeCodeableConcept": {
      "description": "Identifies the outcome at the point when the status of the activity is assessed.  For example,
 the outcome of an education activity could be patient understands (or not).",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "outcomeReference": {
      "description": "Details of the outcome or action resulting from the activity.  The reference to an \"event\"
resource, such as Procedure or Encounter or Observation, is the result/outcome of the activity itself.  The activity
 can be conveyed using CarePlan.activity.detail OR using the CarePlan.activity.reference (a reference to a “request”
 resource).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "progress": {
      "description": "Notes about the adherence/status/progress of the activity.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "reference": {
      "description": "The details of the proposed activity represented in a specific resource.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "detail": {
      "description": "A simple summary of a planned activity suitable for a general care plan system (e.g. form
driven) that doesn\u0027t know about specific resources such as procedure etc.",
      "$ref": "#/definitions/CarePlan_Detail"
    }
  },
  "additionalProperties": false
},
"CarePlan_Detail": {
  "description": "Describes the intention of how one or more practitioners intend to deliver care for a particular
patient, group or community for a period of time, possibly limited to care for a specific condition or set of
conditions.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "kind": {
```

```
      "description": "A description of the kind of resource the in-line definition of a care plan activity is representing.  The CarePlan.activity.detail is an in-line definition when a resource is not referenced using CarePlan.activity.reference.  For example, a MedicationRequest, a ServiceRequest, or a CommunicationRequest.",
      "$ref": "#/definitions/code"
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "instantiatesCanonical": {
      "description": "The URL pointing to a FHIR-defined protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan activity.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "The URL pointing to an externally maintained protocol, guideline, questionnaire or other definition that is adhered to in whole or in part by this CarePlan activity.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "code": {
      "description": "Detailed description of the type of planned activity; e.g. what lab test, what procedure, what kind of encounter.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "reasonCode": {
      "description": "Provides the rationale that drove the inclusion of this particular activity as part of the plan or the reason why the activity was prohibited.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource, such as the health condition(s), whose existence justifies this request and drove the inclusion of this particular activity as part of the plan.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "goal": {
      "description": "Internal reference that identifies the goals that this activity is intended to contribute towards meeting.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "status": {
      "description": "Identifies what progress is being made for the specific activity.",
      "enum": [
        "not-started",
        "scheduled",
        "in-progress",
        "on-hold",
        "completed",
        "cancelled",
        "stopped",
        "unknown",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Provides reason why the activity isn\u0027t yet started, is on hold, was cancelled, etc.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "doNotPerform": {
      "description": "If true, indicates that the described activity is one that must NOT be engaged in when following the plan.  If false, or missing, indicates that the described activity is one that should be engaged in when following the plan.",
      "$ref": "#/definitions/boolean"
    },
```

```
        "_doNotPerform": {
          "description": "Extensions for doNotPerform",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "scheduledTiming": {
          "description": "The period, timing or frequency upon which the described activity is to occur.",
          "$ref": "Timing.schema.json#/definitions/Timing"
        },
        "scheduledPeriod": {
          "description": "The period, timing or frequency upon which the described activity is to occur.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "scheduledString": {
          "description": "The period, timing or frequency upon which the described activity is to occur.",
          "pattern": "^[ \\r\\n\\t\\S]+$",
          "type": "string"
        },
        "_scheduledString": {
          "description": "Extensions for scheduledString",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "location": {
          "description": "Identifies the facility where the activity will occur; e.g. home, hospital, specific clinic,
    etc.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "performer": {
          "description": "Identifies who\u0027s expected to be involved in the activity.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "productCodeableConcept": {
          "description": "Identifies the food, drug or other product to be consumed or supplied in the activity.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "productReference": {
          "description": "Identifies the food, drug or other product to be consumed or supplied in the activity.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "dailyAmount": {
          "description": "Identifies the quantity expected to be consumed in a given day.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "quantity": {
          "description": "Identifies the quantity expected to be supplied, administered or consumed by the subject.",
          "$ref": "Quantity.schema.json#/definitions/Quantity"
        },
        "description": {
          "description": "This provides a textual description of constraints on the intended activity occurrence,
    including relation to other activities.  It may also include objectives, pre-conditions and end-conditions.
    Finally, it may convey specifics about the activity such as body site, method, route, etc.",
          "$ref": "#/definitions/string"
        },
        "_description": {
          "description": "Extensions for description",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


9.5.13 Resource CarePlan - Maturity Dependency View

URL: https://hl7.org/fhir/R4/careplan-dependencies.html

9.5.13 Resource CarePlan - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for CarePlan FMM level 2

Name | Card. | Type | Dependency Analysis
CarePlan | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical(PlanDefinition | Questionnaire | Measure | ActivityDefinition | OperationDefinition)
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(CarePlan)
replaces | 0..* | Reference(CarePlan)
partOf | 0..* | Reference(CarePlan)
status | 1..1 | code
intent | 1..1 | code
category | 0..* | CodeableConcept

title | 0..1 | string
description | 0..1 | string
subject | 1..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
period | 0..1 | Period
created | 0..1 | dateTime
author | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam)
contributor | 0..* | Reference(Patient | Practitioner | PractitionerRole | Device | RelatedPerson | Organization | CareTeam)
careTeam | 0..* | Reference(CareTeam)
addresses | 0..* | Reference(Condition)
supportingInfo | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
goal | 0..* | Reference(Goal)
activity | 0..* | BackboneElement
outcomeCodeableConcept | 0..* | CodeableConcept
outcomeReference | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
progress | 0..* | Annotation
reference | 0..1 | Reference(Appointment | CommunicationRequest | DeviceRequest | MedicationRequest | NutritionOrder | Task | ServiceRequest | VisionPrescription | RequestGroup) | Type Warning: (DeviceRequest = FMM1-Trial Use vs. Element = FMM2-Trial Use)
detail | 0..1 | BackboneElement
kind | 0..1 | code
instantiatesCanonical | 0..* | canonical(PlanDefinition | ActivityDefinition | Questionnaire | Measure | OperationDefinition)
instantiatesUri | 0..* | uri
code | 0..1 | CodeableConcept
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
goal | 0..* | Reference(Goal)
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
doNotPerform | 0..1 | boolean
scheduled[x] | 0..1
scheduledTiming | | Timing
scheduledPeriod | | Period
scheduledString | | string
location | 0..1 | Reference(Location)
performer | 0..* | Reference(Practitioner | PractitionerRole | Organization | RelatedPerson | Patient | CareTeam | HealthcareService | Device)
product[x] | 0..1
productCodeableConcept | | CodeableConcept
productReference | | Reference(Medication | Substance
dailyAmount | 0..1 | SimpleQuantity
quantity | 0..1 | SimpleQuantity
description | 0..1 | string
note | 0..* | Annotation

9.5.14 Value Set Sc-valueset-care-plan-activity-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-care-plan-activity-status.html

9.5.14 Value Set Sc-valueset-care-plan-activity-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.5.15 CarePlanActivityStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-care-plan-activity-status)

Mapping from http://hl7.org/fhir/ValueSet/care-plan-activity-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes that reflect the current state of a care plan activity within its overall life cycle."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
not-started | equivalent | draft
scheduled | equivalent | accepted
in-progress | equivalent | active
on-hold | equivalent | suspended

stopped | equivalent | failed
completed | equivalent | complete
cancelled | equivalent | abandoned
unknown | equivalent | unknown

9.6 Resource Goal - Content

URL: https://hl7.org/fhir/R4/goal.html
Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

Describes the intended objective(s) for a patient, group or organization care, for example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.

9.6.1 Scope and Usage

A Goal in health care services delivery is generally an expressed desired health state to be achieved by a subject of care (or family/group) over a period or at a specific point of time. This desired target health state may be achieved as a result of health care intervention(s) or resulting from natural recovery over time. For example:

• A goal of a plan for a condition such as a diabetes might specify desired outcome(s) (e.g. HgbA1c level =<5.6% in 3 months) as a result of interventions such as medication therapy, nutritional management and/or increase physical activity.

• A goal of a procedure might be to meet the intended objective of the procedure (e.g. wet-dry-dressing changes twice a day; goal: wound healed completely in 2 weeks) or to prevent an unintended complication (e.g. repositioning a patient every two hours: goal to maintain skin integrity)

Goals may address the prevention of illness, cure or mitigation of a condition, prolongation of life, or mitigation of pain and discomfort.

When dealing with groups, goals may also reflect health state, such as a reduction of addiction behaviors. However, they may also reflect population health objectives such as education, screening, etc.

Organizational goals are typically not health state specific but may instead identify measurement targets such as infection control, cost management, patient satisfaction, etc.

9.6.2 Boundaries and Relationships

Goals are typically established in the context of a CarePlan. However, goals may also be directly referenced by request-type resources (e.g. MedicationRequest or ServiceRequest) by using an extension.

A goal represents a specific goal instance for a particular patient, group, etc. It is not intended to be used to define types of potential goals as part of an order set or protocol definition. Protocol definitions and order sets are supported through PlanDefinition. The Goal resource is intended to be used once an order set is instantiated or assigned to a patient, which is when the potential goals become the actual goals, if not changed or deleted.

Goals are often evaluated using Observations.

This resource is referenced by CarePlan

9.6.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
Goal | TU | | DomainResource | Describes the intended objective(s) for a patient, group or organization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External Ids for this goal
lifecycleStatus | ?!Σ | 1..1 | code | proposed | planned | accepted | active | on-hold | completed | cancelled | entered-in-error | rejected GoalLifecycleStatus (Required)
achievementStatus | Σ | 0..1 | CodeableConcept | in-progress | improving | worsening | no-change | achieved | sustaining | not-achieved | no-progress | not-attainable Goal achievement status (Preferred)
category | Σ | 0..* | CodeableConcept | E.g. Treatment, dietary, behavioral, etc. Goal category (Example)
priority | Σ | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Goal priority (Preferred)
description | Σ | 1..1 | CodeableConcept | Code or text describing goal SNOMED CT Clinical Findings (Example)

subject | Σ | 1..1 | Reference(Patient | Group | Organization) | Who this goal is intended for
start[x] | Σ | 0..1 | | When goal pursuit begins Goal start event (Example)
startDate | | | date
startCodeableConcept | | | CodeableConcept
target | I | 0..* | BackboneElement | Target outcome for the goal + Rule: Goal.target.measure is required if Goal.target.detail is populated
measure | ΣI | 0..1 | CodeableConcept | The parameter whose value is being tracked LOINC Codes (Example)
detail[x] | ΣI | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
detailString | | | string
detailBoolean | | | boolean
detailInteger | | | integer
detailRatio | | | Ratio
due[x] | Σ | 0..1 | | Reach goal on or before
dueDate | | | date
dueDuration | | | Duration
statusDate | Σ | 0..1 | date | When goal status took effect
statusReason | | 0..1 | string | Reason for current status
expressedBy | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who's responsible for creating Goal?
addresses | | 0..* | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ServiceRequest | RiskAssessment) | Issues addressed by this goal
note | | 0..* | Annotation | Comments about the goal
outcomeCode | | 0..* | CodeableConcept | What result was achieved regarding the goal? SNOMED CT Clinical Findings (Example)
outcomeReference | | 0..* | Reference(Observation) | Observation that resulted from goal
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Goal xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this goal --></identifier>
 <lifecycleStatus value="[code]"/><!-- 1..1 proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected -->
 <achievementStatus><!-- 0..1 CodeableConcept in-progress | improving | worsening | no-change | achieved | sustaining |
    not-achieved | no-progress | not-attainable --></achievementStatus>
 <category><!-- 0..* CodeableConcept E.g. Treatment, dietary, behavioral, etc. --></category>
 <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
 <description><!-- 1..1 CodeableConcept Code or text describing goal --></description>
 <subject><!-- 1..1 Reference(Patient|Group|Organization) Who this goal is intended for --></subject>
 <start[x]><!-- 0..1 date|CodeableConcept When goal pursuit begins --></start[x]>
 <target>  <!--  0..* Target outcome for the goal -->
  <measure><!--  0..1 CodeableConcept The parameter whose value is being tracked --></measure>
  <detail[x]><!--  0..1 Quantity|Range|CodeableConcept|string|boolean|integer|
    Ratio The target value to be achieved --></detail[x]>
  <due[x]><!-- 0..1 date|Duration Reach goal on or before --></due[x]>
 </target>
 <statusDate value="[date]"/><!-- 0..1 When goal status took effect -->
 <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
 <expressedBy><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) Who's responsible for creating Goal? --></expressedBy>
 <addresses><!-- 0..* Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ServiceRequest|RiskAssessment) Issues addressed by this goal --></addresses>
 <note><!-- 0..* Annotation Comments about the goal --></note>
 <outcomeCode><!-- 0..* CodeableConcept What result was achieved regarding the goal? --></outcomeCode>
 <outcomeReference><!-- 0..* Reference(Observation) Observation that resulted from goal --></outcomeReference>
</Goal>
```

JSON Template

```
{
  "resourceType" : "Goal",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this goal
  "lifecycleStatus" : "<code>", // R!  proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected
  "achievementStatus" : { CodeableConcept }, // in-progress | improving | worsening | no-change | achieved | sustaining
    | not-achieved | no-progress | not-attainable
  "category" : [{ CodeableConcept }], // E.g. Treatment, dietary, behavioral, etc.
  "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
  "description" : { CodeableConcept }, // R!  Code or text describing goal
  "subject" : { Reference(Patient|Group|Organization) }, // R!  Who this goal is intended for
  // start[x]: When goal pursuit begins. One of these 2:
  "startDate" : "<date>",
  "startCodeableConcept" : { CodeableConcept },
  "target" : [{ // C? Target outcome for the goal
    "measure" : { CodeableConcept }, // C? The parameter whose value is being tracked
    // detail[x]: The target value to be achieved. One of these 7:
    "detailQuantity" : { Quantity },
    "detailRange" : { Range },
```

```
    "detailCodeableConcept" : { CodeableConcept },
    "detailString" : "<string>",
    "detailBoolean" : <boolean>,
    "detailInteger" : <integer>,
    "detailRatio" : { Ratio },
    // due[x]: Reach goal on or before. One of these 2:
    "dueDate" : "<date>"
    "dueDuration" : { Duration }
  }],
  "statusDate" : "<date>", // When goal status took effect
  "statusReason" : "<string>", // Reason for current status
  "expressedBy" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) }, // Who's responsible for creating Goal?
  "addresses" : [{ Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ServiceRequest|RiskAssessment) }], // Issues addressed by this goal
  "note" : [{ Annotation }], // Comments about the goal
  "outcomeCode" : [{ CodeableConcept }], // What result was achieved regarding the goal?
  "outcomeReference" : [{ Reference(Observation) }] // Observation that resulted from goal
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Goal;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Goal.identifier [ Identifier ], ... ; # 0..* External Ids for this goal
  fhir:Goal.lifecycleStatus [ code ]; # 1..1 proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected
  fhir:Goal.achievementStatus [ CodeableConcept ]; # 0..1 in-progress | improving | worsening | no-change | achieved |
    sustaining | not-achieved | no-progress | not-attainable
  fhir:Goal.category [ CodeableConcept ], ... ; # 0..* E.g. Treatment, dietary, behavioral, etc.
  fhir:Goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
  fhir:Goal.description [ CodeableConcept ]; # 1..1 Code or text describing goal
  fhir:Goal.subject [ Reference(Patient|Group|Organization) ]; # 1..1 Who this goal is intended for
  # Goal.start[x] : 0..1 When goal pursuit begins. One of these 2
    fhir:Goal.startDate [ date ]
    fhir:Goal.startCodeableConcept [ CodeableConcept ]
  fhir:Goal.target [ # 0..* Target outcome for the goal
    fhir:Goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is being tracked
    # Goal.target.detail[x] : 0..1 The target value to be achieved. One of these 7
      fhir:Goal.target.detailQuantity [ Quantity ]
      fhir:Goal.target.detailRange [ Range ]
      fhir:Goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:Goal.target.detailString [ string ]
      fhir:Goal.target.detailBoolean [ boolean ]
      fhir:Goal.target.detailInteger [ integer ]
      fhir:Goal.target.detailRatio [ Ratio ]
    # Goal.target.due[x] : 0..1 Reach goal on or before. One of these 2
      fhir:Goal.target.dueDate [ date ]
      fhir:Goal.target.dueDuration [ Duration ]
  ], ...;
  fhir:Goal.statusDate [ date ]; # 0..1 When goal status took effect
  fhir:Goal.statusReason [ string ]; # 0..1 Reason for current status
  fhir:Goal.expressedBy [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who's responsible for
     creating Goal?
  fhir:Goal.addresses [ Reference(Condition|Observation|MedicationStatement|NutritionOrder|ServiceRequest|
  RiskAssessment) ], ... ; # 0..* Issues addressed by this goal
  fhir:Goal.note [ Annotation ], ... ; # 0..* Comments about the goal
  fhir:Goal.outcomeCode [ CodeableConcept ], ... ; # 0..* What result was achieved regarding the goal?
  fhir:Goal.outcomeReference [ Reference(Observation) ], ... ; # 0..* Observation that resulted from goal
]
```

Changes since R3

Goal
Goal.lifecycleStatus | Added Mandatory Element
Goal.achievementStatus | Added Element
Goal.subject | Min Cardinality changed from 0 to 1
Goal.target | Max Cardinality changed from 1 to *
Goal.target.detail[x] | Add Types string, boolean, integer, Ratio
Goal.expressedBy | Type Reference: Added Target Type PractitionerRole
Goal.addresses | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest
Goal.status | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

Goal | TU | | DomainResource | Describes the intended objective(s) for a patient, group or organization Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | External Ids for this goal
lifecycleStatus | ?!Σ | 1..1 | code | proposed | planned | accepted | active | on-hold | completed | cancelled | entered-in-error | rejected GoalLifecycleStatus (Required)
achievementStatus | Σ | 0..1 | CodeableConcept | in-progress | improving | worsening | no-change | achieved | sustaining | not-achieved | no-progress | not-attainable Goal achievement status (Preferred)
category | Σ | 0..* | CodeableConcept | E.g. Treatment, dietary, behavioral, etc. Goal category (Example)
priority | Σ | 0..1 | CodeableConcept | high-priority | medium-priority | low-priority Goal priority (Preferred)
description | Σ | 1..1 | CodeableConcept | Code or text describing goal SNOMED CT Clinical Findings (Example)
subject | Σ | 1..1 | Reference(Patient | Group | Organization) | Who this goal is intended for
start[x] | Σ | 0..1 | | When goal pursuit begins Goal start event (Example)
startDate | | | date
startCodeableConcept | | | CodeableConcept
target | I | 0..* | BackboneElement | Target outcome for the goal + Rule: Goal.target.measure is required if Goal.target.detail is populated
measure | ΣI | 0..1 | CodeableConcept | The parameter whose value is being tracked LOINC Codes (Example)
detail[x] | ΣI | 0..1 | | The target value to be achieved
detailQuantity | | | Quantity
detailRange | | | Range
detailCodeableConcept | | | CodeableConcept
detailString | | | string
detailBoolean | | | boolean
detailInteger | | | integer
detailRatio | | | Ratio
due[x] | Σ | 0..1 | | Reach goal on or before
dueDate | | | date
dueDuration | | | Duration
statusDate | Σ | 0..1 | date | When goal status took effect
statusReason | | 0..1 | string | Reason for current status
expressedBy | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who's responsible for creating Goal?
addresses | | 0..* | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ServiceRequest | RiskAssessment) | Issues addressed by this goal
note | | 0..* | Annotation | Comments about the goal
outcomeCode | | 0..* | CodeableConcept | What result was achieved regarding the goal? SNOMED CT Clinical Findings (Example)
outcomeReference | | 0..* | Reference(Observation) | Observation that resulted from goal
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Goal xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this goal --></identifier>
 <lifecycleStatus value="[code]"/><!-- 1..1 proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected -->
 <achievementStatus><!-- 0..1 CodeableConcept in-progress | improving | worsening | no-change | achieved | sustaining |
    not-achieved | no-progress | not-attainable --></achievementStatus>
 <category><!-- 0..* CodeableConcept E.g. Treatment, dietary, behavioral, etc. --></category>
 <priority><!-- 0..1 CodeableConcept high-priority | medium-priority | low-priority --></priority>
 <description><!-- 1..1 CodeableConcept Code or text describing goal --></description>
 <subject><!-- 1..1 Reference(Patient|Group|Organization) Who this goal is intended for --></subject>
 <start[x]><!-- 0..1 date|CodeableConcept When goal pursuit begins --></start[x]>
 <target>  <!--  0..* Target outcome for the goal -->
  <measure><!--  0..1 CodeableConcept The parameter whose value is being tracked --></measure>
  <detail[x]><!--  0..1 Quantity|Range|CodeableConcept|string|boolean|integer|
    Ratio The target value to be achieved --></detail[x]>
  <due[x]><!-- 0..1 date|Duration Reach goal on or before --></due[x]>
 </target>
 <statusDate value="[date]"/><!-- 0..1 When goal status took effect -->
 <statusReason value="[string]"/><!-- 0..1 Reason for current status -->
 <expressedBy><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) Who's responsible for creating Goal? --></expressedBy>
 <addresses><!-- 0..* Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ServiceRequest|RiskAssessment) Issues addressed by this goal --></addresses>
 <note><!-- 0..* Annotation Comments about the goal --></note>
 <outcomeCode><!-- 0..* CodeableConcept What result was achieved regarding the goal? --></outcomeCode>
 <outcomeReference><!-- 0..* Reference(Observation) Observation that resulted from goal --></outcomeReference>
</Goal>
```

JSON Template

```
{
  "resourceType" : "Goal",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this goal
  "lifecycleStatus" : "<code>", // R!  proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected
  "achievementStatus" : { CodeableConcept }, // in-progress | improving | worsening | no-change | achieved | sustaining
    | not-achieved | no-progress | not-attainable
```

```
  "category" : [{ CodeableConcept }], // E.g. Treatment, dietary, behavioral, etc.
  "priority" : { CodeableConcept }, // high-priority | medium-priority | low-priority
  "description" : { CodeableConcept }, // R!  Code or text describing goal
  "subject" : { Reference(Patient|Group|Organization) }, // R!  Who this goal is intended for
  // start[x]: When goal pursuit begins. One of these 2:
  "startDate" : "<date>",
  "startCodeableConcept" : { CodeableConcept },
  "target" : [{ // C? Target outcome for the goal
    "measure" : { CodeableConcept }, // C? The parameter whose value is being tracked
    // detail[x]: The target value to be achieved. One of these 7:
    "detailQuantity" : { Quantity },
    "detailRange" : { Range },
    "detailCodeableConcept" : { CodeableConcept },
    "detailString" : "<string>",
    "detailBoolean" : <boolean>,
    "detailInteger" : <integer>,
    "detailRatio" : { Ratio },
    // due[x]: Reach goal on or before. One of these 2:
    "dueDate" : "<date>"
    "dueDuration" : { Duration }
  }],
  "statusDate" : "<date>", // When goal status took effect
  "statusReason" : "<string>", // Reason for current status
  "expressedBy" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) }, // Who's responsible for creating Goal?
  "addresses" : [{ Reference(Condition|Observation|MedicationStatement|
   NutritionOrder|ServiceRequest|RiskAssessment) }], // Issues addressed by this goal
  "note" : [{ Annotation }], // Comments about the goal
  "outcomeCode" : [{ CodeableConcept }], // What result was achieved regarding the goal?
  "outcomeReference" : [{ Reference(Observation) }] // Observation that resulted from goal
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Goal;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Goal.identifier [ Identifier ], ... ; # 0..* External Ids for this goal
  fhir:Goal.lifecycleStatus [ code ]; # 1..1 proposed | planned | accepted | active | on-hold | completed | cancelled |
    entered-in-error | rejected
  fhir:Goal.achievementStatus [ CodeableConcept ]; # 0..1 in-progress | improving | worsening | no-change | achieved |
    sustaining | not-achieved | no-progress | not-attainable
  fhir:Goal.category [ CodeableConcept ], ... ; # 0..* E.g. Treatment, dietary, behavioral, etc.
  fhir:Goal.priority [ CodeableConcept ]; # 0..1 high-priority | medium-priority | low-priority
  fhir:Goal.description [ CodeableConcept ]; # 1..1 Code or text describing goal
  fhir:Goal.subject [ Reference(Patient|Group|Organization) ]; # 1..1 Who this goal is intended for
  # Goal.start[x] : 0..1 When goal pursuit begins. One of these 2
    fhir:Goal.startDate [ date ]
    fhir:Goal.startCodeableConcept [ CodeableConcept ]
  fhir:Goal.target [ # 0..* Target outcome for the goal
    fhir:Goal.target.measure [ CodeableConcept ]; # 0..1 The parameter whose value is being tracked
    # Goal.target.detail[x] : 0..1 The target value to be achieved. One of these 7
      fhir:Goal.target.detailQuantity [ Quantity ]
      fhir:Goal.target.detailRange [ Range ]
      fhir:Goal.target.detailCodeableConcept [ CodeableConcept ]
      fhir:Goal.target.detailString [ string ]
      fhir:Goal.target.detailBoolean [ boolean ]
      fhir:Goal.target.detailInteger [ integer ]
      fhir:Goal.target.detailRatio [ Ratio ]
    # Goal.target.due[x] : 0..1 Reach goal on or before. One of these 2
      fhir:Goal.target.dueDate [ date ]
      fhir:Goal.target.dueDuration [ Duration ]
  ], ...;
  fhir:Goal.statusDate [ date ]; # 0..1 When goal status took effect
  fhir:Goal.statusReason [ string ]; # 0..1 Reason for current status
  fhir:Goal.expressedBy [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who's responsible for
     creating Goal?
  fhir:Goal.addresses [ Reference(Condition|Observation|MedicationStatement|NutritionOrder|ServiceRequest|
  RiskAssessment) ], ... ; # 0..* Issues addressed by this goal
  fhir:Goal.note [ Annotation ], ... ; # 0..* Comments about the goal
  fhir:Goal.outcomeCode [ CodeableConcept ], ... ; # 0..* What result was achieved regarding the goal?
  fhir:Goal.outcomeReference [ Reference(Observation) ], ... ; # 0..* Observation that resulted from goal
]
```

Changes since Release 3

Goal

Goal.lifecycleStatus | Added Mandatory Element

Goal.achievementStatus | Added Element

Goal.subject | Min Cardinality changed from 0 to 1

Goal.target | Max Cardinality changed from 1 to *

Goal.target.detail[x] | Add Types string, boolean, integer, Ratio

Goal.expressedBy | Type Reference: Added Target Type PractitionerRole

Goal.addresses | Type Reference: Added Target Type ServiceRequest Type Reference: Removed Target Type ProcedureRequest

Goal.status | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.6.3.1 Terminology Bindings

Path | Definition | Type | Reference
Goal.lifecycleStatus | Codes that reflect the current state of a goal and whether the goal is still being targeted. | Required | GoalLifecycleStatus
Goal.achievementStatus | Indicates the progression, or lack thereof, towards the goal against the target. | Preferred | GoalAchievementStatus
Goal.category | Codes for grouping and sorting goals. | Example | GoalCategory
Goal.priority | The level of importance associated with a goal. | Preferred | GoalPriority
Goal.description | Codes providing the details of a particular goal. This will generally be system or implementation guide-specific. In many systems, only the text element will be used. | Example | SNOMEDCTClinicalFindings
Goal.start[x] | Codes describing events that can trigger the initiation of a goal. | Example | GoalStartEvent
Goal.target.measure | Codes to identify the value being tracked, e.g. body weight, blood pressure, or hemoglobin A1c level. | Example | LOINCCodes
Goal.target.detail[x] | Codes to identify the target value of the focus to be achieved to signify the fulfillment of the goal. | Unknown | No details provided yet
Goal.outcomeCode | The result of the goal; e.g. "25% increase in shoulder mobility", "Anxiety reduced to moderate levels". "15 kg weight loss sustained over 6 months". | Example | SNOMEDCTClinicalFindings

9.6.3.2 Constraints

id | Level | Location | Description | Expression
gol-1 | Rule | Goal.target | Goal.target.measure is required if Goal.target.detail is populated | (detail.exists() and measure.exists()) or detail.exists().not()

9.6.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
achievement-status | token | in-progress | improving | worsening | no-change | achieved | sustaining | not-achieved | no-progress | not-attainable | Goal.achievementStatus
category | token | E.g. Treatment, dietary, behavioral, etc. | Goal.category
identifier | token | External Ids for this goal | Goal.identifier | 30 Resources
lifecycle-status | token | proposed | planned | accepted | active | on-hold | completed | cancelled | entered-in-error | rejected | Goal.lifecycleStatus
patient | reference | Who this goal is intended for | Goal.subject.where(resolve() is Patient) (Patient) | 33 Resources
start-date | date | When goal pursuit begins | (Goal.start as date)
subject | reference | Who this goal is intended for | Goal.subject (Group, Organization, Patient)
target-date | date | Reach goal on or before | (Goal.target.due as date)

9.6.5 Resource Goal - Examples

URL: https://hl7.org/fhir/R4/goal-examples.html

9.6.5 Resource Goal - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Example Name | id | Format
Goal of Weight Loss | example | XML | JSON | Turtle
Goal to Stop Smoking | stop-smoking | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.6.6 Resource Goal - Detailed Descriptions

URL: https://hl7.org/fhir/R4/goal-definitions.html

9.6.6 Resource Goal - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Patient

Detailed Descriptions for the elements in the Goal resource.

Goal
Element Id | Goal

Definition | Describes the intended objective(s) for a patient, group or organization care, for example, weight loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process improvement objective, etc.
Cardinality | 0..*
Type | DomainResource
Comments | Goal can be achieving a particular change or merely maintaining a current state or even slowing a decline.
Goal.identifier
Element Id | Goal.identifier
Definition | Business identifiers assigned to this goal by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the goal as it is known by various participating systems and in a way that remains consistent across servers.
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
Goal.lifecycleStatus
Element Id | Goal.lifecycleStatus
Definition | The state of the goal throughout its lifecycle.
Cardinality | 1..1
Terminology Binding | GoalLifecycleStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Requirements | Allows knowing whether goal needs to be further tracked.
Summary | true
Comments | This element is labeled as a modifier because the lifecycleStatus contains codes that mark the resource as not currently valid.
Goal.achievementStatus
Element Id | Goal.achievementStatus
Definition | Describes the progression, or lack thereof, towards the goal against the target.
Cardinality | 0..1
Terminology Binding | Goal achievement status (Preferred)
Type | CodeableConcept
Summary | true
Goal.category
Element Id | Goal.category
Definition | Indicates a category the goal falls within.
Cardinality | 0..*
Terminology Binding | Goal category (Example)
Type | CodeableConcept
Requirements | Allows goals to be filtered and sorted.
Summary | true
Goal.priority
Element Id | Goal.priority
Definition | Identifies the mutually agreed level of importance associated with reaching/sustaining the goal.
Cardinality | 0..1
Terminology Binding | Goal priority (Preferred)
Type | CodeableConcept
Requirements | Used for sorting and presenting goals.
Summary | true
Comments | Extensions are available to track priorities as established by each participant (i.e. Priority from the patient's perspective, different practitioners' perspectives, family member's perspectives) The ordinal extension on Coding can be used to convey a numerically comparable ranking to priority. (Keep in mind that different coding systems may use a "low value=important".
Goal.description
Element Id | Goal.description
Definition | Human-readable and/or coded description of a specific desired objective of care, such as "control blood pressure" or "negotiate an obstacle course" or "dance with child at wedding".
Cardinality | 1..1
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Requirements | Without a description of what's trying to be achieved, element has no purpose.
Summary | true
Comments | If no code is available, use CodeableConcept.text.
Goal.subject
Element Id | Goal.subject
Definition | Identifies the patient, group or organization for whom the goal is being established.
Cardinality | 1..1
Type | Reference(Patient | Group | Organization)
Requirements | Subject is optional to support annonymized reporting.

Summary | true
Goal.start[x]
Element Id | Goal.start[x]
Definition | The date or event after which the goal should begin being pursued.
Cardinality | 0..1
Terminology Binding | Goal start event (Example)
Type | date|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Goals can be established prior to there being an intention to start pursuing them; e.g. Goals for post-surgical recovery established prior to surgery.
Summary | true
Goal.target
Element Id | Goal.target
Definition | Indicates what should be done by when.
Cardinality | 0..*
Requirements | Allows the progress of the goal to be monitored against an observation or due date. Target is 0..* to support Observations with multiple components, such as blood pressure goals with both a systolic and diastolic target.
Comments | When multiple targets are present for a single goal instance, all targets must be met for the overall goal to be met.
Invariants | Defined on this element gol-1 Rule Goal.target.measure is required if Goal.target.detail is populated (detail.exists() and measure.exists()) or detail.exists().not() Affect this element
Defined on this element
gol-1 | Rule | Goal.target.measure is required if Goal.target.detail is populated | (detail.exists() and measure.exists()) or detail.exists().not()
Affect this element
Goal.target.measure
Element Id | Goal.target.measure
Definition | The parameter whose value is being tracked, e.g. body weight, blood pressure, or hemoglobin A1c level.
Cardinality | 0..1
Terminology Binding | LOINC Codes (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element gol-1 Rule Goal.target.measure is required if Goal.target.detail is populated (detail.exists() and measure.exists()) or detail.exists().not()
Affect this element
gol-1 | Rule | Goal.target.measure is required if Goal.target.detail is populated | (detail.exists() and measure.exists()) or detail.exists().not()
Goal.target.detail[x]
Element Id | Goal.target.detail[x]
Definition | The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150 pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is missing, it indicates that the goal is achieved at any focus value at or above the low value.
Cardinality | 0..1
Terminology Binding | GoalTargetDetail:
Type | Quantity|Range|CodeableConcept|string|boolean|integer|Ratio
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | A CodeableConcept with just a text would be used instead of a string if the field was usually coded, or if the type associated with the Goal.target.measure defines a coded value.
Invariants | Affect this element gol-1 Rule Goal.target.measure is required if Goal.target.detail is populated (detail.exists() and measure.exists()) or detail.exists().not()
Affect this element
gol-1 | Rule | Goal.target.measure is required if Goal.target.detail is populated | (detail.exists() and measure.exists()) or detail.exists().not()
Goal.target.due[x]
Element Id | Goal.target.due[x]
Definition | Indicates either the date or the duration after start by which the goal should be met.
Cardinality | 0..1
Type | date|Duration
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Identifies when the goal should be evaluated.
Summary | true
Goal.statusDate
Element Id | Goal.statusDate
Definition | Identifies when the current status. I.e. When initially created, when achieved, when cancelled, etc.
Cardinality | 0..1
Type | date
Summary | true
Comments | To see the date for past statuses, query history.
Goal.statusReason
Element Id | Goal.statusReason
Definition | Captures the reason for the current status.
Cardinality | 0..1

Type | string
Comments | This will typically be captured for statuses such as rejected, on-hold or cancelled, but could be present for others.
Goal.expressedBy
Element Id | Goal.expressedBy
Definition | Indicates whose goal this is - patient goal, practitioner goal, etc.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
Summary | true
Comments | This is the individual responsible for establishing the goal, not necessarily who recorded it. (For that, use the Provenance resource.).
Goal.addresses
Element Id | Goal.addresses
Definition | The identified conditions and other health record elements that are intended to be addressed by the goal.
Cardinality | 0..*
Type | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ServiceRequest | RiskAssessment)
Requirements | Allows specific goals to explicitly linked to the concerns they're dealing with - makes the goal more understandable.
Goal.note
Element Id | Goal.note
Definition | Any comments related to the goal.
Cardinality | 0..*
Type | Annotation
Requirements | There's a need to capture information about the goal that doesn't actually describe the goal.
Comments | May be used for progress notes, concerns or other related information that doesn't actually describe the goal itself.
Goal.outcomeCode
Element Id | Goal.outcomeCode
Definition | Identifies the change (or lack of change) at the point when the status of the goal is assessed.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Requirements | Outcome tracking is a key aspect of care planning.
Comments | Note that this should not duplicate the goal status.
Goal.outcomeReference
Element Id | Goal.outcomeReference
Definition | Details of what's changed (or not changed).
Cardinality | 0..*
Type | Reference(Observation)
Requirements | Outcome tracking is a key aspect of care planning.
Comments | The goal outcome is independent of the outcome of the related activities. For example, if the Goal is to achieve a target body weight of 150 lb and a care plan activity is defined to diet, then the care plan’s activity outcome could be calories consumed whereas goal outcome is an observation for the actual body weight measured.

9.6.7 Resource Goal - Mappings

URL: https://hl7.org/fhir/R4/goal-mappings.html

9.6.7 Resource Goal - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the goal resource (see Mappings to Other Standards for further information & status).

9.6.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Goal
identifier | FiveWs.identifier
lifecycleStatus | FiveWs.status
category | FiveWs.class
priority | FiveWs.grade
description | FiveWs.what[x]
subject | FiveWs.subject[x]
start[x] | FiveWs.planned
due[x] | FiveWs.done[x]
statusDate | FiveWs.recorded
expressedBy | FiveWs.source
addresses | FiveWs.why[x]

9.6.7.2 HL7 v2 Mapping (http://hl7.org/v2)

Goal | GOL.1
identifier
lifecycleStatus | GOL-18-goal life cycle status
achievementStatus
category
priority
description | GOL-3.2-goal ID.text
subject | PID-3-patient ID list
start[x]
target
measure
detail[x]
due[x]
statusDate
statusReason
expressedBy
addresses
note | GOL-16-goal evaluation + NTE?
outcomeCode
outcomeReference

9.6.7.3 RIM Mapping (http://hl7.org/v3)

Goal | .outboundRelationship[typeCode<=OBJ].
identifier | .id
lifecycleStatus | .statusCode in-progress = active (classCode = OBJ) cancelled = aborted
achievementStatus | .statusCode achieved = complete sustaining = active
priority | .priorityCode
description | .text
subject | .participation[typeCode=PAT].role[classCode=PAT]
addresses | .outboundRelationship[typeCode=SUBJ].target[classCode=CONC]
note | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="annotation"].value

9.6.8 Resource Goal - Extensions & Profiles

URL: https://hl7.org/fhir/R4/goal-profiles.html

9.6.8 Resource Goal - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

This table lists profiles and extensions for the Goal resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.6.8.1 Profiles

No Profiles defined for this resource

9.6.8.2 Extensions

goal-acceptance | Goal | acceptance | for FHIR Core Goal Profile
goal-reasonRejected | Goal | reasonRejected | for FHIR Core Goal Profile
goal-relationship | Goal | relationship | for FHIR Core Goal Profile
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

goal-relationship | relationship | for FHIR Core Goal Profile
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.6.8.3 Search Extensions

No Search Extensions defined for this resource

9.6.9 Resource Goal - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/goal-version-maps.html

No validation errors - all conversions are clean

9.6.9 Resource Goal - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient

R3 : R4 Conversion maps for Goal.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.6.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Goal3to4" = "R3 to R4 Conversions for Goal"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Goal" alias GoalR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Goal" alias Goal as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Goal(source src : GoalR3, target tgt : Goal) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status where value in ('proposed' | 'planned' | 'accepted' | 'on-hold' | 'cancelled' | 'entered-in-error' |
    'rejected') -> tgt.lifecycleStatus;
  src.status where value = 'achieved' -> tgt.lifecycleStatus = 'completed';
  src.status as s where value in ('in-progress' | 'sustaining') ->  tgt.lifecycleStatus = 'active',
    tgt.achievementStatus as cc,  cc.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/goal-achievement',
      c.code = s;
  src.status where value = 'on-target' ->  tgt.lifecycleStatus = 'active',  tgt.achievementStatus as cc,  cc.coding as
    c,  c.system = 'http://terminology.hl7.org/CodeSystem/goal-achievement',  c.code = 'sustaining';
  src.status where value = 'ahead-of-target' ->  tgt.lifecycleStatus = 'active',  tgt.achievementStatus as cc,
    cc.coding as c,  c.system = 'http://terminology.hl7.org/CodeSystem/goal-achievement',  c.code = 'improving';
  src.status where value = 'behind-target' ->  tgt.lifecycleStatus = 'active',  tgt.achievementStatus as cc,  cc.coding
    as c,  c.system = 'http://terminology.hl7.org/CodeSystem/goal-achievement',  c.code = 'worsening';
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.start : date as vs -> tgt.start = create('date') as vt then date(vs, vt);
  src.start : CodeableConcept as vs -> tgt.start = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
  src.statusDate -> tgt.statusDate;
  src.statusReason -> tgt.statusReason;
  src.expressedBy -> tgt.expressedBy;
  src.addresses -> tgt.addresses;
  src.note -> tgt.note;
  src.outcomeCode -> tgt.outcomeCode;
  src.outcomeReference -> tgt.outcomeReference;
}

group target(source src, target tgt) extends BackboneElement {
  src.measure -> tgt.measure;
  src.detail : Quantity as vs -> tgt.detail = create('Quantity') as vt then Quantity(vs, vt);
  src.detail : Range as vs -> tgt.detail = create('Range') as vt then Range(vs, vt);
  src.detail : CodeableConcept as vs -> tgt.detail = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.due : date as vs -> tgt.due = create('date') as vt then date(vs, vt);
  src.due : Duration as vs -> tgt.due = create('Duration') as vt then Duration(vs, vt);
}
```

9.6.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Goal4to3" = "R4 to R3 Conversion for Goal"

uses "http://hl7.org/fhir/StructureDefinition/Goal" alias Goal as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Goal" alias GoalR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Goal(source src : Goal, target tgt : GoalR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.lifecycleStatus where value in ('proposed' | 'planned' | 'accepted' | 'on-hold' | 'cancelled' | 'entered-in-error'
     | 'rejected') -> tgt.status;
  src.lifecycleStatus where value = 'completed' -> tgt.status = 'achieved';
  src.achievementStatus where coding.where((system = 'http://terminology.hl7.org/CodeSystem/goal-achievement') and (code
     = 'in-progress')).exists() -> tgt.status = 'in-progress';
  src.achievementStatus where coding.where((system = 'http://terminology.hl7.org/CodeSystem/goal-achievement') and (code
     = 'sustaining')).exists() -> tgt.status = 'sustaining';
  src.achievementStatus where coding.where((system = 'http://terminology.hl7.org/CodeSystem/goal-achievement') and (code
     = 'sustaining')) -> tgt.status = 'on-target';
  src.achievementStatus where coding.where((system = 'http://terminology.hl7.org/CodeSystem/goal-achievement') and (code
     = 'improving')) -> tgt.status = 'ahead-of-target';
  src.achievementStatus where coding.where((system = 'http://terminology.hl7.org/CodeSystem/goal-achievement') and (code
     = 'worsening')) -> tgt.status = 'behind-target';
  src.category -> tgt.category;
  src.priority -> tgt.priority;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.start : date as vs -> tgt.start = create('date') as vt then date(vs, vt);
```

```
  src.start : CodeableConcept as vs -> tgt.start = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
  src.statusDate -> tgt.statusDate;
  src.statusReason -> tgt.statusReason;
  src.expressedBy -> tgt.expressedBy;
  src.addresses -> tgt.addresses;
  src.note -> tgt.note;
  src.outcomeCode -> tgt.outcomeCode;
  src.outcomeReference -> tgt.outcomeReference;
}

group target(source src, target tgt) extends BackboneElement {
  src.measure -> tgt.measure;
  src.detail : Quantity as vs -> tgt.detail = create('Quantity') as vt then Quantity(vs, vt);
  src.detail : Range as vs -> tgt.detail = create('Range') as vt then Range(vs, vt);
  src.detail : CodeableConcept as vs -> tgt.detail = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.due : date as vs -> tgt.due = create('date') as vt then date(vs, vt);
  src.due : Duration as vs -> tgt.due = create('Duration') as vt then Duration(vs, vt);
}
```

9.6.9.3 R4 Validation Errors

9.6.10 FHIR Core Goal Profile

URL: https://hl7.org/fhir/R4/goal-extensions.html

This profile holds core Goal extensions

9.6.10 FHIR Core Goal Profile

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative

9.6.10.1 Content

Extensions:

goal-reasonRejected | reasonRejected : The reason the goal was not accepted. Applies only if the status of the goal is rejected.

goal-relationship | relationship : Establishes a relationship between this goal and other goals.

goal-acceptance | acceptance : Information about the acceptance and relative priority assigned to the goal by the patient, practitioners and other stake-holders.

9.6.11 JSON Schema for Goal

URL: https://hl7.org/fhir/R4/goal.schema.json.html

9.6.11 JSON Schema for Goal

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for Goal, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Goal",
  "$ref": "#/definitions/Goal",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Goal": {
      "description": "Describes the intended objective(s) for a patient, group or organization care, for example, weight
     loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process
    improvement objective, etc.",
      "properties": {
        "resourceType": {
          "description": "This is a Goal resource",
          "const": "Goal"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
```

```
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this goal by the performer or other systems which remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "lifecycleStatus": {
      "description": "The state of the goal throughout its lifecycle.",
      "enum": [
        "proposed",
        "planned",
        "accepted",
        "active",
        "on-hold",
        "completed",
        "cancelled",
        "entered-in-error",
        "rejected"
      ]
    },
    "_lifecycleStatus": {
      "description": "Extensions for lifecycleStatus",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "achievementStatus": {
      "description": "Describes the progression, or lack thereof, towards the goal against the target.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "category": {
      "description": "Indicates a category the goal falls within.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "priority": {
      "description": "Identifies the mutually agreed level of importance associated with reaching/sustaining the goal.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "Human-readable and/or coded description of a specific desired objective of care, such as
```

```
\"control blood pressure\" or \"negotiate an obstacle course\" or \"dance with child at wedding\".",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "Identifies the patient, group or organization for whom the goal is being established.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "startDate": {
      "description": "The date or event after which the goal should begin being pursued.",
      "pattern":
"^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
      "type": "string"
    },
    "_startDate": {
      "description": "Extensions for startDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "startCodeableConcept": {
      "description": "The date or event after which the goal should begin being pursued.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "target": {
      "description": "Indicates what should be done by when.",
      "items": {
        "$ref": "#/definitions/Goal_Target"
      },
      "type": "array"
    },
    "statusDate": {
      "description": "Identifies when the current status.  I.e. When initially created, when achieved, when
cancelled, etc.",
      "$ref": "#/definitions/date"
    },
    "_statusDate": {
      "description": "Extensions for statusDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "statusReason": {
      "description": "Captures the reason for the current status.",
      "$ref": "#/definitions/string"
    },
    "_statusReason": {
      "description": "Extensions for statusReason",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expressedBy": {
      "description": "Indicates whose goal this is - patient goal, practitioner goal, etc.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "addresses": {
      "description": "The identified conditions and other health record elements that are intended to be addressed
by the goal.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Any comments related to the goal.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "outcomeCode": {
      "description": "Identifies the change (or lack of change) at the point when the status of the goal is
assessed.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "outcomeReference": {
      "description": "Details of what\u0027s changed (or not changed).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "description",
    "resourceType"
  ]
},
"Goal_Target": {
  "description": "Describes the intended objective(s) for a patient, group or organization care, for example, weight
```

```
 loss, restoring an activity of daily living, obtaining herd immunity via immunization, meeting a process
improvement objective, etc.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "measure": {
      "description": "The parameter whose value is being tracked, e.g. body weight, blood pressure, or hemoglobin
A1c level.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "detailQuantity": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "detailRange": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "detailCodeableConcept": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "detailString": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_detailString": {
      "description": "Extensions for detailString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "detailBoolean": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_detailBoolean": {
      "description": "Extensions for detailBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "detailInteger": {
      "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
missing, it indicates that the goal is achieved at any focus value at or above the low value.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
```

```
        },
        "_detailInteger": {
          "description": "Extensions for detailInteger",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "detailRatio": {
          "description": "The target value of the focus to be achieved to signify the fulfillment of the goal, e.g. 150
    pounds, 7.0%. Either the high or low or both values of the range can be specified. When a low value is missing, it
    indicates that the goal is achieved at any focus value at or below the high value. Similarly, if the high value is
    missing, it indicates that the goal is achieved at any focus value at or above the low value.",
          "$ref": "Ratio.schema.json#/definitions/Ratio"
        },
        "dueDate": {
          "description": "Indicates either the date or the duration after start by which the goal should be met.",
          "pattern":
    "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-1]))?)?$",
          "type": "string"
        },
        "_dueDate": {
          "description": "Extensions for dueDate",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "dueDuration": {
          "description": "Indicates either the date or the duration after start by which the goal should be met.",
          "$ref": "Duration.schema.json#/definitions/Duration"
        }
      },
      "additionalProperties": false
    }
  }
}
```


9.6.12 Resource Goal - Maturity Dependency View

URL: https://hl7.org/fhir/R4/goal-dependencies.html

9.6.12 Resource Goal - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Dependency Graph for Goal FMM level 2

Name | Card. | Type | Dependency Analysis
Goal | | DomainResource
identifier | 0..* | Identifier
lifecycleStatus | 1..1 | code
achievementStatus | 0..1 | CodeableConcept
category | 0..* | CodeableConcept
priority | 0..1 | CodeableConcept
description | 1..1 | CodeableConcept
subject | 1..1 | Reference(Patient | Group | Organization) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
start[x] | 0..1
startDate | | date
startCodeableConcept | | CodeableConcept
target | 0..* | BackboneElement
measure | 0..1 | CodeableConcept
detail[x] | 0..1
detailQuantity | | Quantity
detailRange | | Range
detailCodeableConcept | | CodeableConcept
detailString | | string
detailBoolean | | boolean
detailInteger | | integer
detailRatio | | Ratio
due[x] | 0..1
dueDate | | date
dueDuration | | Duration
statusDate | 0..1 | date
statusReason | 0..1 | string
expressedBy | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
addresses | 0..* | Reference(Condition | Observation | MedicationStatement | NutritionOrder | ServiceRequest | RiskAssessment) | Type Warning: (RiskAssessment = FMM1-Trial Use vs. Element = FMM2-Trial Use)
note | 0..* | Annotation
outcomeCode | 0..* | CodeableConcept
outcomeReference | 0..* | Reference(Observation)

9.6.13 Value Set Sc-valueset-goal-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-goal-status.html

9.6.13 Value Set Sc-valueset-goal-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.6.14 GoalLifecycleStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-goal-status)

Mapping from http://hl7.org/fhir/ValueSet/goal-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes that reflect the current state of a goal and whether the goal is still being targeted."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
proposed	equivalent	proposed
planned	equivalent	planned
rejected	equivalent	declined
accepted	equivalent	accepted
active	equivalent	active
on-hold	equivalent	suspended
completed	equivalent	complete
cancelled	equivalent	abandoned

9.6.15 Value Set Sc-valueset-goal-achievement - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-goal-achievement.html

9.6.15 Value Set Sc-valueset-goal-achievement - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.6.16 GoalAchievementStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-goal-achievement)

Mapping from http://hl7.org/fhir/ValueSet/goal-achievement to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 International - Patient Care WG (http://www.hl7.org/Special/com...).

Canonical Mapping for "Describes the progression, or lack thereof, towards the goal against the target."

Source Code	Equivalence	Destination Code
in-progress, sustaining, improving, worsening, no-change	equivalent	active
achieved	equivalent	complete
not-attainable	equivalent	abandoned
no-progress, not-achieved	equivalent	not-done

9.7 Resource CareTeam - Content

URL: https://hl7.org/fhir/R4/careteam.html

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

The Care Team includes all the people and organizations who plan to participate in the coordination and delivery of care for a patient.

9.7.1 Scope and Usage

The CareTeam includes all the people, teams, and organizations who plan to participate in the coordination and delivery of care for a single patient or a group (such as a married couple in therapy or a support group). CareTeam can also be organizationally assigned without a subject in context, such as a code blue team or emergency response team. This is not limited to practitioners, but may include other caregivers such as family members, guardians, the patient themself, or others. The Care Team, depending on where used, may include care team members specific to a particular care plan, an episode, an encounter, or may reflect all known team members across these perspectives. An individual's CareTeam can be dynamic over time, such that there can be transience of team members, such as a rehabilitation team.

9.7.2 Boundaries and Relationships

Care Team is distinct from Group. Group is patient-independent and identifies an undifferentiated set of individuals who are intended to be the target of one or more clinical activities (e.g. set of clinical trial participants, set of individuals impacted by or at risk of a public health event, a herd or flock, etc.) The CareTeam resource establishes a set of relationships and roles and is specific to a particular Patient. The actors are the individual members or organized group of individuals. CareTeam can be referenced by EpisodeOfCare, Encounter, or CarePlan to identify the set of individuals (and their respective roles) who are intended to be involved in providing the care defined by those resources.

This resource is referenced by CarePlan, itself, ChargeItem, Communication, CommunicationRequest, Consent, Contract, DeviceRequest, DiagnosticReport, EpisodeOfCare, ImagingStudy, Media, MedicationRequest, Observation, ServiceRequest and Task

9.7.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
CareTeam | TU | | DomainResource | Planned participants in the coordination and delivery of care for a patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this team
status | ?!Σ | 0..1 | code | proposed | active | suspended | inactive | entered-in-error CareTeamStatus (Required)
category | Σ | 0..* | CodeableConcept | Type of team Care Team category (Example)
name | Σ | 0..1 | string | Name of the team, such as crisis assessment team
subject | Σ | 0..1 | Reference(Patient | Group) | Who care team is for
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
period | Σ | 0..1 | Period | Time period team covers
participant | I | 0..* | BackboneElement | Members of the team + Rule: CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner
role | Σ | 0..* | CodeableConcept | Type of involvement Participant Roles (Example)
member | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Organization | CareTeam) | Who is involved
onBehalfOf | Σ | 0..1 | Reference(Organization) | Organization of the practitioner
period | | 0..1 | Period | Time period of participant
reasonCode | | 0..* | CodeableConcept | Why the care team exists SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition) | Why the care team exists
managingOrganization | Σ | 0..* | Reference(Organization) | Organization responsible for the care team
telecom | | 0..* | ContactPoint | A contact detail for the care team (that applies to all members)
note | | 0..* | Annotation | Comments made about the CareTeam
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CareTeam xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this team --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | active | suspended | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Type of team --></category>
 <name value="[string]"/><!-- 0..1 Name of the team, such as crisis assessment team -->
 <subject><!-- 0..1 Reference(Patient|Group) Who care team is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <period><!-- 0..1 Period Time period team covers --></period>
 <participant>  <!--  0..* Members of the team -->
  <role><!-- 0..* CodeableConcept Type of involvement --></role>
  <member><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Organization|CareTeam) Who is involved --></member>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization of the practitioner --></onBehalfOf>
  <period><!-- 0..1 Period Time period of participant --></period>
 </participant>
 <reasonCode><!-- 0..* CodeableConcept Why the care team exists --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition) Why the care team exists --></reasonReference>
 <managingOrganization><!-- 0..* Reference(Organization) Organization responsible for the care team
    --></managingOrganization>
 <telecom><!-- 0..* ContactPoint A contact detail for the care team (that applies to all members) --></telecom>
 <note><!-- 0..* Annotation Comments made about the CareTeam --></note>
</CareTeam>
```

JSON Template

```
{
  "resourceType" : "CareTeam",
```

```
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this team
  "status" : "<code>", // proposed | active | suspended | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Type of team
  "name" : "<string>", // Name of the team, such as crisis assessment team
  "subject" : { Reference(Patient|Group) }, // Who care team is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "period" : { Period }, // Time period team covers
  "participant" : [{ // C? Members of the team
    "role" : [{ CodeableConcept }], // Type of involvement
    "member" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Organization|CareTeam) }, // Who is involved
    "onBehalfOf" : { Reference(Organization) }, // Organization of the practitioner
    "period" : { Period } // Time period of participant
  }],
  "reasonCode" : [{ CodeableConcept }], // Why the care team exists
  "reasonReference" : [{ Reference(Condition) }], // Why the care team exists
  "managingOrganization" : [{ Reference(Organization) }], // Organization responsible for the care team
  "telecom" : [{ ContactPoint }], // A contact detail for the care team (that applies to all members)
  "note" : [{ Annotation }] // Comments made about the CareTeam
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CareTeam;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CareTeam.identifier [ Identifier ], ... ; # 0..* External Ids for this team
  fhir:CareTeam.status [ code ]; # 0..1 proposed | active | suspended | inactive | entered-in-error
  fhir:CareTeam.category [ CodeableConcept ], ... ; # 0..* Type of team
  fhir:CareTeam.name [ string ]; # 0..1 Name of the team, such as crisis assessment team
  fhir:CareTeam.subject [ Reference(Patient|Group) ]; # 0..1 Who care team is for
  fhir:CareTeam.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CareTeam.period [ Period ]; # 0..1 Time period team covers
  fhir:CareTeam.participant [ # 0..* Members of the team
    fhir:CareTeam.participant.role [ CodeableConcept ], ... ; # 0..* Type of involvement
    fhir:CareTeam.participant.member [
    Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Organization|CareTeam) ]; # 0..1 Who is involved
    fhir:CareTeam.participant.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization of the practitioner
    fhir:CareTeam.participant.period [ Period ]; # 0..1 Time period of participant
  ], ...;
  fhir:CareTeam.reasonCode [ CodeableConcept ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.reasonReference [ Reference(Condition) ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.managingOrganization [ Reference(Organization) ], ... ; # 0..* Organization responsible for the care
    team
  fhir:CareTeam.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the care team (that applies to all members)
  fhir:CareTeam.note [ Annotation ], ... ; # 0..* Comments made about the CareTeam
]
```

Changes since R3

CareTeam
CareTeam.status | Change value set from http://hl7.org/fhir/ValueSet/care-team-status to http://hl7.org/fhir/ValueSet/care-team-status|4.0.1
CareTeam.encounter | Added Element
CareTeam.participant.role | Max Cardinality changed from 1 to *
CareTeam.participant.member | Type Reference: Added Target Type PractitionerRole
CareTeam.telecom | Added Element
CareTeam.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
CareTeam | TU | | DomainResource | Planned participants in the coordination and delivery of care for a patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | External Ids for this team
status | ?!Σ | 0..1 | code | proposed | active | suspended | inactive | entered-in-error CareTeamStatus (Required)
category | Σ | 0..* | CodeableConcept | Type of team Care Team category (Example)
name | Σ | 0..1 | string | Name of the team, such as crisis assessment team
subject | Σ | 0..1 | Reference(Patient | Group) | Who care team is for
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
period | Σ | 0..1 | Period | Time period team covers
participant | I | 0..* | BackboneElement | Members of the team + Rule: CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner
role | Σ | 0..* | CodeableConcept | Type of involvement Participant Roles (Example)

member | Σ | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Organization | CareTeam) | Who is involved
onBehalfOf | Σ | 0..1 | Reference(Organization) | Organization of the practitioner
period | | 0..1 | Period | Time period of participant
reasonCode | | 0..* | CodeableConcept | Why the care team exists SNOMED CT Clinical Findings (Example)
reasonReference | | 0..* | Reference(Condition) | Why the care team exists
managingOrganization | Σ | 0..* | Reference(Organization) | Organization responsible for the care team
telecom | | 0..* | ContactPoint | A contact detail for the care team (that applies to all members)
note | | 0..* | Annotation | Comments made about the CareTeam
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CareTeam xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External Ids for this team --></identifier>
 <status value="[code]"/><!-- 0..1 proposed | active | suspended | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Type of team --></category>
 <name value="[string]"/><!-- 0..1 Name of the team, such as crisis assessment team -->
 <subject><!-- 0..1 Reference(Patient|Group) Who care team is for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <period><!-- 0..1 Period Time period team covers --></period>
 <participant>  <!--  0..* Members of the team -->
  <role><!-- 0..* CodeableConcept Type of involvement --></role>
  <member><!-- 0..1 Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Organization|CareTeam) Who is involved --></member>
  <onBehalfOf><!-- 0..1 Reference(Organization) Organization of the practitioner --></onBehalfOf>
  <period><!-- 0..1 Period Time period of participant --></period>
 </participant>
 <reasonCode><!-- 0..* CodeableConcept Why the care team exists --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition) Why the care team exists --></reasonReference>
 <managingOrganization><!-- 0..* Reference(Organization) Organization responsible for the care team
    --></managingOrganization>
 <telecom><!-- 0..* ContactPoint A contact detail for the care team (that applies to all members) --></telecom>
 <note><!-- 0..* Annotation Comments made about the CareTeam --></note>
</CareTeam>
```

JSON Template

```
{
  "resourceType" : "CareTeam",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External Ids for this team
  "status" : "<code>", // proposed | active | suspended | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Type of team
  "name" : "<string>", // Name of the team, such as crisis assessment team
  "subject" : { Reference(Patient|Group) }, // Who care team is for
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "period" : { Period }, // Time period team covers
  "participant" : [{ // C? Members of the team
    "role" : [{ CodeableConcept }], // Type of involvement
    "member" : { Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|
    Organization|CareTeam) }, // Who is involved
    "onBehalfOf" : { Reference(Organization) }, // Organization of the practitioner
    "period" : { Period } // Time period of participant
  }],
  "reasonCode" : [{ CodeableConcept }], // Why the care team exists
  "reasonReference" : [{ Reference(Condition) }], // Why the care team exists
  "managingOrganization" : [{ Reference(Organization) }], // Organization responsible for the care team
  "telecom" : [{ ContactPoint }], // A contact detail for the care team (that applies to all members)
  "note" : [{ Annotation }] // Comments made about the CareTeam
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CareTeam;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CareTeam.identifier [ Identifier ], ... ; # 0..* External Ids for this team
  fhir:CareTeam.status [ code ]; # 0..1 proposed | active | suspended | inactive | entered-in-error
  fhir:CareTeam.category [ CodeableConcept ], ... ; # 0..* Type of team
  fhir:CareTeam.name [ string ]; # 0..1 Name of the team, such as crisis assessment team
  fhir:CareTeam.subject [ Reference(Patient|Group) ]; # 0..1 Who care team is for
  fhir:CareTeam.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:CareTeam.period [ Period ]; # 0..1 Time period team covers
  fhir:CareTeam.participant [ # 0..* Members of the team
    fhir:CareTeam.participant.role [ CodeableConcept ], ... ; # 0..* Type of involvement
    fhir:CareTeam.participant.member [
    Reference(Practitioner|PractitionerRole|RelatedPerson|Patient|Organization|CareTeam) ]; # 0..1 Who is involved
    fhir:CareTeam.participant.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization of the practitioner
    fhir:CareTeam.participant.period [ Period ]; # 0..1 Time period of participant
  ], ...;
```

```
  fhir:CareTeam.reasonCode [ CodeableConcept ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.reasonReference [ Reference(Condition) ], ... ; # 0..* Why the care team exists
  fhir:CareTeam.managingOrganization [ Reference(Organization) ], ... ; # 0..* Organization responsible for the care
    team
  fhir:CareTeam.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the care team (that applies to all members)
  fhir:CareTeam.note [ Annotation ], ... ; # 0..* Comments made about the CareTeam
]
```

Changes since Release 3

CareTeam
CareTeam.status | Change value set from http://hl7.org/fhir/ValueSet/care-team-status to http://hl7.org/fhir/ValueSet/care-team-status|4.0.1
CareTeam.encounter | Added Element
CareTeam.participant.role | Max Cardinality changed from 1 to *
CareTeam.participant.member | Type Reference: Added Target Type PractitionerRole
CareTeam.telecom | Added Element
CareTeam.context | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
CareTeam.status | Indicates the status of the care team. | Required | CareTeamStatus
CareTeam.category | Indicates the type of care team. | Example | CareTeamCategory
CareTeam.participant.role | Indicates specific responsibility of an individual within the care team, such as "Primary physician", "Team coordinator", "Caregiver", etc. | Example | ParticipantRoles
CareTeam.reasonCode | Indicates the reason for the care team. | Example | SNOMEDCTClinicalFindings

9.7.3.2 Constraints

id | Level | Location | Description | Expression
ctm-1 | Rule | CareTeam.participant | CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner | onBehalfOf.exists() implies (member.resolve().iif(empty(), true, ofType(Practitioner).exists()))

9.7.3.3 Notes

The Provenance resource can be used for detailed review information, such as when the care team was last reviewed and by whom.

9.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | Type of team | CareTeam.category
date | date | Time period team covers | CareTeam.period | 17 Resources
encounter | reference | Encounter created as part of | CareTeam.encounter (Encounter)
identifier | token | External Ids for this team | CareTeam.identifier | 30 Resources
participant | reference | Who is involved | CareTeam.participant.member (Practitioner, Organization, CareTeam, Patient, PractitionerRole, RelatedPerson)
patient | reference | Who care team is for | CareTeam.subject.where(resolve() is Patient) (Patient) | 33 Resources
status | token | proposed | active | suspended | inactive | entered-in-error | CareTeam.status
subject | reference | Who care team is for | CareTeam.subject (Group, Patient)

9.7.5 Resource CareTeam - Examples

URL: https://hl7.org/fhir/R4/careteam-examples.html

9.7.5 Resource CareTeam - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Care Team to address obesity | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.7.6 Resource CareTeam - Detailed Descriptions

URL: https://hl7.org/fhir/R4/careteam-definitions.html

9.7.6 Resource CareTeam - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CareTeam resource.

CareTeam
Element Id | CareTeam
Definition | The Care Team includes all the people and organizations who plan to participate in the coordination and delivery of care for a patient.
Cardinality | 0..*
Type | DomainResource
CareTeam.identifier
Element Id | CareTeam.identifier
Definition | Business identifiers assigned to this care team by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the care team as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
CareTeam.status
Element Id | CareTeam.status
Definition | Indicates the current state of the care team.
Cardinality | 0..1
Terminology Binding | CareTeamStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the care team as not currently valid.
CareTeam.category
Element Id | CareTeam.category
Definition | Identifies what kind of team. This is to support differentiation between multiple co-existing teams, such as care plan team, episode of care team, longitudinal care team.
Cardinality | 0..*
Terminology Binding | Care Team category (Example)
Type | CodeableConcept
Requirements | Used for filtering what teams(s) are retrieved and displayed to different types of users.
Summary | true
Comments | There may be multiple axis of categorization and one team may serve multiple purposes.
CareTeam.name
Element Id | CareTeam.name
Definition | A label for human use intended to distinguish like teams. E.g. the "red" vs. "green" trauma teams.
Cardinality | 0..1
Type | string
Summary | true
Comments | The meaning/purpose of the team is conveyed in CareTeam.category. This element may also convey semantics of the team (e.g. "Red trauma team"), but its primary purpose is to distinguish between identical teams in a human-friendly way. ("Team 18735" isn't as friendly.).
CareTeam.subject
Element Id | CareTeam.subject
Definition | Identifies the patient or group whose intended care is handled by the team.
Cardinality | 0..1
Type | Reference(Patient | Group)
Requirements | Allows the team to care for a group (e.g. marriage) therapy. Allows for an organization to designate a team such as the PICC line team.
Alternate Names | patient
Summary | true
CareTeam.encounter
Element Id | CareTeam.encounter
Definition | The Encounter during which this CareTeam was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)

Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter.
CareTeam.period
Element Id | CareTeam.period
Definition | Indicates when the team did (or is intended to) come into effect and end.
Cardinality | 0..1
Type | Period
Requirements | Allows tracking what team(s) are in effect at a particular time.
Summary | true
CareTeam.participant
Element Id | CareTeam.participant
Definition | Identifies all people and organizations who are expected to be involved in the care team.
Cardinality | 0..*
Invariants | Defined on this element ctm-1 Rule CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner onBehalfOf.exists() implies (member.resolve().iif(empty(), true, ofType(Practitioner).exists())) Affect this element
Defined on this element
ctm-1 | Rule | CareTeam.participant.onBehalfOf can only be populated when CareTeam.participant.member is a Practitioner | onBehalfOf.exists() implies (member.resolve().iif(empty(), true, ofType(Practitioner).exists()))
Affect this element
CareTeam.participant.role
Element Id | CareTeam.participant.role
Definition | Indicates specific responsibility of an individual within the care team, such as "Primary care physician", "Trained social worker counselor", "Caregiver", etc.
Cardinality | 0..*
Terminology Binding | Participant Roles (Example)
Type | CodeableConcept
Summary | true
Comments | Roles may sometimes be inferred by type of Practitioner. These are relationships that hold only within the context of the care team. General relationships should be handled as properties of the Patient resource directly.
CareTeam.participant.member
Element Id | CareTeam.participant.member
Definition | The specific person or organization who is participating/expected to participate in the care team.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Organization | CareTeam)
Summary | true
Comments | Patient only needs to be listed if they have a role other than "subject of care". Member is optional because some participants may be known only by their role, particularly in draft plans.
CareTeam.participant.onBehalfOf
Element Id | CareTeam.participant.onBehalfOf
Definition | The organization of the practitioner.
Cardinality | 0..1
Type | Reference(Organization)
Requirements | Practitioners can be associated with multiple organizations. This element indicates which organization they were acting on behalf of.
Summary | true
CareTeam.participant.period
Element Id | CareTeam.participant.period
Definition | Indicates when the specific member or organization did (or is intended to) come into effect and end.
Cardinality | 0..1
Type | Period
CareTeam.reasonCode
Element Id | CareTeam.reasonCode
Definition | Describes why the care team exists.
Cardinality | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
CareTeam.reasonReference
Element Id | CareTeam.reasonReference
Definition | Condition(s) that this care team addresses.
Cardinality | 0..*
Type | Reference(Condition)
CareTeam.managingOrganization
Element Id | CareTeam.managingOrganization
Definition | The organization responsible for the care team.
Cardinality | 0..*
Type | Reference(Organization)
Requirements | Allows for multiple organizations to collaboratively manage cross-organizational, longitudinal care plan.
Summary | true
CareTeam.telecom
Element Id | CareTeam.telecom
Definition | A central contact detail for the care team (that applies to all members).

Cardinality | 0..*
Type | ContactPoint
Comments | The ContactPoint.use code of home is not appropriate to use. These contacts are not the contact details of individual care team members.
CareTeam.note
Element Id | CareTeam.note
Definition | Comments made about the CareTeam.
Cardinality | 0..*
Type | Annotation

9.7.7 Resource CareTeam - Mappings

URL: https://hl7.org/fhir/R4/careteam-mappings.html

9.7.7 Resource CareTeam - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

HL7 v2 Mapping

RIM Mapping

Mappings for the careteam resource (see Mappings to Other Standards for further information & status).

9.7.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CareTeam
identifier | FiveWs.identifier
status | FiveWs.status
category | FiveWs.class
subject | FiveWs.subject[x]
encounter | FiveWs.context
period | FiveWs.init
member | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

9.7.7.2 HL7 v2 Mapping (http://hl7.org/v2)

CareTeam
identifier
status
category
name
subject
encounter
period
participant | REL (REL.4 is always the Patient) (or PRT?)
role | REL.2 (or PRT-4?)
member | REL.5 (or PRT-5 : (PRV-4 {provider participations}) / PRT-5 : (PRV-4 {non-provider person participations}) / PRT-5 : (PRV-4 = (patient non-subject of care)) / PRT-8?)
onBehalfOf
period
reasonCode
reasonReference
managingOrganization
telecom
note

9.7.7.3 RIM Mapping (http://hl7.org/v3)

CareTeam
participant | .participation[typeCode=PRF]
role | .functionCode
member | .role

9.7.8 Resource CareTeam - Extensions & Profiles

URL: https://hl7.org/fhir/R4/careteam-profiles.html

9.7.8 Resource CareTeam - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the CareTeam resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.7.8.1 Profiles

No Profiles defined for this resource

9.7.8.2 Extensions

workflow-episodeOfCare | CareTeam | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.7.8.3 Search Extensions

No Search Extensions defined for this resource

9.7.9 Resource CareTeam - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/careteam-version-maps.html

No validation errors - all conversions are clean

9.7.9 Resource CareTeam - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for CareTeam.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

9.7.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CareTeam3to4" = "R3 to R4 Conversions for CareTeam"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CareTeam" alias CareTeamR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CareTeam" alias CareTeam as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CareTeam(source src : CareTeamR3, target tgt : CareTeam) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.name -> tgt.name;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.period -> tgt.period;
  src.participant as vs0 -> tgt.participant as vt0 then participant(vs0, vt0);
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.managingOrganization -> tgt.managingOrganization;
  src.note -> tgt.note;
}

group participant(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.member -> tgt.member;
  src.onBehalfOf -> tgt.onBehalfOf;
  src.period -> tgt.period;
}
```

9.7.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CareTeam4to3" = "R4 to R3 Conversion for CareTeam"

uses "http://hl7.org/fhir/StructureDefinition/CareTeam" alias CareTeam as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CareTeam" alias CareTeamR3 as target
```

```
imports "http://hl7.org/fhir/StructureMap/*4to3"

group CareTeam(source src : CareTeam, target tgt : CareTeamR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.category;
  src.name -> tgt.name;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.period -> tgt.period;
  src.participant as vs0 -> tgt.participant as vt0 then participant(vs0, vt0);
  src.reasonCode -> tgt.reasonCode;
  src.reasonReference -> tgt.reasonReference;
  src.managingOrganization -> tgt.managingOrganization;
  src.note -> tgt.note;
}

group participant(source src, target tgt) extends BackboneElement {
  src.role -> tgt.role;
  src.member -> tgt.member;
  src.onBehalfOf -> tgt.onBehalfOf;
  src.period -> tgt.period;
}
```

9.7.9.3 R4 Validation Errors

9.7.10 JSON Schema for CareTeam

URL: https://hl7.org/fhir/R4/careteam.schema.json.html

9.7.10 JSON Schema for CareTeam

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for CareTeam, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CareTeam",
  "$ref": "#/definitions/CareTeam",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CareTeam": {
      "description": "The Care Team includes all the people and organizations who plan to participate in the
    coordination and delivery of care for a patient.",
      "properties": {
        "resourceType": {
          "description": "This is a CareTeam resource",
          "const": "CareTeam"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
```

```
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this care team by the performer or other systems which remain
 constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the current state of the care team.",
      "enum": [
        "proposed",
        "active",
        "suspended",
        "inactive",
        "entered-in-error"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "Identifies what kind of team.  This is to support differentiation between multiple co-existing
 teams, such as care plan team, episode of care team, longitudinal care team.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "name": {
      "description": "A label for human use intended to distinguish like teams.  E.g. the \"red\" vs. \"green\"
trauma teams.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "Identifies the patient or group whose intended care is handled by the team.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which this CareTeam was created or to which the creation of this record
is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "period": {
      "description": "Indicates when the team did (or is intended to) come into effect and end.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "participant": {
      "description": "Identifies all people and organizations who are expected to be involved in the care team.",
      "items": {
        "$ref": "#/definitions/CareTeam_Participant"
      },
      "type": "array"
    },
```

```
    "reasonCode": {
      "description": "Describes why the care team exists.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Condition(s) that this care team addresses.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "managingOrganization": {
      "description": "The organization responsible for the care team.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "telecom": {
      "description": "A central contact detail for the care team (that applies to all members).",
      "items": {
        "$ref": "ContactPoint.schema.json#/definitions/ContactPoint"
      },
      "type": "array"
    },
    "note": {
      "description": "Comments made about the CareTeam.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"CareTeam_Participant": {
  "description": "The Care Team includes all the people and organizations who plan to participate in the
coordination and delivery of care for a patient.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "role": {
      "description": "Indicates specific responsibility of an individual within the care team, such as \"Primary
care physician\", \"Trained social worker counselor\", \"Caregiver\", etc.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "member": {
      "description": "The specific person or organization who is participating/expected to participate in the care
team.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "onBehalfOf": {
      "description": "The organization of the practitioner.",
```

```
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "period": {
          "description": "Indicates when the specific member or organization did (or is intended to) come into effect
    and end.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false
    }
  }
}
```

9.7.11 Resource CareTeam - Maturity Dependency View

URL: https://hl7.org/fhir/R4/careteam-dependencies.html

9.7.11 Resource CareTeam - Maturity Dependency View

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for CareTeam FMM level 2

Name | Card. | Type | Dependency Analysis
CareTeam | | DomainResource
identifier | 0..* | Identifier
status | 0..1 | code
category | 0..* | CodeableConcept
name | 0..1 | string
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
period | 0..1 | Period
participant | 0..* | BackboneElement
role | 0..* | CodeableConcept
member | 0..1 | Reference(Practitioner | PractitionerRole | RelatedPerson | Patient | Organization | CareTeam)
onBehalfOf | 0..1 | Reference(Organization)
period | 0..1 | Period
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition)
managingOrganization | 0..* | Reference(Organization)
telecom | 0..* | ContactPoint
note | 0..* | Annotation

9.7.12 Value Set Sc-valueset-care-team-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-care-team-status.html

9.7.12 Value Set Sc-valueset-care-team-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.7.13 CareTeamStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-care-team-status)

Mapping from http://hl7.org/fhir/ValueSet/care-team-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Indicates the status of the care team."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
proposed | equivalent | proposed
active | equivalent | active
suspended | equivalent | suspended
inactive | equivalent | inactive

9.8 Resource ClinicalImpression - Content

URL: https://hl7.org/fhir/R4/clinicalimpression.html
Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

A record of a clinical assessment performed to determine what problem(s) may affect the patient and before planning the treatments or management strategies that are best to manage a patient's condition. Assessments are often 1:1 with a clinical consultation / encounter, but this varies greatly depending on the clinical workflow. This resource is called "ClinicalImpression" rather than "ClinicalAssessment" to avoid confusion with the recording of assessment tools such as Apgar score.

9.8.1 Scope and Usage

Performing a clinical assessment is a fundamental part of a clinician's workflow, performed repeatedly throughout the day. In spite of this - or perhaps, because of it - there is wide variance in how clinical impressions are recorded. Some clinical assessments simply result in an impression recorded as a single text note in the patient 'record' (e.g. "Progress satisfactory, continue with treatment"), while others are associated with careful, detailed record keeping of the evidence gathered and the reasoning leading to a differential diagnosis, and there is a continuum between these. This resource is intended to be used to cover all these use cases.

The assessment is intimately linked to the process of care. It may occur in the context of a care plan, and it very often results in a new (or revised) care plan. Normally, clinical assessments are part of an ongoing process of care, and the patient will be re-assessed repeatedly. For this reason, the clinical impression can explicitly reference both care plans (preceding and resulting) and reference a previous impression that this impression follows.

An impression is a clinical summation of information and/or an opinion formed, which is the outcome of the clinical assessment process. The ClinicalImpression may lead to a statement of a Condition about a patient.

In FHIR, an assessment is typically an instrument or tool used to collect information about a patient.

Trial-Use Note: Unlike many other resources, there is little prior art with regard to exchanging records of clinical assessments. For this reason, this resource should be regarded as particularly prone to ongoing revision. In terms of scope and usage, the Patient Care workgroup wishes to draw the attention of reviewers and implementers to the following issues:

• When is an existing clinical impression revised, rather than a new one created (that references the existing one)? How does that affect the status? what's the interplay between the status of the diagnosis and the status of the impression? (e.g. for a 'provisional' impression, which bit is provisional?)

• This structure doesn't differentiate between a working and a final diagnosis. Given an answer to the previous question, should it?

• Further clarify around the relationship between care plan and impression is needed. Both answers to the previous questions and ongoing discussions around revisions to the care plan will influence the design of clinical impression

• Should prognosis be represented, and if so, how much structure should it have?

• Should an impression reference other impressions that are related? (how related?)

• Investigations - the specification needs a good value set for the code for the group, and will be considering the name "investigations" further

Feedback is welcome here .

9.8.2 Boundaries and Relationships

ClinicalImpression is the equivalent of the "A" in Weed SOAP. It is the outcome of the clinical assessment process. The ClinicalImpression may lead to a statement of a Condition about a patient. There is another related clinical concept often called an "assessment": assessment Tools such as Apgar (also known as "Assessment Scales"). This is not what the ClinicalImpression resource is about; assessment tools such as Apgar are represented as Observations, and Questionnaires may be used to help generate these. Clinical Impressions may refer to these assessment tools as one of the investigations that was performed during the assessment process.

9.8.3 Background and Context

An important background to understanding this resource is the FHIR Confluence page for clinical assessment . In particular, the storyboards there drove the design of the resource, and will be the basis for all examples created.

PLANNED CHANGE:

ClinicalImpression is one of the Event resources in the FHIR Workflow specification. As such, it is expected to be adjusted to align with the Event workflow pattern which will involve adding a number of additional data elements and potentially renaming a few elements. Any concerns about performing such alignment are welcome as ballot comments and/or tracker items.

This resource is referenced by itself and Condition

9.8.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ClinicalImpression | TU | | DomainResource | A clinical assessment performed when planning treatments and management strategies for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | in-progress | completed | entered-in-error Clinical Impression Status (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status
code | Σ | 0..1 | CodeableConcept | Kind of assessment performed
description | Σ | 0..1 | string | Why/how the assessment was performed
subject | Σ | 1..1 | Reference(Patient | Group) | Patient or group assessed
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
effective[x] | Σ | 0..1 | | Time of assessment
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
date | Σ | 0..1 | dateTime | When the assessment was documented
assessor | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | The clinician performing the assessment
previous | | 0..1 | Reference(ClinicalImpression) | Reference to last assessment
problem | Σ | 0..* | Reference(Condition | AllergyIntolerance) | Relevant impressions of patient state
investigation | | 0..* | BackboneElement | One or more sets of investigations (signs, symptoms, etc.)
code | | 1..1 | CodeableConcept | A name/code for the set Investigation Type (Example)
item | | 0..* | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy | Media) | Record of a specific investigation
protocol | | 0..* | uri | Clinical Protocol followed
summary | | 0..1 | string | Summary of the assessment
finding | | 0..* | BackboneElement | Possible or likely findings and diagnoses
itemCodeableConcept | | 0..1 | CodeableConcept | What was found Condition/Problem/Diagnosis Codes (Example)
itemReference | | 0..1 | Reference(Condition | Observation | Media) | What was found
basis | | 0..1 | string | Which investigations support finding
prognosisCodeableConcept | | 0..* | CodeableConcept | Estimate of likely outcome Clinical Impression Prognosis (Example)
prognosisReference | | 0..* | Reference(RiskAssessment) | RiskAssessment expressing likely outcome
supportingInfo | | 0..* | Reference(Any) | Information supporting the clinical impression
note | | 0..* | Annotation | Comments made about the ClinicalImpression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ClinicalImpression xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 in-progress | completed | entered-in-error -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <code><!-- 0..1 CodeableConcept Kind of assessment performed --></code>
 <description value="[string]"/><!-- 0..1 Why/how the assessment was performed -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient or group assessed --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period Time of assessment --></effective[x]>
 <date value="[dateTime]"/><!-- 0..1 When the assessment was documented -->
 <assessor><!-- 0..1 Reference(Practitioner|PractitionerRole) The clinician performing the assessment --></assessor>
 <previous><!-- 0..1 Reference(ClinicalImpression) Reference to last assessment --></previous>
 <problem><!-- 0..* Reference(Condition|AllergyIntolerance) Relevant impressions of patient state --></problem>
 <investigation>  <!-- 0..* One or more sets of investigations (signs, symptoms, etc.) -->
  <code><!-- 1..1 CodeableConcept A name/code for the set --></code>
  <item><!-- 0..* Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy|Media) Record of a specific investigation --></item>
 </investigation>
 <protocol value="[uri]"/><!-- 0..* Clinical Protocol followed -->
 <summary value="[string]"/><!-- 0..1 Summary of the assessment -->
 <finding>  <!-- 0..* Possible or likely findings and diagnoses -->
  <itemCodeableConcept><!-- 0..1 CodeableConcept What was found --></itemCodeableConcept>
  <itemReference><!-- 0..1 Reference(Condition|Observation|Media) What was found --></itemReference>
  <basis value="[string]"/><!-- 0..1 Which investigations support finding -->
 </finding>
 <prognosisCodeableConcept><!-- 0..* CodeableConcept Estimate of likely outcome --></prognosisCodeableConcept>
 <prognosisReference><!-- 0..* Reference(RiskAssessment) RiskAssessment expressing likely outcome
    --></prognosisReference>
 <supportingInfo><!-- 0..* Reference(Any) Information supporting the clinical impression --></supportingInfo>
```

```
 <note><!-- 0..* Annotation Comments made about the ClinicalImpression --></note>
</ClinicalImpression>
```

JSON Template

```
{
  "resourceType" : "ClinicalImpression",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  in-progress | completed | entered-in-error
  "statusReason" : { CodeableConcept }, // Reason for current status
  "code" : { CodeableConcept }, // Kind of assessment performed
  "description" : "<string>", // Why/how the assessment was performed
  "subject" : { Reference(Patient|Group) }, // R!  Patient or group assessed
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // effective[x]: Time of assessment. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "date" : "<dateTime>", // When the assessment was documented
  "assessor" : { Reference(Practitioner|PractitionerRole) }, // The clinician performing the assessment
  "previous" : { Reference(ClinicalImpression) }, // Reference to last assessment
  "problem" : [{ Reference(Condition|AllergyIntolerance) }], // Relevant impressions of patient state
  "investigation" : [{ // One or more sets of investigations (signs, symptoms, etc.)
    "code" : { CodeableConcept }, // R!  A name/code for the set
    "item" : [{ Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy|Media) }] // Record of a specific investigation
  }],
  "protocol" : ["<uri>"], // Clinical Protocol followed
  "summary" : "<string>", // Summary of the assessment
  "finding" : [{ // Possible or likely findings and diagnoses
    "itemCodeableConcept" : { CodeableConcept }, // What was found
    "itemReference" : { Reference(Condition|Observation|Media) }, // What was found
    "basis" : "<string>" // Which investigations support finding
  }],
  "prognosisCodeableConcept" : [{ CodeableConcept }], // Estimate of likely outcome
  "prognosisReference" : [{ Reference(RiskAssessment) }], // RiskAssessment expressing likely outcome
  "supportingInfo" : [{ Reference(Any) }], // Information supporting the clinical impression
  "note" : [{ Annotation }] // Comments made about the ClinicalImpression
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ClinicalImpression;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ClinicalImpression.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ClinicalImpression.status [ code ]; # 1..1 in-progress | completed | entered-in-error
  fhir:ClinicalImpression.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:ClinicalImpression.code [ CodeableConcept ]; # 0..1 Kind of assessment performed
  fhir:ClinicalImpression.description [ string ]; # 0..1 Why/how the assessment was performed
  fhir:ClinicalImpression.subject [ Reference(Patient|Group) ]; # 1..1 Patient or group assessed
  fhir:ClinicalImpression.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # ClinicalImpression.effective[x] : 0..1 Time of assessment. One of these 2
    fhir:ClinicalImpression.effectiveDateTime [ dateTime ]
    fhir:ClinicalImpression.effectivePeriod [ Period ]
  fhir:ClinicalImpression.date [ dateTime ]; # 0..1 When the assessment was documented
  fhir:ClinicalImpression.assessor [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The clinician performing the
    assessment
  fhir:ClinicalImpression.previous [ Reference(ClinicalImpression) ]; # 0..1 Reference to last assessment
  fhir:ClinicalImpression.problem [ Reference(Condition|AllergyIntolerance) ], ... ; # 0..* Relevant impressions of
    patient state
  fhir:ClinicalImpression.investigation [ # 0..* One or more sets of investigations (signs, symptoms, etc.)
    fhir:ClinicalImpression.investigation.code [ CodeableConcept ]; # 1..1 A name/code for the set
    fhir:ClinicalImpression.investigation.item [
    Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|DiagnosticReport|
  RiskAssessment|ImagingStudy|Media) ], ... ; # 0..* Record of a specific investigation
  ], ...;
  fhir:ClinicalImpression.protocol [ uri ], ... ; # 0..* Clinical Protocol followed
  fhir:ClinicalImpression.summary [ string ]; # 0..1 Summary of the assessment
  fhir:ClinicalImpression.finding [ # 0..* Possible or likely findings and diagnoses
    fhir:ClinicalImpression.finding.itemCodeableConcept [ CodeableConcept ]; # 0..1 What was found
    fhir:ClinicalImpression.finding.itemReference [ Reference(Condition|Observation|Media) ]; # 0..1 What was found
    fhir:ClinicalImpression.finding.basis [ string ]; # 0..1 Which investigations support finding
  ], ...;
  fhir:ClinicalImpression.prognosisCodeableConcept [ CodeableConcept ], ... ; # 0..* Estimate of likely outcome
  fhir:ClinicalImpression.prognosisReference [ Reference(RiskAssessment) ], ... ; # 0..* RiskAssessment expressing
    likely outcome
  fhir:ClinicalImpression.supportingInfo [ Reference(Any) ], ... ; # 0..* Information supporting the clinical impression
  fhir:ClinicalImpression.note [ Annotation ], ... ; # 0..* Comments made about the ClinicalImpression
]
```

Changes since R3

ClinicalImpression

ClinicalImpression.status | Change value set from http://hl7.org/fhir/ValueSet/clinical-impression-status to http://hl7.org/fhir/ValueSet/clinicalimpression-status|4.0.1
ClinicalImpression.statusReason | Added Element
ClinicalImpression.encounter | Added Element
ClinicalImpression.assessor | Type Reference: Added Target Type PractitionerRole
ClinicalImpression.investigation.item | Type Reference: Added Target Type Media
ClinicalImpression.finding.itemCodeableConcept | Added Element
ClinicalImpression.finding.itemReference | Added Element
ClinicalImpression.supportingInfo | Added Element
ClinicalImpression.context | deleted
ClinicalImpression.finding.item[x] | deleted
ClinicalImpression.action | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
ClinicalImpression | TU | | DomainResource | A clinical assessment performed when planning treatments and management strategies for a patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
status | ?!Σ | 1..1 | code | in-progress | completed | entered-in-error Clinical Impression Status (Required)
statusReason | | 0..1 | CodeableConcept | Reason for current status
code | Σ | 0..1 | CodeableConcept | Kind of assessment performed
description | Σ | 0..1 | string | Why/how the assessment was performed
subject | Σ | 1..1 | Reference(Patient | Group) | Patient or group assessed
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
effective[x] | Σ | 0..1 | | Time of assessment
effectiveDateTime | | | dateTime
effectivePeriod | | | Period
date | Σ | 0..1 | dateTime | When the assessment was documented
assessor | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | The clinician performing the assessment
previous | | 0..1 | Reference(ClinicalImpression) | Reference to last assessment
problem | Σ | 0..* | Reference(Condition | AllergyIntolerance) | Relevant impressions of patient state
investigation | | 0..* | BackboneElement | One or more sets of investigations (signs, symptoms, etc.)
code | | 1..1 | CodeableConcept | A name/code for the set Investigation Type (Example)
item | | 0..* | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy | Media) | Record of a specific investigation
protocol | | 0..* | uri | Clinical Protocol followed
summary | | 0..1 | string | Summary of the assessment
finding | | 0..* | BackboneElement | Possible or likely findings and diagnoses
itemCodeableConcept | | 0..1 | CodeableConcept | What was found Condition/Problem/Diagnosis Codes (Example)
itemReference | | 0..1 | Reference(Condition | Observation | Media) | What was found
basis | | 0..1 | string | Which investigations support finding
prognosisCodeableConcept | | 0..* | CodeableConcept | Estimate of likely outcome Clinical Impression Prognosis (Example)
prognosisReference | | 0..* | Reference(RiskAssessment) | RiskAssessment expressing likely outcome
supportingInfo | | 0..* | Reference(Any) | Information supporting the clinical impression
note | | 0..* | Annotation | Comments made about the ClinicalImpression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ClinicalImpression xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 in-progress | completed | entered-in-error -->
 <statusReason><!-- 0..1 CodeableConcept Reason for current status --></statusReason>
 <code><!-- 0..1 CodeableConcept Kind of assessment performed --></code>
 <description value="[string]"/><!-- 0..1 Why/how the assessment was performed -->
 <subject><!-- 1..1 Reference(Patient|Group) Patient or group assessed --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <effective[x]><!-- 0..1 dateTime|Period Time of assessment --></effective[x]>
 <date value="[dateTime]"/><!-- 0..1 When the assessment was documented -->
 <assessor><!-- 0..1 Reference(Practitioner|PractitionerRole) The clinician performing the assessment --></assessor>
 <previous><!-- 0..1 Reference(ClinicalImpression) Reference to last assessment --></previous>
 <problem><!-- 0..* Reference(Condition|AllergyIntolerance) Relevant impressions of patient state --></problem>
 <investigation>  <!-- 0..* One or more sets of investigations (signs, symptoms, etc.) -->
  <code><!-- 1..1 CodeableConcept A name/code for the set --></code>
  <item><!-- 0..* Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy|Media) Record of a specific investigation --></item>
 </investigation>
 <protocol value="[uri]"/><!-- 0..* Clinical Protocol followed -->
 <summary value="[string]"/><!-- 0..1 Summary of the assessment -->
```

```
 <finding>  <!-- 0..* Possible or likely findings and diagnoses -->
  <itemCodeableConcept><!-- 0..1 CodeableConcept What was found --></itemCodeableConcept>
  <itemReference><!-- 0..1 Reference(Condition|Observation|Media) What was found --></itemReference>
  <basis value="[string]"/><!-- 0..1 Which investigations support finding -->
 </finding>
 <prognosisCodeableConcept><!-- 0..* CodeableConcept Estimate of likely outcome --></prognosisCodeableConcept>
 <prognosisReference><!-- 0..* Reference(RiskAssessment) RiskAssessment expressing likely outcome
    --></prognosisReference>
 <supportingInfo><!-- 0..* Reference(Any) Information supporting the clinical impression --></supportingInfo>
 <note><!-- 0..* Annotation Comments made about the ClinicalImpression --></note>
</ClinicalImpression>
```

JSON Template

```
{
  "resourceType" : "ClinicalImpression",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  in-progress | completed | entered-in-error
  "statusReason" : { CodeableConcept }, // Reason for current status
  "code" : { CodeableConcept }, // Kind of assessment performed
  "description" : "<string>", // Why/how the assessment was performed
  "subject" : { Reference(Patient|Group) }, // R!  Patient or group assessed
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  // effective[x]: Time of assessment. One of these 2:
  "effectiveDateTime" : "<dateTime>",
  "effectivePeriod" : { Period },
  "date" : "<dateTime>", // When the assessment was documented
  "assessor" : { Reference(Practitioner|PractitionerRole) }, // The clinician performing the assessment
  "previous" : { Reference(ClinicalImpression) }, // Reference to last assessment
  "problem" : [{ Reference(Condition|AllergyIntolerance) }], // Relevant impressions of patient state
  "investigation" : [{ // One or more sets of investigations (signs, symptoms, etc.)
    "code" : { CodeableConcept }, // R!  A name/code for the set
    "item" : [{ Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|
    DiagnosticReport|RiskAssessment|ImagingStudy|Media) }] // Record of a specific investigation
  }],
  "protocol" : ["<uri>"], // Clinical Protocol followed
  "summary" : "<string>", // Summary of the assessment
  "finding" : [{ // Possible or likely findings and diagnoses
    "itemCodeableConcept" : { CodeableConcept }, // What was found
    "itemReference" : { Reference(Condition|Observation|Media) }, // What was found
    "basis" : "<string>" // Which investigations support finding
  }],
  "prognosisCodeableConcept" : [{ CodeableConcept }], // Estimate of likely outcome
  "prognosisReference" : [{ Reference(RiskAssessment) }], // RiskAssessment expressing likely outcome
  "supportingInfo" : [{ Reference(Any) }], // Information supporting the clinical impression
  "note" : [{ Annotation }] // Comments made about the ClinicalImpression
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ClinicalImpression;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ClinicalImpression.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:ClinicalImpression.status [ code ]; # 1..1 in-progress | completed | entered-in-error
  fhir:ClinicalImpression.statusReason [ CodeableConcept ]; # 0..1 Reason for current status
  fhir:ClinicalImpression.code [ CodeableConcept ]; # 0..1 Kind of assessment performed
  fhir:ClinicalImpression.description [ string ]; # 0..1 Why/how the assessment was performed
  fhir:ClinicalImpression.subject [ Reference(Patient|Group) ]; # 1..1 Patient or group assessed
  fhir:ClinicalImpression.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  # ClinicalImpression.effective[x] : 0..1 Time of assessment. One of these 2
    fhir:ClinicalImpression.effectiveDateTime [ dateTime ]
    fhir:ClinicalImpression.effectivePeriod [ Period ]
  fhir:ClinicalImpression.date [ dateTime ]; # 0..1 When the assessment was documented
  fhir:ClinicalImpression.assessor [ Reference(Practitioner|PractitionerRole) ]; # 0..1 The clinician performing the
    assessment
  fhir:ClinicalImpression.previous [ Reference(ClinicalImpression) ]; # 0..1 Reference to last assessment
  fhir:ClinicalImpression.problem [ Reference(Condition|AllergyIntolerance) ], ... ; # 0..* Relevant impressions of
    patient state
  fhir:ClinicalImpression.investigation [ # 0..* One or more sets of investigations (signs, symptoms, etc.)
    fhir:ClinicalImpression.investigation.code [ CodeableConcept ]; # 1..1 A name/code for the set
    fhir:ClinicalImpression.investigation.item [
    Reference(Observation|QuestionnaireResponse|FamilyMemberHistory|DiagnosticReport|
  RiskAssessment|ImagingStudy|Media) ], ... ; # 0..* Record of a specific investigation
  ], ...;
  fhir:ClinicalImpression.protocol [ uri ], ... ; # 0..* Clinical Protocol followed
  fhir:ClinicalImpression.summary [ string ]; # 0..1 Summary of the assessment
  fhir:ClinicalImpression.finding [ # 0..* Possible or likely findings and diagnoses
    fhir:ClinicalImpression.finding.itemCodeableConcept [ CodeableConcept ]; # 0..1 What was found
    fhir:ClinicalImpression.finding.itemReference [ Reference(Condition|Observation|Media) ]; # 0..1 What was found
    fhir:ClinicalImpression.finding.basis [ string ]; # 0..1 Which investigations support finding
  ], ...;
  fhir:ClinicalImpression.prognosisCodeableConcept [ CodeableConcept ], ... ; # 0..* Estimate of likely outcome
  fhir:ClinicalImpression.prognosisReference [ Reference(RiskAssessment) ], ... ; # 0..* RiskAssessment expressing
```

```
    likely outcome
  fhir:ClinicalImpression.supportingInfo [ Reference(Any) ], ... ; # 0..* Information supporting the clinical impression
  fhir:ClinicalImpression.note [ Annotation ], ... ; # 0..* Comments made about the ClinicalImpression
]
```

Changes since Release 3

ClinicalImpression
ClinicalImpression.status | Change value set from http://hl7.org/fhir/ValueSet/clinical-impression-status to http://hl7.org/fhir/ValueSet/clinicalimpression-status|4.0.1
ClinicalImpression.statusReason | Added Element
ClinicalImpression.encounter | Added Element
ClinicalImpression.assessor | Type Reference: Added Target Type PractitionerRole
ClinicalImpression.investigation.item | Type Reference: Added Target Type Media
ClinicalImpression.finding.itemCodeableConcept | Added Element
ClinicalImpression.finding.itemReference | Added Element
ClinicalImpression.supportingInfo | Added Element
ClinicalImpression.context | deleted
ClinicalImpression.finding.item[x] | deleted
ClinicalImpression.action | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.8.4.1 Terminology Bindings

Path | Definition | Type | Reference
ClinicalImpression.status | The workflow state of a clinical impression. | Required | ClinicalImpressionStatus
ClinicalImpression.statusReason | Codes identifying the reason for the current state of a clinical impression. | Unknown | No details provided yet
ClinicalImpression.code | Identifies categories of clinical impressions. This is a place-holder only. It may be removed. | Unknown | No details provided yet
ClinicalImpression.investigation.code | A name/code for a set of investigations. | Example | InvestigationType
ClinicalImpression.finding.itemCodeableConcept | Identification of the Condition or diagnosis. | Example | Condition/Problem/DiagnosisCodes
ClinicalImpression.prognosisCodeableConcept | Prognosis or outlook findings. | Example | ClinicalImpressionPrognosis

9.8.4.2 Known Issue

A known issue exists with circular references between Condition and ClinicalImpression, which is due to the low maturity level of ClinicalImpression. The Patient Care work group intends to address this issue when ClinicalImpression is considered substantially complete and ready for implementation.

9.8.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
assessor | reference | The clinician performing the assessment | ClinicalImpression.assessor (Practitioner, PractitionerRole)
date | date | When the assessment was documented | ClinicalImpression.date | 17 Resources
encounter | reference | Encounter created as part of | ClinicalImpression.encounter (Encounter)
finding-code | token | What was found | ClinicalImpression.finding.itemCodeableConcept
finding-ref | reference | What was found | ClinicalImpression.finding.itemReference (Condition, Observation, Media)
identifier | token | Business identifier | ClinicalImpression.identifier
investigation | reference | Record of a specific investigation | ClinicalImpression.investigation.item (RiskAssessment, FamilyMemberHistory, Observation, Media, DiagnosticReport, ImagingStudy, QuestionnaireResponse)
patient | reference | Patient or group assessed | ClinicalImpression.subject.where(resolve() is Patient) (Patient) | 33 Resources
previous | reference | Reference to last assessment | ClinicalImpression.previous (ClinicalImpression)
problem | reference | Relevant impressions of patient state | ClinicalImpression.problem (Condition, AllergyIntolerance)
status | token | in-progress | completed | entered-in-error | ClinicalImpression.status
subject | reference | Patient or group assessed | ClinicalImpression.subject (Group, Patient)
supporting-info | reference | Information supporting the clinical impression | ClinicalImpression.supportingInfo (Any)

9.8.6 Resource ClinicalImpression - Examples

URL: https://hl7.org/fhir/R4/clinicalimpression-examples.html

9.8.6 Resource ClinicalImpression - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Example Name | id | Format
Example of clinicalimpression | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.8.7 Resource ClinicalImpression - Detailed Descriptions

URL: https://hl7.org/fhir/R4/clinicalimpression-definitions.html

9.8.7 Resource ClinicalImpression - Detailed Descriptions

Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Detailed Descriptions for the elements in the ClinicalImpression resource.

ClinicalImpression
Element Id | ClinicalImpression
Definition | A record of a clinical assessment performed to determine what problem(s) may affect the patient and before planning the treatments or management strategies that are best to manage a patient's condition. Assessments are often 1:1 with a clinical consultation / encounter, but this varies greatly depending on the clinical workflow. This resource is called "ClinicalImpression" rather than "ClinicalAssessment" to avoid confusion with the recording of assessment tools such as Apgar score.
Cardinality | 0..*
Type | DomainResource
ClinicalImpression.identifier
Element Id | ClinicalImpression.identifier
Definition | Business identifiers assigned to this clinical impression by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the clinical impression as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
ClinicalImpression.status
Element Id | ClinicalImpression.status
Definition | Identifies the workflow status of the assessment.
Cardinality | 1..1
Terminology Binding | Clinical Impression Status (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains the code entered-in-error that marks the clinical impression as not currently valid.
ClinicalImpression.statusReason
Element Id | ClinicalImpression.statusReason
Definition | Captures the reason for the current state of the ClinicalImpression.
Cardinality | 0..1
Terminology Binding | ClinicalImpressionStatusReason:
Type | CodeableConcept
Alternate Names | Suspended Reason; Cancelled Reason
Comments | This is generally only used for "exception" statuses such as "not-done", "suspended" or "cancelled". [distinct reason codes for different statuses can be enforced using invariants if they are universal bindings].
ClinicalImpression.code
Element Id | ClinicalImpression.code
Definition | Categorizes the type of clinical assessment performed.
Cardinality | 0..1
Terminology Binding | ClinicalImpressionCode:
Type | CodeableConcept
Alternate Names | type
Summary | true
Comments | This is present as a place-holder only and may be removed based on feedback/work group opinion.
ClinicalImpression.description
Element Id | ClinicalImpression.description
Definition | A summary of the context and/or cause of the assessment - why / where it was performed, and what patient events/status prompted it.
Cardinality | 0..1
Type | string
Summary | true
ClinicalImpression.subject

Element Id | ClinicalImpression.subject
Definition | The patient or group of individuals assessed as part of this record.
Cardinality | 1..1
Type | Reference(Patient | Group)
Requirements | Group is typically for veterinary and/or public health purposes.
Summary | true
ClinicalImpression.encounter
Element Id | ClinicalImpression.encounter
Definition | The Encounter during which this ClinicalImpression was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter.
ClinicalImpression.effective[x]
Element Id | ClinicalImpression.effective[x]
Definition | The point in time or period over which the subject was assessed.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | This SHOULD be accurate to at least the minute, though some assessments only have a known date.
ClinicalImpression.date
Element Id | ClinicalImpression.date
Definition | Indicates when the documentation of the assessment was complete.
Cardinality | 0..1
Type | dateTime
Summary | true
ClinicalImpression.assessor
Element Id | ClinicalImpression.assessor
Definition | The clinician performing the assessment.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
ClinicalImpression.previous
Element Id | ClinicalImpression.previous
Definition | A reference to the last assessment that was conducted on this patient. Assessments are often/usually ongoing in nature; a care provider (practitioner or team) will make new assessments on an ongoing basis as new data arises or the patient's conditions changes.
Cardinality | 0..1
Type | Reference(ClinicalImpression)
Comments | It is always likely that multiple previous assessments exist for a patient. The point of quoting a previous assessment is that this assessment is relative to it (see resolved).
ClinicalImpression.problem
Element Id | ClinicalImpression.problem
Definition | A list of the relevant problems/conditions for a patient.
Cardinality | 0..*
Type | Reference(Condition | AllergyIntolerance)
Summary | true
Comments | e.g. The patient is a pregnant, has congestive heart failure, has an Adenocarcinoma, and is allergic to penicillin.
ClinicalImpression.investigation
Element Id | ClinicalImpression.investigation
Definition | One or more sets of investigations (signs, symptoms, etc.). The actual grouping of investigations varies greatly depending on the type and context of the assessment. These investigations may include data generated during the assessment process, or data previously generated and recorded that is pertinent to the outcomes.
Cardinality | 0..*
ClinicalImpression.investigation.code
Element Id | ClinicalImpression.investigation.code
Definition | A name/code for the group ("set") of investigations. Typically, this will be something like "signs", "symptoms", "clinical", "diagnostic", but the list is not constrained, and others such groups such as (exposure|family|travel|nutritional) history may be used.
Cardinality | 1..1
Terminology Binding | Investigation Type (Example)
Type | CodeableConcept
ClinicalImpression.investigation.item
Element Id | ClinicalImpression.investigation.item
Definition | A record of a specific investigation that was undertaken.
Cardinality | 0..*
Type | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy | Media)
Comments | Most investigations are observations of one kind or another but some other specific types of data collection resources can also be used.

ClinicalImpression.protocol
Element Id | ClinicalImpression.protocol
Definition | Reference to a specific published clinical protocol that was followed during this assessment, and/or that provides evidence in support of the diagnosis.
Cardinality | 0..*
Type | uri
ClinicalImpression.summary
Element Id | ClinicalImpression.summary
Definition | A text summary of the investigations and the diagnosis.
Cardinality | 0..1
Type | string
ClinicalImpression.finding
Element Id | ClinicalImpression.finding
Definition | Specific findings or diagnoses that were considered likely or relevant to ongoing treatment.
Cardinality | 0..*
ClinicalImpression.finding.itemCodeableConcept
Element Id | ClinicalImpression.finding.itemCodeableConcept
Definition | Specific text or code for finding or diagnosis, which may include ruled-out or resolved conditions.
Cardinality | 0..1
Terminology Binding | Condition/Problem/Diagnosis Codes (Example)
Type | CodeableConcept
ClinicalImpression.finding.itemReference
Element Id | ClinicalImpression.finding.itemReference
Definition | Specific reference for finding or diagnosis, which may include ruled-out or resolved conditions.
Cardinality | 0..1
Type | Reference(Condition | Observation | Media)
ClinicalImpression.finding.basis
Element Id | ClinicalImpression.finding.basis
Definition | Which investigations support finding or diagnosis.
Cardinality | 0..1
Type | string
ClinicalImpression.prognosisCodeableConcept
Element Id | ClinicalImpression.prognosisCodeableConcept
Definition | Estimate of likely outcome.
Cardinality | 0..*
Terminology Binding | Clinical Impression Prognosis (Example)
Type | CodeableConcept
ClinicalImpression.prognosisReference
Element Id | ClinicalImpression.prognosisReference
Definition | RiskAssessment expressing likely outcome.
Cardinality | 0..*
Type | Reference(RiskAssessment)
ClinicalImpression.supportingInfo
Element Id | ClinicalImpression.supportingInfo
Definition | Information supporting the clinical impression.
Cardinality | 0..*
Type | Reference(Any)
ClinicalImpression.note
Element Id | ClinicalImpression.note
Definition | Commentary about the impression, typically recorded after the impression itself was made, though supplemental notes by the original author could also appear.
Cardinality | 0..*
Type | Annotation
Comments | Don't use this element for content that should more properly appear as one of the specific elements of the impression.

9.8.8 Resource ClinicalImpression - Mappings

URL: https://hl7.org/fhir/R4/clinicalimpression-mappings.html

9.8.8 Resource ClinicalImpression - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the clinicalimpression resource (see Mappings to Other Standards for further information & status).

9.8.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ClinicalImpression
identifier | FiveWs.identifier
status | FiveWs.status
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
effective[x] | FiveWs.done[x]
date | FiveWs.recorded
assessor | FiveWs.author
problem | FiveWs.why[x]

9.8.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ClinicalImpression | Event
identifier | Event.identifier
status | Event.status
statusReason | Event.statusReason
code | Event.code
subject | Event.subject
encounter | Event.context
effective[x] | Event.occurrence[x]
assessor | Event.performer.actor
note | Event.note

9.8.8.3 HL7 v2 Mapping (http://hl7.org/v2)

ClinicalImpression | Partial mapping for problem evaluation
identifier | PID-3
status | PRB-14
statusReason
code
description
subject | PID-3
encounter
effective[x] | PRB-2
date
assessor | ROL-4
previous
problem | PRB-3 / IAM-7
investigation
code
item | OBX-21
protocol
summary | OBX
finding
itemCodeableConcept | OBX
itemReference | OBX
basis
prognosisCodeableConcept | PRB-22
prognosisReference
supportingInfo
note

9.8.8.4 RIM Mapping (http://hl7.org/v3)

ClinicalImpression
statusReason | .inboundRelationship[typeCode=SUBJ].source[classCode=CACT, moodCode=EVN].reasonCOde

9.8.9 Resource ClinicalImpression - Extensions & Profiles

URL: https://hl7.org/fhir/R4/clinicalimpression-profiles.html

9.8.9 Resource ClinicalImpression - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

This table lists profiles and extensions for the ClinicalImpression resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.8.9.1 Profiles

No Profiles defined for this resource

9.8.9.2 Extensions

workflow-episodeOfCare | ClinicalImpression | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.8.9.3 Search Extensions

No Search Extensions defined for this resource

9.8.10 Resource ClinicalImpression - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/clinicalimpression-version-maps.html

No validation errors - all conversions are clean

9.8.10 Resource ClinicalImpression - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Encounter, Patient, Practitioner

R3 : R4 Conversion maps for ClinicalImpression.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.8.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ClinicalImpression3to4" = "R3 to R4 Conversions for ClinicalImpression"

conceptmap "ClinicalImpressionStatus" {
  prefix s = "http://hl7.org/fhir/clinical-impression-status"
  prefix t = "http://hl7.org/fhir/event-status"

  s:draft - t:"'in-progress'"
  s:completed - t:completed
  s:"'entered-in-error'" - t:"'entered-in-error'"
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/ClinicalImpression" alias ClinicalImpressionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ClinicalImpression" alias ClinicalImpression as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ClinicalImpression(source src : ClinicalImpressionR3, target tgt : ClinicalImpression) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status as v -> tgt.status = translate(v, '#ClinicalImpressionStatus', 'code');
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.date -> tgt.date;
  src.assessor -> tgt.assessor;
  src.previous -> tgt.previous;
  src.problem -> tgt.problem;
  src.investigation as vs0 -> tgt.investigation as vt0 then investigation(vs0, vt0);
  src.protocol -> tgt.protocol;
  src.summary -> tgt.summary;
  src.finding as vs0 -> tgt.finding as vt0 then finding(vs0, vt0);
  src.prognosisCodeableConcept -> tgt.prognosisCodeableConcept;
  src.prognosisReference -> tgt.prognosisReference;
  src.action as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ClinicalImpression.action',  vt.value = v;
  src.note -> tgt.note;
}

group investigation(source src, target tgt) extends BackboneElement {
```

```
  src.code -> tgt.code;
  src.item -> tgt.item;
}

group finding(source src, target tgt) extends BackboneElement {
  src.item : CodeableConcept as vs -> tgt.itemCodeableConcept as vt then CodeableConcept(vs, vt);
  src.item : Reference as vs -> tgt.itemReference as vt then Reference(vs, vt);
  src.basis -> tgt.basis;
}
```

9.8.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ClinicalImpression4to3" = "R4 to R3 Conversion for ClinicalImpression"

conceptmap "ClinicalImpressionStatus" {
  prefix s = "http://hl7.org/fhir/event-status"
  prefix t = "http://hl7.org/fhir/clinical-impression-status"

  s:"'in-progress'" - t:draft
  s:completed - t:completed
  s:"'entered-in-error'" - t:"'entered-in-error'"
}

uses "http://hl7.org/fhir/StructureDefinition/ClinicalImpression" alias ClinicalImpression as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ClinicalImpression" alias ClinicalImpressionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ClinicalImpression(source src : ClinicalImpression, target tgt : ClinicalImpressionR3) extends DomainResource
    <<type+>> {
  src.identifier -> tgt.identifier;
  src.status as v -> tgt.status = translate(v, '#ClinicalImpressionStatus', 'code');
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.effective : dateTime as vs -> tgt.effective = create('dateTime') as vt then dateTime(vs, vt);
  src.effective : Period as vs -> tgt.effective = create('Period') as vt then Period(vs, vt);
  src.date -> tgt.date;
  src.assessor -> tgt.assessor;
  src.previous -> tgt.previous;
  src.problem -> tgt.problem;
  src.investigation as vs0 -> tgt.investigation as vt0 then investigation(vs0, vt0);
  src.protocol -> tgt.protocol;
  src.summary -> tgt.summary;
  src.finding as vs0 -> tgt.finding as vt0 then finding(vs0, vt0);
  src.prognosisCodeableConcept -> tgt.prognosisCodeableConcept;
  src.prognosisReference -> tgt.prognosisReference;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-ClinicalImpression.action' then
    {
    e.value -> tgt.action;
  };
  src.note -> tgt.note;
}

group investigation(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.item -> tgt.item;
}

group finding(source src, target tgt) extends BackboneElement {
  src.itemCodeableConcept as vs -> tgt.item = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.itemReference as vs -> tgt.item = create('Reference') as vt then Reference(vs, vt);
  src.basis -> tgt.basis;
}
```

9.8.10.3 R4 Validation Errors

9.8.11 JSON Schema for ClinicalImpression

URL: https://hl7.org/fhir/R4/clinicalimpression.schema.json.html

9.8.11 JSON Schema for ClinicalImpression

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is a representation of the json schema for ClinicalImpression, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ClinicalImpression",
  "$ref": "#/definitions/ClinicalImpression",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ClinicalImpression": {
      "description": "A record of a clinical assessment performed to determine what problem(s) may affect the patient
    and before planning the treatments or management strategies that are best to manage a patient\u0027s condition.
    Assessments are often 1:1 with a clinical consultation / encounter,  but this varies greatly depending on the
    clinical workflow. This resource is called \"ClinicalImpression\" rather than \"ClinicalAssessment\" to avoid
```

```
confusion with the recording of assessment tools such as Apgar score.",
  "properties": {
    "resourceType": {
      "description": "This is a ClinicalImpression resource",
      "const": "ClinicalImpression"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifiers assigned to this clinical impression by the performer or other systems
which remain constant as the resource is updated and propagates from server to server.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Identifies the workflow status of the assessment.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "statusReason": {
      "description": "Captures the reason for the current state of the ClinicalImpression.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "code": {
      "description": "Categorizes the type of clinical assessment performed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "A summary of the context and/or cause of the assessment - why / where it was performed, and
what patient events/status prompted it.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subject": {
      "description": "The patient or group of individuals assessed as part of this record.",
      "$ref": "Reference.schema.json#/definitions/Reference"

    },
    "encounter": {
      "description": "The Encounter during which this ClinicalImpression was created or to which the creation of
this record is tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "effectiveDateTime": {
      "description": "The point in time or period over which the subject was assessed.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_effectiveDateTime": {
      "description": "Extensions for effectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The point in time or period over which the subject was assessed.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "date": {
      "description": "Indicates when the documentation of the assessment was complete.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "assessor": {
      "description": "The clinician performing the assessment.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "previous": {
      "description": "A reference to the last assessment that was conducted on this patient. Assessments are
often/usually ongoing in nature; a care provider (practitioner or team) will make new assessments on an ongoing
basis as new data arises or the patient\u0027s conditions changes.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "problem": {
      "description": "A list of the relevant problems/conditions for a patient.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "investigation": {
      "description": "One or more sets of investigations (signs, symptoms, etc.). The actual grouping of
investigations varies greatly depending on the type and context of the assessment. These investigations may include
data generated during the assessment process, or data previously generated and recorded that is pertinent to the
outcomes.",
      "items": {
        "$ref": "#/definitions/ClinicalImpression_Investigation"
      },
      "type": "array"
    },
    "protocol": {
      "description": "Reference to a specific published clinical protocol that was followed during this assessment,
and/or that provides evidence in support of the diagnosis.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_protocol": {
      "description": "Extensions for protocol",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
```

```
    },
    "summary": {
      "description": "A text summary of the investigations and the diagnosis.",
      "$ref": "#/definitions/string"
    },
    "_summary": {
      "description": "Extensions for summary",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "finding": {
      "description": "Specific findings or diagnoses that were considered likely or relevant to ongoing treatment.",
      "items": {
        "$ref": "#/definitions/ClinicalImpression_Finding"
      },
      "type": "array"
    },
    "prognosisCodeableConcept": {
      "description": "Estimate of likely outcome.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "prognosisReference": {
      "description": "RiskAssessment expressing likely outcome.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "supportingInfo": {
      "description": "Information supporting the clinical impression.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Commentary about the impression, typically recorded after the impression itself was made,
though supplemental notes by the original author could also appear.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"ClinicalImpression_Investigation": {
  "description": "A record of a clinical assessment performed to determine what problem(s) may affect the patient
and before planning the treatments or management strategies that are best to manage a patient\u0027s condition.
Assessments are often 1:1 with a clinical consultation / encounter,  but this varies greatly depending on the
clinical workflow. This resource is called \"ClinicalImpression\" rather than \"ClinicalAssessment\" to avoid
confusion with the recording of assessment tools such as Apgar score.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
```

```
      "description": "A name/code for the group (\"set\") of investigations. Typically, this will be something like \"signs\", \"symptoms\", \"clinical\", \"diagnostic\", but the list is not constrained, and others such groups such as (exposure|family|travel|nutritional) history may be used.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "item": {
      "description": "A record of a specific investigation that was undertaken.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "code"
  ]
},
"ClinicalImpression_Finding": {
  "description": "A record of a clinical assessment performed to determine what problem(s) may affect the patient and before planning the treatments or management strategies that are best to manage a patient\u0027s condition. Assessments are often 1:1 with a clinical consultation / encounter,  but this varies greatly depending on the clinical workflow. This resource is called \"ClinicalImpression\" rather than \"ClinicalAssessment\" to avoid confusion with the recording of assessment tools such as Apgar score.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "itemCodeableConcept": {
      "description": "Specific text or code for finding or diagnosis, which may include ruled-out or resolved conditions.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "itemReference": {
      "description": "Specific reference for finding or diagnosis, which may include ruled-out or resolved conditions.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "basis": {
      "description": "Which investigations support finding or diagnosis.",
      "$ref": "#/definitions/string"
    },
    "_basis": {
      "description": "Extensions for basis",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
}
  }
}
```


9.8.12 Resource ClinicalImpression - Maturity Dependency View

URL: https://hl7.org/fhir/R4/clinicalimpression-dependencies.html

9.8.12 Resource ClinicalImpression - Maturity Dependency View

Patient Care Work Group | Maturity Level: 0 | Trial Use | Compartments: Encounter, Patient, Practitioner

Dependency Graph for ClinicalImpression FMM level 0

Name | Card. | Type | Dependency Analysis
ClinicalImpression | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
statusReason | 0..1 | CodeableConcept
code | 0..1 | CodeableConcept
description | 0..1 | string
subject | 1..1 | Reference(Patient | Group)
encounter | 0..1 | Reference(Encounter)
effective[x] | 0..1
effectiveDateTime | | dateTime
effectivePeriod | | Period
date | 0..1 | dateTime
assessor | 0..1 | Reference(Practitioner | PractitionerRole)
previous | 0..1 | Reference(ClinicalImpression)
problem | 0..* | Reference(Condition | AllergyIntolerance)
investigation | 0..* | BackboneElement
code | 1..1 | CodeableConcept
item | 0..* | Reference(Observation | QuestionnaireResponse | FamilyMemberHistory | DiagnosticReport | RiskAssessment | ImagingStudy | Media)
protocol | 0..* | uri
summary | 0..1 | string
finding | 0..* | BackboneElement
itemCodeableConcept | 0..1 | CodeableConcept
itemReference | 0..1 | Reference(Condition | Observation | Media)
basis | 0..1 | string
prognosisCodeableConcept | 0..* | CodeableConcept
prognosisReference | 0..* | Reference(RiskAssessment)
supportingInfo | 0..* | Reference(Any)
note | 0..* | Annotation

9.8.13 Value Set Sc-valueset-clinicalimpression-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-clinicalimpression-status.html

9.8.13 Value Set Sc-valueset-clinicalimpression-status - Status Code Mapping

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Encounter, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.8.14 ClinicalImpressionStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-clinicalimpression-status)

Mapping from http://hl7.org/fhir/ValueSet/clinicalimpression-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 International - Patient Care WG (http://www.hl7.org/Special/com...).

Canonical Mapping for "Codes that reflect the current state of a clinical impression within its overall lifecycle."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
preparation | equivalent | planned
in-progress | equivalent | active
on-hold | equivalent | suspended
stopped | equivalent | failed
completed | equivalent | complete
not-done | equivalent | abandoned
unknown | equivalent | unknown

9.9 Resource AdverseEvent - Content

URL: https://hl7.org/fhir/R4/adverseevent.html
Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Actual or potential/avoided event causing unintended physical injury resulting from or contributed to by medical care, a research study or other healthcare setting factors that requires additional monitoring, treatment, or hospitalization, or that results in death.

9.9.1 Scope and Usage

AdverseEvent is an event resource from a FHIR workflow perspective - see Workflow Event

This resource applies to events that occur during the course of medical care or medical research which may impact an individual as the recipient of care or the participant in a research study. There are also events that occur within a care setting that might or might not impact an individual but had the potential to cause an adverse event. Health care organizations monitor and report both adverse events as well as events that had the potential to cause patient harm. Data are often aggregated for reporting purposes.

An adverse event is the result of an intervention that caused unintentional harm to a specific subject or group of subjects. Examples of adverse events include the administration of an incorrect drug or an incorrect dose of a drug causing an adverse reaction, the use of an implanted device that causes an infection, or a biologic used during a research study that causes unanticipated renal failure. These events are characterized by the need to capture cause and effect (although they might not be known at the time of the event), severity, and outcome.

The context of an adverse event is also important. A subject may have condition(s) or current treatments (medications, diet, devices) that impact their response to a newly introduced medication, device or procedure. Knowledge of these variables is essential in establishing a cause and effect relationship for an adverse event.

A potential adverse event may also be called a near miss or an error. These are also events but because they were detected did not cause harm to a subject. Examples of potential adverse events include a product problem such as a faulty pacemaker that is detected prior implantation, a doctor working simultaneously on two electronic health records realizing the order for a drug was entered on the incorrect patient and then canceling the order, or a patient with a peanut allergy notices that his hospital dinner tray includes peanuts, and he does not eat the peanuts.

9.9.2 Boundaries and Relationships

The AdverseEvent resource is designed to represent events that have a harmful impact on a subject, or had the potential to cause harm to a subject but were avoided. In the course of medical care there are many actions that may impact how a subject responds to a particular treatment impacting patient safety. Therefore the AdverseEvent resource may reference multiple other resources to represent the context or details of an adverse event including but not limited to Observation, Condition, MedicationAdminsitration, Immunization, Procedure, or ResearchStudy.

A DetectedIssue reference is also related to the context of an AdverseEvent to the extent that a known risk for a potential issue such as a drug-drug interaction is documented. If in the context of a known issue, and adverse event occurs, citing this relationship is important for preventing such an occurrence in the future.

The AdverseEvent resource should not be used when a more specific resource exists.

• The AllergyIntolerance resource is a case specific means of capturing the condition of an allergy or intolerance and the criticality (or potential for future harm) based on the response of a particular individual.

• The Clinical Reasoning module provides resources and operations to enable the representation, distribution, and evaluation of clinical knowledge artifacts such as clinical decision support rules, quality measures, order sets, and protocols. The suite of resources within the clinical reasoning module should be used to capture clinical quality measures and clinical protocols that help drive clinical best practices.

• The AdverseEvent resource is not intended to be used to capture potential subject risk in a prospective manner. A more appropriate resource for this purpose would be Risk Assessment which captures predicted outcomes for a patient or population on the basis of source information. Examples include a prognosis statement for a particular condition, risk of health outcome (heart attack, particular type of cancer) on the basis of lifestyle factors and/or family history or list of potential health risks based on a patient's genetic analysis.

9.9.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

AdverseEvent | ΣTU | | DomainResource | Medical care, research study or other healthcare event causing physical injury Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..1 | Identifier | Business identifier for the event

actuality | ?!Σ | 1..1 | code | actual | potential AdverseEventActuality (Required)
category | Σ | 0..* | CodeableConcept | product-problem | product-quality | product-use-error | wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment AdverseEventCategory (Extensible)
event | Σ | 0..1 | CodeableConcept | Type of the event itself in relation to the subject SNOMED CT Clinical Findings (Example)
subject | Σ | 1..1 | Reference(Patient | Group | Practitioner | RelatedPerson) | Subject impacted by event
encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of
date | Σ | 0..1 | dateTime | When the event occurred
detected | Σ | 0..1 | dateTime | When the event was detected
recordedDate | Σ | 0..1 | dateTime | When the event was recorded
resultingCondition | Σ | 0..* | Reference(Condition) | Effect on the subject due to this event
location | Σ | 0..1 | Reference(Location) | Location where adverse event occurred
seriousness | Σ | 0..1 | CodeableConcept | Seriousness of the event AdverseEventSeriousness (Example)
severity | Σ | 0..1 | CodeableConcept | mild | moderate | severe AdverseEventSeverity (Required)
outcome | Σ | 0..1 | CodeableConcept | resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown AdverseEventOutcome (Required)
recorder | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who recorded the adverse event
contributor | Σ | 0..* | Reference(Practitioner | PractitionerRole | Device) | Who was involved in the adverse event or the potential adverse event
suspectEntity | Σ | 0..* | BackboneElement | The suspected agent causing the adverse event
instance | Σ | 1..1 | Reference(Immunization | Procedure | Substance | Medication | MedicationAdministration | MedicationStatement | Device) | Refers to the specific entity that caused the adverse event
causality | Σ | 0..* | BackboneElement | Information on the possible cause of the event
assessment | Σ | 0..1 | CodeableConcept | Assessment of if the entity caused the event AdverseEventCausalityAssessment (Example)
productRelatedness | Σ | 0..1 | string | AdverseEvent.suspectEntity.causalityProductRelatedness
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | AdverseEvent.suspectEntity.causalityAuthor
method | Σ | 0..1 | CodeableConcept | ProbabilityScale | Bayesian | Checklist AdverseEventCausalityMethod (Example)
subjectMedicalHistory | Σ | 0..* | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure | Media | DocumentReference) | AdverseEvent.subjectMedicalHistory
referenceDocument | Σ | 0..* | Reference(DocumentReference) | AdverseEvent.referenceDocument
study | Σ | 0..* | Reference(ResearchStudy) | AdverseEvent.study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AdverseEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier for the event --></identifier>
 <actuality value="[code]"/><!-- 1..1 actual | potential -->
 <category><!-- 0..* CodeableConcept product-problem | product-quality | product-use-error | wrong-dose | incorrect-
    prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time
     | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment
    --></category>
 <event><!-- 0..1 CodeableConcept Type of the event itself in relation to the subject --></event>
 <subject><!-- 1..1 Reference(Patient|Group|Practitioner|RelatedPerson) Subject impacted by event --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <date value="[dateTime]"/><!-- 0..1 When the event occurred -->
 <detected value="[dateTime]"/><!-- 0..1 When the event was detected -->
 <recordedDate value="[dateTime]"/><!-- 0..1 When the event was recorded -->
 <resultingCondition><!-- 0..* Reference(Condition) Effect on the subject due to this event --></resultingCondition>
 <location><!-- 0..1 Reference(Location) Location where adverse event occurred --></location>
 <seriousness><!-- 0..1 CodeableConcept Seriousness of the event --></seriousness>
 <severity><!-- 0..1 CodeableConcept mild | moderate | severe --></severity>
 <outcome><!-- 0..1 CodeableConcept resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
    --></outcome>
 <recorder><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) Who recorded the adverse event --></recorder>
 <contributor><!-- 0..* Reference(Practitioner|PractitionerRole|Device) Who  was involved in the adverse event or the
    potential adverse event --></contributor>
 <suspectEntity>  <!-- 0..* The suspected agent causing the adverse event -->
  <instance><!-- 1..1 Reference(Immunization|Procedure|Substance|Medication|
    MedicationAdministration|MedicationStatement|Device) Refers to the specific entity that caused the adverse event
    --></instance>
  <causality>  <!-- 0..* Information on the possible cause of the event -->
   <assessment><!-- 0..1 CodeableConcept Assessment of if the entity caused the event --></assessment>
   <productRelatedness value="[string]"/><!-- 0..1 AdverseEvent.suspectEntity.causalityProductRelatedness -->
   <author><!-- 0..1 Reference(Practitioner|PractitionerRole) AdverseEvent.suspectEntity.causalityAuthor --></author>
   <method><!-- 0..1 CodeableConcept ProbabilityScale | Bayesian | Checklist --></method>
  </causality>
 </suspectEntity>
 <subjectMedicalHistory><!-- 0..* Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure|Media|
   DocumentReference) AdverseEvent.subjectMedicalHistory --></subjectMedicalHistory>
 <referenceDocument><!-- 0..* Reference(DocumentReference) AdverseEvent.referenceDocument --></referenceDocument>
 <study><!-- 0..* Reference(ResearchStudy) AdverseEvent.study --></study>
</AdverseEvent>
```

JSON Template

```
{
  "resourceType" : "AdverseEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier for the event
  "actuality" : "<code>", // R!  actual | potential
  "category" : [{ CodeableConcept }], // product-problem | product-quality | product-use-error | wrong-dose | incorrect-
    prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time
     | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment
  "event" : { CodeableConcept }, // Type of the event itself in relation to the subject
  "subject" : { Reference(Patient|Group|Practitioner|RelatedPerson) }, // R!  Subject impacted by event
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "date" : "<dateTime>", // When the event occurred
  "detected" : "<dateTime>", // When the event was detected
  "recordedDate" : "<dateTime>", // When the event was recorded
  "resultingCondition" : [{ Reference(Condition) }], // Effect on the subject due to this event
  "location" : { Reference(Location) }, // Location where adverse event occurred
  "seriousness" : { CodeableConcept }, // Seriousness of the event
  "severity" : { CodeableConcept }, // mild | moderate | severe
  "outcome" : { CodeableConcept }, // resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
  "recorder" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) }, // Who recorded the adverse event
  "contributor" : [{ Reference(Practitioner|PractitionerRole|Device) }], // Who  was involved in the adverse event or
    the potential adverse event
  "suspectEntity" : [{ // The suspected agent causing the adverse event
    "instance" : { Reference(Immunization|Procedure|Substance|Medication|
    MedicationAdministration|MedicationStatement|Device) }, // R!  Refers to the specific entity that caused the adverse
     event
    "causality" : [{ // Information on the possible cause of the event
      "assessment" : { CodeableConcept }, // Assessment of if the entity caused the event
      "productRelatedness" : "<string>", // AdverseEvent.suspectEntity.causalityProductRelatedness
      "author" : { Reference(Practitioner|PractitionerRole) }, // AdverseEvent.suspectEntity.causalityAuthor
      "method" : { CodeableConcept } // ProbabilityScale | Bayesian | Checklist
    }]
  }],
  "subjectMedicalHistory" : [{ Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure|Media|
   DocumentReference) }], // AdverseEvent.subjectMedicalHistory
  "referenceDocument" : [{ Reference(DocumentReference) }], // AdverseEvent.referenceDocument
  "study" : [{ Reference(ResearchStudy) }] // AdverseEvent.study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AdverseEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AdverseEvent.identifier [ Identifier ]; # 0..1 Business identifier for the event
  fhir:AdverseEvent.actuality [ code ]; # 1..1 actual | potential
  fhir:AdverseEvent.category [ CodeableConcept ], ... ; # 0..* product-problem | product-quality | product-use-error |
    wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate |
    wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-
    physical-environment
  fhir:AdverseEvent.event [ CodeableConcept ]; # 0..1 Type of the event itself in relation to the subject
  fhir:AdverseEvent.subject [ Reference(Patient|Group|Practitioner|RelatedPerson) ]; # 1..1 Subject impacted by event
  fhir:AdverseEvent.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:AdverseEvent.date [ dateTime ]; # 0..1 When the event occurred
  fhir:AdverseEvent.detected [ dateTime ]; # 0..1 When the event was detected
  fhir:AdverseEvent.recordedDate [ dateTime ]; # 0..1 When the event was recorded
  fhir:AdverseEvent.resultingCondition [ Reference(Condition) ], ... ; # 0..* Effect on the subject due to this event
  fhir:AdverseEvent.location [ Reference(Location) ]; # 0..1 Location where adverse event occurred
  fhir:AdverseEvent.seriousness [ CodeableConcept ]; # 0..1 Seriousness of the event
  fhir:AdverseEvent.severity [ CodeableConcept ]; # 0..1 mild | moderate | severe
  fhir:AdverseEvent.outcome [ CodeableConcept ]; # 0..1 resolved | recovering | ongoing | resolvedWithSequelae | fatal |
     unknown
  fhir:AdverseEvent.recorder [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who recorded the
     adverse event
  fhir:AdverseEvent.contributor [ Reference(Practitioner|PractitionerRole|Device) ], ... ; # 0..* Who  was involved in
    the adverse event or the potential adverse event
  fhir:AdverseEvent.suspectEntity [ # 0..* The suspected agent causing the adverse event
    fhir:AdverseEvent.suspectEntity.instance [
    Reference(Immunization|Procedure|Substance|Medication|MedicationAdministration|
  MedicationStatement|Device) ]; # 1..1 Refers to the specific entity that caused the adverse event
    fhir:AdverseEvent.suspectEntity.causality [ # 0..* Information on the possible cause of the event
      fhir:AdverseEvent.suspectEntity.causality.assessment [ CodeableConcept ]; # 0..1 Assessment of if the entity
    caused the event
      fhir:AdverseEvent.suspectEntity.causality.productRelatedness [ string ]; # 0..1
    AdverseEvent.suspectEntity.causalityProductRelatedness
      fhir:AdverseEvent.suspectEntity.causality.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1
    AdverseEvent.suspectEntity.causalityAuthor
      fhir:AdverseEvent.suspectEntity.causality.method [ CodeableConcept ]; # 0..1 ProbabilityScale | Bayesian |
    Checklist
    ], ...;
  ], ...;
  fhir:AdverseEvent.subjectMedicalHistory [
```

```
    Reference(Condition|Observation|AllergyIntolerance|FamilyMemberHistory|Immunization|
  Procedure|Media|DocumentReference) ], ... ; # 0..* AdverseEvent.subjectMedicalHistory
  fhir:AdverseEvent.referenceDocument [ Reference(DocumentReference) ], ... ; # 0..* AdverseEvent.referenceDocument
  fhir:AdverseEvent.study [ Reference(ResearchStudy) ], ... ; # 0..* AdverseEvent.study
]
```

Changes since R3

AdverseEvent

AdverseEvent.actuality | Added Mandatory Element

AdverseEvent.category | Max Cardinality changed from 1 to * Type changed from code to CodeableConcept Change binding strength from required to extensible

AdverseEvent.event | Added Element

AdverseEvent.subject | Min Cardinality changed from 0 to 1 Type Reference: Added Target Types Group, Practitioner, RelatedPerson Type Reference: Removed Target Types ResearchSubject, Medication, Device

AdverseEvent.encounter | Added Element

AdverseEvent.detected | Added Element

AdverseEvent.recordedDate | Added Element

AdverseEvent.resultingCondition | Added Element

AdverseEvent.severity | Added Element

AdverseEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/adverse-event-outcome to http://hl7.org/fhir/ValueSet/adverse-event-outcome|4.0.1

AdverseEvent.recorder | Type Reference: Added Target Type PractitionerRole

AdverseEvent.contributor | Added Element

AdverseEvent.suspectEntity.instance | Type Reference: Added Target Types Immunization, Procedure

AdverseEvent.suspectEntity.causality | Max Cardinality changed from 1 to * Type changed from code to BackboneElement Remove Binding http://hl7.org/fhir/ValueSet/adverse-event-causality (required)

AdverseEvent.suspectEntity.causality.assessment | Added Element

AdverseEvent.suspectEntity.causality.productRelatedness | Added Element

AdverseEvent.suspectEntity.causality.author | Added Element

AdverseEvent.suspectEntity.causality.method | Added Element

AdverseEvent.subjectMedicalHistory | Type Reference: Added Target Types Media, DocumentReference

AdverseEvent.type | deleted

AdverseEvent.reaction | deleted

AdverseEvent.eventParticipant | deleted

AdverseEvent.description | deleted

AdverseEvent.suspectEntity.causalityAssessment | deleted

AdverseEvent.suspectEntity.causalityProductRelatedness | deleted

AdverseEvent.suspectEntity.causalityMethod | deleted

AdverseEvent.suspectEntity.causalityAuthor | deleted

AdverseEvent.suspectEntity.causalityResult | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

AdverseEvent | ΣTU | | DomainResource | Medical care, research study or other healthcare event causing physical injury Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..1 | Identifier | Business identifier for the event

actuality | ?!Σ | 1..1 | code | actual | potential AdverseEventActuality (Required)

category | Σ | 0..* | CodeableConcept | product-problem | product-quality | product-use-error | wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment AdverseEventCategory (Extensible)

event | Σ | 0..1 | CodeableConcept | Type of the event itself in relation to the subject SNOMED CT Clinical Findings (Example)

subject | Σ | 1..1 | Reference(Patient | Group | Practitioner | RelatedPerson) | Subject impacted by event

encounter | Σ | 0..1 | Reference(Encounter) | Encounter created as part of

date | Σ | 0..1 | dateTime | When the event occurred

detected | Σ | 0..1 | dateTime | When the event was detected

recordedDate | Σ | 0..1 | dateTime | When the event was recorded

resultingCondition | Σ | 0..* | Reference(Condition) | Effect on the subject due to this event

location | Σ | 0..1 | Reference(Location) | Location where adverse event occurred

seriousness | Σ | 0..1 | CodeableConcept | Seriousness of the event AdverseEventSeriousness (Example)

severity | Σ | 0..1 | CodeableConcept | mild | moderate | severe AdverseEventSeverity (Required)

outcome | Σ | 0..1 | CodeableConcept | resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown AdverseEventOutcome (Required)

recorder | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson) | Who recorded the adverse event

contributor | Σ | 0..* | Reference(Practitioner | PractitionerRole | Device) | Who was involved in the adverse event or the potential adverse event

suspectEntity | Σ | 0..* | BackboneElement | The suspected agent causing the adverse event

instance | Σ | 1..1 | Reference(Immunization | Procedure | Substance | Medication | MedicationAdministration | MedicationStatement | Device) | Refers to the specific entity that caused the adverse event

causality | Σ | 0..* | BackboneElement | Information on the possible cause of the event
assessment | Σ | 0..1 | CodeableConcept | Assessment of if the entity caused the event AdverseEventCausalityAssessment (Example)
productRelatedness | Σ | 0..1 | string | AdverseEvent.suspectEntity.causalityProductRelatedness
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole) | AdverseEvent.suspectEntity.causalityAuthor
method | Σ | 0..1 | CodeableConcept | ProbabilityScale | Bayesian | Checklist AdverseEventCausalityMethod (Example)
subjectMedicalHistory | Σ | 0..* | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure | Media | DocumentReference) | AdverseEvent.subjectMedicalHistory
referenceDocument | Σ | 0..* | Reference(DocumentReference) | AdverseEvent.referenceDocument
study | Σ | 0..* | Reference(ResearchStudy) | AdverseEvent.study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AdverseEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business identifier for the event --></identifier>
 <actuality value="[code]"/><!-- 1..1 actual | potential -->
 <category><!-- 0..* CodeableConcept product-problem | product-quality | product-use-error | wrong-dose | incorrect-
    prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time
     | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment
    --></category>
 <event><!-- 0..1 CodeableConcept Type of the event itself in relation to the subject --></event>
 <subject><!-- 1..1 Reference(Patient|Group|Practitioner|RelatedPerson) Subject impacted by event --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter created as part of --></encounter>
 <date value="[dateTime]"/><!-- 0..1 When the event occurred -->
 <detected value="[dateTime]"/><!-- 0..1 When the event was detected -->
 <recordedDate value="[dateTime]"/><!-- 0..1 When the event was recorded -->
 <resultingCondition><!-- 0..* Reference(Condition) Effect on the subject due to this event --></resultingCondition>
 <location><!-- 0..1 Reference(Location) Location where adverse event occurred --></location>
 <seriousness><!-- 0..1 CodeableConcept Seriousness of the event --></seriousness>
 <severity><!-- 0..1 CodeableConcept mild | moderate | severe --></severity>
 <outcome><!-- 0..1 CodeableConcept resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
    --></outcome>
 <recorder><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) Who recorded the adverse event --></recorder>
 <contributor><!-- 0..* Reference(Practitioner|PractitionerRole|Device) Who  was involved in the adverse event or the
    potential adverse event --></contributor>
 <suspectEntity>  <!-- 0..* The suspected agent causing the adverse event -->
  <instance><!-- 1..1 Reference(Immunization|Procedure|Substance|Medication|
    MedicationAdministration|MedicationStatement|Device) Refers to the specific entity that caused the adverse event
    --></instance>
  <causality>  <!-- 0..* Information on the possible cause of the event -->
   <assessment><!-- 0..1 CodeableConcept Assessment of if the entity caused the event --></assessment>
   <productRelatedness value="[string]"/><!-- 0..1 AdverseEvent.suspectEntity.causalityProductRelatedness -->
   <author><!-- 0..1 Reference(Practitioner|PractitionerRole) AdverseEvent.suspectEntity.causalityAuthor --></author>
   <method><!-- 0..1 CodeableConcept ProbabilityScale | Bayesian | Checklist --></method>
  </causality>
 </suspectEntity>
 <subjectMedicalHistory><!-- 0..* Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure|Media|
   DocumentReference) AdverseEvent.subjectMedicalHistory --></subjectMedicalHistory>
 <referenceDocument><!-- 0..* Reference(DocumentReference) AdverseEvent.referenceDocument --></referenceDocument>
 <study><!-- 0..* Reference(ResearchStudy) AdverseEvent.study --></study>
</AdverseEvent>
```

JSON Template

```
{
  "resourceType" : "AdverseEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business identifier for the event
  "actuality" : "<code>", // R!  actual | potential
  "category" : [{ CodeableConcept }], // product-problem | product-quality | product-use-error | wrong-dose | incorrect-
    prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time
     | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment
  "event" : { CodeableConcept }, // Type of the event itself in relation to the subject
  "subject" : { Reference(Patient|Group|Practitioner|RelatedPerson) }, // R!  Subject impacted by event
  "encounter" : { Reference(Encounter) }, // Encounter created as part of
  "date" : "<dateTime>", // When the event occurred
  "detected" : "<dateTime>", // When the event was detected
  "recordedDate" : "<dateTime>", // When the event was recorded
  "resultingCondition" : [{ Reference(Condition) }], // Effect on the subject due to this event
  "location" : { Reference(Location) }, // Location where adverse event occurred
  "seriousness" : { CodeableConcept }, // Seriousness of the event
  "severity" : { CodeableConcept }, // mild | moderate | severe
  "outcome" : { CodeableConcept }, // resolved | recovering | ongoing | resolvedWithSequelae | fatal | unknown
  "recorder" : { Reference(Patient|Practitioner|PractitionerRole|
   RelatedPerson) }, // Who recorded the adverse event
  "contributor" : [{ Reference(Practitioner|PractitionerRole|Device) }], // Who  was involved in the adverse event or
    the potential adverse event
  "suspectEntity" : [{ // The suspected agent causing the adverse event
    "instance" : { Reference(Immunization|Procedure|Substance|Medication|
    MedicationAdministration|MedicationStatement|Device) }, // R!  Refers to the specific entity that caused the adverse
     event
```

```
    "causality" : [{ // Information on the possible cause of the event
      "assessment" : { CodeableConcept }, // Assessment of if the entity caused the event
      "productRelatedness" : "<string>", // AdverseEvent.suspectEntity.causalityProductRelatedness
      "author" : { Reference(Practitioner|PractitionerRole) }, // AdverseEvent.suspectEntity.causalityAuthor
      "method" : { CodeableConcept } // ProbabilityScale | Bayesian | Checklist
    }]
  }],
  "subjectMedicalHistory" : [{ Reference(Condition|Observation|
   AllergyIntolerance|FamilyMemberHistory|Immunization|Procedure|Media|
   DocumentReference) }], // AdverseEvent.subjectMedicalHistory
  "referenceDocument" : [{ Reference(DocumentReference) }], // AdverseEvent.referenceDocument
  "study" : [{ Reference(ResearchStudy) }] // AdverseEvent.study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AdverseEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AdverseEvent.identifier [ Identifier ]; # 0..1 Business identifier for the event
  fhir:AdverseEvent.actuality [ code ]; # 1..1 actual | potential
  fhir:AdverseEvent.category [ CodeableConcept ], ... ; # 0..* product-problem | product-quality | product-use-error |
    wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate |
    wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-
    physical-environment
  fhir:AdverseEvent.event [ CodeableConcept ]; # 0..1 Type of the event itself in relation to the subject
  fhir:AdverseEvent.subject [ Reference(Patient|Group|Practitioner|RelatedPerson) ]; # 1..1 Subject impacted by event
  fhir:AdverseEvent.encounter [ Reference(Encounter) ]; # 0..1 Encounter created as part of
  fhir:AdverseEvent.date [ dateTime ]; # 0..1 When the event occurred
  fhir:AdverseEvent.detected [ dateTime ]; # 0..1 When the event was detected
  fhir:AdverseEvent.recordedDate [ dateTime ]; # 0..1 When the event was recorded
  fhir:AdverseEvent.resultingCondition [ Reference(Condition) ], ... ; # 0..* Effect on the subject due to this event
  fhir:AdverseEvent.location [ Reference(Location) ]; # 0..1 Location where adverse event occurred
  fhir:AdverseEvent.seriousness [ CodeableConcept ]; # 0..1 Seriousness of the event
  fhir:AdverseEvent.severity [ CodeableConcept ]; # 0..1 mild | moderate | severe
  fhir:AdverseEvent.outcome [ CodeableConcept ]; # 0..1 resolved | recovering | ongoing | resolvedWithSequelae | fatal |
     unknown
  fhir:AdverseEvent.recorder [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson) ]; # 0..1 Who recorded the
     adverse event
  fhir:AdverseEvent.contributor [ Reference(Practitioner|PractitionerRole|Device) ], ... ; # 0..* Who  was involved in
    the adverse event or the potential adverse event
  fhir:AdverseEvent.suspectEntity [ # 0..* The suspected agent causing the adverse event
    fhir:AdverseEvent.suspectEntity.instance [
    Reference(Immunization|Procedure|Substance|Medication|MedicationAdministration|
  MedicationStatement|Device) ]; # 1..1 Refers to the specific entity that caused the adverse event
    fhir:AdverseEvent.suspectEntity.causality [ # 0..* Information on the possible cause of the event
      fhir:AdverseEvent.suspectEntity.causality.assessment [ CodeableConcept ]; # 0..1 Assessment of if the entity
    caused the event
      fhir:AdverseEvent.suspectEntity.causality.productRelatedness [ string ]; # 0..1
    AdverseEvent.suspectEntity.causalityProductRelatedness
      fhir:AdverseEvent.suspectEntity.causality.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1
    AdverseEvent.suspectEntity.causalityAuthor
      fhir:AdverseEvent.suspectEntity.causality.method [ CodeableConcept ]; # 0..1 ProbabilityScale | Bayesian |
    Checklist
    ], ...;
  ], ...;
  fhir:AdverseEvent.subjectMedicalHistory [
    Reference(Condition|Observation|AllergyIntolerance|FamilyMemberHistory|Immunization|
  Procedure|Media|DocumentReference) ], ... ; # 0..* AdverseEvent.subjectMedicalHistory
  fhir:AdverseEvent.referenceDocument [ Reference(DocumentReference) ], ... ; # 0..* AdverseEvent.referenceDocument
  fhir:AdverseEvent.study [ Reference(ResearchStudy) ], ... ; # 0..* AdverseEvent.study
]
```

Changes since Release 3

AdverseEvent

AdverseEvent.actuality | Added Mandatory Element

AdverseEvent.category | Max Cardinality changed from 1 to * Type changed from code to CodeableConcept Change binding strength from required to extensible

AdverseEvent.event | Added Element

AdverseEvent.subject | Min Cardinality changed from 0 to 1 Type Reference: Added Target Types Group, Practitioner, RelatedPerson Type Reference: Removed Target Types ResearchSubject, Medication, Device

AdverseEvent.encounter | Added Element

AdverseEvent.detected | Added Element

AdverseEvent.recordedDate | Added Element

AdverseEvent.resultingCondition | Added Element

AdverseEvent.severity | Added Element

AdverseEvent.outcome | Change value set from http://hl7.org/fhir/ValueSet/adverse-event-outcome to http://hl7.org/fhir/ValueSet/adverse-event-outcome|4.0.1

AdverseEvent.recorder | Type Reference: Added Target Type PractitionerRole

AdverseEvent.contributor | Added Element

AdverseEvent.suspectEntity.instance | Type Reference: Added Target Types Immunization, Procedure

AdverseEvent.suspectEntity.causality | Max Cardinality changed from 1 to * Type changed from code to BackboneElement Remove Binding http://hl7.org/fhir/ValueSet/adverse-event-causality (required)
AdverseEvent.suspectEntity.causality.assessment | Added Element
AdverseEvent.suspectEntity.causality.productRelatedness | Added Element
AdverseEvent.suspectEntity.causality.author | Added Element
AdverseEvent.suspectEntity.causality.method | Added Element
AdverseEvent.subjectMedicalHistory | Type Reference: Added Target Types Media, DocumentReference
AdverseEvent.type | deleted
AdverseEvent.reaction | deleted
AdverseEvent.eventParticipant | deleted
AdverseEvent.description | deleted
AdverseEvent.suspectEntity.causalityAssessment | deleted
AdverseEvent.suspectEntity.causalityProductRelatedness | deleted
AdverseEvent.suspectEntity.causalityMethod | deleted
AdverseEvent.suspectEntity.causalityAuthor | deleted
AdverseEvent.suspectEntity.causalityResult | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.9.3.1 Terminology Bindings

Path | Definition | Type | Reference
AdverseEvent.actuality | Overall nature of the adverse event, e.g. real or potential. | Required | AdverseEventActuality
AdverseEvent.category | Overall categorization of the event, e.g. product-related or situational. | Extensible | AdverseEventCategory
AdverseEvent.event | Detailed type of event. | Example | SNOMEDCTClinicalFindings
AdverseEvent.seriousness | Overall seriousness of this event for the patient. | Example | AdverseEventSeriousness
AdverseEvent.severity | The severity of the adverse event itself, in direct relation to the subject. | Required | AdverseEventSeverity
AdverseEvent.outcome | TODO (and should this be required?). | Required | AdverseEventOutcome
AdverseEvent.suspectEntity.causality.assessment | Codes for the assessment of whether the entity caused the event. | Example | AdverseEventCausalityAssessment
AdverseEvent.suspectEntity.causality.method | TODO. | Example | AdverseEventCausalityMethod

9.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actuality | token | actual | potential | AdverseEvent.actuality
category | token | product-problem | product-quality | product-use-error | wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment | AdverseEvent.category
date | date | When the event occurred | AdverseEvent.date
event | token | Type of the event itself in relation to the subject | AdverseEvent.event
location | reference | Location where adverse event occurred | AdverseEvent.location (Location)
recorder | reference | Who recorded the adverse event | AdverseEvent.recorder (Practitioner, Patient, PractitionerRole, RelatedPerson)
resultingcondition | reference | Effect on the subject due to this event | AdverseEvent.resultingCondition (Condition)
seriousness | token | Seriousness of the event | AdverseEvent.seriousness
severity | token | mild | moderate | severe | AdverseEvent.severity
study | reference | AdverseEvent.study | AdverseEvent.study (ResearchStudy)
subject | reference | Subject impacted by event | AdverseEvent.subject (Practitioner, Group, Patient, RelatedPerson)
substance | reference | Refers to the specific entity that caused the adverse event | AdverseEvent.suspectEntity.instance (Immunization, Device, Medication, Procedure, Substance, MedicationAdministration, MedicationStatement)

9.9.5 Resource AdverseEvent - Examples

URL: https://hl7.org/fhir/R4/adverseevent-examples.html

9.9.5 Resource AdverseEvent - Examples

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Example Name | id | Format
Example of adverseevent | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.9.6 Resource AdverseEvent - Detailed Descriptions

URL: https://hl7.org/fhir/R4/adverseevent-definitions.html

9.9.6 Resource AdverseEvent - Detailed Descriptions

Patient Care Work Group | Maturity Level: 0 | Trial Use | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the AdverseEvent resource.

AdverseEvent
Element Id | AdverseEvent
Definition | Actual or potential/avoided event causing unintended physical injury resulting from or contributed to by medical care, a research study or other healthcare setting factors that requires additional monitoring, treatment, or hospitalization, or that results in death.
Cardinality | 0..*
Type | DomainResource
Alternate Names | AE
Summary | true
AdverseEvent.identifier
Element Id | AdverseEvent.identifier
Definition | Business identifiers assigned to this adverse event by the performer or other systems which remain constant as the resource is updated and propagates from server to server.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Requirements | Allows identification of the adverse event as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
Comments | This is a business identifier, not a resource identifier (see discussion). It is best practice for the identifier to only appear on a single resource instance, however business practices may occasionally dictate that multiple resource instances with the same identifier can exist - possibly even with different resource types. For example, multiple Patient and a Person resource instance might share the same social insurance number.
AdverseEvent.actuality
Element Id | AdverseEvent.actuality
Definition | Whether the event actually happened, or just had the potential to. Note that this is independent of whether anyone was affected or harmed or how severely.
Cardinality | 1..1
Terminology Binding | AdverseEventActuality (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it has a potential code that indicates the adverse event did not actually happen.)
Summary | true
AdverseEvent.category
Element Id | AdverseEvent.category
Definition | The overall type of event, intended for search and filtering purposes.
Cardinality | 0..*
Terminology Binding | AdverseEventCategory (Extensible)
Type | CodeableConcept
Summary | true
AdverseEvent.event
Element Id | AdverseEvent.event
Definition | This element defines the specific type of event that occurred or that was prevented from occurring.
Cardinality | 0..1
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.subject
Element Id | AdverseEvent.subject
Definition | This subject or group impacted by the event.
Cardinality | 1..1
Type | Reference(Patient | Group | Practitioner | RelatedPerson)
Requirements | Allows for exposure of biohazard (such as legionella) to a group of individuals in a hospital.
Alternate Names | patient
Summary | true
Comments | If AdverseEvent.resultingCondition differs among members of the group, then use Patient as the subject.
AdverseEvent.encounter
Element Id | AdverseEvent.encounter
Definition | The Encounter during which AdverseEvent was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official completion of an encounter but still be tied to the context of the encounter. For example, if a medication administration

was considered an adverse event because it resulted in a rash, then the encounter when the medication administration was given is the context. If the patient reports the AdverseEvent during a second encounter, that second encounter is not the context.
AdverseEvent.date
Element Id | AdverseEvent.date
Definition | The date (and perhaps time) when the adverse event occurred.
Cardinality | 0..1
Type | dateTime
Summary | true
AdverseEvent.detected
Element Id | AdverseEvent.detected
Definition | Estimated or actual date the AdverseEvent began, in the opinion of the reporter.
Cardinality | 0..1
Type | dateTime
Summary | true
AdverseEvent.recordedDate
Element Id | AdverseEvent.recordedDate
Definition | The date on which the existence of the AdverseEvent was first recorded.
Cardinality | 0..1
Type | dateTime
Summary | true
Comments | The recordedDate represents the date when this particular AdverseEvent record was created in the system, not the date of the most recent update. The date of the last record modification can be retrieved from the resource metadata.
AdverseEvent.resultingCondition
Element Id | AdverseEvent.resultingCondition
Definition | Includes information about the reaction that occurred as a result of exposure to a substance (for example, a drug or a chemical).
Cardinality | 0..*
Type | Reference(Condition)
Summary | true
AdverseEvent.location
Element Id | AdverseEvent.location
Definition | The information about where the adverse event occurred.
Cardinality | 0..1
Type | Reference(Location)
Summary | true
AdverseEvent.seriousness
Element Id | AdverseEvent.seriousness
Definition | Assessment whether this event was of real importance.
Cardinality | 0..1
Terminology Binding | AdverseEventSeriousness (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.severity
Element Id | AdverseEvent.severity
Definition | Describes the severity of the adverse event, in relation to the subject. Contrast to AdverseEvent.seriousness - a severe rash might not be serious, but a mild heart problem is.
Cardinality | 0..1
Terminology Binding | AdverseEventSeverity (Required)
Type | CodeableConcept
Summary | true
AdverseEvent.outcome
Element Id | AdverseEvent.outcome
Definition | Describes the type of outcome from the adverse event.
Cardinality | 0..1
Terminology Binding | AdverseEventOutcome (Required)
Type | CodeableConcept
Summary | true
AdverseEvent.recorder
Element Id | AdverseEvent.recorder
Definition | Information on who recorded the adverse event. May be the patient or a practitioner.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
Summary | true
AdverseEvent.contributor
Element Id | AdverseEvent.contributor
Definition | Parties that may or should contribute or have contributed information to the adverse event, which can consist of one or more activities. Such information includes information leading to the decision to perform the activity and how to perform the activity (e.g. consultant), information that the activity itself seeks to reveal (e.g. informant of clinical history), or information about what activity was performed (e.g. informant witness).
Cardinality | 0..*
Type | Reference(Practitioner | PractitionerRole | Device)
Summary | true
AdverseEvent.suspectEntity

Element Id | AdverseEvent.suspectEntity
Definition | Describes the entity that is suspected to have caused the adverse event.
Cardinality | 0..*
Summary | true
AdverseEvent.suspectEntity.instance
Element Id | AdverseEvent.suspectEntity.instance
Definition | Identifies the actual instance of what caused the adverse event. May be a substance, medication, medication administration, medication statement or a device.
Cardinality | 1..1
Type | Reference(Immunization | Procedure | Substance | Medication | MedicationAdministration | MedicationStatement | Device)
Alternate Names | Substance or Medication or Device
Summary | true
AdverseEvent.suspectEntity.causality
Element Id | AdverseEvent.suspectEntity.causality
Definition | Information on the possible cause of the event.
Cardinality | 0..*
Summary | true
To Do | is this meant to be a backbone element?
AdverseEvent.suspectEntity.causality.assessment
Element Id | AdverseEvent.suspectEntity.causality.assessment
Definition | Assessment of if the entity caused the event.
Cardinality | 0..1
Terminology Binding | AdverseEventCausalityAssessment (Example)
Type | CodeableConcept
Summary | true
To Do | what is difference between assessment and result?
AdverseEvent.suspectEntity.causality.productRelatedness
Element Id | AdverseEvent.suspectEntity.causality.productRelatedness
Definition | AdverseEvent.suspectEntity.causalityProductRelatedness.
Cardinality | 0..1
Type | string
Summary | true
AdverseEvent.suspectEntity.causality.author
Element Id | AdverseEvent.suspectEntity.causality.author
Definition | AdverseEvent.suspectEntity.causalityAuthor.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)
Summary | true
AdverseEvent.suspectEntity.causality.method
Element Id | AdverseEvent.suspectEntity.causality.method
Definition | ProbabilityScale | Bayesian | Checklist.
Cardinality | 0..1
Terminology Binding | AdverseEventCausalityMethod (Example)
Type | CodeableConcept
Summary | true
AdverseEvent.subjectMedicalHistory
Element Id | AdverseEvent.subjectMedicalHistory
Definition | AdverseEvent.subjectMedicalHistory.
Cardinality | 0..*
Type | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure | Media | DocumentReference)
Summary | true
AdverseEvent.referenceDocument
Element Id | AdverseEvent.referenceDocument
Definition | AdverseEvent.referenceDocument.
Cardinality | 0..*
Type | Reference(DocumentReference)
Summary | true
AdverseEvent.study
Element Id | AdverseEvent.study
Definition | AdverseEvent.study.
Cardinality | 0..*
Type | Reference(ResearchStudy)
Summary | true

9.9.7 Resource AdverseEvent - Mappings

URL: https://hl7.org/fhir/R4/adverseevent-mappings.html

9.9.7 Resource AdverseEvent - Mappings

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Mappings for the adverseevent resource (see Mappings to Other Standards for further information & status).

9.9.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

AdverseEvent
actuality | FiveWs.class
category | FiveWs.class
event | FiveWs.what[x]
subject | FiveWs.who
encounter | FiveWs.context
date | FiveWs.init
recordedDate | FiveWs.recorded
recorder | FiveWs.author

9.9.8 Resource AdverseEvent - Extensions & Profiles

URL: https://hl7.org/fhir/R4/adverseevent-profiles.html

9.9.8 Resource AdverseEvent - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the AdverseEvent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.9.8.1 Profiles

No Profiles defined for this resource

9.9.8.2 Extensions

workflow-episodeOfCare | AdverseEvent | episodeOfCare | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.9.8.3 Search Extensions

No Search Extensions defined for this resource

9.9.9 Resource AdverseEvent - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/adverseevent-version-maps.html

No validation errors - all conversions are clean

9.9.9 Resource AdverseEvent - R3/R4 Conversions

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for AdverseEvent.

Functional status for this map: 1 test that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

9.9.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/AdverseEvent3to4" = "R3 to R4 Conversions for AdverseEvent"

uses "http://hl7.org/fhir/3.0/StructureDefinition/AdverseEvent" alias AdverseEventR3 as source
uses "http://hl7.org/fhir/StructureDefinition/AdverseEvent" alias AdverseEvent as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group AdverseEvent(source src : AdverseEventR3, target tgt : AdverseEvent) extends DomainResource <<type+>> {
```

```
  src.identifier -> tgt.identifier;
  src.actuality -> tgt.actuality;
  src.category as vs ->  tgt.category as vt,  vt.coding as c,  c.system =
    'http://terminology.hl7.org/CodeSystem/adverse-event-category',  c.code = vs;
  src.type -> tgt.event;
  src.description as vs where src.type.exists().not() ->  tgt.event as vt,  vt.text = vs "use description in leiu of
    type for event";
  src.subject -> tgt.subject;
  src.date -> tgt.date;
  src.detected -> tgt.detected;
  src.recordedDate -> tgt.recordedDate;
  src.resultingCondition -> tgt.resultingCondition;
  src.location -> tgt.location;
  src.seriousness -> tgt.seriousness;
  src.severity -> tgt.severity;
  src.outcome -> tgt.outcome;
  src.recorder -> tgt.recorder;
  src.contributor -> tgt.contributor;
  src.suspectEntity as s -> tgt.suspectEntity as t then AdverseEventSuspectEntity(s, t);
  src.subjectMedicalHistory -> tgt.subjectMedicalHistory;
  src.referenceDocument -> tgt.referenceDocument;
  src.study -> tgt.study;
}

group AdverseEventSuspectEntity(source src, target tgt) extends BackboneElement {
  src.instance -> tgt.instance;
  src.causality as s -> tgt.causality as t then AdverseEventSuspectEntityCausality(s, t);
}

group AdverseEventSuspectEntityCausality(source src, target tgt) extends BackboneElement {
  src.assessment -> tgt.assessment;
  src.productRelatedness -> tgt.productRelatedness;
  src.author -> tgt.author;
  src.method -> tgt.method;
}
```

9.9.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/AdverseEvent4to3" = "R4 to R3 Conversion for AdverseEvent"

uses "http://hl7.org/fhir/StructureDefinition/AdverseEvent" alias AdverseEvent as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/AdverseEvent" alias AdverseEventR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group AdverseEvent(source src : AdverseEventR3, target tgt : AdverseEvent) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.actuality -> tgt.actuality;
  src.category as vs then {
    vs.coding as c where system = 'http://terminology.hl7.org/CodeSystem/adverse-event-category' then {
      c.code -> tgt.category;
    };
  };
  src.event -> tgt.type;
  src.subject -> tgt.subject;
  src.date -> tgt.date;
  src.detected -> tgt.detected;
  src.recordedDate -> tgt.recordedDate;
  src.resultingCondition -> tgt.resultingCondition;
  src.location -> tgt.location;
  src.seriousness -> tgt.seriousness;
  src.severity -> tgt.severity;
  src.outcome -> tgt.outcome;
  src.recorder -> tgt.recorder;
  src.contributor -> tgt.contributor;
  src.suspectEntity as s -> tgt.suspectEntity as t then AdverseEventSuspectEntity(s, t);
  src.subjectMedicalHistory -> tgt.subjectMedicalHistory;
  src.referenceDocument -> tgt.referenceDocument;
  src.study -> tgt.study;
}

group AdverseEventSuspectEntity(source src, target tgt) extends BackboneElement {
  src.instance -> tgt.instance;
  src.causality as s -> tgt.causality as t then AdverseEventSuspectEntityCausality(s, t);
}

group AdverseEventSuspectEntityCausality(source src, target tgt) extends BackboneElement {
  src.assessment -> tgt.assessment;
  src.productRelatedness -> tgt.productRelatedness;
  src.author -> tgt.author;
  src.method -> tgt.method;
}
```

9.9.9.3 R4 Validation Errors

9.9.10 JSON Schema for AdverseEvent

URL: https://hl7.org/fhir/R4/adverseevent.schema.json.html

9.9.10 JSON Schema for AdverseEvent

Patient Care Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

This is a representation of the json schema for AdverseEvent, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/AdverseEvent",
  "$ref": "#/definitions/AdverseEvent",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "AdverseEvent": {
      "description": "Actual or  potential/avoided event causing unintended physical injury resulting from or
    contributed to by medical care, a research study or other healthcare setting factors that requires additional
    monitoring, treatment, or hospitalization, or that results in death.",
      "properties": {
        "resourceType": {
          "description": "This is a AdverseEvent resource",
          "const": "AdverseEvent"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "identifier": {
```

```
      "description": "Business identifiers assigned to this adverse event by the performer or other systems which
remain constant as the resource is updated and propagates from server to server.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "actuality": {
      "description": "Whether the event actually happened, or just had the potential to. Note that this is
independent of whether anyone was affected or harmed or how severely.",
      "enum": [
        "actual",
        "potential"
      ]
    },
    "_actuality": {
      "description": "Extensions for actuality",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "category": {
      "description": "The overall type of event, intended for search and filtering purposes.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "event": {
      "description": "This element defines the specific type of event that occurred or that was prevented from
occurring.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "This subject or group impacted by the event.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The Encounter during which AdverseEvent was created or to which the creation of this record is
 tightly associated.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date (and perhaps time) when the adverse event occurred.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "detected": {
      "description": "Estimated or actual date the AdverseEvent began, in the opinion of the reporter.",
      "$ref": "#/definitions/dateTime"
    },
    "_detected": {
      "description": "Extensions for detected",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "recordedDate": {
      "description": "The date on which the existence of the AdverseEvent was first recorded.",
      "$ref": "#/definitions/dateTime"
    },
    "_recordedDate": {
      "description": "Extensions for recordedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resultingCondition": {
      "description": "Includes information about the reaction that occurred as a result of exposure to a substance
(for example, a drug or a chemical).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "location": {
      "description": "The information about where the adverse event occurred.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "seriousness": {
      "description": "Assessment whether this event was of real importance.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "severity": {
      "description": "Describes the severity of the adverse event, in relation to the subject. Contrast to
AdverseEvent.seriousness - a severe rash might not be serious, but a mild heart problem is.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "outcome": {
      "description": "Describes the type of outcome from the adverse event.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "recorder": {
      "description": "Information on who recorded the adverse event.  May be the patient or a practitioner.",
      "$ref": "Reference.schema.json#/definitions/Reference"
```

```
    },
    "contributor": {
      "description": "Parties that may or should contribute or have contributed information to the adverse event,
which can consist of one or more activities.  Such information includes information leading to the decision to
perform the activity and how to perform the activity (e.g. consultant), information that the activity itself seeks
to reveal (e.g. informant of clinical history), or information about what activity was performed (e.g. informant
witness).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "suspectEntity": {
      "description": "Describes the entity that is suspected to have caused the adverse event.",
      "items": {
        "$ref": "#/definitions/AdverseEvent_SuspectEntity"
      },
      "type": "array"
    },
    "subjectMedicalHistory": {
      "description": "AdverseEvent.subjectMedicalHistory.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "referenceDocument": {
      "description": "AdverseEvent.referenceDocument.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "study": {
      "description": "AdverseEvent.study.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"AdverseEvent_SuspectEntity": {
  "description": "Actual or  potential/avoided event causing unintended physical injury resulting from or
contributed to by medical care, a research study or other healthcare setting factors that requires additional
monitoring, treatment, or hospitalization, or that results in death.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "instance": {
      "description": "Identifies the actual instance of what caused the adverse event.  May be a substance,
medication, medication administration, medication statement or a device.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "causality": {
      "description": "Information on the possible cause of the event.",
      "items": {
```

```
          "$ref": "#/definitions/AdverseEvent_Causality"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "instance"
    ]
  },
  "AdverseEvent_Causality": {
    "description": "Actual or  potential/avoided event causing unintended physical injury resulting from or contributed to by medical care, a research study or other healthcare setting factors that requires additional monitoring, treatment, or hospitalization, or that results in death.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part  of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "assessment": {
        "description": "Assessment of if the entity caused the event.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "productRelatedness": {
        "description": "AdverseEvent.suspectEntity.causalityProductRelatedness.",
        "$ref": "#/definitions/string"
      },
      "_productRelatedness": {
        "description": "Extensions for productRelatedness",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "author": {
        "description": "AdverseEvent.suspectEntity.causalityAuthor.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "method": {
        "description": "ProbabilityScale | Bayesian | Checklist.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      }
    },
    "additionalProperties": false
  }
 }
}
```


9.9.11 Resource AdverseEvent - Maturity Dependency View

URL: https://hl7.org/fhir/R4/adverseevent-dependencies.html

9.9.11 Resource AdverseEvent - Maturity Dependency View

Patient Care Work Group | Maturity Level: 0 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Dependency Graph for AdverseEvent FMM level 0

Name | Card. | Type | Dependency Analysis
AdverseEvent | | DomainResource
identifier | 0..1 | Identifier
actuality | 1..1 | code
category | 0..* | CodeableConcept
event | 0..1 | CodeableConcept
subject | 1..1 | Reference(Patient | Group | Practitioner | RelatedPerson)

encounter | 0..1 | Reference(Encounter)
date | 0..1 | dateTime
detected | 0..1 | dateTime
recordedDate | 0..1 | dateTime
resultingCondition | 0..* | Reference(Condition)
location | 0..1 | Reference(Location)
seriousness | 0..1 | CodeableConcept
severity | 0..1 | CodeableConcept
outcome | 0..1 | CodeableConcept
recorder | 0..1 | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson)
contributor | 0..* | Reference(Practitioner | PractitionerRole | Device)
suspectEntity | 0..* | BackboneElement
instance | 1..1 | Reference(Immunization | Procedure | Substance | Medication | MedicationAdministration | MedicationStatement | Device)
causality | 0..* | BackboneElement
assessment | 0..1 | CodeableConcept
productRelatedness | 0..1 | string
author | 0..1 | Reference(Practitioner | PractitionerRole)
method | 0..1 | CodeableConcept
subjectMedicalHistory | 0..* | Reference(Condition | Observation | AllergyIntolerance | FamilyMemberHistory | Immunization | Procedure | Media | DocumentReference)
referenceDocument | 0..* | Reference(DocumentReference)
study | 0..* | Reference(ResearchStudy)

9.10 Resource DetectedIssue - Content

URL: https://hl7.org/fhir/R4/detectedissue.html
Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.

9.10.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow, specifically Event.

This resource applies to various circumstances where there is a concern about an existing or proposed set of clinical activity. The issue could relate to single, proposed, or multiple actions. It does not apply to technical issues (e.g. lack of user permissions) but could relate to violation of patient consent limitations. Examples include:

- Drug-drug interactions
- Inappropriate therapy (wrong dose, frequency, body site)
- Duplicate therapy

This resource represents a specific instance of a potential issue for a particular patient. It is not intended to represent general patient-independent knowledge. This resource is also not intended to be used in defining general prohibitions on actions such as "No NSAIDs", "No solid oral dose forms" or "No MRIs - metallic tattoos". These guidelines can be captured using the AllergyIntolerance, and/or Flag resources. Similarly, this resource is not to be used to capture clinical facts that may imply contraindications such as pregnancy, breast feeding, patient preferences, past procedures, etc. These would be represented using Condition, Procedure or other resources.

9.10.2 Boundaries and Relationships

This resource only applies to documenting a risk associated with a specific planned or ongoing action, or the lack of an action which should be planned - not a general propensity to risk. The latter would be handled using AllergyIntolerance for substance-specific issues or Flag for other types of issues. In addition, the resource represents patient-specific and time-bound risk manifestations, not generic knowledge statements about risks that can exist.

This resource is limited to clinical issues associated with a proposed or ongoing action. It does not cover technical issues such as lack of permission, duplicate identifiers, insufficient data, and other business rule violations. Technical issues are conveyed using the OperationOutcome resource. It is possible to have both OperationOutcome and DetectedIssue together, where the OperationOutcome might indicate that a requested action was rejected due to a clinical issue and the DetectedIssue provides the details of the issue.

9.10.3 Background and Context

Detected issues are typically identified by decision support systems. However, they may also be captured directly by clinicians. The latter typically happens for one of two reasons:

• A clinician wishes to communicate an issue to another clinician whose responsibility would be to resolve it (e.g. a pharmacist identifying an issue with a prescription prior to putting it on hold)

• A clinician wishes to pre-emptively identify that an issue is known and is being managed (to avoid red flags being raised as part of downstream workflow); e.g. Submitting a new order and including a link to a "duplicate therapy" issue with mitigation indicating that the therapy is not considered to be duplicate.

Decision-support generated issues can result from calling a decision-support engine directly (e.g. via a custom OperationDefinition) or as part of an attempt to perform some other function (creating an order, submitting an insurance claim, capturing a medication list). When the issues are generated as a by-product of performing some other sort of action, they may be included in the "response" to the requested action in the same manner as an OperationOutcome. In fact, both may be present - the OperationOutcome indicating that there was a warning or error associated with the request and a DetectedIssue providing the clinical details. (The OperationOutcome could point to the DetectedIssue via an extension.)

In those circumstances where requested operations are rejected as a result of a detected issue, the workflow may support allowing the operation to be re-tried, provided that the identified issue is included as part of the submission (possibly also including a mitigation). In doing so, the sender acknowledges the issue and takes responsibility for it, thus allowing the requested operation to proceed. See Linking to Detected Issues for guidance on how a DetectedIssue instance might be included as part of another operation.

Systems that require such workflows should document expected behavior as part of their CapabilityStatement declarations.

This resource is referenced by MedicationDispense, MedicationKnowledge and MedicationRequest

9.10.4 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name	Flags	Card.	Type	Description & Constraints
DetectedIssue	TU		DomainResource	Clinical issue with action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Unique id for the detected issue
status	?!Σ	1..1	code	registered | preliminary | final | amended + ObservationStatus (Required)
code	Σ	0..1	CodeableConcept	Issue Category, e.g. drug-drug, duplicate therapy, etc. Detected Issue Category (Preferred)
severity	Σ	0..1	code	high | moderate | low DetectedIssueSeverity (Required)
patient	Σ	0..1	Reference(Patient)	Associated patient
identified[x]	Σ	0..1		When identified
identifiedDateTime			dateTime	
identifiedPeriod			Period	
author	Σ	0..1	Reference(Practitioner | PractitionerRole | Device)	The provider or device that identified the issue
implicated	Σ	0..*	Reference(Any)	Problem resource
evidence		0..*	BackboneElement	Supporting evidence
code		0..*	CodeableConcept	Manifestation Manifestation and Symptom Codes (Example)
detail		0..*	Reference(Any)	Supporting information
detail		0..1	string	Description and context
reference		0..1	uri	Authority for issue
mitigation		0..*	BackboneElement	Step taken to address
action		1..1	CodeableConcept	What mitigation? Detected Issue Mitigation Action (Preferred)
date		0..1	dateTime	Date committed
author		0..1	Reference(Practitioner | PractitionerRole)	Who is committing?

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DetectedIssue xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id for the detected issue --></identifier>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <code><!-- 0..1 CodeableConcept Issue Category, e.g. drug-drug, duplicate therapy, etc. --></code>
 <severity value="[code]"/><!-- 0..1 high | moderate | low -->
 <patient><!-- 0..1 Reference(Patient) Associated patient --></patient>
 <identified[x]><!-- 0..1 dateTime|Period When identified --></identified[x]>
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Device) The provider or device that identified the issue
    --></author>
 <implicated><!-- 0..* Reference(Any) Problem resource --></implicated>
```

```
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!-- 0..* CodeableConcept Manifestation --></code>
  <detail><!-- 0..* Reference(Any) Supporting information --></detail>
 </evidence>
 <detail value="[string]"/><!-- 0..1 Description and context -->
 <reference value="[uri]"/><!-- 0..1 Authority for issue -->
 <mitigation>  <!-- 0..* Step taken to address -->
  <action><!-- 1..1 CodeableConcept What mitigation? --></action>
  <date value="[dateTime]"/><!-- 0..1 Date committed -->
  <author><!-- 0..1 Reference(Practitioner|PractitionerRole) Who is committing? --></author>
 </mitigation>
</DetectedIssue>
```

JSON Template

```
{
  "resourceType" : "DetectedIssue",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id for the detected issue
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "code" : { CodeableConcept }, // Issue Category, e.g. drug-drug, duplicate therapy, etc.
  "severity" : "<code>", // high | moderate | low
  "patient" : { Reference(Patient) }, // Associated patient
  // identified[x]: When identified. One of these 2:
  "identifiedDateTime" : "<dateTime>",
  "identifiedPeriod" : { Period },
  "author" : { Reference(Practitioner|PractitionerRole|Device) }, // The provider or device that identified the issue
  "implicated" : [{ Reference(Any) }], // Problem resource
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // Manifestation
    "detail" : [{ Reference(Any) }] // Supporting information
  }],
  "detail" : "<string>", // Description and context
  "reference" : "<uri>", // Authority for issue
  "mitigation" : [{ // Step taken to address
    "action" : { CodeableConcept }, // R!  What mitigation?
    "date" : "<dateTime>", // Date committed
    "author" : { Reference(Practitioner|PractitionerRole) } // Who is committing?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DetectedIssue;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DetectedIssue.identifier [ Identifier ], ... ; # 0..* Unique id for the detected issue
  fhir:DetectedIssue.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:DetectedIssue.code [ CodeableConcept ]; # 0..1 Issue Category, e.g. drug-drug, duplicate therapy, etc.
  fhir:DetectedIssue.severity [ code ]; # 0..1 high | moderate | low
  fhir:DetectedIssue.patient [ Reference(Patient) ]; # 0..1 Associated patient
  # DetectedIssue.identified[x] : 0..1 When identified. One of these 2
    fhir:DetectedIssue.identifiedDateTime [ dateTime ]
    fhir:DetectedIssue.identifiedPeriod [ Period ]
  fhir:DetectedIssue.author [ Reference(Practitioner|PractitionerRole|Device) ]; # 0..1 The provider or device that
    identified the issue
  fhir:DetectedIssue.implicated [ Reference(Any) ], ... ; # 0..* Problem resource
  fhir:DetectedIssue.evidence [ # 0..* Supporting evidence
    fhir:DetectedIssue.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation
    fhir:DetectedIssue.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information
  ], ...;
  fhir:DetectedIssue.detail [ string ]; # 0..1 Description and context
  fhir:DetectedIssue.reference [ uri ]; # 0..1 Authority for issue
  fhir:DetectedIssue.mitigation [ # 0..* Step taken to address
    fhir:DetectedIssue.mitigation.action [ CodeableConcept ]; # 1..1 What mitigation?
    fhir:DetectedIssue.mitigation.date [ dateTime ]; # 0..1 Date committed
    fhir:DetectedIssue.mitigation.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who is committing?
  ], ...;
]
```

Changes since R3

DetectedIssue

DetectedIssue.identifier | Max Cardinality changed from 1 to *

DetectedIssue.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1

DetectedIssue.code | Renamed from category to code

DetectedIssue.severity | Change value set from http://hl7.org/fhir/ValueSet/detectedissue-severity to http://hl7.org/fhir/ValueSet/detectedissue-severity|4.0.1

DetectedIssue.identified[x] | Added Element

DetectedIssue.author | Type Reference: Added Target Type PractitionerRole

DetectedIssue.evidence | Added Element

DetectedIssue.evidence.code | Added Element

DetectedIssue.evidence.detail | Added Element
DetectedIssue.mitigation.author | Type Reference: Added Target Type PractitionerRole
DetectedIssue.date | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints
DetectedIssue | TU | | DomainResource | Clinical issue with action Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique id for the detected issue
status | ?!Σ | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
code | Σ | 0..1 | CodeableConcept | Issue Category, e.g. drug-drug, duplicate therapy, etc. Detected Issue Category (Preferred)
severity | Σ | 0..1 | code | high | moderate | low DetectedIssueSeverity (Required)
patient | Σ | 0..1 | Reference(Patient) | Associated patient
identified[x] | Σ | 0..1 | | When identified
identifiedDateTime | | | dateTime
identifiedPeriod | | | Period
author | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Device) | The provider or device that identified the issue
implicated | Σ | 0..* | Reference(Any) | Problem resource
evidence | | 0..* | BackboneElement | Supporting evidence
code | | 0..* | CodeableConcept | Manifestation Manifestation and Symptom Codes (Example)
detail | | 0..* | Reference(Any) | Supporting information
detail | | 0..1 | string | Description and context
reference | | 0..1 | uri | Authority for issue
mitigation | | 0..* | BackboneElement | Step taken to address
action | | 1..1 | CodeableConcept | What mitigation? Detected Issue Mitigation Action (Preferred)
date | | 0..1 | dateTime | Date committed
author | | 0..1 | Reference(Practitioner | PractitionerRole) | Who is committing?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DetectedIssue xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id for the detected issue --></identifier>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <code><!-- 0..1 CodeableConcept Issue Category, e.g. drug-drug, duplicate therapy, etc. --></code>
 <severity value="[code]"/><!-- 0..1 high | moderate | low -->
 <patient><!-- 0..1 Reference(Patient) Associated patient --></patient>
 <identified[x]><!-- 0..1 dateTime|Period When identified --></identified[x]>
 <author><!-- 0..1 Reference(Practitioner|PractitionerRole|Device) The provider or device that identified the issue
    --></author>
 <implicated><!-- 0..* Reference(Any) Problem resource --></implicated>
 <evidence>  <!-- 0..* Supporting evidence -->
  <code><!-- 0..* CodeableConcept Manifestation --></code>
  <detail><!-- 0..* Reference(Any) Supporting information --></detail>
 </evidence>
 <detail value="[string]"/><!-- 0..1 Description and context -->
 <reference value="[uri]"/><!-- 0..1 Authority for issue -->
 <mitigation>  <!-- 0..* Step taken to address -->
  <action><!-- 1..1 CodeableConcept What mitigation? --></action>
  <date value="[dateTime]"/><!-- 0..1 Date committed -->
  <author><!-- 0..1 Reference(Practitioner|PractitionerRole) Who is committing? --></author>
 </mitigation>
</DetectedIssue>
```

JSON Template

```
{
  "resourceType" : "DetectedIssue",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id for the detected issue
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "code" : { CodeableConcept }, // Issue Category, e.g. drug-drug, duplicate therapy, etc.
  "severity" : "<code>", // high | moderate | low
  "patient" : { Reference(Patient) }, // Associated patient
  // identified[x]: When identified. One of these 2:
  "identifiedDateTime" : "<dateTime>",
  "identifiedPeriod" : { Period },
  "author" : { Reference(Practitioner|PractitionerRole|Device) }, // The provider or device that identified the issue
  "implicated" : [{ Reference(Any) }], // Problem resource
  "evidence" : [{ // Supporting evidence
    "code" : [{ CodeableConcept }], // Manifestation
    "detail" : [{ Reference(Any) }] // Supporting information
  }],
  "detail" : "<string>", // Description and context
```

```
  "reference" : "<uri>", // Authority for issue
  "mitigation" : [{ // Step taken to address
    "action" : { CodeableConcept }, // R!  What mitigation?
    "date" : "<dateTime>", // Date committed
    "author" : { Reference(Practitioner|PractitionerRole) } // Who is committing?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DetectedIssue;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DetectedIssue.identifier [ Identifier ], ... ; # 0..* Unique id for the detected issue
  fhir:DetectedIssue.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:DetectedIssue.code [ CodeableConcept ]; # 0..1 Issue Category, e.g. drug-drug, duplicate therapy, etc.
  fhir:DetectedIssue.severity [ code ]; # 0..1 high | moderate | low
  fhir:DetectedIssue.patient [ Reference(Patient) ]; # 0..1 Associated patient
  # DetectedIssue.identified[x] : 0..1 When identified. One of these 2
    fhir:DetectedIssue.identifiedDateTime [ dateTime ]
    fhir:DetectedIssue.identifiedPeriod [ Period ]
  fhir:DetectedIssue.author [ Reference(Practitioner|PractitionerRole|Device) ]; # 0..1 The provider or device that
    identified the issue
  fhir:DetectedIssue.implicated [ Reference(Any) ], ... ; # 0..* Problem resource
  fhir:DetectedIssue.evidence [ # 0..* Supporting evidence
    fhir:DetectedIssue.evidence.code [ CodeableConcept ], ... ; # 0..* Manifestation
    fhir:DetectedIssue.evidence.detail [ Reference(Any) ], ... ; # 0..* Supporting information
  ], ...;
  fhir:DetectedIssue.detail [ string ]; # 0..1 Description and context
  fhir:DetectedIssue.reference [ uri ]; # 0..1 Authority for issue
  fhir:DetectedIssue.mitigation [ # 0..* Step taken to address
    fhir:DetectedIssue.mitigation.action [ CodeableConcept ]; # 1..1 What mitigation?
    fhir:DetectedIssue.mitigation.date [ dateTime ]; # 0..1 Date committed
    fhir:DetectedIssue.mitigation.author [ Reference(Practitioner|PractitionerRole) ]; # 0..1 Who is committing?
  ], ...;
]
```

Changes since Release 3

DetectedIssue
DetectedIssue.identifier | Max Cardinality changed from 1 to *
DetectedIssue.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1
DetectedIssue.code | Renamed from category to code
DetectedIssue.severity | Change value set from http://hl7.org/fhir/ValueSet/detectedissue-severity to http://hl7.org/fhir/ValueSet/detectedissue-severity|4.0.1
DetectedIssue.identified[x] | Added Element
DetectedIssue.author | Type Reference: Added Target Type PractitionerRole
DetectedIssue.evidence | Added Element
DetectedIssue.evidence.code | Added Element
DetectedIssue.evidence.detail | Added Element
DetectedIssue.mitigation.author | Type Reference: Added Target Type PractitionerRole
DetectedIssue.date | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.10.4.1 Terminology Bindings

Path | Definition | Type | Reference
DetectedIssue.status | Indicates the status of the identified issue. | Required | ObservationStatus
DetectedIssue.code | Codes identifying the general type of detected issue; e.g. Drug-drug interaction, Timing issue, Duplicate therapy, etc. | Preferred | DetectedIssueCategory
DetectedIssue.severity | Indicates the potential degree of impact of the identified issue on the patient. | Required | DetectedIssueSeverity
DetectedIssue.evidence.code | Codes that describes the types of evidence for a detected issue. | Example | ManifestationAndSymptomCodes
DetectedIssue.mitigation.action | Codes describing steps taken to resolve the issue or other circumstances that mitigate the risk associated with the issue; e.g. 'added concurrent therapy', 'prior therapy documented', etc. | Preferred | DetectedIssueMitigationAction

9.10.5 Linking to Detected Issues

DetectedIssue follows the pattern of linking from the resource created "second". As DetectedIssue originates in response to one or more other existing records, it points to those records rather than being pointed to from them.

In some cases, a detected issue might be associated with a single record. When this occurs, it may be stored as a contained resource within the implicated resource provided that there is no expected need to search for the detected issue directly. However, with detected issues that implicate multiple records, containment is more problematic. In some workflows, a detected issue might be deemed to be "owned" by the record whose creation triggers the contraindication being created - i.e. the "second" or "last" record. However, where multiple actions are proposed as part of a single submission, there can be no single owner and containment will not be feasible.

If there is a strong need to point from an implicated resource to DetectedIssue and containment is not appropriate, an extension can be used.

9.10.6 Workflow Challenges

DetectedIssue is a resource that is frequently associated with workflow challenges where frequent alerts that are not clinically relevant result in clinicians tuning out (or turning off) the content and thus missing relevant alerts. Give consideration to this issue before making heavy use of this resource.

9.10.7 Open Issues

• Are author, reference and/or mitigation (and its various parts) all part of the 80%?

9.10.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The provider or device that identified the issue | DetectedIssue.author (Practitioner, Device, PractitionerRole)
code | token | Issue Category, e.g. drug-drug, duplicate therapy, etc. | DetectedIssue.code
identified | date | When identified | DetectedIssue.identified
identifier | token | Unique id for the detected issue | DetectedIssue.identifier | 30 Resources
implicated | reference | Problem resource | DetectedIssue.implicated (Any)
patient | reference | Associated patient | DetectedIssue.patient (Patient) | 33 Resources

9.10.9 Resource DetectedIssue - Examples

URL: https://hl7.org/fhir/R4/detectedissue-examples.html

9.10.9 Resource DetectedIssue - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Example Name | id | Format
Drug-drug interaction between prescription and medication statement | ddi | XML | JSON | Turtle
Diagnostic order is placed when a recent image of the same body site already exists | duplicate | XML | JSON | Turtle
Patient is allergic to an inactive ingredient in an ordered medication | allergy | XML | JSON | Turtle
Lab order is contraindicated based on a patient's existing medication | lab | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.10.10 Resource DetectedIssue - Detailed Descriptions

URL: https://hl7.org/fhir/R4/detectedissue-definitions.html

9.10.10 Resource DetectedIssue - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the DetectedIssue resource.

DetectedIssue
Element Id | DetectedIssue
Definition | Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.
Cardinality | 0..*
Type | DomainResource
Alternate Names | DDI; drug-drug interaction; Contraindication; Alert
DetectedIssue.identifier
Element Id | DetectedIssue.identifier
Definition | Business identifier associated with the detected issue record.
Note | This is a business identifier, not a resource identifier (see discussion)

Cardinality | 0..*
Type | Identifier
Requirements | Allows linking instances of the same detected issue found on different servers.
Summary | true
DetectedIssue.status
Element Id | DetectedIssue.status
Definition | Indicates the status of the detected issue.
Cardinality | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Alternate Names | status
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes cancelled and entered-in-error that mark the issue as not currently valid.
DetectedIssue.code
Element Id | DetectedIssue.code
Definition | Identifies the general type of issue identified.
Cardinality | 0..1
Terminology Binding | Detected Issue Category (Preferred)
Type | CodeableConcept
Alternate Names | type
Summary | true
DetectedIssue.severity
Element Id | DetectedIssue.severity
Definition | Indicates the degree of importance associated with the identified issue based on the potential impact on the patient.
Cardinality | 0..1
Terminology Binding | DetectedIssueSeverity (Required)
Type | code
Alternate Names | severity
Summary | true
DetectedIssue.patient
Element Id | DetectedIssue.patient
Definition | Indicates the patient whose record the detected issue is associated with.
Cardinality | 0..1
Type | Reference(Patient)
Requirements | While the subject could be inferred by tracing the subject of the implicated resources, it's useful to have a direct link for query purposes.
Summary | true
DetectedIssue.identified[x]
Element Id | DetectedIssue.identified[x]
Definition | The date or period when the detected issue was initially identified.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | No-one can be responsible for mitigation prior to the issue being identified.
Summary | true
DetectedIssue.author
Element Id | DetectedIssue.author
Definition | Individual or device responsible for the issue being raised. For example, a decision support application or a pharmacist conducting a medication review.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Device)
Summary | true
DetectedIssue.implicated
Element Id | DetectedIssue.implicated
Definition | Indicates the resource representing the current activity or proposed activity that is potentially problematic.
Cardinality | 0..*
Type | Reference(Any)
Alternate Names | cause
Summary | true
Comments | There's an implicit constraint on the number of implicated resources based on DetectedIssue.type; e.g. For drug-drug, there would be more than one. For timing, there would typically only be one.
DetectedIssue.evidence
Element Id | DetectedIssue.evidence
Definition | Supporting evidence or manifestations that provide the basis for identifying the detected issue such as a GuidanceResponse or MeasureReport.
Cardinality | 0..*
DetectedIssue.evidence.code
Element Id | DetectedIssue.evidence.code
Definition | A manifestation that led to the recording of this detected issue.
Cardinality | 0..*

Terminology Binding | Manifestation and Symptom Codes (Example)
Type | CodeableConcept
DetectedIssue.evidence.detail
Element Id | DetectedIssue.evidence.detail
Definition | Links to resources that constitute evidence for the detected issue such as a GuidanceResponse or MeasureReport.
Cardinality | 0..*
Type | Reference(Any)
DetectedIssue.detail
Element Id | DetectedIssue.detail
Definition | A textual explanation of the detected issue.
Cardinality | 0..1
Type | string
Comments | Should focus on information not covered elsewhere as discrete data - no need to duplicate the narrative.
DetectedIssue.reference
Element Id | DetectedIssue.reference
Definition | The literature, knowledge-base or similar reference that describes the propensity for the detected issue identified.
Cardinality | 0..1
Type | uri
DetectedIssue.mitigation
Element Id | DetectedIssue.mitigation
Definition | Indicates an action that has been taken or is committed to reduce or eliminate the likelihood of the risk identified by the detected issue from manifesting. Can also reflect an observation of known mitigating factors that may reduce/eliminate the need for any action.
Cardinality | 0..*
DetectedIssue.mitigation.action
Element Id | DetectedIssue.mitigation.action
Definition | Describes the action that was taken or the observation that was made that reduces/eliminates the risk associated with the identified issue.
Cardinality | 1..1
Terminology Binding | Detected Issue Mitigation Action (Preferred)
Type | CodeableConcept
Comments | The "text" component can be used for detail or when no appropriate code exists.
DetectedIssue.mitigation.date
Element Id | DetectedIssue.mitigation.date
Definition | Indicates when the mitigating action was documented.
Cardinality | 0..1
Type | dateTime
Comments | This might not be the same as when the mitigating step was actually taken.
DetectedIssue.mitigation.author
Element Id | DetectedIssue.mitigation.author
Definition | Identifies the practitioner who determined the mitigation and takes responsibility for the mitigation step occurring.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole)

9.10.11 Resource DetectedIssue - Mappings

URL: https://hl7.org/fhir/R4/detectedissue-mappings.html

9.10.11 Resource DetectedIssue - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the detectedissue resource (see Mappings to Other Standards for further information & status).

9.10.11.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

DetectedIssue
identifier | FiveWs.identifier
status | FiveWs.status
code | FiveWs.class
severity | FiveWs.grade
patient | FiveWs.subject[x]
identified[x] | FiveWs.recorded
author | FiveWs.author
evidence | FiveWs.why[x]
code | FiveWs.why[x]

detail | FiveWs.why[x]

9.10.11.2 Workflow Pattern (http://hl7.org/fhir/workflow)

DetectedIssue | Event
identifier | Event.identifier
status | Event.status
code | Event.code
patient | Event.subject
identified[x] | Event.occurrence[x]
author | Event.performer.actor
implicated | Event.reasonReference
detail | Event.note

9.10.11.3 HL7 v2 Mapping (http://hl7.org/v2)

DetectedIssue | Partial mapping from messages containing the AL1 or IAM segment, such as the ADT^A01, ADT^A08, and ADT^A60.
identifier | IAM-7
status
code | AL1-2 / IAM-9
severity | AL1-4/ IAR-2
patient | PID-3
identified[x] | ALI-6 / IAM-11
author | IAM-14 / IAM-16
implicated
evidence
code
detail
detail | AL1-5 / IAM-8
reference
mitigation
action
date
author

9.10.11.4 RIM Mapping (http://hl7.org/v3)

DetectedIssue | Observation[classCode=ALRT, moodCode=EVN)
identifier | .id
code | .code
severity | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code=SEV].value
patient | .participation[typeCode=SBJ].role[classCode=PAT]
identified[x] | .effectiveTime
author | .participation[typeCode=AUT].role[classCode=ASSIGNED]
implicated | .outboundRelationship[typeCode=SUBJ].target[moodCode=EVN, INT]
detail | .text
reference | .outboundRelationship[typeCode=INST].target[classCode=ALRT, moodCode=DEFN].text.uri
mitigation | .outboundRelationship[typeCode=MITG].target[classCode=ACT, moodCode=EVN]
action | .code
date | .effectiveTime
author | .participation[typeCode=AUT].role[classCode=ASSIGNED]

9.10.12 Resource DetectedIssue - Extensions & Profiles

URL: https://hl7.org/fhir/R4/detectedissue-profiles.html

9.10.12 Resource DetectedIssue - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the DetectedIssue resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.10.12.1 Profiles

No Profiles defined for this resource

9.10.12.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

operationoutcome-detectedIssue | detectedIssue | for Standard OperationOutcome Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.10.12.3 Search Extensions

No Search Extensions defined for this resource

9.10.13 Resource DetectedIssue - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/detectedissue-version-maps.html

No validation errors - all conversions are clean

9.10.13 Resource DetectedIssue - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

R3 : R4 Conversion maps for DetectedIssue.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

9.10.13.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/DetectedIssue3to4" = "R3 to R4 Conversions for DetectedIssue"

uses "http://hl7.org/fhir/3.0/StructureDefinition/DetectedIssue" alias DetectedIssueR3 as source
uses "http://hl7.org/fhir/StructureDefinition/DetectedIssue" alias DetectedIssue as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group DetectedIssue(source src : DetectedIssueR3, target tgt : DetectedIssue) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.category -> tgt.code;
  src.severity -> tgt.severity;
  src.patient -> tgt.patient;
  src.date as vs -> tgt.identified = create('dateTime') as vt then dateTime(vs, vt);
  src.author -> tgt.author;
  src.implicated -> tgt.implicated;
  src.detail -> tgt.detail;
  src.reference -> tgt.reference;
  src.mitigation as vs0 -> tgt.mitigation as vt0 then mitigation(vs0, vt0);
}

group mitigation(source src, target tgt) extends BackboneElement {
  src.action -> tgt.action;
  src.date -> tgt.date;
  src.author -> tgt.author;
}
```

9.10.13.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/DetectedIssue4to3" = "R4 to R3 Conversion for DetectedIssue"

uses "http://hl7.org/fhir/StructureDefinition/DetectedIssue" alias DetectedIssue as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/DetectedIssue" alias DetectedIssueR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group DetectedIssue(source src : DetectedIssue, target tgt : DetectedIssueR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.code -> tgt.category;
  src.severity -> tgt.severity;
  src.patient -> tgt.patient;
  src.identified : dateTime as vs -> tgt.date as vt then dateTime(vs, vt);
  src.author -> tgt.author;
  src.implicated -> tgt.implicated;
  src.detail -> tgt.detail;
  src.reference -> tgt.reference;
  src.mitigation as vs0 -> tgt.mitigation as vt0 then mitigation(vs0, vt0);
}

group mitigation(source src, target tgt) extends BackboneElement {
  src.action -> tgt.action;
  src.date -> tgt.date;
  src.author -> tgt.author;
```

}

9.10.13.3 R4 Validation Errors

9.10.14 JSON Schema for DetectedIssue

URL: https://hl7.org/fhir/R4/detectedissue.schema.json.html

9.10.14 JSON Schema for DetectedIssue

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for DetectedIssue, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/DetectedIssue",
  "$ref": "#/definitions/DetectedIssue",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "DetectedIssue": {
      "description": "Indicates an actual or potential clinical issue with or between one or more active or proposed
    clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition
    conflict, etc.",
      "properties": {
        "resourceType": {
          "description": "This is a DetectedIssue resource",
          "const": "DetectedIssue"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
```

```
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifier associated with the detected issue record.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "Indicates the status of the detected issue.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Identifies the general type of issue identified.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "severity": {
      "description": "Indicates the degree of importance associated with the identified issue based on the potential
 impact on the patient.",
      "enum": [
        "high",
        "moderate",
        "low"
      ]
    },
    "_severity": {
      "description": "Extensions for severity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "patient": {
      "description": "Indicates the patient whose record the detected issue is associated with.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "identifiedDateTime": {
      "description": "The date or period when the detected issue was initially identified.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_identifiedDateTime": {
      "description": "Extensions for identifiedDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifiedPeriod": {
      "description": "The date or period when the detected issue was initially identified.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "author": {
      "description": "Individual or device responsible for the issue being raised.  For example, a decision support
application or a pharmacist conducting a medication review.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "implicated": {
      "description": "Indicates the resource representing the current activity or proposed activity that is
potentially problematic.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "evidence": {
      "description": "Supporting evidence or manifestations that provide the basis for identifying the detected
issue such as a GuidanceResponse or MeasureReport.",
      "items": {
        "$ref": "#/definitions/DetectedIssue_Evidence"
      },
      "type": "array"
    },
    "detail": {
      "description": "A textual explanation of the detected issue.",
      "$ref": "#/definitions/string"
    },
    "_detail": {
      "description": "Extensions for detail",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reference": {
```

```
      "description": "The literature, knowledge-base or similar reference that describes the propensity for the detected issue identified.",
      "$ref": "#/definitions/uri"
    },
    "_reference": {
      "description": "Extensions for reference",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "mitigation": {
      "description": "Indicates an action that has been taken or is committed to reduce or eliminate the likelihood of the risk identified by the detected issue from manifesting.  Can also reflect an observation of known mitigating factors that may reduce/eliminate the need for any action.",
      "items": {
        "$ref": "#/definitions/DetectedIssue_Mitigation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"DetectedIssue_Evidence": {
  "description": "Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A manifestation that led to the recording of this detected issue.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "detail": {
      "description": "Links to resources that constitute evidence for the detected issue such as a GuidanceResponse or MeasureReport.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"DetectedIssue_Mitigation": {
  "description": "Indicates an actual or potential clinical issue with or between one or more active or proposed clinical actions for a patient; e.g. Drug-drug interaction, Ineffective treatment frequency, Procedure-condition conflict, etc.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "action": {
          "description": "Describes the action that was taken or the observation that was made that reduces/eliminates
    the risk associated with the identified issue.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "date": {
          "description": "Indicates when the mitigating action was documented.",
          "$ref": "#/definitions/dateTime"
        },
        "_date": {
          "description": "Extensions for date",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "author": {
          "description": "Identifies the practitioner who determined the mitigation and takes responsibility for the
    mitigation step occurring.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "action"
      ]
    }
  }
}
```


9.10.15 Resource DetectedIssue - Maturity Dependency View

URL: https://hl7.org/fhir/R4/detectedissue-dependencies.html

9.10.15 Resource DetectedIssue - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for DetectedIssue FMM level 1

Name | Card. | Type | Dependency Analysis
DetectedIssue | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
code | 0..1 | CodeableConcept
severity | 0..1 | code
patient | 0..1 | Reference(Patient)
identified[x] | 0..1
identifiedDateTime | | dateTime
identifiedPeriod | | Period
author | 0..1 | Reference(Practitioner | PractitionerRole | Device)
implicated | 0..* | Reference(Any)
evidence | 0..* | BackboneElement
code | 0..* | CodeableConcept
detail | 0..* | Reference(Any)
detail | 0..1 | string
reference | 0..1 | uri
mitigation | 0..* | BackboneElement
action | 1..1 | CodeableConcept
date | 0..1 | dateTime
author | 0..1 | Reference(Practitioner | PractitionerRole)

9.11 Resource RiskAssessment - Content

URL: https://hl7.org/fhir/R4/riskassessment.html

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of each outcome.

9.11.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow, specifically Event.

This resource captures predicted outcomes for a patient or population on the basis of source information. Examples include:

• A prognosis statement for a particular condition

• Risk of health outcome (heart attack, particular type of cancer) on the basis of lifestyle factors and/or family history

• List of potential health risks based on a patient's genetic analysis

• A prediction of outbreak infection rates within a geography based on immunization rates

This resource can be used to represent the results of formal scoring/decision support tools that evaluate risk. It can also be used to capture a practitioner's subjective assessment of the patient's risk based on existing knowledge and previous experience.

9.11.2 Boundaries and Relationships

Risk assessments are a specialized type of observation. A specialized resource is used to provide a simpler mechanism to capture of a series of risks and to associate those risks with time-ranges, probabilities, etc. All risk assertions are captured at one time based on a single set of source inputs. Capture of a single risk MAY be done using the Observation class, particularly in circumstances where it's treated as a generic observation, but for consistency, all risk assessments and prognosis SHOULD be captured using RiskAssessment.

Risk assessments may be based on a variety of factors, including:

• Basic demographic information from the Patient or Group resources

• Various Observations including vital signs, lab information, assessments, genetic information, etc.

• FamilyMemberHistory

• Current, past and proposed therapies Immunization, Procedure, CarePlan, etc.

Because so many resources can potentially be used, no limit is placed on what resources can be sent in RiskAssessment.basis. However, some resources would be nonsensical for this use (e.g. MessageHeader).

AllergyIntolerance and RiskAssessment

AllergyIntolerance describes a specific type of risk - propensity to reaction to a substance while RiskAssessment describes general risks to a subject, not generally based on a reaction.

9.11.3 Background and Context

Risk Assessments can be determined manually by health-care providers based on their professional expertise, by using an algorithmic scoring system or through a combination. Risk assessments can also be computed by decision support systems. OperationDefinitions may be defined by servers to generate risk assessments based on particular combinations of source resources or other inputs.

This resource is referenced by ClinicalImpression and Goal

9.11.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
RiskAssessment | TU | | DomainResource | Potential outcomes for a subject with likelihood Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Unique identifier for the assessment
basedOn | | 0..1 | Reference(Any) | Request fulfilled by this assessment
parent | | 0..1 | Reference(Any) | Part of this occurrence
status | Σ | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
method | Σ | 0..1 | CodeableConcept | Evaluation mechanism
code | Σ | 0..1 | CodeableConcept | Type of assessment

subject | Σ | 1..1 | Reference(Patient | Group) | Who/what does assessment apply to?
encounter | Σ | 0..1 | Reference(Encounter) | Where was assessment performed?
occurrence[x] | Σ | 0..1 | | When was assessment made?
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
condition | Σ | 0..1 | Reference(Condition) | Condition assessed
performer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Device) | Who did assessment?
reasonCode | | 0..* | CodeableConcept | Why the assessment was necessary?
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the assessment was necessary?
basis | | 0..* | Reference(Any) | Information used in assessment
prediction | I | 0..* | BackboneElement | Outcome predicted + Rule: Must be <= 100
outcome | | 0..1 | CodeableConcept | Possible outcome for the subject
probability[x] | I | 0..1 | | Likelihood of specified outcome + Rule: low and high must be percentages, if present
probabilityDecimal | | | decimal
probabilityRange | | | Range
qualitativeRisk | | 0..1 | CodeableConcept | Likelihood of specified outcome as a qualitative value Risk Probability (Example)
relativeRisk | | 0..1 | decimal | Relative likelihood
when[x] | | 0..1 | | Timeframe or age range
whenPeriod | | | Period
whenRange | | | Range
rationale | | 0..1 | string | Explanation of prediction
mitigation | | 0..1 | string | How to reduce risk
note | | 0..* | Annotation | Comments on the risk assessment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskAssessment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier for the assessment --></identifier>
 <basedOn><!-- 0..1 Reference(Any) Request fulfilled by this assessment --></basedOn>
 <parent><!-- 0..1 Reference(Any) Part of this occurrence --></parent>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <method><!-- 0..1 CodeableConcept Evaluation mechanism --></method>
 <code><!-- 0..1 CodeableConcept Type of assessment --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who/what does assessment apply to? --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Where was assessment performed? --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period When was assessment made? --></occurrence[x]>
 <condition><!-- 0..1 Reference(Condition) Condition assessed --></condition>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Device) Who did assessment? --></performer>
 <reasonCode><!-- 0..* CodeableConcept Why the assessment was necessary? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why the assessment was necessary? --></reasonReference>
 <basis><!-- 0..* Reference(Any) Information used in assessment --></basis>
 <prediction>  <!-- 0..* Outcome predicted -->
  <outcome><!-- 0..1 CodeableConcept Possible outcome for the subject --></outcome>
  <probability[x]><!-- 0..1 decimal|Range Likelihood of specified outcome --></probability[x]>
  <qualitativeRisk><!-- 0..1 CodeableConcept Likelihood of specified outcome as a qualitative value
    --></qualitativeRisk>
  <relativeRisk value="[decimal]"/><!-- 0..1 Relative likelihood -->
  <when[x]><!-- 0..1 Period|Range Timeframe or age range --></when[x]>
  <rationale value="[string]"/><!-- 0..1 Explanation of prediction -->
 </prediction>
 <mitigation value="[string]"/><!-- 0..1 How to reduce risk -->
 <note><!-- 0..* Annotation Comments on the risk assessment --></note>
</RiskAssessment>
```

JSON Template

```
{
  "resourceType" : "RiskAssessment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier for the assessment
  "basedOn" : { Reference(Any) }, // Request fulfilled by this assessment
  "parent" : { Reference(Any) }, // Part of this occurrence
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "method" : { CodeableConcept }, // Evaluation mechanism
  "code" : { CodeableConcept }, // Type of assessment
  "subject" : { Reference(Patient|Group) }, // R!  Who/what does assessment apply to?
  "encounter" : { Reference(Encounter) }, // Where was assessment performed?
  // occurrence[x]: When was assessment made?. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "condition" : { Reference(Condition) }, // Condition assessed
  "performer" : { Reference(Practitioner|PractitionerRole|Device) }, // Who did assessment?
  "reasonCode" : [{ CodeableConcept }], // Why the assessment was necessary?
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why the assessment was necessary?
  "basis" : [{ Reference(Any) }], // Information used in assessment
  "prediction" : [{ // Outcome predicted
```

```
    "outcome" : { CodeableConcept }, // Possible outcome for the subject
    // probability[x]: Likelihood of specified outcome. One of these 2:
    "probabilityDecimal" : <decimal>,
    "probabilityRange" : { Range },
    "qualitativeRisk" : { CodeableConcept }, // Likelihood of specified outcome as a qualitative value
    "relativeRisk" : <decimal>, // Relative likelihood
    // when[x]: Timeframe or age range. One of these 2:
    "whenPeriod" : { Period },
    "whenRange" : { Range },
    "rationale" : "<string>" // Explanation of prediction
  }],
  "mitigation" : "<string>", // How to reduce risk
  "note" : [{ Annotation }] // Comments on the risk assessment
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskAssessment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskAssessment.identifier [ Identifier ], ... ; # 0..* Unique identifier for the assessment
  fhir:RiskAssessment.basedOn [ Reference(Any) ]; # 0..1 Request fulfilled by this assessment
  fhir:RiskAssessment.parent [ Reference(Any) ]; # 0..1 Part of this occurrence
  fhir:RiskAssessment.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:RiskAssessment.method [ CodeableConcept ]; # 0..1 Evaluation mechanism
  fhir:RiskAssessment.code [ CodeableConcept ]; # 0..1 Type of assessment
  fhir:RiskAssessment.subject [ Reference(Patient|Group) ]; # 1..1 Who/what does assessment apply to?
  fhir:RiskAssessment.encounter [ Reference(Encounter) ]; # 0..1 Where was assessment performed?
  # RiskAssessment.occurrence[x] : 0..1 When was assessment made?. One of these 2
    fhir:RiskAssessment.occurrenceDateTime [ dateTime ]
    fhir:RiskAssessment.occurrencePeriod [ Period ]
  fhir:RiskAssessment.condition [ Reference(Condition) ]; # 0..1 Condition assessed
  fhir:RiskAssessment.performer [ Reference(Practitioner|PractitionerRole|Device) ]; # 0..1 Who did assessment?
  fhir:RiskAssessment.reasonCode [ CodeableConcept ], ... ; # 0..* Why the assessment was necessary?
  fhir:RiskAssessment.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why the assessment was necessary?
  fhir:RiskAssessment.basis [ Reference(Any) ], ... ; # 0..* Information used in assessment
  fhir:RiskAssessment.prediction [ # 0..* Outcome predicted
    fhir:RiskAssessment.prediction.outcome [ CodeableConcept ]; # 0..1 Possible outcome for the subject
    # RiskAssessment.prediction.probability[x] : 0..1 Likelihood of specified outcome. One of these 2
      fhir:RiskAssessment.prediction.probabilityDecimal [ decimal ]
      fhir:RiskAssessment.prediction.probabilityRange [ Range ]
    fhir:RiskAssessment.prediction.qualitativeRisk [ CodeableConcept ]; # 0..1 Likelihood of specified outcome as a
    qualitative value
    fhir:RiskAssessment.prediction.relativeRisk [ decimal ]; # 0..1 Relative likelihood
    # RiskAssessment.prediction.when[x] : 0..1 Timeframe or age range. One of these 2
      fhir:RiskAssessment.prediction.whenPeriod [ Period ]
      fhir:RiskAssessment.prediction.whenRange [ Range ]
    fhir:RiskAssessment.prediction.rationale [ string ]; # 0..1 Explanation of prediction
  ], ...;
  fhir:RiskAssessment.mitigation [ string ]; # 0..1 How to reduce risk
  fhir:RiskAssessment.note [ Annotation ], ... ; # 0..* Comments on the risk assessment
]
```

Changes since R3

RiskAssessment

RiskAssessment.identifier | Max Cardinality changed from 1 to *

RiskAssessment.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1

RiskAssessment.subject | Min Cardinality changed from 0 to 1

RiskAssessment.encounter | Added Element

RiskAssessment.performer | Type Reference: Added Target Type PractitionerRole

RiskAssessment.reasonCode | Added Element

RiskAssessment.reasonReference | Added Element

RiskAssessment.prediction.outcome | Min Cardinality changed from 1 to 0

RiskAssessment.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation

RiskAssessment.context | deleted

RiskAssessment.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

RiskAssessment | TU | | DomainResource | Potential outcomes for a subject with likelihood Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Unique identifier for the assessment
basedOn | | 0..1 | Reference(Any) | Request fulfilled by this assessment
parent | | 0..1 | Reference(Any) | Part of this occurrence
status | Σ | 1..1 | code | registered | preliminary | final | amended + ObservationStatus (Required)
method | Σ | 0..1 | CodeableConcept | Evaluation mechanism
code | Σ | 0..1 | CodeableConcept | Type of assessment
subject | Σ | 1..1 | Reference(Patient | Group) | Who/what does assessment apply to?
encounter | Σ | 0..1 | Reference(Encounter) | Where was assessment performed?
occurrence[x] | Σ | 0..1 | | When was assessment made?
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
condition | Σ | 0..1 | Reference(Condition) | Condition assessed
performer | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Device) | Who did assessment?
reasonCode | | 0..* | CodeableConcept | Why the assessment was necessary?
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the assessment was necessary?
basis | | 0..* | Reference(Any) | Information used in assessment
prediction | I | 0..* | BackboneElement | Outcome predicted + Rule: Must be <= 100
outcome | | 0..1 | CodeableConcept | Possible outcome for the subject
probability[x] | I | 0..1 | | Likelihood of specified outcome + Rule: low and high must be percentages, if present
probabilityDecimal | | | decimal
probabilityRange | | | Range
qualitativeRisk | | 0..1 | CodeableConcept | Likelihood of specified outcome as a qualitative value Risk Probability (Example)
relativeRisk | | 0..1 | decimal | Relative likelihood
when[x] | | 0..1 | | Timeframe or age range
whenPeriod | | | Period
whenRange | | | Range
rationale | | 0..1 | string | Explanation of prediction
mitigation | | 0..1 | string | How to reduce risk
note | | 0..* | Annotation | Comments on the risk assessment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskAssessment xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier for the assessment --></identifier>
 <basedOn><!-- 0..1 Reference(Any) Request fulfilled by this assessment --></basedOn>
 <parent><!-- 0..1 Reference(Any) Part of this occurrence --></parent>
 <status value="[code]"/><!-- 1..1 registered | preliminary | final | amended + -->
 <method><!-- 0..1 CodeableConcept Evaluation mechanism --></method>
 <code><!-- 0..1 CodeableConcept Type of assessment --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Who/what does assessment apply to? --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Where was assessment performed? --></encounter>
 <occurrence[x]><!-- 0..1 dateTime|Period When was assessment made? --></occurrence[x]>
 <condition><!-- 0..1 Reference(Condition) Condition assessed --></condition>
 <performer><!-- 0..1 Reference(Practitioner|PractitionerRole|Device) Who did assessment? --></performer>
 <reasonCode><!-- 0..* CodeableConcept Why the assessment was necessary? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why the assessment was necessary? --></reasonReference>
 <basis><!-- 0..* Reference(Any) Information used in assessment --></basis>
 <prediction>  <!-- 0..* Outcome predicted -->
  <outcome><!-- 0..1 CodeableConcept Possible outcome for the subject --></outcome>
  <probability[x]><!-- 0..1 decimal|Range Likelihood of specified outcome --></probability[x]>
  <qualitativeRisk><!-- 0..1 CodeableConcept Likelihood of specified outcome as a qualitative value
    --></qualitativeRisk>
  <relativeRisk value="[decimal]"/><!-- 0..1 Relative likelihood -->
  <when[x]><!-- 0..1 Period|Range Timeframe or age range --></when[x]>
  <rationale value="[string]"/><!-- 0..1 Explanation of prediction -->
 </prediction>
 <mitigation value="[string]"/><!-- 0..1 How to reduce risk -->
 <note><!-- 0..* Annotation Comments on the risk assessment --></note>
</RiskAssessment>
```

JSON Template

```
{
  "resourceType" : "RiskAssessment",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier for the assessment
  "basedOn" : { Reference(Any) }, // Request fulfilled by this assessment
  "parent" : { Reference(Any) }, // Part of this occurrence
  "status" : "<code>", // R!  registered | preliminary | final | amended +
  "method" : { CodeableConcept }, // Evaluation mechanism
  "code" : { CodeableConcept }, // Type of assessment
  "subject" : { Reference(Patient|Group) }, // R!  Who/what does assessment apply to?
  "encounter" : { Reference(Encounter) }, // Where was assessment performed?
  // occurrence[x]: When was assessment made?. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
```

```
  "condition" : { Reference(Condition) }, // Condition assessed
  "performer" : { Reference(Practitioner|PractitionerRole|Device) }, // Who did assessment?
  "reasonCode" : [{ CodeableConcept }], // Why the assessment was necessary?
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why the assessment was necessary?
  "basis" : [{ Reference(Any) }], // Information used in assessment
  "prediction" : [{ // Outcome predicted
    "outcome" : { CodeableConcept }, // Possible outcome for the subject
    // probability[x]: Likelihood of specified outcome. One of these 2:
    "probabilityDecimal" : <decimal>,
    "probabilityRange" : { Range },
    "qualitativeRisk" : { CodeableConcept }, // Likelihood of specified outcome as a qualitative value
    "relativeRisk" : <decimal>, // Relative likelihood
    // when[x]: Timeframe or age range. One of these 2:
    "whenPeriod" : { Period },
    "whenRange" : { Range },
    "rationale" : "<string>" // Explanation of prediction
  }],
  "mitigation" : "<string>", // How to reduce risk
  "note" : [{ Annotation }] // Comments on the risk assessment
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskAssessment;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskAssessment.identifier [ Identifier ], ... ; # 0..* Unique identifier for the assessment
  fhir:RiskAssessment.basedOn [ Reference(Any) ]; # 0..1 Request fulfilled by this assessment
  fhir:RiskAssessment.parent [ Reference(Any) ]; # 0..1 Part of this occurrence
  fhir:RiskAssessment.status [ code ]; # 1..1 registered | preliminary | final | amended +
  fhir:RiskAssessment.method [ CodeableConcept ]; # 0..1 Evaluation mechanism
  fhir:RiskAssessment.code [ CodeableConcept ]; # 0..1 Type of assessment
  fhir:RiskAssessment.subject [ Reference(Patient|Group) ]; # 1..1 Who/what does assessment apply to?
  fhir:RiskAssessment.encounter [ Reference(Encounter) ]; # 0..1 Where was assessment performed?
  # RiskAssessment.occurrence[x] : 0..1 When was assessment made?. One of these 2
    fhir:RiskAssessment.occurrenceDateTime [ dateTime ]
    fhir:RiskAssessment.occurrencePeriod [ Period ]
  fhir:RiskAssessment.condition [ Reference(Condition) ]; # 0..1 Condition assessed
  fhir:RiskAssessment.performer [ Reference(Practitioner|PractitionerRole|Device) ]; # 0..1 Who did assessment?
  fhir:RiskAssessment.reasonCode [ CodeableConcept ], ... ; # 0..* Why the assessment was necessary?
  fhir:RiskAssessment.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why the assessment was necessary?
  fhir:RiskAssessment.basis [ Reference(Any) ], ... ; # 0..* Information used in assessment
  fhir:RiskAssessment.prediction [ # 0..* Outcome predicted
    fhir:RiskAssessment.prediction.outcome [ CodeableConcept ]; # 0..1 Possible outcome for the subject
    # RiskAssessment.prediction.probability[x] : 0..1 Likelihood of specified outcome. One of these 2
      fhir:RiskAssessment.prediction.probabilityDecimal [ decimal ]
      fhir:RiskAssessment.prediction.probabilityRange [ Range ]
    fhir:RiskAssessment.prediction.qualitativeRisk [ CodeableConcept ]; # 0..1 Likelihood of specified outcome as a
    qualitative value
    fhir:RiskAssessment.prediction.relativeRisk [ decimal ]; # 0..1 Relative likelihood
    # RiskAssessment.prediction.when[x] : 0..1 Timeframe or age range. One of these 2
      fhir:RiskAssessment.prediction.whenPeriod [ Period ]
      fhir:RiskAssessment.prediction.whenRange [ Range ]
    fhir:RiskAssessment.prediction.rationale [ string ]; # 0..1 Explanation of prediction
  ], ...;
  fhir:RiskAssessment.mitigation [ string ]; # 0..1 How to reduce risk
  fhir:RiskAssessment.note [ Annotation ], ... ; # 0..* Comments on the risk assessment
]
```

Changes since Release 3

RiskAssessment
RiskAssessment.identifier | Max Cardinality changed from 1 to *
RiskAssessment.status | Change value set from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/observation-status|4.0.1
RiskAssessment.subject | Min Cardinality changed from 0 to 1
RiskAssessment.encounter | Added Element
RiskAssessment.performer | Type Reference: Added Target Type PractitionerRole
RiskAssessment.reasonCode | Added Element
RiskAssessment.reasonReference | Added Element
RiskAssessment.prediction.outcome | Min Cardinality changed from 1 to 0
RiskAssessment.note | Renamed from comment to note Max Cardinality changed from 1 to * Type changed from string to Annotation
RiskAssessment.context | deleted
RiskAssessment.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

9.11.4.1 Terminology Bindings

Path | Definition | Type | Reference
RiskAssessment.status | The status of the risk assessment; e.g. preliminary, final, amended, etc. | Required | ObservationStatus
RiskAssessment.method | The mechanism or algorithm used to make the assessment; e.g. TIMI, PRISM, Cardiff Type 2 diabetes, etc. | Unknown | No details provided yet
RiskAssessment.prediction.outcome | The condition or other outcome; e.g. death, remission, amputation, infection, etc. | Unknown | No details provided yet
RiskAssessment.prediction.qualitativeRisk | The likelihood of the occurrence of a specified outcome. | Example | RiskProbability

9.11.4.2 Constraints

id | Level | Location | Description | Expression
ras-1 | Rule | RiskAssessment.prediction.probability[x] | low and high must be percentages, if present | (low.empty() or ((low.code = '%') and (low.system = %ucum))) and (high.empty() or ((high.code = '%') and (high.system = %ucum)))
ras-2 | Rule | RiskAssessment.prediction | Must be <= 100 | probability is decimal implies (probability as decimal) <= 100

9.11.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
condition | reference | Condition assessed | RiskAssessment.condition (Condition)
date | date | When was assessment made? | (RiskAssessment.occurrence as dateTime) | 17 Resources
encounter | reference | Where was assessment performed? | RiskAssessment.encounter (Encounter) | 12 Resources
identifier | token | Unique identifier for the assessment | RiskAssessment.identifier | 30 Resources
method | token | Evaluation mechanism | RiskAssessment.method
patient | reference | Who/what does assessment apply to? | RiskAssessment.subject.where(resolve() is Patient) (Patient) | 33 Resources
performer | reference | Who did assessment? | RiskAssessment.performer (Practitioner, Device, PractitionerRole)
probability | number | Likelihood of specified outcome | RiskAssessment.prediction.probability
risk | token | Likelihood of specified outcome as a qualitative value | RiskAssessment.prediction.qualitativeRisk
subject | reference | Who/what does assessment apply to? | RiskAssessment.subject (Group, Patient)

9.11.6 Resource RiskAssessment - Examples

URL: https://hl7.org/fhir/R4/riskassessment-examples.html

9.11.6 Resource RiskAssessment - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Example Name | id | Format
Genetic risk assessment | genetic | XML | JSON | Turtle
Patient prognosis | prognosis | XML | JSON | Turtle
Cardiac risk assessment | cardiac | XML | JSON | Turtle
Public health population risk assessment | population | XML | JSON | Turtle
Breast cancer risk assessment | breastcancer-risk | XML | JSON | Turtle
Example risk assessment | riskexample | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

9.11.7 Resource RiskAssessment - Detailed Descriptions

URL: https://hl7.org/fhir/R4/riskassessment-definitions.html

9.11.7 Resource RiskAssessment - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Security Category: Patient | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the RiskAssessment resource.

RiskAssessment
Element Id | RiskAssessment
Definition | An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of each outcome.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Prognosis
RiskAssessment.identifier

Element Id | RiskAssessment.identifier
Definition | Business identifier assigned to the risk assessment.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
RiskAssessment.basedOn
Element Id | RiskAssessment.basedOn
Definition | A reference to the request that is fulfilled by this risk assessment.
Cardinality | 0..1
Type | Reference(Any)
RiskAssessment.parent
Element Id | RiskAssessment.parent
Definition | A reference to a resource that this risk assessment is part of, such as a Procedure.
Cardinality | 0..1
Type | Reference(Any)
RiskAssessment.status
Element Id | RiskAssessment.status
Definition | The status of the RiskAssessment, using the same statuses as an Observation.
Cardinality | 1..1
Terminology Binding | ObservationStatus (Required)
Type | code
Summary | true
RiskAssessment.method
Element Id | RiskAssessment.method
Definition | The algorithm, process or mechanism used to evaluate the risk.
Cardinality | 0..1
Terminology Binding | RiskAssessmentMethod:
Type | CodeableConcept
Requirements | The method can influence the results of the assessment.
Summary | true
RiskAssessment.code
Element Id | RiskAssessment.code
Definition | The type of the risk assessment performed.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
RiskAssessment.subject
Element Id | RiskAssessment.subject
Definition | The patient or group the risk assessment applies to.
Cardinality | 1..1
Type | Reference(Patient | Group)
Summary | true
RiskAssessment.encounter
Element Id | RiskAssessment.encounter
Definition | The encounter where the assessment was performed.
Cardinality | 0..1
Type | Reference(Encounter)
Summary | true
RiskAssessment.occurrence[x]
Element Id | RiskAssessment.occurrence[x]
Definition | The date (and possibly time) the risk assessment was performed.
Cardinality | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The assessment results lose validity the more time elapses from when they are first made.
Summary | true
RiskAssessment.condition
Element Id | RiskAssessment.condition
Definition | For assessments or prognosis specific to a particular condition, indicates the condition being assessed.
Cardinality | 0..1
Type | Reference(Condition)
Summary | true
RiskAssessment.performer
Element Id | RiskAssessment.performer
Definition | The provider or software application that performed the assessment.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Device)
Summary | true
RiskAssessment.reasonCode
Element Id | RiskAssessment.reasonCode
Definition | The reason the risk assessment was performed.

Cardinality | 0..*
Type | CodeableConcept
RiskAssessment.reasonReference
Element Id | RiskAssessment.reasonReference
Definition | Resources supporting the reason the risk assessment was performed.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
RiskAssessment.basis
Element Id | RiskAssessment.basis
Definition | Indicates the source data considered as part of the assessment (for example, FamilyHistory, Observations, Procedures, Conditions, etc.).
Cardinality | 0..*
Type | Reference(Any)
RiskAssessment.prediction
Element Id | RiskAssessment.prediction
Definition | Describes the expected outcome for the subject.
Cardinality | 0..*
Comments | Multiple repetitions can be used to identify the same type of outcome in different timeframes as well as different types of outcomes.
Invariants | Defined on this element ras-2 Rule Must be <= 100 probability is decimal implies (probability as decimal) <= 100
Defined on this element
ras-2 | Rule | Must be <= 100 | probability is decimal implies (probability as decimal) <= 100
RiskAssessment.prediction.outcome
Element Id | RiskAssessment.prediction.outcome
Definition | One of the potential outcomes for the patient (e.g. remission, death, a particular condition).
Cardinality | 0..1
Terminology Binding | RiskAssessmentOutcome:
Type | CodeableConcept
RiskAssessment.prediction.probability[x]
Element Id | RiskAssessment.prediction.probability[x]
Definition | Indicates how likely the outcome is (in the specified timeframe).
Cardinality | 0..1
Type | decimal|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If range is used, it represents the lower and upper bounds of certainty; e.g. 40-60% Decimal values are expressed as percentages as well (max = 100).
Invariants | Defined on this element ras-1 Rule low and high must be percentages, if present (low.empty() or ((low.code = '%') and (low.system = %ucum))) and (high.empty() or ((high.code = '%') and (high.system = %ucum)))
Defined on this element
ras-1 | Rule | low and high must be percentages, if present | (low.empty() or ((low.code = '%') and (low.system = %ucum))) and (high.empty() or ((high.code = '%') and (high.system = %ucum)))
RiskAssessment.prediction.qualitativeRisk
Element Id | RiskAssessment.prediction.qualitativeRisk
Definition | Indicates how likely the outcome is (in the specified timeframe), expressed as a qualitative value (e.g. low, medium, or high).
Cardinality | 0..1
Terminology Binding | Risk Probability (Example)
Type | CodeableConcept
RiskAssessment.prediction.relativeRisk
Element Id | RiskAssessment.prediction.relativeRisk
Definition | Indicates the risk for this particular subject (with their specific characteristics) divided by the risk of the population in general. (Numbers greater than 1 = higher risk than the population, numbers less than 1 = lower risk.).
Cardinality | 0..1
Type | decimal
Requirements | Absolute risk is less meaningful than relative risk.
RiskAssessment.prediction.when[x]
Element Id | RiskAssessment.prediction.when[x]
Definition | Indicates the period of time or age range of the subject to which the specified probability applies.
Cardinality | 0..1
Type | Period|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | If not specified, the risk applies "over the subject's lifespan".
RiskAssessment.prediction.rationale
Element Id | RiskAssessment.prediction.rationale
Definition | Additional information explaining the basis for the prediction.
Cardinality | 0..1
Type | string
RiskAssessment.mitigation
Element Id | RiskAssessment.mitigation
Definition | A description of the steps that might be taken to reduce the identified risk(s).
Cardinality | 0..1
Type | string
Requirements | One of the main reasons for assessing risks is to identify whether interventional steps are needed to reduce risk.

RiskAssessment.note
Element Id | RiskAssessment.note
Definition | Additional comments about the risk assessment.
Cardinality | 0..*
Type | Annotation

9.11.8 Resource RiskAssessment - Mappings

URL: https://hl7.org/fhir/R4/riskassessment-mappings.html

9.11.8 Resource RiskAssessment - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Open EHR Archetype Mapping

Mappings for the riskassessment resource (see Mappings to Other Standards for further information & status).

9.11.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

RiskAssessment
identifier | FiveWs.identifier
status | FiveWs.status
method | FiveWs.class
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
occurrence[x] | FiveWs.done[x]
performer | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

9.11.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

RiskAssessment | Event
identifier | Event.identifier
basedOn | Event.basedOn
parent | Event.partOf
status | Event.status
code | Event.code
subject | Event.subject
encounter | Event.encounter
occurrence[x] | Event.occurrence[x]
performer | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

9.11.8.3 HL7 v2 Mapping (http://hl7.org/v2)

RiskAssessment | N/A

9.11.8.4 RIM Mapping (http://hl7.org/v3)

RiskAssessment | Observation[classCode=OBS, moodCode=EVN, code="Risk Assessment"]
identifier | .id
method | .methodCode
subject | .participation[typeCode=SBJ].role
encounter | .inboundRelationship[typeCode=COMP].source[classCode=ENC, moodCode=EVN]
occurrence[x] | .effectiveTime
condition | .outboundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN]
performer | .participation[typeCode=PFM].role[classCode=ASSIGNED]
basis | .outboundRelationship[typeCode=DRIV].target
prediction | .outboundRelationship[typeCode=VALUE].target[classCod=OBS, moodCode=RISK]
outcome | .value
probability[x] | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="probability"].value
relativeRisk | .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="relative risk"].value
when[x] | .effectiveTime or .inboundRelationship[typeCode=SUBJ].source[classCode=OBS, moodCode=EVN, code="age"].value
rationale | .reasonCode.text

mitigation | .outboundRelationship[typeCode=MTGT].target[classCode=ACT, moodCode=EVN].text

9.11.8.5 Open EHR Archetype Mapping (http://openehr.org)

RiskAssessment | OpenEHR-EHR-EVALUATION.risk.v1
identifier | Workflow Identifier
basedOn
parent
status
method
code
subject | Subject of Care
encounter
occurrence[x]
condition
performer | Information Provider
reasonCode
reasonReference
basis | Risk Factor; Link to Evidence
prediction
outcome | Risk
probability[x] | Category
qualitativeRisk
relativeRisk
when[x]
rationale | Rationale
mitigation
note

9.11.9 Resource RiskAssessment - Extensions & Profiles

URL: https://hl7.org/fhir/R4/riskassessment-profiles.html

9.11.9 Resource RiskAssessment - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Device, Patient, Practitioner

This table lists profiles and extensions for the RiskAssessment resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

9.11.9.1 Profiles

No Profiles defined for this resource

9.11.9.2 Extensions

workflow-researchStudy | RiskAssessment | researchStudy | for Workflow Pattern HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

diagnosticReport-risk | risk | for DiagnosticReport HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

9.11.9.3 Search Extensions

No Search Extensions defined for this resource

9.11.10 Resource RiskAssessment - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/riskassessment-version-maps.html

No validation errors - all conversions are clean

9.11.10 Resource RiskAssessment - R3/R4 Conversions

R3 : R4 Conversion maps for RiskAssessment.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

9.11.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/RiskAssessment3to4" = "R3 to R4 Conversions for RiskAssessment"

uses "http://hl7.org/fhir/3.0/StructureDefinition/RiskAssessment" alias RiskAssessmentR3 as source
uses "http://hl7.org/fhir/StructureDefinition/RiskAssessment" alias RiskAssessment as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group RiskAssessment(source src : RiskAssessmentR3, target tgt : RiskAssessment) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.parent -> tgt.parent;
  src.status -> tgt.status;
  src.method -> tgt.method;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.condition -> tgt.condition;
  src.performer -> tgt.performer;
  src.reason : CodeableConcept as vs -> tgt.reasonCode as vt then CodeableConcept(vs, vt);
  src.reason : Reference as vs -> tgt.reasonReference as vt then Reference(vs, vt);
  src.basis -> tgt.basis;
  src.prediction as vs0 -> tgt.prediction as vt0 then prediction(vs0, vt0);
  src.mitigation -> tgt.mitigation;
  src.comment as v ->  tgt.note as n,  n.text = v;
}

group prediction(source src, target tgt) extends BackboneElement {
  src.outcome -> tgt.outcome;
  src.probability : decimal as vs -> tgt.probability = create('decimal') as vt then decimal(vs, vt);
  src.probability : Range as vs -> tgt.probability = create('Range') as vt then Range(vs, vt);
  src.qualitativeRisk -> tgt.qualitativeRisk;
  src.relativeRisk -> tgt.relativeRisk;
  src.when : Period as vs -> tgt.when = create('Period') as vt then Period(vs, vt);
  src.when : Range as vs -> tgt.when = create('Range') as vt then Range(vs, vt);
  src.rationale -> tgt.rationale;
}
```

9.11.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/RiskAssessment4to3" = "R4 to R3 Conversion for RiskAssessment"

uses "http://hl7.org/fhir/StructureDefinition/RiskAssessment" alias RiskAssessment as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/RiskAssessment" alias RiskAssessmentR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group RiskAssessment(source src : RiskAssessment, target tgt : RiskAssessmentR3) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.basedOn -> tgt.basedOn;
  src.parent -> tgt.parent;
  src.status -> tgt.status;
  src.method -> tgt.method;
  src.code -> tgt.code;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.occurrence : dateTime as vs -> tgt.occurrence = create('dateTime') as vt then dateTime(vs, vt);
  src.occurrence : Period as vs -> tgt.occurrence = create('Period') as vt then Period(vs, vt);
  src.condition -> tgt.condition;
  src.performer -> tgt.performer;
  src.reasonCode as vs -> tgt.reason = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.reasonReference as vs -> tgt.reason = create('Reference') as vt then Reference(vs, vt);
  src.basis -> tgt.basis;
  src.prediction as vs0 -> tgt.prediction as vt0 then prediction(vs0, vt0);
  src.mitigation -> tgt.mitigation;
  src.note as n then {
    n.text -> tgt.comment;
  };
}

group prediction(source src, target tgt) extends BackboneElement {
  src.outcome -> tgt.outcome;
  src.probability : decimal as vs -> tgt.probability = create('decimal') as vt then decimal(vs, vt);
  src.probability : Range as vs -> tgt.probability = create('Range') as vt then Range(vs, vt);
  src.qualitativeRisk -> tgt.qualitativeRisk;
  src.relativeRisk -> tgt.relativeRisk;
  src.when : Period as vs -> tgt.when = create('Period') as vt then Period(vs, vt);
  src.when : Range as vs -> tgt.when = create('Range') as vt then Range(vs, vt);
  src.rationale -> tgt.rationale;
```

}

9.11.10.3 R4 Validation Errors

9.11.11 JSON Schema for RiskAssessment

URL: https://hl7.org/fhir/R4/riskassessment.schema.json.html

9.11.11 JSON Schema for RiskAssessment

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is a representation of the json schema for RiskAssessment, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/RiskAssessment",
  "$ref": "#/definitions/RiskAssessment",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "RiskAssessment": {
      "description": "An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of
     each outcome.",
      "properties": {
        "resourceType": {
          "description": "This is a RiskAssessment resource",
          "const": "RiskAssessment"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Business identifier assigned to the risk assessment.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A reference to the request that is fulfilled by this risk assessment.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "parent": {
      "description": "A reference to a resource that this risk assessment is part of, such as a Procedure.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "status": {
      "description": "The status of the RiskAssessment, using the same statuses as an Observation.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "method": {
      "description": "The algorithm, process or mechanism used to evaluate the risk.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "code": {
      "description": "The type of the risk assessment performed.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The patient or group the risk assessment applies to.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "The encounter where the assessment was performed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "occurrenceDateTime": {
      "description": "The date (and possibly time) the risk assessment was performed.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_occurrenceDateTime": {
      "description": "Extensions for occurrenceDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "occurrencePeriod": {
      "description": "The date (and possibly time) the risk assessment was performed.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "condition": {
      "description": "For assessments or prognosis specific to a particular condition, indicates the condition being
 assessed.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "performer": {
      "description": "The provider or software application that performed the assessment.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "The reason the risk assessment was performed.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Resources supporting the reason the risk assessment was performed.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "basis": {
      "description": "Indicates the source data considered as part of the assessment (for example, FamilyHistory,
Observations, Procedures, Conditions, etc.).",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
```

```
      },
      "type": "array"
    },
    "prediction": {
      "description": "Describes the expected outcome for the subject.",
      "items": {
        "$ref": "#/definitions/RiskAssessment_Prediction"
      },
      "type": "array"
    },
    "mitigation": {
      "description": "A description of the steps that might be taken to reduce the identified risk(s).",
      "$ref": "#/definitions/string"
    },
    "_mitigation": {
      "description": "Extensions for mitigation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "Additional comments about the risk assessment.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "subject",
    "resourceType"
  ]
},
"RiskAssessment_Prediction": {
  "description": "An assessment of the likely outcome(s) for a patient or other subject as well as the likelihood of
 each outcome.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "outcome": {
      "description": "One of the potential outcomes for the patient (e.g. remission, death,  a particular
condition).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "probabilityDecimal": {
      "description": "Indicates how likely the outcome is (in the specified timeframe).",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_probabilityDecimal": {
      "description": "Extensions for probabilityDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "probabilityRange": {
      "description": "Indicates how likely the outcome is (in the specified timeframe).",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "qualitativeRisk": {
      "description": "Indicates how likely the outcome is (in the specified timeframe), expressed as a qualitative
value (e.g. low, medium, or high).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "relativeRisk": {
```

```
        "description": "Indicates the risk for this particular subject (with their specific characteristics) divided by the risk of the population in general.  (Numbers greater than 1 \u003d higher risk than the population, numbers less than 1 \u003d lower risk.).",
        "$ref": "#/definitions/decimal"
      },
      "_relativeRisk": {
        "description": "Extensions for relativeRisk",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "whenPeriod": {
        "description": "Indicates the period of time or age range of the subject to which the specified probability applies.",
        "$ref": "Period.schema.json#/definitions/Period"
      },
      "whenRange": {
        "description": "Indicates the period of time or age range of the subject to which the specified probability applies.",
        "$ref": "Range.schema.json#/definitions/Range"
      },
      "rationale": {
        "description": "Additional information explaining the basis for the prediction.",
        "$ref": "#/definitions/string"
      },
      "_rationale": {
        "description": "Extensions for rationale",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false
  }
 }
}
```


9.11.12 Resource RiskAssessment - Maturity Dependency View

URL: https://hl7.org/fhir/R4/riskassessment-dependencies.html

9.11.12 Resource RiskAssessment - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Dependency Graph for RiskAssessment FMM level 1

Name | Card. | Type | Dependency Analysis
RiskAssessment | | DomainResource
identifier | 0..* | Identifier
basedOn | 0..1 | Reference(Any)
parent | 0..1 | Reference(Any)
status | 1..1 | code
method | 0..1 | CodeableConcept
code | 0..1 | CodeableConcept
subject | 1..1 | Reference(Patient | Group)
encounter | 0..1 | Reference(Encounter)
occurrence[x] | 0..1
occurrenceDateTime | | dateTime
occurrencePeriod | | Period
condition | 0..1 | Reference(Condition)
performer | 0..1 | Reference(Practitioner | PractitionerRole | Device)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
basis | 0..* | Reference(Any)
prediction | 0..* | BackboneElement
outcome | 0..1 | CodeableConcept
probability[x] | 0..1
probabilityDecimal | | decimal
probabilityRange | | Range
qualitativeRisk | 0..1 | CodeableConcept
relativeRisk | 0..1 | decimal
when[x] | 0..1
whenPeriod | | Period
whenRange | | Range
rationale | 0..1 | string
mitigation | 0..1 | string
note | 0..* | Annotation

9.11.13 Value Set Sc-valueset-observation-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-observation-status.html

9.11.13 Value Set Sc-valueset-observation-status - Status Code Mapping

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Patient, Practitioner

This is the narrative for the resource. See also the XML, JSON or Turtle format.

9.11.14 ObservationStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-observation-status)

Mapping from http://hl7.org/fhir/ValueSet/observation-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "Codes providing the status of an observation."

Source Code	Equivalence	Destination Code
entered-in-error	equivalent	error
preliminary	equivalent	draft
registered	equivalent	received
corrected, amended	equivalent	replaced
final	equivalent	complete
cancelled	equivalent	abandoned
unknown	equivalent	unknown